To my mother.

Nay, if you read this line, remember not
The hand that writ it; for I love you so,
That I in your sweet thoughts would be forgot,
If thinking on me then should make you woe.

William Shakespeare, The Sonnets

Resist much, obey little.

…

If you want me again look for me under your boot-soles.

Walt Whitman, Leaves of Grass

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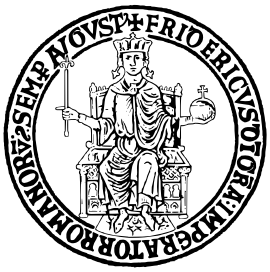

Scuola Politecnica e delle Scienze di Base

Dipartimento di Ingegneria Industriale

Dottorato di ricerca in Ingegneria Aerospaziale

Digital Certification Technologies: An MBSE approach

Relatori
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Candidato
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Anno Accademico 2026–2027

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Abstract

This thesis investigates digital certification technologies for aeronautical products through a Model-Based Systems Engineering (MBSE) approach. The work addresses the limitations of document-based certification by developing digital toolchains that encode airworthiness requirements, substantiation standards, aircraft data, calculation methods, and certification artifacts within structured and traceable workflows. The research focuses primarily on EASA CS-23 and FAA FAR Part 23 aircraft, with emphasis on Subpart C (Structure) and flight-loads substantiation. Three implementations are presented. The first combines an Eclipse Papyrus SysML model with MATLAB/Simulink routines to parse certification information, execute substantiation calculations, and generate an EASA-compliant *Flight Loads* certification report. The second uses CATIA Magic Systems of Systems Architect to demonstrate automation through SysML Activity Diagrams and external C++ routines. The third, developed in collaboration with the Georgia Institute of Technology, extends a larger FAA FAR Part 23 SysML model with ASTM International standards and uses Python routines to extract certification data and generate compliance artifacts. The results show that MBSE can improve traceability, repeatability, and automation in certification workflows. The thesis also discusses practical limitations, including model-development effort, tool integration, organizational adoption, and the need for industrial validation.

Sommario

Questa tesi studia le tecnologie di certificazione digitale per prodotti aeronautici attraverso un approccio basato sulla Model-Based Systems Engineering (MBSE). Il lavoro affronta i limiti della certificazione documentale sviluppando toolchain digitali capaci di codificare requisiti di aeronavigabilità, standard di sostanziazione, dati del velivolo, metodi di calcolo e artefatti di certificazione all'interno di workflow strutturati e tracciabili. La ricerca si concentra principalmente su velivoli certificabili secondo EASA CS-23 e FAA FAR Part 23, con particolare attenzione alla Subpart C (Structure) e alla sostanziazione dei carichi di volo. Sono presentate tre implementazioni. La prima combina un modello SysML sviluppato in Eclipse Papyrus con routine MATLAB/Simulink per estrarre informazioni di certificazione, eseguire calcoli di sostanziazione e generare un rapporto *Flight Loads* conforme alle pratiche EASA. La seconda utilizza CATIA Magic Systems of Systems Architect per dimostrare l'automazione del workflow mediante SysML Activity Diagrams e routine C++ esterne. La terza, sviluppata in collaborazione con il Georgia Institute of Technology, estende un più ampio modello SysML della FAA FAR Part 23 con standard ASTM International e utilizza routine Python per estrarre dati di certificazione e generare artefatti di conformità. I risultati mostrano che l'MBSE può migliorare tracciabilità, ripetibilità e automazione nei processi di certificazione. La tesi discute inoltre alcune limitazioni pratiche, tra cui lo sviluppo del modello, l'integrazione degli strumenti, l'adozione organizzativa e la necessità di validazione industriale.

List of Publications

This thesis is based in part on the publications listed below, which were prepared during the doctoral program. Publications P1 to P5 report work presented in this thesis; publications P6 and P7 were produced during the same period on related topics and are not part of the dissertation. The name of the candidate is shown in bold.

Journal articles

[P1] **Claudio Mirabella**, Michele Tuccillo, and Pierluigi Della Vecchia. "A Model-Based Systems Engineering Digital Certification Framework for General Aviation Aircraft: C. Mirabella et al." In: *Aerotecnica Missili & Spazio* 104.5 (2025), pp. 371–386.

[P2] Pierluigi Della Vecchia, **Claudio Mirabella**, Michele Tuccillo, Carlo Emanuele Riboldi, Marco Fioriti, Fixherald Shahini, and Francesca Roncolini. "A SysML-based framework towards EASA CS-23 digitalization: An MBSE approach". In: *Transportation Research Interdisciplinary Perspectives* 36 (2026), p. 101810. ISSN: 2590-1982. DOI: 10.1016/j.trip.2025.101810. URL: https://www.sciencedirect.com/science/article/pii/S2590198225004890.

[P3] **Claudio Mirabella**, Michele Tuccillo, and Pierluigi Della Vecchia. "Evaluating MBSE Approaches and Tools for Aircraft Design and Certification: A Comparative Perspective". In: *Systems* 14.5 (2026). Accessed 2026-05-23, p. 482. DOI: 10.3390/systems14050482. URL: https://www.mdpi.com/2079-8954/14/5/482.

Conference papers

[P4] **Claudio Mirabella**, Michele Tuccillo, and Pierluigi Della Vecchia. "A Model-Based System Engineering Approach Towards Aircraft Digital Certification". In: *Proceedings of the International Council of the Aeronautical Sciences (ICAS)*. Accessed 2026-05-23. 2024. URL: https://www.icas.org/icas_archive/icas2024/data/papers/icas2024_0603_paper.pdf.

Preprints

[P5] **Claudio Mirabella**, Elena Garcia, Adam T. Baker, and Pierluigi Della Vecchia. *A SysML Modeling Strategy to Digitalize FAR 23 Substantiation Methods*. Preprint, Preprints.org. Accessed 2026-10-21. 2026. DOI: 10.20944/preprints202609.2735.v1. URL: https://doi.org/10.20944/preprints202609.2735.v1.

Other publications

[P6] Pierluigi Della Vecchia, Vincenzo Cusati, and **Claudio Mirabella**. "Flight Loads Evaluation and Airworthiness Compliance for the V-Tail of a Medium-Altitude Long-Endurance Unmanned Platform". In: *Drones* 9.12 (2025). Accessed 2026-10-21, p. 835. DOI: 10.3390/drones9120835. URL: https://www.mdpi.com/2504-446X/9/12/835.

[P7] Gabriele Morra, **Claudio Mirabella**, Fabrizio Nicolosi, and Pierluigi Della Vecchia. "Lateral-directional trim for air and ground minimum control speeds evaluation in conceptual design". In: *Aerospace Science and Technology* 168 (2026). Accessed 2026-10-21, p. 110856. DOI: 10.1016/j.ast.2025.110856.

Contribution of the candidate

The candidate's contribution to each publication related to this thesis is stated below, using the CRediT contributor roles, together with the parts of the thesis that draw on it.

[P1] *Used in:* Section 3.1 and Appendix C.
Contribution: Conceptualization; methodologies; data gathering and curation; code and SysML modelization; original draft.

[P2] *Used in:* Section 3.1.1, Figures 3.1 and 3.2, and Table B.4.
Contribution: Writing original draft; review and editing; supervision and coordination; software; resources; methodology; investigation; formal analysis; data curation; conceptualization.

[P3] *Used in:* Section 2.5 and Figures 1.2 and 2.1.
Contribution: Conceptualization; methodology; validation; formal analysis; investigation; data curation; original draft preparation, review, and editing; visualization; supervision.

[P4] *Used in:* Sections 2.3 and 3.1.
Contribution: Original draft preparation, review, and editing; conceptualization and methodology; investigation; SysML modelization; code design and preparation; input/output curation and visualization.

[P5] *Used in:* Section 3.3.
Contribution: Conceptualization; methodology; software; validation; formal analysis; investigation; data curation; writing-original draft preparation; writing-review and editing; visualization.

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Nomenclature

α	Angle of attack referred to the x-body axes
α_S	Stabilizer angle of attack
α_{0_L}	Zero-lift angle of attack
α_{ht}	Horizontal tailplane angle of attack
α_{wb}	Angle of attack referred to the x-body axes of the wing-body configuration
β	Sideslip angle
$\ddot{\phi}$	Yawing angular acceleration measured in $\text{rad}\,\text{s}^{-2}$
$\ddot{\theta}$	Pitching angular acceleration measured in $\text{rad}\,\text{s}^{-2}$
ΔC_m	Aileron moment coefficient increment
ΔL_{ht}	Incremental horizontal tailplane load due to gust
Δv	Velocity increase due to pitching control deflection
$\delta_{\text{a, max}}$	Maximum aileron deflection angle
$\delta_{\text{e, max}}$	Maximum pitch control deflection angle
$\delta_{\text{r, max}}$	Maximum rudder deflection angle
$\dot{\phi}$	Yawing angular speed measured in $\text{rad}\,\text{s}^{-1}$
$\dot{\psi}$	Heading rate
$\dot{\theta}$	Pitching angular speed measured in $\text{rad}\,\text{s}^{-1}$
$\dot{q}$	Pitching angular speed measured in $\text{rad}\,\text{s}^{-1}$
ϵ	Downwash angle
$\frac{W}{S}$	Wing loading measured in $\text{N}\,\text{m}^{-2}$

$\hat{q}$ Reduced pitch rate

$\hat{r}$ Reduced yaw rate

μ_g Aircraft mass factor

ω Control angular deflection speed measured in rad s^{-1}

ω_1 Propeller rotation speed measured in rad s^{-1}

ω_2 Pitch or Yaw angular speed measured in rad s^{-1}

$\overline{c}$ Mean aerodynamic chord

$\overline{c}_{ht}$ Horizontal tailplane mean aerodynamic chord

$\overline{c}_{vt}$ Vertical tailplane mean aerodynamic chord

ϕ Yawing angle measured in rad

ψ Heading angle

ρ Standard atmosphere air density measured in kg m^{-3}

ρ_0 Standard atmosphere air density measured at sea level in kg m^{-3}

RPM_{MC} Engine rotation speed at maximum continuous power

AM Analysis mass

BEM Basic empty mass

$CG_{aft,\ MTOM}$ Most aft position of the CG at the MTOM

$CG_{aft,\ OEM}$ Most aft position of the CG at the OEM

$CG_{fwd,\ MidM}$ Most forward position of the CG at the MidM

$CG_{fwd,\ MTOM}$ Most forward position of the CG at the MTOM

$CG_{fwd,\ MZFM}$ Most forward position of the CG at the MZFM

$CG_{fwd,\ OEM}$ Most forward position of the CG at the OEM

EM Empty mass

MidM Transitional mass on the CG envelope diagram

MZFM Maximum zero-fuel mass

OEM Operational empty mass

UM Useful mass

θ Pitching angle measured in rad

$A_{\text{propeller}}$ Propeller disk area

A_{spinner} Propeller spinner area

b Main wing span

b_{ht} Horizontal tailplane span

b_{vt} Vertical tailplane span

C_L Aircraft lift coefficient

$c_{\text{root, ht}}$ Horizontal tailplane root chord

$c_{\text{root, vt}}$ Vertical tailplane root chord

c_{root} Main wing root chord

$c_{\text{tip, ht}}$ Horizontal tailplane tip chord

$c_{\text{tip, vt}}$ Vertical tailplane tip chord

c_{tip} Main wing tip chord

C_{D_0} Aircraft zero-lift drag coefficient

$C_d(y)$ Spanwise drag coefficient distribution

$C_{L_{\alpha,\text{ fv}}}$ Full vehicle lift curve slope

$C_{L_{\alpha,\text{ ht}}}$ Horizontal tailplane lift curve slope

$C_{L_{\text{max}}}$ Maximum lift coefficient

$C_{L_{\text{min}}}$ Minimum lift coefficient

$C_l(y)$ Spanwise lift coefficient distribution

$C_m(y)$ Spanwise pitching moment coefficient distribution

C_{N_β} Yawing moment coefficient per unit of sideslip angle β

$C_{N_{\delta_r}}$ Yawing moment coefficient per unit of δ_r

$C_{N_{\delta_{r,\text{ tab}}}}$ Yawing moment coefficient per unit of $\delta_{r,\text{ tab}}$

C_{N_A} Maximum Normal Force Coefficient

C_{N_r} Yawing moment coefficient per unit of yawing angular speed r

e_{fv} Full-vehicle Oswald efficiency factor

f Equivalent parasite area measured in m^2

g_0 Standard gravitational acceleration

h Above sea-level height

I_p Propeller polar moment of inertia

$I_{xx,\,\text{aft}}$ Principal moment of inertia along the x-axis with the CG at most aft position

$I_{xx,\,\text{fwd}}$ Principal moment of inertia along the x-axis with the CG at most forward position

$I_{yy,\,\text{aft}}$ Principal moment of inertia along the y-axis with the CG at most aft position

$I_{yy,\,\text{fwd}}$ Principal moment of inertia along the y-axis with the CG at most forward position

$I_{zz,\,\text{aft}}$ Principal moment of inertia along the z-axis with the CG at most aft position

$I_{zz,\,\text{fwd}}$ Principal moment of inertia along the z-axis with the CG at most forward position

I_{zz} Principal moment of Inertia along the z-body axes

K Radius of gyration in yaw measured in m

k_g Gust alleviation factor

L_{α_S} Horizontal tailplane load due to unit of α_S measured in $\text{N}\,\text{deg}^{-1}$

L_{δ_e} Horizontal tailplane load due to unit of δ_e measured in $\text{N}\,\text{deg}^{-1}$

L_{camber} Horizontal tailplane load due to unit of camber measured in N

$l_{\text{ht,\,aft}}$ Horizontal tailplane moment arm with the CG at most aft position

$l_{\text{ht,\,fwd}}$ Horizontal tailplane moment arm with the CG at most forward position

L_{tail} Horizontal tailplane load measured in N

$l_{\text{vt,\,aft}}$ Vertical tailplane moment arm with the CG at most aft position

$l_{\text{vt,\,fwd}}$ Vertical tailplane moment arm with the CG at most forward position

M Mach number

$M(y)$ Spanwise bending moment distribution

M_r Induced rolling moment at vertical tailplane

M_{α_S} Horizontal tailplane pitching moment due to unit of α_S measured in $\mathrm{N\,m\,deg^{-1}}$

M_{δ_e} Horizontal tailplane pitching moment due to unit of δ_e measured in $\mathrm{N\,m\,deg^{-1}}$

M_{camber} Horizontal tailplane pitching moment due to unit of camber measured in N m

M_{Engine} Engine mass per single engine

M_{gyro} Pitch or Yaw gyroscopic couple

M_{tail} Horizontal tailplane pitching moment about $0.25 \cdot \overline{c}$ measured in N m

n Limit load factor for establishing the Maneuvering and Gust Flight Envelope

$N(y)$ Spanwise normal force distribution

n_m Reference (positive) limit maneuvering load factor

N_{Cylinder} Number of cylinders

N_{Eng} Number of installed engines

n_{in} Total inertial limit load factor acting on the powerplant structures during a pitching manevuer

n_{rot} Rotational limit load factor during a pitching maneuver

n_{sideload} Limit powerplant sideload factor

P_{MC} Maximum continuous power

q Dynamic pressure

r Yawing angular speed measured in $\mathrm{rad\,s^{-1}}$

S Main wing surface

$S(y)$ Spanwise shear distribution

S_{ht} Horizontal tailplane surface

S_{vt} Vertical tailplane surface

$T(y)$ Spanwise torsion couple distribution

$t_{\mathrm{deflection}}$ Control surface deflection time interval

U_{de} Gust intensity

V_A Design Maneuvering Speed

V_B Design Speed for Maximum Gust Intensity

V_C Design Cruising Speed

V_D Design Dive Speed

V_H Maximum Horizontal Speed at Maximum Continuous Power and Sea Level

V_{MCA} Minimum control speed in air

V_{MCG} Minimum control speed on ground

V_{S_1} Clean Design Stalling Speed at standard 1 g acceleration at MTOW

V_{S_F} Design Stalling Speed at standard 1 g acceleration with Flaps fully extended at MTOW

W Aircraft weight

w Side-displacement

W_{TO} Maximum Takeoff Weight

x_{mL} Main Left landing gear

x_{mR} Main Right landing gear

x_n Nose landing gear

x_{ac} Dimensional position of the aerodynamic center from the given datum measured in m

x_{cg} Dimensional position of the CG from the given datum measured in m

y_e Critical engine yaw moment arm

$y_{\text{a, in}}$ Inner position of the aileron along the wing span

$y_{\text{a, out}}$ Outer position of the aileron along the wing span

y_{MAC} Spanwise location of the mean aerodynamic chord

AARC Aircraft Airworthiness Review Certificate

AC Advisory Circular

AC Aircraft

ac Aerodynamic Center

AD Activity Diagram

AD Advisory Directive

AFL Approval of Flight Conditions

AFM Aircraft Flight Manual

AGL Above Ground Level

AI Artificial Intelligence

AIP Aeronautical Information Publication

AMC Acceptable means of compliance

AMM Aircraft Maintenance Manual

API Application Programming Interface

ARM Airworthiness Review Certificate

ASDL Aerospace Systems Design Laboratory

ASTM American Society for Testing and Materials

BL Balancing Loads

CAA Civil Aviation Authority

CAD Computer-aided Design

CAM Computer-aided Manufacturing

CAS Calibrated Airspeed

CATIA Computer-Aided Three-dimensional Interactive Application

CAW Continuous Airworthiness

CCC Compliance Certification Checklist

CCD Compliance Certification Data

CDA Compliance Demonstration Activities

CDD Compliance Demonstration Data

CDI Compliance Demonstration Items

CFD Computational Fluidmechanics

CFR Code of Federal Regulations

CG Center of gravity

CS Certification Specifications

CS-23 Certification Specifications for Normal Category Aeroplanes

CS-25 Certification Specifications and Acceptable Means of Compliance for Large Commercial Aeroplanes

DA Damping angle

DB Document Based

DEP Distributed electric propulsion

DF Damping factor

DIGACE Digital General Aviation Certification

DOA Design Organization Approval

EAB EASA Advisory Board

EASA European Union Aviation Safety Agency

ELOS Equivalent Level of Safety

EMB EASA Management Board

EOL Epsilon Object Language

EUROCAE European Organization for Civil Aviation Equipment

FAA Federal Aviation Administration

FAR Federal Aviation Regulations

FCOM Flight Crew Operating Manual

FEM Finite Element Methods

GA General Aviation

GR Gear ratio

HIRF High-Intensity Radiated Fields

IATA International Air Transport Association

IAW Initial Airworthiness

ICA Instructions for Continued Airworthiness

ICAO International Civil Aviation Organization

INCOSE International Council on Systems Engineering

ISA International Standard Atmosphere

JAA Joint Aviation Authorities

JAR Joint Aviation Requirements

JSON Javascript Object Notation

KerML Kernel Modeling Languages

KPI Key Performance Indicator

LLM Large Language Model

LOA Letter of Agreement

LOI Level of Involvement

LSA Light Sport Aircraft

LT Engine limit torque

M2M Model-to-Model Transformation

M2O Model-to-Object Transformation

MAB Member State Advisory Board

MAC Mean aerodynamic chord

MATLAB Matrix Laboratory

MBSE Model-based Systems Engineering

MCP Maximum Continuous Power

MF Multiplying Factor

MOA Maintenance Organization Approval

MSoSA Magic Systems of Systems Architect

MT Engine mean torque

MTO Maintenance Training Organization

MTOW Maximum Takeoff Weight

NAA National Aviation Authority

NCOSE National Council on Systems Engineering

NOTAM Notice to Airmen or Air Mission

OMG Object Management Group

PDF Portable Document Format

POA Production Organization Approval

POLIMI Politecnico di Milano

POLITO Politecnico di Torino

RIA Regulatory Impact Assessment

RTCA Radio Technical Commission for Aeronautics

SAE Society of Automotive Engineers

SARP Standard and Recommended Practice

SC Special conditions

SIB Safety Information Bulletin

SIC Safety Information Communication

SL Sideload

SMS Safety Management System

SP Safety Program

SRM Safety Risk Management

SSoT Single Source of Truth

STC Supplemental Type Certificate

STOL Short Take Off and Landing

SysML Systems Modeling Language

TC Type Certificate

TCCA Transport Canada Civil Organization

TI Technical Instruction

TOA Training Organization Approval

TR Type Rating

TSO Technical Standard Order

UA Unmanned Aircraft

UAM Urban Air Mobility

UAS Unmanned Aircraft System

UAV Unmanned Aircraft Vehicle

UML Unified Modeling Language

UNINA University of Naples, Federico II

V&V Verification and Validation

VLA Very Light Aircraft

VSP Vehicle Sketch Pad

WG Working Group

WI Work Instructions

XMI XML Metadata Interchange

XML Extensible Markup Language

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This Chapter outlines the content and structure of the thesis. Section 1.1 describes the current regulatory framework adopted by major aviation authorities and certification agencies. It also examines relevant trends and the challenges that hinder the development and introduction of new aircraft configurations incorporating advanced technologies that may enhance the value of aeronautical products. Section 1.2 introduces MBSE-based digital approaches and methods that can reduce the certification burden on aviation stakeholders, including approved design and production organizations. Finally, Section 1.8 presents the organization of the thesis and summarizes the content of each Chapter.

1.1 Background and Challenges in Aircraft Certification

Certification of engineered products is a complex [1, 2], mandatory process that can substantially influence performance [3], life-cycle cost, and time to market [4, 5, 6]. Regulatory authorities oversee this process by assessing whether a given product conforms

to applicable requirements and regulations [7], which may include standards developed by third-party organizations [8]. The process is typically multistage and resource-intensive [9, 10] and includes activities such as the following [11]:

- The manufacturer should establish verification and validation procedures [12, 13, 14], documented in a *Compliance Certification Checklist* [15, 16], together with a robust and maintainable database for collecting certification data, documents, and artifacts. Collectively, these records constitute the *Compliance Certification Items* [17].

- The manufacturer and the relevant regulatory authority should exchange data using an agreed data standard [18, 19, 20, 21]. The exchanged information should include the tests performed to demonstrate compliance, the data-acquisition equipment, the tools used, and the personnel involved [22, 23, 24]. A formal review and revision process should be defined and executed within an established quality management system [25].

- Throughout the product life cycle [26], certification information should remain accessible through an efficient retrieval mechanism that supports modifications [27, 28], corrective actions, and progressive product improvements, ensuring the quality and safety of the aeronautical products over time [29, 30, 31, 32, 33, 34, 35, 36].

The *International Civil Aviation Organization* (ICAO) [37, 38] and major aviation regulators are working to reduce the burden of certification on industry stakeholders while maintaining established safety standards [39, 40, 41, 42]. One important enabler is the use of performance-based requirements [43, 44] that are comprehensive and broadly applicable. This approach can facilitate the use of consensus standards across jurisdictions and may improve harmonization of certification outcomes [45]. For example, when consensus standards such as those issued by ASTM International are accepted by multiple authorities (e.g., the FAA and EASA), manufacturers can, in principle, reduce duplication of compliance efforts and limit the need for costly and time-consuming re-certification when seeking approvals in different regions [46, 47].

Additional information concerning the certification landscape in aeronautics may be found in Appendix D. From the applicant's standpoint, some degree of commonality and reusability exists across certification programs for different type certificates [48]. However, because the process remains strongly document-based, applicants often need to rebuild substantial portions of the certification baseline for each program [49]. This typically leads to high costs, driven largely by the engineering labor required to develop, track, and update certification items, and it can extend program schedules [50]. In addition, the iterative nature of compliance planning and evidence generation can result in costly revisions and rework as issues are identified and resolved during certification activities [51]. Therefore, the above-mentioned concerns about errors, inconsistencies, and parallel versions of the same document are practical and recurring challenges in aeronautical certification. The outcome of the aviation certification process is the *Type Certificate* [52], issued by a

regulatory authority to confirm that an aeronautical product (e.g., a propeller, an engine, or an airframe) complies with all applicable requirements.

1.2 The Digital Certification Process

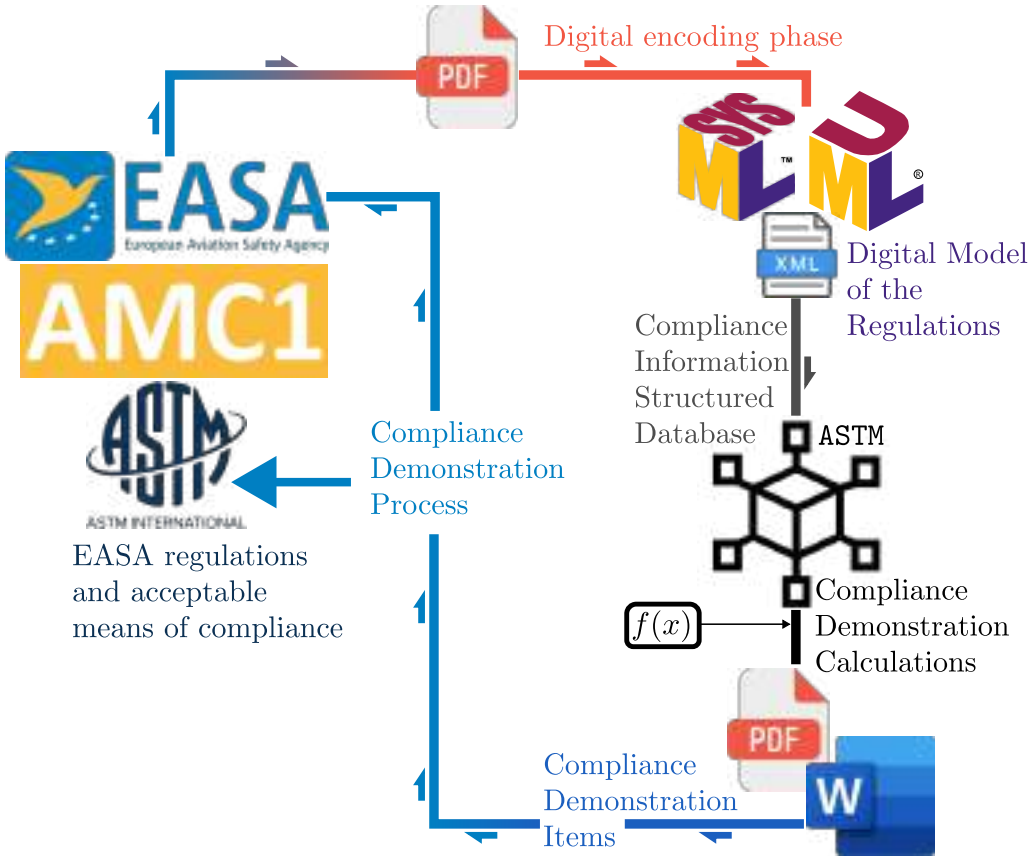

Figure 1.1: A schematic representation of the digital certification framework. Adapted from [53] for illustrative purposes. Compliance calculations also draw on the aircraft design data.

This thesis aims to present a digital alternative to the conventional document-based certification process [54]. Figure 1.1 presents the conceptual structure of the digital certification framework implemented in this thesis. The figure illustrates its most likely hybrid configuration, in which the digital model is used to generate compliance-demonstration items, including technical reports and other supporting documents. Historically, the adoption of new tools and methods in the aeronautical industry has proceeded slowly, and a noticeable gap often separates research developments from their implementation in operational practice [55]. Nevertheless, digitally managed processes are already well established in aviation, particularly in areas such as simulation, computer-aided design

(CAD), computational fluid dynamics (CFD), and finite element methods (FEM), and other advanced computer-based tools [56, 57]. These tools are routinely used to support engineering analyses and to generate the certification artifacts required for compliance demonstrations.

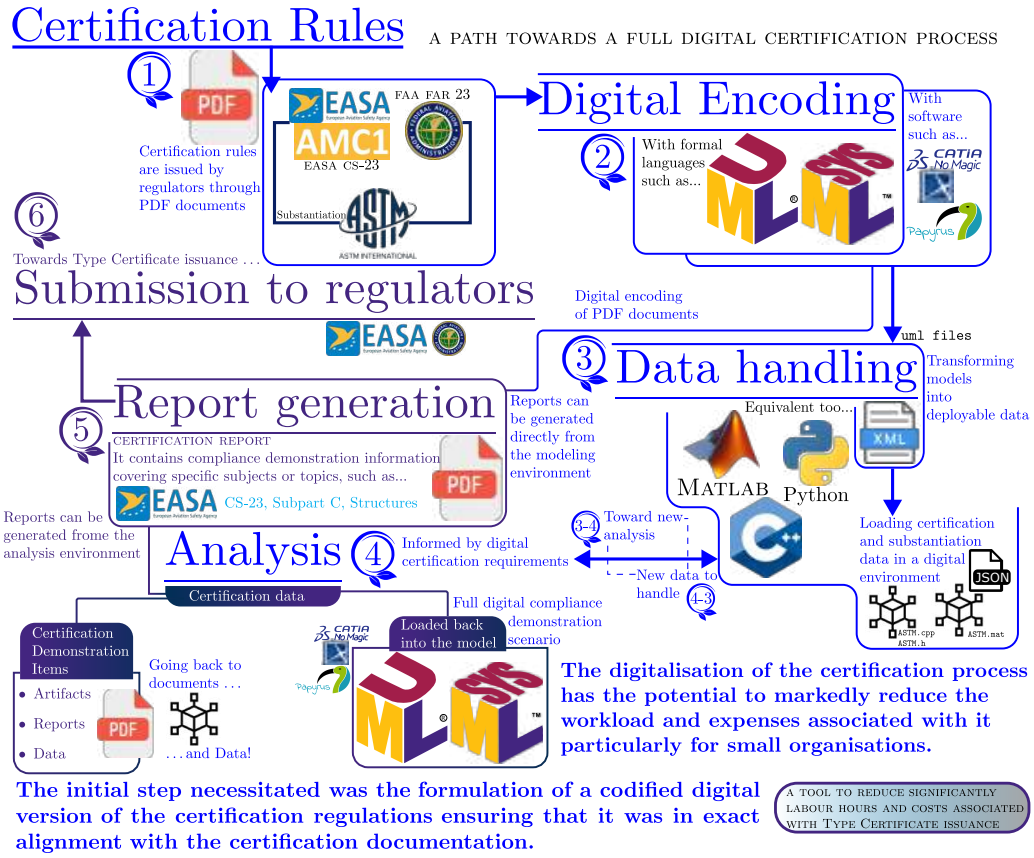

Figure 1.2: The general conceptualization of digital certification.

The certification substantiation process requires the applicant to collect evidence for each *Compliance Demonstration Item* (CDI) and to compile the associated certification artifacts, typically within technical substantiation reports [58]. Over the life cycle of a type certificate—including amendments and post-certification changes—the certification database may grow to hundreds of such reports. For this reason, it is desirable to consolidate these records within a single digital framework that can generate certification artifacts on demand [59, 60, 61].

Under this approach, the certification process can be represented as a reusable model or template that is instantiated for each type-certificate program and adapted to the specific aircraft configuration, including geometric features, powerplant, and other certification-relevant characteristics [62]. Report preparation can then be simplified because the certification model can be queried and parsed to generate a specific report when needed or, when appropriate, to extract only the artifacts associated with a single compliance item. In turn, the information stored in a digital certification model can be more consistent

and easier to maintain, particularly when the model is hosted in an online, collaborative environment with appropriate access control and configuration management [63, 64]. Engineers can define standard information-retrieval queries and execute them on demand to support reviews, updates, and change impact assessments. In the most integrated scenario, both the applicant and the regulatory authority could access the same model and monitor progress in near real time, reducing the latency associated with assembling and transmitting complete document packages. Such an approach could accelerate compliance activities and reduce reliance on static reports by enabling incremental review of certification evidence as it becomes available [53].

Although the most optimistic scenario may be unlikely in the near term, a hybrid digital certification framework capable of generating certification artifacts and supporting documentation remains feasible [65]. This direction is consistent with ongoing discussions among aviation stakeholders, both for large commercial aircraft and for the general aviation sector. The need to transition toward digitally managed certification processes is also reflected in research initiatives worldwide that seek to identify effective tools, methods, and best practices for digital certification [66, 67, 68].

Figure 1.2 shows the general conceptualization, at least at a theoretical level, of a digital approach to aeronautical certification. Several phases may be identified, following the flow diagram:

1. The starting point is the certification material, including the consensus standards. Usually, this information is available as PDF documents.

2. The digitalization process encodes certification and substantiation information within a digital environment. The flow diagram presents an implementation based on the UML/SysML family of modeling languages, as supported by Eclipse Papyrus and CATIA MSoSA.

3. Once the modeling phase is complete, a mechanism must be established for accessing and processing the encoded certification data. Several strategies are available, some of which are investigated in this thesis. Data handling can be performed directly within the modeling environment. Alternatively, the model can be exported in a machine-readable format and processed by external code suites.

4. The data-handling phase extends throughout the execution of the digital certification framework and is essential because it encompasses the collection, management, and reuse of data generated during the analysis phase. In Figure 1.2, this continuous exchange is represented by a bidirectional arrow connecting the data-handling and analysis phases. The analysis phase produces not only numerical results but also certification artifacts, including tables, figures, and technical documents.

5. Automated generation of technical reports is a central capability of the digital certification framework developed in this thesis. The data and certification artifacts produced during the analysis phase are assembled into a well-structured and readable technical report tailored to the certification task under consideration. Within

the scope of this thesis, this capability is demonstrated through the automated generation of a *Flight Loads* certification report.

6. The final step is the submission of the substantiation material to the regulatory authority. Although all certification activities could, in principle, be performed within a shared digital environment accessible to the relevant stakeholders, this scenario is unlikely to be implemented in the near term. A more realistic approach is for applicants to use the digital certification framework to produce the required data, documents, and certification artifacts; organize these outputs into a submission package; and submit the package to the regulatory authority through established procedures. This approach represents a hybrid adoption of MBSE methodologies, combining model-based data management and analysis with conventional document-based submission practices.

The complexity of aeronautical certification creates a need for a rigorous approach to digitally encoding the information required to initiate and manage the process [69, 70]. An effective encoding procedure must transform certification documents into information that can be processed reliably. In particular, paragraphs and subparagraphs that contain formulas, numerical limits, ordered requirements, and other constraints must be captured in a standard and repeatable form. Although many digital standards have emerged with advances in information technology, only a subset can support the deterministic and predictable behavior required for certification activities. One viable approach to this nontrivial problem is the use of formal languages within established systems engineering practices [71]. Systems engineering is a well-established discipline used in the development of complex products, including chemical and nuclear facilities, satellites, and aircraft. Industry has applied several systems engineering paradigms successfully, and engineers have accumulated substantial experience in adopting these methods in practice. The expansion of digitally managed engineering environments has created demand for approaches that better align design and manufacturing processes with digital tools and data. In this context, model-based systems engineering (MBSE) [72, 73, 74] emerged as a major focus within the systems engineering community, including organizations such as the *National Council on Systems Engineering* (NCOSE), which later evolved into the *International Council on Systems Engineering* (INCOSE) [75, 76, 77, 78, 79, 80]. MBSE promotes the use of models to represent requirements and system architecture during conceptual design and to support analysis, verification, and validation across the system life cycle [81]. In principle, MBSE-based frameworks can improve communication and data sharing within and across organizations and can support earlier detection of design issues and compliance inconsistencies [82, 83]. A central objective is the creation of a model that consolidates product knowledge in a form that can be reused across programs and learned efficiently by personnel. When implemented with appropriate governance, the approach can be standardized and applied to subsequent products, leveraging lessons learned, past experiences, and organizational knowledge [84, 85].

To create a model effectively, an encoding language must be defined and applied consistently. Early model-based approaches in software engineering were commonly

expressed using the *Unified Modeling Language* (UML) [86], which is standardized by the *Object Management Group* (OMG), an international, open-membership, not-for-profit standards consortium [87]. MBSE itself is tool-agnostic: it provides principles and practices for managing complex product development, independent of any specific software platform [88].

Because physical systems introduce concerns that are not central in software modeling (e.g., continuous dynamics, energy flows, and physical interfaces), OMG extended UML to define the *Systems Modeling Language* (SysML) [89, 90, 91]. SysML version 1.x is widely used across industries and is based on UML concepts and extensions [92]. A newer generation, SysML v2, is defined on top of the *Kernel Modeling Language* (KerML) and introduces a standardized textual notation alongside graphical notation, together with interfaces intended to improve tool interoperability [93, 94]. Textual modeling can facilitate automation, version control, and the development of more reproducible modeling workflows, and it may also support emerging forms of computer-assisted modeling [95].

SysML provides mechanisms for extending the language through user-defined stereotypes and profiles, as well as a broad set of diagram types for representing structure, behavior, and constraints. Properly designed stereotypes can capture certification-relevant requirements and attributes of physical systems by introducing domain-specific properties and semantics while retaining the underlying SysML formalism [96]. In effect, these extensions allow the modeler to tailor the language to the needs of the application, provided that the extensions are defined consistently and remain aligned with the intended modeling purpose.

When aeronautical certification is treated as a modeling problem, one part of the model represents the certification and substantiation requirements, while another represents the aircraft architecture and its defining data [97]. Once the SysML model includes both the regulatory content and the design information, it can be instantiated to support certification-related operations, including calculations, traceability checks, and verification and validation activities. These compliance-demonstration procedures may be implemented directly within the modeling environment when the platform provides adequate execution and automation capabilities [98]. Otherwise, the model can be exported and processed by an external companion code suite, which parses the model according to predefined exchange formats and data-extraction rules that are tailored to the specific certification activity. The research conducted produced an initial digital representation of the EASA CS-23 high-level requirements, including the associated AMC1 references, using the Eclipse Papyrus modeling environment [99]. Papyrus supports the definition of profiles, user-defined stereotypes, and other domain-specific modeling elements [100]. However, it does not provide the same level of built-in workflow execution and automation available in more integrated environments, such as *Dassault Systèmès* CATIA *Magic Systems of Systems Architect* (MSoSA) [101]; therefore, automation must be implemented through external mechanisms, for example, by using model-transformation and scripting frameworks such as the Epsilon programming language. In practice, developing and deploying bespoke automation features can require substantial effort, and the resulting solutions are often specialized to a narrow use case, which can limit reuse and long-term

maintainability. Nevertheless, Papyrus remains suitable for defining custom stereotypes and diagrams that encode key features of airworthiness rules, including formulas, numerical parameters, constraints, and other structured information. The scope of this thesis is limited to EASA CS-23 Subparts B (Flight), C (Structure), and F (Systems and equipment), using AMC1 as the substantiation basis. This thesis draws on, but is not limited to, the work carried out within a collaboration with other Italian universities, such as the *Politecnico di Milano* (POLIMI) and *Politecnico di Torino* (POLITO). Its scope also includes research conducted in collaboration with colleagues at the Georgia Institute of Technology in Atlanta, Georgia, United States. The author benefited from access to methods and tools developed by these researchers and from the partnerships established by UNINA.

Collectively, the research focused on the following EASA CS-23 subparts:

- Subpart B, Flight, modeled by POLIMI—not included within the scope of this thesis.
- Subpart C, Structure, modeled by UNINA.
- Subpart F, Systems and Equipment, modeled by POLITO—not included within the scope of this thesis.

Accordingly, this thesis focuses primarily on Subpart C and presents the applications developed in the context of the *Flight Loads* certification report. UNINA developed a digital certification framework to generate the corresponding certification artifacts in an automated manner by integrating EASA CS-23 requirements, ASTM International standards, and aircraft input data. The framework performs the required calculations and produces a technical report suitable for submission to EASA. The next Section describes the state of the art of MBSE strategies applied to the problem of aeronautical certification.

1.3 The state of the art

Industrial and research organizations are applying MBSE techniques to aeronautical certification. Many industrial initiatives focus on certification-basis definition and *Certification by Analysis* [102, 103] (CbA). This approach improves traceability among requirements, aircraft data, analytical procedures, and certification artifacts. The use of SysML/UML formalisms may help address incomplete traceability, inconsistent standards, and regulatory gaps [104, 105]. Current initiatives involving EASA and major industrial stakeholders indicate growing interest in digital certification [106, 107]. This represents a critical point, especially for new design architectures embodying new technological features and complex systems without a consolidated operational history [108, 109, 110, 111, 112]. Although further independent evaluation is needed, these developments represent an important step toward digital aircraft certification. The general goal is to harmonize the guidelines and best practices [113, 114] for current methodologies and include them in the development of new paradigms.

Although MBSE can improve collaboration, traceability, and information consistency [115, 116, 117, 118], its adoption may also increase process complexity and require new organizational competencies [119, 120]. The resulting SysML model is exported in a structured UML-based interchange format and processed by code routines that perform certification calculations and generate compliance artifacts [121]. Thorough investigation should be conducted to study the connection between formally encoded certification information and executable substantiation methods [122]. Many studies discussed technological and industrial factors that hindered the adoption of MBSE methodologies in specific industrial tasks, proposing new approaches and frameworks to overcome present limitations [123].

Advancements in large language models (LLMs) encourage engineers to use these tools to interpret requirements, but aviation certification experts currently consider the practice hazardous and unreliable [124]. Due to the lack of interpretability of results—a problem that can negate benefits in deploying AI tools in the certification loop—applicants and regulators do not use LLMs in establishing certification bases or in generating the certification artifacts. In practice, implementing AI-based instruments can hinder the entire certification process and jeopardize Type Certification because of mandatory validation and verification procedures for the new tools. This problem is well-known in the aviation industry as the *projects-within-projects* effect [125]; in the best-case scenario, the certification standards impose on industrial practice an increase in time and costs for accepting new tools and techniques, reducing products profitability. AI-based aviation systems are not currently certifiable; many studies have addressed this problem [126, 127], exploring problems unrelated to aircraft operation and certification compliance demonstration. The certification process must be improved to ensure continuous airworthiness with the same assured level of safety [128, 129, 130, 131].

The consensus for integrating generative artificial intelligence (AI) into aviation design advises an incremental approach. In the short term, generative AI can support non-safety-critical activities, such as document analysis and requirements management once the certification material and artifacts are available [132]. Human oversight can mitigate compliance risks in these activities. As research, standards, and assurance methods mature, the use of generative AI may gradually extend to higher-assurance tasks through hybrid approaches combining deterministic algorithms and AI-based components. The long-term objective is to integrate generative AI throughout the system life cycle, supported by harmonized standards and validated assurance methods. This could enable the safe, efficient, and compliant automation of future aviation systems [133].

The conceptualization of the methodologies presented in this thesis and the development of the companion code suites were not developed with AI assistants, except for the Python code presented in Section 3.4.2, which was partially developed using AI coding assistants.

At the time of writing, several research groups had addressed the representation of certification requirements and documents in digital environments. Researchers at the Georgia Institute of Technology focused on developing tools and methodologies that enable certification-domain experts to establish an appropriate certification basis within

a complex SysML model. Their approach also provides warnings about critical issues, including gaps in the certification requirements for novel design architectures incorporating emerging technologies, and supports the traceability of requirements to the corresponding systems, creating the Certification Compliance Checklist automatically [134]. Researchers at Concordia University, by contrast, focused on encoding the knowledge required to identify applicable certification requirements [135, 136, 137] in a reliable single source of truth. Their work involved developing digital environments in which this knowledge can be readily accessed and retrieved. The resulting certification database records and justifies design decisions, supports the implementation and certification of minor and major changes to an established Type Certificate, and communicates accumulated knowledge about the final product to relevant stakeholders throughout its life cycle. Together, these strategies have significantly advanced the treatment of certification as a complex industrial process using MBSE techniques and methodologies, especially by improving traceability, compliance understandability, and the formulation of the complete certification scenario for a specific design architecture.

This thesis focuses on closing the gap between addressing the applicable certification and substantiation requirements for a given design architecture, including special conditions, and executing the certification tasks. This thesis will consider EASA CS-23, Subpart C, although the conceptual approach is applicable to other Subparts requiring the additional step of verification and validation. Subpart C can be considered fulfilled once the applicants follow the established certification basis agreed with the regulators. We can list two exceptions that still require testing and verification and validation procedures:

- Paragraph CS 23.2235 (*Structural strength*). This requirement usually requires a proper ground testing campaign with repeated audits and verification and validation for each test article used.

- Paragraph CS 23.2245 (*Aeroelasticity*) usually requires establishing ground and flight testing campaigns.

However, the *Flight Loads* certification requirements and the corresponding substantiation methods represent a use case scenario where the process of showing compliance coincides with drafting a technical certification report, i.e., a single document. The digital certification capabilities developed and demonstrated in this thesis include the following:

- Automated substantiation calculations for Subpart C (Structure), with the generation of a comprehensive certification database containing the outcomes of the calculations.

- Automatic generation of a certification-artifact database, including figures, tables, and associated metadata.

- In the Eclipse Papyrus/MATLAB implementation, automatic generation of a complete *Flight Loads* certification report, i.e., the document discussed above.

A potential extension of these capabilities is the introduction of verification and validation procedures applied to flight-test data. Such functionality could support compliance demonstration with the requirements of Subpart B (Flight). In addition, for industrial applications, the use of environments such as CATIA MSoSA can further extend the framework by leveraging integrated modeling and user-interface capabilities. Eclipse Papyrus can likewise be extended to support additional framework functions, but such extensions typically require the development and maintenance of bespoke plug-ins, as discussed in Section 2.2.1.

1.4 Research Gaps and Motivation

This Section outlines the research gaps that this thesis addresses. The critical step addressed by this PhD thesis is to enable the execution of digital models encoding certification rules and consensus standards material to generate certification artifacts; the compliance demonstration material obtained by executing the MBSE framework will therefore be verifiable and traced to the corresponding high-level certification requirements. Figure 1.3 summarizes their formal definition, given as follows:

Research Gap 1

There are no applications of MBSE methodologies to the compliance substantiation process in the field of aviation certification. The methodologies presented in this doctoral dissertation demonstrate how to use SysML models to demonstrate compliance with high-level requirements, using EASA CS-23, Subpart C, as an example.

Research Gap 2

There are no applications in the literature that enable the automatic drafting of technical reports for compliance demonstration. The applications described in this thesis propose a possible MBSE approach to implementing these functionalities.

Figure 1.3: The two research gaps delineated in this thesis.

Research Gap 1: There are no research studies in the literature about establishing compliance demonstration procedures and verification and validation tools informed by certification and substantiation materials. This thesis presents use case scenarios and applications that demonstrate the feasibility of creating Model-Based Systems Engineering (MBSE) digital certification frameworks that can generate certification artifacts valuable for demonstrating compliance to regulators.

Research Gap 2: Consequently, there are no research studies in the literature about

generating appropriately formatted compliance documentation from a digital SysML model, such as a technical certification report, from performed certification and substantiation procedures. This thesis demonstrates the feasibility of generating compliance demonstration reports, limiting the scope to flight loads as treated by EASA CS-23, Subpart C.

This thesis discusses strategies to address these gaps; however, the methodologies and tools have not been tested in a practical setting. Therefore, they can be considered an initial proof of concept for MBSE techniques applied to aircraft certification. To quantify the benefits of implementing these techniques, collaboration is required among domain experts, regulators, and aviation industry stakeholders. These stakeholders must adopt a common set of KPIs to measure costs, labor effort, time to market, and other quantitative variables.

1.5 Research Objectives and Questions

Research Question 1

How can EASA CS-23 and FAA Part 23 requirements and their associated substantiation provisions be represented in a SysML-based model and linked to executable calculation routines while preserving traceability to the applicable regulatory sources?

Research Question 2

What trade-offs emerge among the investigated MBSE implementation architectures in terms of workflow automation, data extraction, maintainability, regulatory-change sensitivity, and generation of compliance evidence?

Research Question 3

To what extent do the developed prototypes demonstrate the feasibility of model-based substantiation workflows, and which measurable indicators should be used in future industrial assessments of their effectiveness?

Figure 1.4: The research questions addressed in this thesis.

This Section outlines the purpose of this thesis and states it in terms of three research questions. An abstract formalization of the purpose of this thesis is the following:

> To develop and evaluate model-based workflows that connect SysML representations of airworthiness requirements and their means of compliance to

executable substantiation routines and certification artifacts for EASA CS-23 and FAA Part 23, Subpart C flight loads.

Since the selected formal language for this application is SysML, conceptual development design rules must be established to create the digital model. Therefore, this discussion can lead to the first research question, answered by this thesis:

Research Question 1 (Digital representation and execution)**.** *How can EASA CS-23 and FAA Part 23 requirements and their associated substantiation provisions be represented in a SysML-based model and linked to executable calculation routines while preserving traceability to the applicable regulatory sources?*

As with many design problems, designers have many options to explore and evaluate. There is no single Model-Based Systems Engineering (MBSE) digital certification framework that optimally addresses all requirements simultaneously. This can lead to the identification of the following research question:

Research Question 2 (Architectural trade-offs)**.** *What trade-offs emerge among the investigated MBSE implementation architectures in terms of workflow automation, data extraction, maintainability, regulatory-change sensitivity, and generation of compliance evidence?*

Finally, conducting a preliminary qualitative evaluation of the developed methodologies is a useful way to guide and direct future development. This leads to the following research question:

Research Question 3 (Measuring impact)**.** *To what extent do the developed prototypes demonstrate the feasibility of model-based substantiation workflows, and which measurable indicators should be used in future industrial assessments of their effectiveness?*

Figure 1.4 summarizes the research questions outlined in this Section.

1.6 Contributions and publications

This Section lists some of the bibliographical resources relevant to the scope of this PhD thesis. A resource outside the scope of the thesis has been included for completeness and because the topic covered by Della Vecchia, Cusati, and Mirabella gives interesting perspectives on loads acting on non-conventional empennages. Table 1.1 summarizes the contributions published by the author of this thesis, whether in scientific journals or presented at conferences. An additional resource is included, although at the time of writing this thesis, it is still under review. The author listed only the contributions more relevant to the research study presented in this thesis. All the articles and papers published by the author of this thesis are available online in open access form. For additional bibliographical resources, the interested reader should consult the Bibliography included in this thesis.

Table 1.1: List of publications and contributions.

Publication type	Article	Cited in thesis
Journal	**P1.** Mirabella, Tuccillo, and Della Vecchia, "A Model-Based Systems Engineering Digital Certification Framework for General Aviation Aircraft," *Aerotecnica Missili & Spazio* (2025), doi: 10.1007/s42496-024-00241-5	Yes [112] Sec. 3.1
	P2. Della Vecchia, Mirabella, Tuccillo, Riboldi, Fioriti, Shahini, and Roncolini, "A SysML-based framework towards EASA CS-23 digitalization: An MBSE approach," *Transportation Research Interdisciplinary Perspectives*, Vol. 36 (2026), Article 101810	Yes [111] Sec. 3.1.1
	P3. Mirabella, Tuccillo, and Della Vecchia, "Evaluating MBSE Approaches and Tools for Aircraft Design and Certification: A Comparative Perspective," *Systems*, Vol. 14, No. 5 (2026), Article 482	Yes [97] Sec. 2.5
	P6. Della Vecchia, Cusati, and Mirabella, "Flight Loads Evaluation and Airworthiness Compliance for the V-Tail of a Medium-Altitude Long-Endurance Unmanned Platform," *Drones* (2025), doi: 10.3390/drones9120835	No Related work
	P7. Morra, Mirabella, Nicolosi, and Della Vecchia, "Lateral-directional trim for air and ground minimum control speeds evaluation in conceptual design," *Aerospace Science and Technology* (2026), doi: 10.1016/j.ast.2025.110856	No Outside scope
Conference	**P3.** Mirabella, Tuccillo, and Della Vecchia, "A Model-Based System Engineering Approach Towards Aircraft Digital Certification," *ICAS 2024*	Yes [95] Chs. 1–3
Preprint	**P5.** Mirabella, Garcia, Baker, and Della Vecchia, "A SysML Modeling Strategy to Digitalize FAR 23 Substantiation Methods," *Preprints.org* (2026), doi: 10.20944/preprints202609.2735.v1	No Sec. 3.3

1.7 Proposed Approaches

This thesis examines the development of a digital certification framework with limited applicability to EASA CS-23, Subpart C. The framework is conceived as a working paradigm within systems-of-systems engineering; accordingly, multiple implementation strategies are feasible in practice, depending on the selected tools and architectural choices. The work presented in this thesis is organized into three strands:

- In collaboration with researchers from POLITO and POLIMI, the author developed a SysML model in Eclipse Papyrus to encode EASA requirements and ASTM International standards [139, 140]. A MATLAB code suite parsed the encoded information, performed Subpart C calculations, generated certification artifacts, and produced a *Flight Loads* report. The principal limitation was the use of regular expressions, which made the parser sensitive to changes in regulatory text [141, 142, 143]. Subsequent implementations adopted a more structured extraction strategy based on model properties. The author developed both the parser and the Subpart C calculation routines without AI-assisted development tools, which were not available at the time.

- A second, limited-scope application was developed in the Dassault Systèmès CATIA

MSoSA environment. SysML Activity Diagrams automated model parsing, data transfer, and execution of an external C++ certification suite [144, 145]. The resulting artifacts could be assembled into a *Flight Loads* report using the built-in report wizard [146]. Unlike the earlier regular-expression approach, this implementation extracts certification data from structured stereotype properties, improving robustness across regulatory amendments. The author developed this specific application autonomously, from conceptualization to the companion C++ code suite, again without AI-assisted development tools.

- Finally, the author collaborated with researchers at the Georgia Institute of Technology on a use-case application based on a broader CATIA MSoSA digital certification infrastructure [147, 148, 149]. Python routines parsed the SysML model, exported it as a UML interchange file, and generated structured inputs for reusable calculations and certification artifacts. Structured stereotype properties supported consistent data extraction, while the CATIA MSoSA report wizard offered a potential mechanism for automated report preparation [150]. Although Activity Diagrams did not orchestrate the parsing process, the workflow remained accessible to users familiar with CATIA MSoSA. The author expanded the original Georgia Tech SysML infrastructure with new Stereotypes, modeling the ASTM International standards; furthermore, the author developed the Python companion code suite using AI assistants.

1.8 Structure of this thesis

The structure of this thesis is the following:

1. Chapter 1, the present Chapter, illustrates the topics and describes the reference engineering landscape; Section 1.1 provides a thorough description of the challenges in aeronautical certification regulations addressed by this thesis. Section 1.2 introduces the conceptual approaches used to build the MBSE framework. Section 1.3 presents methodologies, tools, and applications concerning the digitalization of certification requirements that may be found in the literature. Section 1.4 outlines the research gaps addressed by this thesis. Section 1.5 introduces the research questions derived from the research gaps. Section 1.6 provides a list of contributions and publications closely related to the thesis topic and authored by the PhD candidate. Section 1.7 lists the applications presented in this thesis and describes the PhD candidate's role in their development. Finally, Section 1.8 concludes this Chapter, describing the structure of this PhD thesis.

2. Chapter 2 illustrates the tools and the methodologies applied for building the SysML digital regulatory infrastructure, summarizing the MBSE principles that guided its development. Section 2.1 gives a thorough introductory discussion, specifying advantages and possible limitations of the approaches introduced by this PhD thesis.

Section 2.2 describes the tools and programming languages studied within the scope of the digital certification frameworks built and presented in this PhD thesis. Section 2.3 describes the methodologies and the rationale for building the SysML infrastructure in detail. Section 2.4 describes the code routines used for executing the certification tasks, enabling the creation of a complete certification artifacts database and the drafting of a certification flight loads report with the certification compliance material. Section 2.5 concludes the Chapter, briefly outlining and summarizing its content.

3. Chapter 3 describes the three digital certification frameworks presented in this PhD thesis. Section 3.2 illustrates the implementation developed in collaboration with POLIMI and POLITO, built using Eclipse Papyrus to create and export the SysML model encoding the regulatory information, and MATLAB/Simulink for parsing the model, loading the aircraft design architecture information, and executing the certification tasks. At the end of its execution, the most relevant output is a complete certification flight loads report. Section 3.3 presents a limited application of a more industry-oriented, practical automation approach exploiting the Activity Diagrams available in CATIA MSoSA, for executing the model without explicit user inputs. The execution still relies on external C++ routines, although it runs through commands sent during the execution of the Activity Diagrams included in the model. Section 3.4 describes the use case scenario built using a larger-in-scope SysML model, created by Georgia Tech researchers. The application efficiently exploits the certification information encoded by the model via an external Python substantiation code library of methods and classes, creating high-quality certification artifacts. Section 3.5 concludes this Chapter, describing briefly the three applications presented and their limitations.

4. Chapter 4 concludes this PhD thesis, providing a deep discussion of the advantages, the limitations, and possible future developments of the work presented in this thesis. Section 4.1 summarizes the research study described in this thesis; although no quantitative evaluation has been conducted in such an industry-oriented paradigm, a qualitative assessment has been included for completeness. Section 4.2 assesses contributions and limitations of the research works presented in this PhD thesis. Section 4.3 adds concluding considerations about the concept of digital certification and compliance demonstration.

5. Finally, several appendices have been included in this thesis. Appendix A describes the mathematical background of certification and substantiation tasks. Appendix B includes a selection of input data concerning the reference design architectures examined with the MBSE digital techniques presented in this thesis; it also presents a comparison between results obtained with different implementations of the MBSE digital certification conceptual framework, and a Table comparing the AMC provided by EASA CS-23, Amendment 6, and the analysis conducted in Eclipse Papyrus during the drafting of the Requirement Diagrams. Appendix C presents excerpts

from the certification flight loads report, generated via the MATLAB/Simulink digital certification framework. Appendix D provides additional information about the aircraft certification process, the certification material issued by regulators, and the consensus standards generally recognized by aviation industry stakeholders.

–2–
Tools, Methodologies, and Code

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This Chapter outlines the tools, methodologies, and companion code suites developed as part of the author's research. Section 2.1 describes the aeronautical certification process and the challenges associated with obtaining a new Type Certificate (TC). Section 2.2 introduces the tools used to build the SysML infrastructure. Subsection 2.3.1 summarizes the MBSE principles that guide the conceptual design of the digital certification framework. This Section presents a high-level, tool-agnostic description of the framework and defines its scope. Section 2.3 describes the proposed approaches and their practical implementation for aeronautical digital certification. Section 2.4 discusses the certification-calculation code suites that interface with the SysML infrastructure to form the overall digital certification toolchain. Section 2.5 concludes the Chapter by summarizing its main content.

2.1 Introduction

This Chapter introduces the tools, methodologies, and programming languages used to build the digital certification toolchain. Section 2.2 presents the selected SysML modeling environment used to develop the certification model, while Subsection 2.3.1 outlines the MBSE principles that inform the approaches described in Section 2.3. This Chapter also provides a high-level overview of the digital certification toolchains developed in this thesis, specifying the properties required for practical adoption and identifying the main limitations that may constrain their use as industry-grade certification tools [151].

The research conducted provided a foundation for defining the needs and requirements of an efficient digital certification framework [152], at least at the conceptual level:

- The digital framework should encapsulate certification information as comprehensively as possible by interpreting and encoding the relevant certification documents.
- Once certification information is encoded in a digital model, the toolchain should be able to parse and deploy that content into a programming environment, so that constraints, numerical data, and formulas can inform the certification calculation methods [153].
- The digital certification toolchain should execute certification operations, including calculations, verification, and validation activities [154]. During this phase, it should generate certification artifacts and supporting data and store them in a structured and accessible form.
- Finally, the toolchain should compile the certification artifacts into an appropriately formatted certification report. Although the report is generated as an output, it remains a document-based output; this reflects a hybrid implementation that departs from a fully model-centric workflow while retaining the advantages of model-based data management.

Automatic generation of a certification report is a valuable capability because it allows the user to review, correct, and supplement the output as needed. A fully model-based certification process would require the following conditions [155]:

1. The applicant and the regulator adopt a standardized method for encoding the reference certification documentation [156].
2. The applicant and the regulator agree on a data format for representing the aircraft configuration, the applicant organization and personnel, the list of certification demonstration items, and other relevant metadata. This format must remain compatible with the encoding approach described in item 1.
3. The applicant and the regulator monitor certification status and conformance with the agreed certification program by accessing the digital model directly, using the model as the authoritative representation of the process.

4. The applicant and the regulator rely on the certification artifacts stored within the model following execution, within the scope of the applicable rules. No separate documents are generated; certification evidence is retained in the model [157].

The developed capabilities increased the coherence and consistency of the certification database, also increasing the speed and the understandability of how the data are produced, their relations with the corresponding airworthiness and substantiation regulations, and potential errors in the outcome. These considerations represent an initial step toward developing technical solutions to RQ 1 and RQ 2, although the results remain limited to a research setting. The above discussion describes a novel application of MBSE methodologies to the process of substantiating requirements, i.e., producing the certification material needed to show compliance.

In practice, achieving this fully model-based scenario is unlikely in the near term [158]. A hybrid implementation that uses MBSE to manage certification data while still producing document-based outputs can provide a pragmatic compromise in an industry where safety, quality assurance, and standardization are often prioritized over process innovations aimed at reducing time and cost. For many small design and manufacturing organizations in general aviation, the risk and investment required to adopt new tools, skills, and workflows can outweigh the perceived benefits, leading them to accept less efficient certification processes for obtaining a new type certificate [159].

Accordingly, during the conceptual design of the MBSE infrastructure developed to enable the digital certification toolchain, the capability to generate certification documents in a partially automated fashion was retained. This decision reflects the continued reliance on document-based evidence in the current aeronautical certification landscape and the practical value of producing regulator-ready reports.

Researchers have previously addressed the digital encoding of airworthiness requirements and related certification processes using model-based techniques. In most cases, the formalization phase relies on a standardized modeling language. SysML is often selected for this purpose because it is extensible enough to represent certification-relevant structures and workflows while remaining sufficiently rigorous for capturing requirements and supporting verification and validation activities. For these reasons, SysML has proved useful for assessing requirement applicability to complex systems, establishing certification bases for technologically novel products, and structuring verification and validation frameworks [160]. In these contexts, it can support the development of digital toolchains for managing certification processes and associated compliance operations [161].

Despite these capabilities, scholars continue to debate best practices for defining standardized procedures, rules, requirements, and performance parameters for the digital models used in aeronautical certification [162]. Formal models can link domain knowledge—captured through the expertise of engineers and regulators—to a structured representation of the certification process. The modeling phase can also help identify an appropriate certification basis and clarify how requirements apply to a given design architecture. However, it remains difficult to trace a digital certification workflow back to the corresponding source documents with sufficient rigor. Because certification must be deter-

ministic, verifiable, validated, and repeatable, a trustworthy digital certification toolchain must provide clear and auditable links between requirements and the certification activities that demonstrate compliance [163]. If airworthiness and substantiation information were made available to applicants as digital models, the knowledge gap—and the associated risk of errors when determining requirement applicability—could be reduced, and the tracking verifications straightforward [164]. In practice, however, regulators typically publish certification specifications as PDF documents, and the same is generally true for standards, advisory circulars, and other guidance material [165].

In response to these considerations, the digital toolchain developed in this thesis aims to create a structured digital representation of airworthiness and substantiation rules that can be interpreted by dedicated code suites [166]. The textual content is digitally encoded within the model and can be exported and parsed to support subsequent certification and substantiation activities. The example shown is taken from the Eclipse Papyrus modeling environment, and it is representative of the general conceptual approach followed by the methodologies developed in this study. Once the appropriate set of SysML stereotypes has been defined, the modeling phase can begin. The resulting model can be exported in a machine-readable interchange format, typically UML/XMI/XML, and processed using search algorithms and parsing routines. The structure of the exported file preserves the logical organization and relationships of the certification material. This capability provides an initial, concrete answer to the research question stated in RQ 1: formal system models can transform certification requirements into structured digital information that can be interpreted by programmable certification and substantiation routines. This thesis demonstrates the approach through its application to EASA CS-23, Subpart C, described in Chapter 3. However, the proposed digital certification paradigm can also be adapted to other subparts, including Subpart F, which addresses onboard systems and equipment, and Subpart B, which addresses flight test requirements. During parsing, these routines generate structured variables that encode the relevant certification information. This information is then used to construct executable functions, thereby linking regulatory provisions directly to the certification calculations performed by the toolchain. This approach provides a practical strategy for improving the interpretability and traceability of certification operations to the source documents without requiring extensive manual investigation. The selected encoding strategy relies on the SysML formal language, which is widely used in model-based systems engineering.

In practical terms, the implementation can be summarized as follows:

1. Define SysML stereotypes that match the scope of the certification application and are sufficiently general to support verification and validation activities when required.

2. Specify a consistent representation for numerical data, formulas, constraints, and other certification-relevant information.

3. Select a programming environment to execute certification operations (e.g., calculations, checks, and artifact generation) and to interface with the exported model

data.

Constructing a digital model that encodes certification information typically requires translating the relevant paragraphs and subparagraphs of EASA and FAA requirements into model elements and populating their properties with numerical values, formulas, constraints, and related metadata [167]. As a consequence, this encoding phase is labor-intensive and time-consuming; nevertheless, it is a necessary step for transforming document-based requirements into a structured digital representation that can be processed by the toolchain.

At the current stage of development, this thesis does not rely on artificial-intelligence methods to automate the encoding task [168]. Two considerations motivate this choice. First, translating regulatory language into a formal, machine-executable representation requires consistent semantics and unambiguous interpretation, which is difficult to guarantee from natural-language input alone. Second, large language models typically produce probabilistic outputs that may vary with prompt phrasing and context; therefore, without a robust verification strategy, their use could reduce determinism and auditability in a certification-oriented workflow [169].

Artificial intelligence may nonetheless support and expedite the creation of a complex SysML representation of certification requirements when it is integrated into the modeling environment, for example, in commercial, licensed software that adopts SysML v2, such as the CATIA MSoSA environment [170]. In such settings, the embedded AI assistant can be configured to work with certification document repositories and to support interpretation during model-building operations, exploiting the KerML scripting features [171]. Even in these scenarios, however, concerns and limitations remain, and caution must be exercised when using AI during the modeling phase [172].

For small organizations, acquiring and tailoring an AI capability to a specific certification application may be impractical, particularly when using less integrated, open-source modeling environments (e.g., Eclipse Papyrus or Modelio). In this thesis, it is therefore argued that smaller organizations are more likely to rely on general-purpose AI tools, used externally and in a supporting role, rather than investing in a dedicated, domain-specific AI system [173]. In such a scenario, AI would more plausibly assist the construction of model's transformation infrastructure using tools such as the Epsilon Transformation Language for model-to-model or model-to-object transformations [174, 175], enhancing the execution capabilities of the model itself [176]; another possible task where AI can support digital certification activities is the generation of supporting documentation [177].

A general consideration applies to all the applications explored in this thesis. Large language models (LLMs) are now widely used in both research and industrial settings. Nevertheless, they are not yet suitable for directly interpreting airworthiness requirements for several important reasons:

- Airworthiness requirements rely on highly specialized terminology and domain-specific context. The effective use of LLMs would require a dedicated AI system trained on consensus standards, regulatory documents, certification reports, and

other relevant certification material. In practice, developing and maintaining such a domain-specific system can be prohibitively expensive.

- Even if an AI-based tool for interpreting requirements were available, its use could still be problematic. In particular, outputs generated in response to prompts requesting certification-basis assessments, document preparation, or other complex certification tasks are not inherently deterministic or repeatable. As a result, it would be difficult to justify the adoption of a specific certification basis without extensive verification and validation, thereby undermining the intended efficiency benefits of using AI.

- More broadly, establishing a certification process is a complex problem of knowledge synthesis. It requires evaluation of requirement applicability, identification of acceptable substantiation strategies, and selection of appropriate methods for demonstrating compliance. In this context, it is not sufficient to produce an answer; it is equally necessary to explain how that answer was derived and what conceptual framework underlies it. For this reason, AI-based tools would need to be coupled with additional mechanisms for traceability and explanation, which would further increase implementation cost and complexity.

- Finally, the creation of a SysML model can be regarded as a necessary step in the development of digital certification tools that may subsequently be integrated with other digital and computational tools. A well-structured, sufficiently rich SysML model of the airworthiness and consensus-standard documents could support and constrain AI-based tasks more rigorously, at least in initially limited-scope applications.

Overall, AI-assisted tools are likely to be introduced first on the modeling side, particularly in the creation and maintenance of large SysML representations of airworthiness rules for a wide range of certification applications. Future work will examine the use of SysML v2 features, including the scripting capabilities associated with its KerML-based core. Script-based model creation may be accelerated by LLM-assisted coding, given that assisted programming practices are already widely adopted in several engineering and industrial contexts.

In this thesis, constructing the SysML infrastructure is the primary task in establishing the digital certification framework. Therefore, execution capabilities are implemented downstream in the companion certification code suites. The main exception is the CATIA MSoSA application, which uses Activity Diagrams to automate the execution of a limited set of certification calculations and the generation of the corresponding certification artifacts, illustrated in detail later in this thesis.

Section 2.2 explains the rationale for selecting specific software tools and modeling environments for the digital encoding phase. With a focus on developing the SysML infrastructure, the research presented in this thesis draws primarily on experience with two OMG-compliant modeling environments:

- *Eclipse Papyrus*, an open-source UML and SysML modeling environment that supports the definition and extension of custom stereotypes. However, it provides limited native workflow-execution capabilities; therefore, automated verification and validation procedures typically require external automation mechanisms [178].
- Dassault Systèmes CATIA *Magic Systems of Systems Architect*, a commercial modeling environment that supports several formal languages, including SysML, and provides integrated capabilities for building a digital certification infrastructure. These capabilities include workflow orchestration, automated generation of certification artifacts and reports, and support for verification and validation activities.

Although the two environments offer comparable core SysML modeling capabilities, CATIA MSoSA provides a more integrated user experience and stronger built-in automation support, which can reduce implementation effort [179]. However, licensing costs may limit accessibility for smaller general aviation manufacturers. Open-source platforms can mitigate acquisition costs and remain valuable for research and prototyping, but practical deployment may require additional investment in integration, process governance, and long-term maintenance to achieve the robustness typically expected in industrial settings [180]. Section 2.3 elaborates on the MBSE methods developed in this study and explains how the selected tools were applied to implement them. Section 2.4 then describes how the resulting digital certification model is used to execute certification activities.

2.2 Tools

This Section describes the modeling environments selected for the conduct of this research study. Licensed tools, such as CATIA MSoSA, Sparx, and IBM Rhapsody, are extremely capable and relatively easy to include in existing industrial contexts and workflows. However, the costs associated with acquiring the licenses are very high and often not sustainable over time. Open source alternatives, although limited in many aspects, can be selected among software such as Eclipse SysOn, Eclipse Papyrus, Modelio, and Eclipse Capella. The deployability and integrability of this software into a stable industrial process are questionable; its use is often limited to prototyping new toolchains, introducing new digital tools and techniques [181].

The applications presented in this thesis use two of the environments introduced above: Eclipse Papyrus and CATIA MSoSA. The following Subsections describe the main characteristics of each platform.

2.2.1 Eclipse Papyrus

Eclipse Papyrus is an OMG-compliant modeling environment that supports the UML and SysML modeling languages [182]. It provides facilities for defining custom stereotypes and creating diagrams, thereby extending the expressiveness of the digital model. At

the time of writing, Eclipse Papyrus is also available in a web-based form that supports collaborative, cloud-based modeling; however, this capability is currently limited to UML use cases. SysML modeling is supported in the desktop (installed) version. In both cases, the environment provides a containment-tree view that allows users to inspect and navigate the model's hierarchical structure.

Papyrus is distributed through the Eclipse ecosystem and is released as open-source software. Key features include cross-platform portability, rapid deployment, support for user-defined plug-ins, and integration with other open-source tools. In principle, Papyrus can be used alongside other modeling environments, such as Eclipse Capella and Modelio, when additional modeling features or execution capabilities are required. In practice, however, integrating multiple tools into a single, reliable workflow can be complex and time-consuming, and it often requires expertise that is more common in software engineering than in typical design organizations. The AGILE consortium has reported work on integrated conceptual aircraft design workflows based on such toolchains, with results that support feasibility under constrained use cases [183].

Additional capabilities can be introduced through model-transformation and code-generation languages, such as Acceleo [184] and Epsilon [185]. These tools support model-to-model and model-to-text transformations and can be used to interpret model content and extract information according to user-defined rules [186]. Although such transformation infrastructures can automate portions of certification procedures, developing and maintaining complex, bespoke plug-ins is often time-consuming. For relatively small-scope applications, this effort is frequently difficult to justify. As a result, Papyrus is often used primarily for modeling, with limited in-environment execution and automation.

In the MBSE-based digital certification framework developed in this thesis, the SysML model is exported as a UML interchange file (e.g., XMI), which is structurally similar to an XML document [187]. Dedicated companion code suites can then parse the exported file using predefined extraction rules tailored to the certification activity of interest. This thesis adopts that approach: parsing the model and using its content to support certification operations. The details of the parsing workflow and the associated execution logic are presented in Sections 2.3 and 2.4

2.2.2 CATIA Magic Systems of Systems Architect

Dassault Systèmes is a well-established company that provides software widely used in aerospace design and manufacturing. It offers an integrated ecosystem of tools within a cloud-based, collaborative environment known as 3DEXPERIENCE. This platform supports multiple phases of the product-development workflow, from computer-aided design to manufacturing simulation and product life-cycle management. Its primary advantage is the consolidation of processes, data, and teams within a single environment.

At the same time, this level of integration can introduce practical constraints. Organizations may feel pressure to adopt a broader tool suite than their immediate needs require. In addition, once workflows and data management are tightly coupled to a single ecosystem, transitioning to alternative tools can involve nontrivial cost and effort.

Within this broader ecosystem, the aerospace users are well-accustomed to the CATIA portfolio, which includes software suites that support a range of activities, including computer-aided design (CAD), concept and ontology modeling, enterprise architecture and management, and, most relevant to this thesis, model-based systems engineering (MBSE). In 2018, Dassault Systèmes acquired No Magic, thereby strengthening its systems engineering offering within the CATIA and 3DEXPERIENCE environment. This integration aims to reinforce continuity between the system model and the physical product within its operational context.

Since the acquisition of No Magic, Dassault Systèmes has continued to expand the CATIA Magic product line—now marketed as CATIA *Magic Systems of Systems Architect* (MSoSA)—and to extend the tool ecosystem that supports model-based systems engineering. Recent releases provide support for SysML v2, including synchronized graphical and script-based modeling views [188, 189]. The graphical notation can improve the readability and interpretability of the model structure, whereas the underlying programming language, based on the newly introduced KerML infrastructure, can accelerate the creation and maintenance of large models that include hundreds of elements and relationships. In addition, the vendor has promoted AI-assisted capabilities within its SysML v2 ecosystem (e.g., for model querying and automation) with solid training on requirements and regulations interpretation, although such features still require careful governance when used for certification-related modeling activities [190].

The applications presented in this thesis were developed using multiple releases of the software, specifically the 2022 and 2025 versions. The author also obtained access to the Georgia Institute of Technology digital certification model through the courtesy of the researchers involved in that effort. This model supports ongoing work on MBSE-based approaches intended to address practical challenges in the United States aeronautical certification context. The relevant standards and methodological assumptions are introduced in the following Sections and Subsections.

2.3 Methodologies

This Section presents the model-based approaches and methodologies developed in this thesis for implementing a digital certification toolchain. The conceptual design phase of the MBSE certification framework is critical because it determines whether the toolchain can support a sufficiently broad and reusable set of capabilities for aeronautical certification. The model should be rigorous and detailed enough to encode high-level requirements, substantiation standards, and other certification-relevant information efficiently. At the same time, it should remain flexible, so that it can accommodate the range of engineering topics typically addressed during certification and, where appropriate, incorporate verification and validation activities.

The substantiation procedures in the consensus standards are typically algebraically simple [191, 192]; the primary challenge is not the mathematics but the consistent digital encoding and automated application of the prescribed rules [193]. Two examples illustrate

this point. First, the maneuvering limit load factor—the same formula is reported in Figure 3.3 as it is encoded in the SysML model—can be expressed as:

$$n = \begin{cases} 2.1 + \frac{24000}{W+10000} \leq 3.8 \;\; \text{with W measured in lb} \;\; (6.0 \text{ for aerobatics}) \\ 0.4 \cdot n \text{ on the negative side} \;\; (0.5 \cdot n \text{ for aerobatics}) \end{cases} \tag{2.1}$$

Second, the gust limit load factor can be evaluated as [194, 195]:

$$n = 1 \pm \frac{k_g \cdot U_{\text{de}} \cdot V \cdot C_{L_\alpha}}{498 \cdot \frac{W}{S}} \tag{2.2}$$

with

$$k_g = \frac{0.88 \cdot \mu_g}{5.3 + \mu_g} = \text{Gust alleviation factor} \tag{2.3}$$

$$\mu_g = \frac{2 \cdot \frac{W}{S}}{\rho \cdot \bar{c} \cdot C_{L_\alpha} \cdot g} = \text{Airplane mass ratio} \tag{2.4}$$

Here, U_{de} and V are in ft s^{-1}, ρ is in slug ft^{-3}, $\frac{W}{S}$ is in lb ft^{-2}, $\bar{c}$ is in ft, and g is in ft s^{-2}. The slope C_{L_α} denotes the slope of the wing-body configuration lift curve, measured in rad^{-1}. The standards specify gust-design values of U_{de} at key speeds, including the design cruise speed V_C, the design dive speed V_D, and the rough-air design speed V_B (for Level 4, high-performance aircraft). Although the functional relationship between n and V is straightforward, the central challenge in applying MBSE to these procedures is to encode the rules in a digital form and execute them accordingly, with the appropriate scope, conditions, and limits.

This observation applies to many calculation items in the consensus standards and extends to the verification and validation activities used to assemble auditable evidence of compliance [196]. Appendix A provides a more thorough description of these standard substantiation methodologies.

Subsection 2.3.1 introduces the MBSE methodologies considered in this thesis and their application to aeronautical certification. Subsection 2.3.2 describes the implementation of the digital certification framework, including the user-defined stereotypes and their properties, while Subsection 2.3.3 describes the data formats and exchange standards used for input/output handling within and outside the framework.

2.3.1 MBSE Techniques

Model-based systems engineering (MBSE) encourages engineers and domain experts to formalize a system's characteristics and properties in a model, typically developed and maintained in a digital environment. A principal motivation for MBSE is to reduce reliance on large, document-centric baselines by capturing requirements, design decisions, assumptions, and interfaces within a coherent model [197]. When maintained with

appropriate governance, the model can serve as an authoritative representation of the system and its development process and is often described in industry as a *Single Source of Truth* (SSoT) [198, 199].

In broad terms, MBSE frameworks rest on three core elements:

- A shared modeling *language* and vocabulary for expressing system information in a consistent form. SysML is a widely used example of such a language.
- One or more *modeling environments* in which the model is created, managed, and accessed. In more mature applications, these environments may be cloud-based and collaborative, supporting multidisciplinary development and controlled information sharing.
- A reference *methodology* that defines modeling objectives, best practices, and rules for creating, reviewing, versioning, and maintaining model content and associated digital artifacts [200].

Because this thesis adopts SysML as the reference modeling language, the formal application of MBSE requires explicit specification of system behavior and of the minimum set of requirements the system must satisfy. In addition, the model architecture and its parametric properties must be designed carefully so that the model remains consistent with the physical system and its intended operational context.

Developing a SysML model that can encode high-level airworthiness requirements (e.g., EASA CS-23 or FAA FAR Part 23) together with supporting consensus standards requires the introduction of user-defined stereotypes [201]. In SysML, stereotypes are defined through *Profile Diagrams*, which provide the mechanism for extending the language to capture domain-specific concepts. Within a profile, the modeler can define new stereotype elements and assign them properties by reusing attributes of existing SysML metaclasses and, when necessary, introducing additional properties to represent certification-relevant data and constraints.

At a higher level of abstraction, two user-defined stereotypes are sufficient to address this task:

1. A stereotype that encodes information from the regulatory document paragraphs and subparagraphs. For EASA certification specifications, this stereotype can be named *CS*.
2. A stereotype that encodes information from the consensus standards documents. In the applications discussed in this thesis, the model scope is limited to ASTM International standards; therefore, this stereotype can be named *ASTM*.

Defining an appropriate set of properties for these stereotypes can be subject to debate because the selection depends on the level of detail and rigor required by the certification application. Both stereotypes inherit from the SysML *Requirement* metaclass to preserve standard requirement semantics and traceability mechanisms. This choice supports the

consistent linking of certification statements to related model elements, including system capabilities, design features, constraints, and verification and validation evidence that must be addressed during the development and certification of the physical system.

The conceptual design of user-defined modeling elements is guided by the needs of the specific application and by the modeler's design choices [202]. In the Georgia Institute of Technology use-case scenario, the stereotypes used to represent the structure of the airworthiness rules were organized across multiple layers of abstraction and implemented with a higher degree of sophistication. In particular, several stereotypes and metaclass extensions were applied concurrently to elements that model the structure of FAA FAR Part 23 [203].

In that model, the root metaclass for the customized elements is the SysML *Block*, which is commonly used to represent physical systems but can also be applied to abstract artifacts such as documents. Multiple user-defined properties were then added to capture regulatory content, yielding a structured digital representation of the document. This design choice reflects the original purpose of the Georgia Tech model, which was developed to identify gaps in the regulatory framework, especially for the certification of novel powerplant technologies and propulsion architectures. Such gap analysis requires a more detailed model with richer semantics and greater navigability [204]. By contrast, the information-extraction workflows addressed in this thesis impose less stringent modeling requirements and therefore can be achieved with a simpler modeling phase.

For EASA certification specifications, one possible set of properties is as follows:

- A property named *certification level*, indicating the certification level of the aircraft design. This indicator captures the maximum number of occupants, from 0 to 19.
- A property named *performance level*, indicating the aircraft performance category. It can assume two values: *low* for aircraft with a never-exceed speed less than or equal to 250 kn or a maximum operating Mach number less than or equal to 0.6, and *high* for aircraft exceeding either threshold.
- A property named *aerobatics*, indicating whether the aircraft is approved for aerobatic operations.
- A property named *amendment*, identifying the applicable amendment of the certification specifications.
- A property named *engine type*, capturing the main technological characteristics of the powerplant, such as a turbopropeller or electrical installed powerplant.
- Additional auxiliary properties capturing other certification-relevant attributes, such as the number of installed engines.
- One or more properties named *TestValue1*, *TestValue2*, ... to store numerical data, constraints, boundaries, and formulas. In the initial implementation, this field was constrained by the capabilities of the Eclipse Papyrus environment and was

defined as a string or double vector with zero-to-one multiplicity, which limits its suitability for representing heterogeneous paragraph content. A more flexible definition, discussed in subsequent Sections, adopts a zero-to-many multiplicity with a single collector named *TestValue*.

- Finally, an additional set of properties (at least two) named *TestExpression1*, *TestExpression2*, ... to store mathematical expressions, including equations and ordered relationships. The underlying type for these properties is a string.

A similar set of properties can be used to represent paragraphs and subparagraphs from consensus standards, with the addition of fields that support document identification and navigation. For example:

- A property named *StandardID*, defined as a string, to identify the applicable standard.
- A property named *SectionID*, defined as a string, to identify the relevant Section of the standard.

2.3.2 Digital Certification

The SysML model consolidates certification information in a convenient *Single Source of Truth* (SSoT) and, once it is available, enables the start of digital certification operations [205]. From the perspective of the model itself, it is essential to define the input–output data flows and the data types accepted by the digital certification framework, particularly for aircraft configuration data and the set of applicable requirements. The ideal scenario is to access and execute the workflow entirely within the modeling environment, thereby avoiding the need to translate information across multiple data standards during certification tasks. In practice, however, it is not always feasible to rely on a single, all-encompassing environment. A robust digital certification framework should therefore provide explicit mechanisms for exchanging and managing data across heterogeneous formats and interfaces [206].

A primary objective of a digital certification framework is to reduce human intervention in the certification process to the extent that it can be done without compromising accountability and regulatory rigor [207]. Because certification and substantiation activities must follow a deterministic, auditable, and repeatable workflow, they should not depend on probabilistic large language model outputs as the primary automation mechanism [208]. The central issue is not the mathematical complexity of the calculations but the need for controlled execution, standardized data handling, and clear traceability from inputs and requirements to outputs [209]. For this reason, certification tasks are automated by coupling a highly formalized digital model with explicitly defined calculation routines, so that each result can be reproduced and explained in terms of the governing requirements and the executed procedures.

The conceptual design of the applications illustrated in this thesis explored three different case studies:

1. The first case study, developed in the Eclipse Papyrus modeling environment, relies on exporting certification information as a standard UML interchange file and parsing it with an external tool. In this application, the parsing strategy is a key enabling factor, because the stability and flexibility of the MBSE framework depend on how effectively the required information is extracted. Once extracted, the information must also inform calculation methods that substantiate the airworthiness requirements and generate the corresponding *Compliance Demonstration Items* needed to show compliance. The exported UML representation is structurally comparable to an XML document, with additional fields and elements typical of a highly structured, machine-readable format. This application formed the core of this research effort and used MATLAB as the external parsing and execution environment, with the modeling environment and the computational workflow implemented as separate tools.

2. The second case study examines an automated implementation of the MBSE concept for a digital certification framework. Although the SysML model is narrower in scope, it includes the minimum information required to initiate certification operations, in this case, the automated drafting of the maneuvering and gust flight envelope [210]. The workflow was developed using the CATIA MSoSA modeling environment in combination with external code suites. Activity Diagrams are embedded in the model to orchestrate the execution of certification calculations and to trigger external companion code suites written in C++ and Python. Input datasets are generated and passed to the calculation routines automatically. The resulting certification artifacts are then available for inclusion in the model and can be used to assemble a certification compliance report or other compliance demonstration items, such as the calculation results. In this case study, the workflow is largely contained within a single environment, in the sense that execution is orchestrated from within the model and the generated outputs can be managed as model-associated artifacts.

3. The third case study uses a large digital certification framework developed by researchers at the Georgia Institute of Technology using the CATIA MSoSA modeling environment. The main goal of this application was to incorporate a set of substantiation documents issued by ASTM International and to perform the calculations required to generate the maneuvering and gust flight envelope. Introducing these standards required new stereotypes to capture the relevant certification information; the availability of stereotypes with application-specific properties enabled the parsing phase, which was implemented using external Python code suites. The model was exported in a UML interchange format and parsed by code that scans for elements with non-empty properties and collects the extracted information into a JavaScript Object Notation (JSON) file. JSON was selected because it supports structured data representation and can be integrated readily into Python workflows as structured variables. Although this framework does not embed the calculation code within the model, it provides a flexible and robust extraction strategy, provided that the model elements preserve the required stereotype structure and populate

the corresponding fields.

Each implementation presented in this thesis serves as a proof of concept for the proposed MBSE digital certification. The first implementation offers comparatively extensive automated report-generation capabilities, but it is also the least stable during execution and is best viewed as a demonstrative prototype. Although the present thesis emphasizes Subpart C, the broader research effort also produced results relevant to Subparts B, C, and F [211, 212].

Within this workflow, the *Flight Loads* certification report represents the principal output. It provides an automated mechanism for generating successive report drafts, evaluating the effect of changes in aircraft geometric, aerodynamic, and powerplant characteristics, and validating the underlying certification calculations, within the flight loads calculations required by the regulators [213]. By contrast, assessing alternative certification bases is more challenging because, as discussed above, this toolchain is more sensitive to revisions and amendments in the underlying airworthiness rules [214].

From an industrial perspective, the second implementation of the MBSE-based certification toolchain is particularly attractive because it supports a higher degree of workflow automation [209]. The CATIA MSoSA environment provides *Activity Diagrams* that can be used not only to represent execution logic within the model but also to orchestrate multi-step workflows that coordinate data exchange and external computation. Once the infrastructure is in place, users can invoke selected Activity Diagrams to execute certification operations either individually or as part of a larger sequence, thereby populating a certification-artifact database with minimal manual intervention.

The use of a C++ companion code suite is motivated by the need for robust, maintainable calculation libraries that can be executed efficiently and consistently. In this configuration, CATIA MSoSA acts as the orchestration layer, while the C++ routines implement the substantiation calculations and generate the associated outputs for subsequent inclusion in certification reports and compliance documentation. C++ is often selected for safety-critical aerospace software because it supports fine-grained control of performance and resource management; however, Python has become increasingly common in aerospace engineering workflows, particularly for analysis, automation, and tool integration. This case study demonstrates that, even with a complex SysML infrastructure, Activity Diagrams can add workflow automation and execution capabilities on top of the model's core features and can connect the model to externally defined code libraries in a seamless manner. The third application presented in this thesis is compatible with the development of similar orchestration and automation capabilities [215].

Finally, the third implementation provides an extensive set of capabilities enabled by multiple SysML mechanisms. Its original purpose was to investigate gaps in the airworthiness requirements, including internal inconsistencies and cases of limited applicability under specific operating conditions. The model also represents the use of special conditions and incorporates automated scanning mechanisms for comparing aircraft configurations, functional characteristics, and domain-specific requirements and identifying where the certification basis is incomplete, insufficient, or not applicable. The

use case presented in this thesis leverages SysML's extensibility to support the extraction of certification data from structured model content, including text fields and tabulated information, a capability not implemented since the model's inception.

Although report generation was not implemented in this instance, future work could employ the CATIA MSoSA report wizard to produce certification documentation. While the initial development of report templates can be time-consuming, once a template is established, it can be maintained and extended with comparatively limited effort. In this way, the report-generation capability can support a hybrid implementation of the MBSE-based digital certification toolchain suitable for practical deployment.

The versatility of the digital certification framework can increase the productivity of certification personnel. Although the workflow can be designed to remain controllable and deterministic, it also introduces several practical limitations:

- **Adoption costs and organizational change.** Upfront costs—including personnel training, acquisition of facilities, software licensing, and the organizational change required to integrate the toolchain—can be a barrier to adoption [216]. This barrier may be reduced if regulators publish airworthiness requirements in machine-readable, model-based form using standardized (and preferably open) mechanisms. When authoritative models are issued by regulators, industry is more likely to invest in the capabilities needed to use them.

- **Model development and update effort.** Developing and maintaining the SysML infrastructure is not conducive to rapid deployment. When a certification task depends on standards that are not yet represented in the model, those standards must be encoded as SysML elements, and the execution routines must be updated accordingly. Sustaining this capability requires specialized skills within the organization (or reliable external support); otherwise, extending the model and toolchain within practical time constraints becomes difficult.

- **Toolchain integration and data governance.** Integrating the components of an MBSE-based toolchain is inherently complex. A data management system is required to control inputs and outputs, maintain configuration integrity, and make evidence accessible to all stakeholders. In parallel, organizations typically need to strengthen internal communication practices, because the mechanisms for accessing model-based information differ substantially from those used in document-centric repositories.

- **Hybrid workflow complexity.** Maintaining a hybrid approach—model-based data management combined with document-based deliverables—can introduce additional process overhead. If not carefully designed, this overhead may offset some of the efficiency gains expected from digitalization.

Several of these limitations reflect the current maturity of the tools and workflows and should be assessed through industry-oriented validation, including structured feedback from domain experts, certification practitioners, and regulatory personnel. A final concern

is the security and segregation of sensitive information from malicious threats, especially when enabled through an AI agent swarm; cloud-based, collaborative, digital environments are intrinsically more vulnerable to these hazards, and countermeasures may be too costly to acquire and maintain over time for small to medium design and manufacturing organizations. Section 2.5 provides a qualitative partial evaluation of these matters with a limited set of KPIs.

2.3.3 Data management system

This Section clarifies the information flows and data-exchange mechanisms through which MBSE tools and methods are applied to design the digital certification process. Because certification activities draw on multiple data sources, the framework must support consistent and traceable information exchange. To simplify integration across the digital certification infrastructure, the applications presented in this thesis rely on established, widely supported data formats.

With reference to aircraft architecture data, the input data handling can be summarized as follows:

- For the Eclipse Papyrus/MATLAB implementation, aircraft input data are stored in multiple Excel files organized across folders that correspond to standalone components of the MATLAB certification code suite. This structure supports a comprehensive end-to-end workflow while preserving the ability to execute individual certification tasks in isolation, for example, for verification and validation of previously obtained results. Full execution of the workflow typically requires less than ten minutes.

- For the CATIA MSoSA implementation, together with Python or C++, aircraft input data are stored in a structured format such as JSON. JSON files can be parsed with standard libraries and loaded as structured objects (e.g., dictionaries), preserving data structure while supporting interoperability across software platforms. In addition, import and export of these datasets can be automated within CATIA MSoSA through appropriately defined Activity Diagrams.

Certification information is encoded directly within the SysML model. However, enabling its use across the broader MBSE digital certification framework requires exporting this information into interoperable data formats. In practice, the SysML model is associated with a UML text-based interchange file that can be parsed using standard techniques. The parsing process produces structured data representations, either as in-memory objects or as auxiliary files, typically in JSON format.

For example, within the MATLAB-based implementation, the parsing routines generate structured MATLAB objects that contain the extracted certification data. In other applications, the parsing process produces external JSON files that can be read by subsequent routines, thereby enabling the creation of structured variables for executing certification calculations.

Certification artifacts and outputs can likewise be encoded using standard data formats. The applications discussed in this thesis primarily adopt JSON for this purpose, as it facilitates data exchange between the certification calculation routines and the CATIA MSoSA modeling environment while maintaining compatibility with commonly used software tools.

2.3.4 Final considerations

To conclude this Section, the following considerations summarize how the proposed digital certification framework may extend to broader certification contexts in aerospace and other complex-system domains:

- In principle, the MBSE architecture presented in this thesis can be applied beyond CS-23 Subpart C to other portions of the aerospace certification landscape. Extensions to Subparts such as F (Systems and Equipment) and E (Powerplants) suggest that a conceptually similar toolchain could support certification tasks for other complex systems (e.g., satellite ground communication equipment, environmental control systems, or satellite power-management systems). The main limiting factor is the need to extend the analysis and substantiation capabilities of the companion code suites so that they can interpret additional rules and generate the corresponding certification artifacts. In practice, this requires: (i) expanding the digitalized rule base and (ii) implementing the computational and documentation facilities needed to address new certification items.

- This thesis focuses on small, privately operated, normal-category general aviation aircraft. Application to transport-category aircraft is feasible; however, Part 25 certification typically introduces greater variability in acceptable substantiation approaches and a larger supporting standards base. Accordingly, the digital certification framework would need: (i) increased flexibility in rules and data representation and (ii) a substantially expanded repository of digitalized standards and guidance. In addition, the framework would need to support alternative substantiation pathways for the same certification item and provide a structured way to compare candidate approaches for a given aircraft configuration and certification objective.

- The framework developed in this thesis is oriented toward civil aerospace applications. The underlying logic could also be adapted to defense-oriented products, provided that the toolchain incorporates appropriate safeguards, security requirements, and process controls consistent with military-grade standards [217, 218, 219]. Such an adaptation would require strengthening the framework's governance, configuration control, and cybersecurity posture so that it can operate within procurement and assurance regimes applicable to defense programs [220]. In this context, Eclipse Papyrus can, in principle, be extended through custom mechanisms (e.g., safety- and security-oriented plug-ins) to support controlled access and data

protection [221]. By contrast, CATIA MSoSA is delivered as an integrated environment that typically includes built-in capabilities for access control and collaborative model management. When deployed in a cloud-based setting, these capabilities can support restricted editing privileges and improved containment of model data.

Difficulties can arise when representing the geometric characteristics of an aircraft configuration, particularly for non-standard layouts and unconventional arrangements of lifting surfaces. For example, horizontal and vertical tailplanes with atypical planforms or installations may require dedicated data structures and specialized procedures to compute areas and other geometric descriptors. Similar considerations apply to control surfaces beyond conventional ailerons, elevators, and rudders.

Although the current codebase supports limited variation in aircraft input data, the companion calculation routines must be extended to perform additional preprocessing steps before certification calculations can begin. Because aircraft geometry spans a very broad design space, comprehensive coverage is not practical; instead, the framework should define explicit scope boundaries and adopt extensible mechanisms for incorporating additional geometric cases as needed [222].

Another source of variability arises from certification and compliance requirements for unconventional empennage configurations, such as a V-tail. These designs can require specialized substantiation methods and calculations that are not always consolidated in a single reference but are instead distributed across multiple documents, including advisory circulars and, when applicable, special conditions. This dispersion complicates the identification and traceability of all required compliance activities and, in a document-based process, can contribute to gaps in the evidence base used to demonstrate compliance with high-level, performance-based requirements.

When generating a flight-loads certification report for a V-tail, the tail surfaces are often represented through projections onto the aircraft symmetry planes. The resulting projected areas and geometric parameters can trigger additional requirements and checks that depend on the configuration and on the derived numerical values. More detailed treatments of these issues are commonly discussed in the specialized literature on unmanned aerial vehicle (UAV) configurations, which frequently employ unconventional empennage arrangements.

In conclusion, the MBSE approach described in this thesis can be understood as a set of tools, methods, and practices that support communication and coordination among organizations involved in the certification of complex industrial products [223]. From both the regulator's and the applicant's perspectives, digitally interpreting and substantiating requirements can streamline the overall process, support safer and less burdensome introduction of new technologies, and enable a more adaptable certification workflow in which changes and modifications can be managed without disproportionate increases in cost or time to market [224]. More broadly, centralized management of such complex processes aligns with common innovation objectives pursued by public research programs and research centers worldwide, including reducing environmental impacts, improving operational safety, and developing digital capabilities that enhance the effectiveness of

public and private organizations [225].

2.4 Companion code suites

This Section introduces the substantiation and certification calculation suites developed to generate the compliance-item database required to demonstrate compliance. Three use-case scenarios are described in this thesis:

1. The first use-case scenario follows a relatively straightforward architecture. The modeling phase is conducted in the Eclipse Papyrus environment, whereas certification, substantiation, verification, and validation tasks are executed through external tools. The companion code suites were developed in the MATLAB/Simulink environment using an object-oriented design. The codebase is organized into packages that group methods and functions in a logical manner. Each package is designed to support standalone execution, enabling the user to analyze a specific certification task in isolation when needed. The codebase includes (i) parsing routines; (ii) preliminary calculations to define aircraft geometry and mass properties; (iii) flight-envelope determination; (iv) balancing-load calculations for the horizontal tailplane at the flight-envelope boundaries; (v) internal wing-load calculations at selected flight-envelope points, including panel-method evaluation of spanwise airloads on the main wing; (vi) maneuvering and gust load calculations for the empennages; (vii) powerplant-load calculations, when applicable; and (viii) report-generation routines that produce the principal certification artifacts generated by the framework. The *Flight Loads* certification report compiles the certification items generated during a single execution of the toolchain. The whole architecture is relatively simple and accepts the input data through a set of conveniently defined Excel files, providing an intuitive interface for personnel accustomed to a document-based approach.

2. The second use-case scenario leverages the execution and automation capabilities available in the CATIA MSoSA modeling environment. As a result, the supporting code suites are more complex and adopt a multi-component architecture implemented in more than one programming language. The certification application considered here—namely, the assessment of aircraft characteristics and design airspeeds and the generation of the maneuvering and gust flight envelope—is intentionally limited in scope. The primary aim of this case study is to demonstrate CATIA MSoSA automation features through the use of Activity Diagrams. In this workflow, Activity Diagrams trigger an automated export phase implemented through an external Python parsing routine, which generates JSON input files tailored to a C++ certification and substantiation code suite. Execution of the C++ routines is then orchestrated through additional Activity Diagrams. Overall, the supporting software can be understood as a modular toolchain in which Python manages model parsing and data preparation, while C++ implements the core certification calculations. All the calculation functions are thoroughly informed by certification data and

information extracted from the regulatory SysML model. This division of responsibilities is consistent with common aerospace and defense software-development practice, where strongly typed, compiled languages are often preferred for stable computational backends and where scripting languages are used for orchestration, automation, and data handling. In this implementation, Activity Diagrams and a minimal Python layer manage data flows, with the aim of maximizing automation while limiting development and maintenance effort. In the longer term, the objective is to extend the CATIA MSoSA-based workflow to encompass a broader set of substantiation operations, including automated assembly of a complete *Flight Loads* certification report.

3. The third use-case scenario resulted from a six-month collaboration with the Georgia Institute of Technology. It builds on the SysML-based digital representation of FAA FAR Part 23 developed by Georgia Tech researchers and was extended in this thesis with ASTM International standards through additional, custom stereotypes. In this case study, Python was used as the sole programming language; specifically, Python 3.9 was selected to maintain compatibility with the existing toolchain at Georgia Tech. Certification and substantiation execution were separated from the SysML environment: the model was exported as a UML interchange file and parsed during execution of the Python libraries. Certification artifacts were therefore generated entirely within the Python workflow, without requiring further interaction with the SysML model or the CATIA MSoSA environment. In future work, this workflow could be extended to reintegrate the generated compliance items and artifacts into the SysML model and to use the built-in report-generation facilities to produce a complete *Flight Loads* substantiation report. At present, the Python libraries cover the maneuvering and gust flight envelope and the balancing loads acting on the horizontal tailplane at the flight-envelope boundaries. Whereas the two previous implementations relied on conventional parsing approaches, a distinguishing feature of this implementation is that calculation routines are derived programmatically from the encoded rule content using established scientific and symbolic-computation libraries. In subsequent Sections and Subsections of this thesis, these features will be described in more detail. Standard Python data and visualization packages (e.g., `pandas`) are then used to generate certification artifacts, including tables and figures, which can support subsequent report assembly.

The resulting MBSE-based digital frameworks are sufficiently mature to support initial testing in a practical setting. Such deployment can proceed by integrating the toolchains with established document-based workflows, comparing their effectiveness and efficiency against legacy practices, and collecting evidence to quantify benefits. This process can also provide structured feedback on limitations, missing capabilities, and priorities for improvement.

2.5 Conclusions

This Chapter has presented the methodological approach used to implement an MBSE-based digital certification framework for small, privately operated Part 23 aircraft. It introduced the conceptual basis for a digital approach to aeronautical certification, identified key needs and requirements, and discussed potential limitations associated with this application of MBSE. The Chapter also examined the digitalization of the regulatory framework and provided an initial description of the framework components that implement certification and substantiation workflows. The Sections that follow, which address each MBSE implementation discussed in this thesis, provide additional detail on the supporting code libraries and their execution mechanisms.

Figure 2.1 presents a possible set of *key performance indicators* (KPIs) for assessing, at least qualitatively, the performance of the use cases examined in this thesis. This assessment also provides an initial, although partial, answer to the research question stated in RQ 3 and introduced in Section 1.8. The proposed methodology evaluates each use case according to five KPIs, using a three-level rating scale: *Low*, *Medium*, and *High*. The meaning of each KPI is explained below:

- The first KPI is **Adoption and Feasibility**. This indicator assesses the practical feasibility of implementing each of the three proposed methodologies. It considers the time, labor, expertise, and resources required to encode the certification and substantiation rules, develop the digital certification framework, and export and parse the digital model using a companion code suite.

- The second KPI is **Operational Efficiency**. This indicator assesses the level of automation achieved by the digital certification framework and the corresponding reduction in manual effort. The objective is not to eliminate human oversight, but to reduce the need for repetitive manual operations while preserving expert supervision of safety-critical decisions. A high level of operational efficiency therefore characterizes a framework that can establish and maintain the certification database, execute certification activities, and support the exchange of certification information with the competent authorities with minimal human intervention.

- The third KPI is **Robustness and Correctness**. First, this indicator assesses the framework's ability to remain operational and reliable when the regulatory material changes, for example following the publication of new amendments or revisions to substantiation documents. Second, it evaluates the accuracy, consistency, and repeatability with which the framework generates the minimum set of certification artifacts required to demonstrate compliance. Because certification activities are subject to rigorous regulatory review and may be repeated under different operating conditions, a robust and correct framework should produce traceable and reproducible results while minimizing the risk of errors.

- When applying MBSE methodologies, developers must first define the scope of the proposed solution. An excessively broad SysML model can become difficult

and costly to maintain. The **Maintainability** KPI therefore provides a qualitative assessment of the labor and resources required to update the model, preserve its consistency, and maintain the operational capabilities of an MBSE-based certification tool over time.

- From the applicant's perspective, it is important to determine the extent to which the proposed methodologies support the compliance-demonstration process. Key capabilities include maintaining traceability among requirements, organizing certification artifacts within a technical report, and presenting the generation of compliance evidence in a clear and understandable manner. These capabilities are essential for managing certification effectively and within acceptable cost and schedule constraints. The **Compliance Evidence** KPI therefore provides a qualitative assessment of the extent to which each proposed methodology supports the generation, organization, traceability, and communication of the evidence required to demonstrate compliance.

Figure 2.1: Proposed qualitative evaluation with a limited set of *key performance indicators* (KPIs). Adapted from [53] for illustrative purposes.

This set of KPIs is not intended to provide an exhaustive qualitative or quantitative evaluation of MBSE methodologies applied to certification. The discussion in this Chapter has highlighted the uncertainties associated with the industrial deployment of the methodologies developed in this study. A rigorous and scientifically sound evaluation would require controlled industrial trials involving certification practitioners, domain experts, and regulatory personnel. These trials should provide structured feedback and

quantitative measures for determining whether, and to what extent, the proposed methodologies reduce the certification burden. The evaluation of development, deployment, and maintenance costs may be performed by applying well-established paradigms from the literature [226]. At the time of writing, the available evidence is therefore insufficient to provide a quantitative answer to the research question stated in RQ 3; however, this thesis included a qualitative evaluation that is grounded in the experience and the labour effort involved with developing the methodologies, the tools, and the certification and substantiation infrastructure involved in establishing a digital certification framework. A broader discussion of this topic is provided in [53].

The three use cases presented in this thesis received different ratings in the qualitative assessment. The first digital certification framework, developed using Eclipse Papyrus and MATLAB, is a functional, research-oriented solution.

Although the SysML infrastructure is relatively straightforward to implement, the framework's operational efficiency, robustness, and correctness may not yet satisfy the requirements of industrial deployment. Eclipse Papyrus provides limited built-in automation capabilities, and the parsing procedure relies on regular expressions. This approach can introduce instability at the interfaces among the SysML model, the parsing routines, the automated extraction of certification information, the creation of executable functions, and the generation of certification artifacts. Consequently, changes in the regulatory material may significantly disrupt the Eclipse Papyrus/MATLAB digital certification framework. Although the framework is limited to EASA CS-23, Subpart C, maintainability remains a concern because changes to the SysML model often require corresponding modifications to the MATLAB parsing code. Nevertheless, the compliance material generated by this implementation is broadly aligned with the expectations of regulatory authorities. A significant factor affecting its adoption is the cost of acquiring and maintaining MATLAB/Simulink licenses, which may represent a substantial barrier for small companies operating in the general aviation sector. Further details on this implementation are provided in Section 3.2.

The other two use cases, both developed in the CATIA MSoSA SysML modeling environment, are conceptually similar but differ in their implementation. The CATIA/C++ framework uses the built-in automation capabilities of CATIA MSoSA to access certification and substantiation information without requiring the user to export the SysML model manually. The extracted data are transferred to an external companion code suite, which performs the certification analyses. The resulting certification artifacts are then imported into the SysML model and prepared for inclusion in a technical certification report using the integrated report wizard. Ideally, this implementation is attractive because it allows users to work within a single, integrated environment. However, several barriers to adoption must be considered:

1. The acquisition and maintenance costs associated with CATIA MSoSA can be substantial, particularly when personnel training is also considered. These costs may exceed the financial and organizational resources available to small design and manufacturing organizations.

2. Although C++ is a widely available, capable, and reliable programming language, developing a companion certification code suite in C++ requires specialized expertise and considerable development time. The associated personnel requirements and costs may therefore represent an additional barrier for small design and manufacturing organizations.

These considerations explain why this use case has received such a high score, despite the high costs associated with adopting and deploying the tools before they can be used successfully and practically. This use case is illustrated in Section 3.3.

The CATIA MSoSA/Python implementation addresses some of these limitations, although with a modest reduction in operational efficiency and robustness. Section 3.4 describes this implementation in detail. In this workflow, the SysML model is exported manually, while the certification and substantiation analyses and the generation of certification artifacts are performed externally. However, because the extraction process relies on structured stereotype properties rather than on textual pattern matching, changes in the certification material have a limited effect on the companion code suite. The manual export procedure slightly reduces operational efficiency, but experienced personnel can manage this limitation by supervising the execution of the digital certification framework.

At the time of writing, the MATLAB/Simulink implementation is the only digital certification framework presented in this thesis that can digitally generate a complete *Flight Loads* certification report. To conclude this Chapter, several theoretical considerations should be added regarding the limitations and expected operational outcomes of the methodologies presented in this thesis. Formal modeling languages such as SysML, supported by an appropriate modeling strategy, can provide a satisfactory digital representation of complex certification information, providing a first step towards answering RQ 2. However, the development of a digital certification infrastructure should also improve the accessibility, interpretability, and traceability of the processes and information involved. Further investigation is required to determine whether the methodologies presented in this thesis can fully achieve these objectives. Such an assessment should incorporate the experience, opinions, and evaluations of stakeholders involved in different phases of the general aviation certification process. One important theoretical consideration concerns the ontological interpretation of certification requirements from different stakeholder perspectives. Encoding these interpretations in a digital model can be particularly difficult when applicants and regulators must establish the appropriate certification basis for a new design. This challenge is especially significant for novel propulsion systems and aircraft configurations that incorporate advanced features not yet demonstrated under representative operating conditions.

–3–
MBSE Applications and Results

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This Chapter provides a detailed description of each MBSE-based digital certification pipeline presented in this thesis. Section 3.2 describes the tools and methods developed within the UNINA research program in collaboration with researchers at POLIMI and POLITO. Section 3.3 presents the second implementation, which introduces automation mechanisms for executing small-airplane certification tasks related to compliance demonstration and substantiation by introducing Activity Diagrams into the digital certification pipeline. Section 3.4 describes a third use case based on a larger SysML model of FAA FAR Part 23 regulations built by researchers at the Georgia Institute of Technology and

its use for automated execution of selected certification tasks. Section 3.5 concludes the Chapter by summarizing the main content and discussing the strengths and limitations of the proposed implementations.

3.1 Introduction

The application examined in this Chapter are briefly summarized in this Section as follows:

- Within the collaboration with researchers from POLITO and POLIMI, a SysML model was developed in the Eclipse Papyrus environment [139]. The author developed the SysML model in its entirety, from conceptualization to final use. Custom stereotypes were introduced to capture EASA high-level airworthiness requirements and the corresponding paragraphs and subparagraphs of ASTM International standards. For completeness, the model was designed to encompass a broad set of ASTM standards [140]. After the encoding phase, a MATLAB code suite was developed to parse certification-relevant information [227], use it in the subsequent calculations, and then generate the certification report. The author developed the parser and the Subpart C calculation subroutines; at the time of development, AI assistants were not available. The parser generated structured data in the MATLAB environment, which supported certification calculations and the automated creation of the required certification artifacts. The workflow culminated in a *Flight Loads* certification report, with numerical results presented as equations, tables, and graphs. The report structure can be extended to include bibliographic references, supporting information, and selected excerpts from the applicable airworthiness and substantiation documents. A key limitation of this implementation is the parsing strategy, which relies on regular expressions [141, 228]. Although regular expressions are practical and relatively easy to implement, this approach is sensitive to changes in the textual structure of regulations and standards [142, 143]. Consequently, the workflow is not robust across multiple revisions and amendments of the source documents. Subsequent implementations address this limitation by adopting a more structured extraction strategy.

- A second, more limited application was developed in the Dassault Systèmes CATIA *Magic Systems of Systems Architect* (MSoSA) environment. A conceptually similar SysML model was constructed, but it was restricted to a selected subset of paragraphs and subparagraphs from EASA CS-23 and the relevant ASTM documents. This investigation focused on using SysML Activity Diagrams to automate the execution of the digital certification workflow [144]. Specifically, Activity Diagrams were used to trigger model parsing, generate an intermediate file containing the extracted information, and pass the resulting dataset to a companion C++ code suite [145]. Additional Activity Diagrams orchestrated the execution of the C++ routines, which generated certification artifacts and the associated data products [229]. This application illustrates how CATIA MSoSA can support a highly automated

workflow that coordinates data flows, information exchange, and certification calculations. Furthermore, using the CATIA MSoSA built-in report wizard, the final output may be organized as a *Flight Loads* certification report suitable for compliance submission [146]. In this implementation, regulatory content is extracted from structured stereotype properties rather than through text-pattern matching. Because the properties store certification information in a standardized form, the extraction strategy is less sensitive to changes in document wording and supports improved robustness across amendments. The same logic is adopted in the final framework, applied at a broader scale.

- Finally, the author collaborated with researchers at the Georgia Institute of Technology (Atlanta, Georgia, United States). The Georgia Tech team developed a digital certification infrastructure in CATIA MSoSA by extending SysML with custom stereotypes [147, 148, 149]. Although this larger SysML infrastructure was not originally intended to execute certification calculations, it was used here in a use-case scenario to extract FAR Part 23 substantiation content and to inform a companion Python code suite. The Python routines parsed the SysML model exported as a UML file and generated input datasets containing formulas, numerical values, constraints, and boundary conditions. A set of Python classes then used these inputs to construct reusable calculation functions and to generate certification artifacts that can be assembled into a certification report. In addition, the CATIA MSoSA report wizard can support automated report preparation by integrating generated tables and figures into document templates [150]. Although the initial configuration of report templates can require substantial effort, the resulting workflow can be expanded and maintained with direct linkage to the SysML model. In this implementation, the use of structured stereotype properties supports stable extraction of certification data across certification scenarios. While parsing was not orchestrated through Activity Diagrams in this case, the overall workflow remains straightforward for users familiar with the CATIA MSoSA environment. The author acknowledges the value of this collaboration and the support provided by the Georgia Tech researchers.

3.2 The Eclipse Papyrus/MATLAB digital certification framework

This research program investigated an initial, comprehensive MBSE-based digital certification pipeline. At the core of this implementation is a concise yet robust SysML model that encodes EASA airworthiness requirements together with the relevant ASTM International consensus standards. The scope of this thesis allows only for describing the EASA CS-23, Subpart C, digital encoding procedure. Subpart B and Subpart F were addressed by POLIMI and POLITO researchers, respectively.

Although this research-oriented tool demonstrates the feasibility of the MBSE digital certification framework, it addresses the research question stated in RQ 2, and it represents

a useful reference or baseline to develop the other two MBSE applications presented in 3.3 and 3.4.

This Section describes the main features of the SysML model and the approach used to encode certification information. It then explains how the model is interpreted by the MATLAB/Simulink parsing routines developed for this work. Finally, it provides an overview of the certification tasks performed within the framework and describes how the report-generation component produces a *Flight Loads* certification report.

Subsection 3.2.1 describes the SysML airworthiness-rules model developed in the Eclipse Papyrus environment. Subsection 3.2.2 presents the execution routines implemented in MATLAB/Simulink. Subsection 3.2.3 then describes the main output of the MBSE digital certification toolchain: a complete *Flight Loads* report generated at the end of its execution. Finally, Subsection 3.2.4 presents a representative subset of the certification artifacts produced during execution and summarizes the framework's key characteristics.

3.2.1 The Eclipse Papyrus EASA CS-23 SysML model

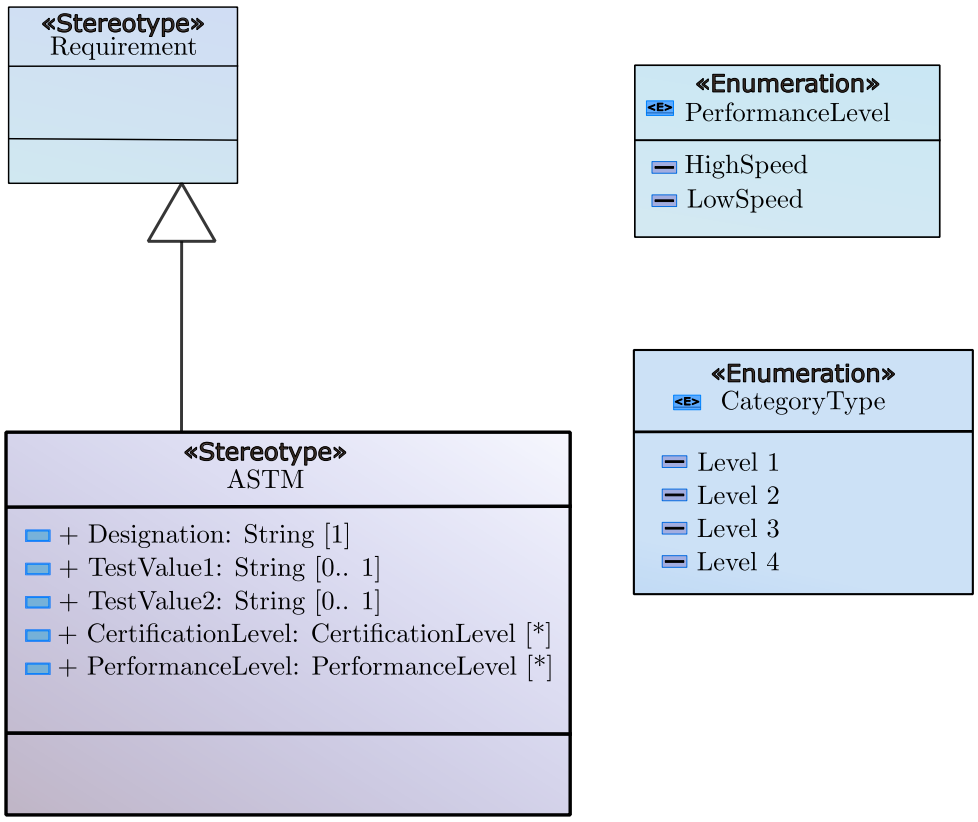


Figure 3.1: The first implementation of the Profile Diagram concerning the applicable certification rules. Adapted from [67] for illustrative purposes.

The Eclipse Papyrus SysML model adopts a hierarchical structure that can be examined through the containment tree. This structure mirrors the organization of the regulatory and consensus-standard documents, following the arrangement of paragraphs and subparagraphs. SysML supports this modeling logic through the *Package* element, which enables modular organization of the model and can also be used to separate distinct regulatory baselines (e.g., different amendments) within a single repository. In the present application, only CS-23, Amendment 6, is represented. The identifiers and titles of the regulatory documents are used to define the corresponding model elements, providing a consistent mapping between the source documents and their digital representation.

At the core of this SysML implementation are two stereotypes:

- The *CS* stereotype encodes the paragraphs and subparagraphs of EASA CS-23, Amendment 6, establishing a one-to-one correspondence.
- The *ASTM* stereotype models the paragraphs and subparagraphs of consensus standards published and maintained by ASTM International.

These stereotypes extend the standard SysML *Requirement* element with user-defined attributes and properties. Figure 3.1 presents the conceptual SysML Profile Diagram used to define the digital representation of an ASTM paragraph or subparagraph, as a representative example. The `ASTM` new SysML element inherits all the characteristics of the SysML *Requirement* standard block, with new properties designed for encoding certification information:

- `TestValue` is a property with zero-to-one multiplicity which may contain any numerical or textual information relevant to the subsequent compliance operations associated with that specific certification rule. It is defined as a standard string.
- `CertificationLevel` is a zero-to-many multiplicity property that denotes the EASA/ASTM certification level, based on the maximum number of occupants of the new design. It is defined as a standard string, assuming values from *Level 1* to *Level 4*.
- Finally, `PerformanceLevel` is a zero-to-many multiplicity property defined as a standard string, assuming two possible values, *High-* or *Low-Speed*.

For the *CS* stereotype, the following properties are defined:

Amendment: An integer that identifies the selected amendment. This attribute can be useful when more than one regulatory baseline must be referenced, for example in the context of major or minor changes to an existing Type Certificate (TC). In practice, applicants who obtained a TC under a given version of the airworthiness rules may need to apply that version throughout the product life cycle and to coordinate with the authority regarding the substantiation and compliance items required for subsequent changes. This property demonstrates that multiple amendments

can be represented within a single digital model. In practical contexts, the situation may quickly become chaotic using a document-based approach, and a TC can present hundreds and hundreds of modifications, changing design characteristics such as the CG-mass allowable combinations, maximum takeoff weight, and the allowed systems and equipment configuration. However, this modeling approach was considered overly broad in this work, because it increases the maintenance burden and can reduce model versatility and practical applicability. Therefore, only Amendment 6 was considered in detail.

Category: This attribute is a string that indicates whether the aircraft design is approved for aerobatic maneuvers. The MATLAB certification-calculation routines use this flag to distinguish between aerobatic and non-aerobatic cases and to apply the corresponding coefficients and limit boundaries when constructing the maneuvering and gust flight envelope.

Engine Category: This attribute is a string that indicates whether the aircraft design uses a piston engine or a turbine engine. Because the installed engine category affects powerplant loads and related load cases, the certification calculations must account for the selected engine type.

Constraint: This attribute can be used to encode constraints and boundary conditions associated with the certification item described in the modeled paragraph or subparagraph. In practical terms, it captures relationships between quantities that are expressed in regulatory prose through comparative language such as *no greater than*, *no less than*, and similar formulations.

TestValue: This attribute, implemented as *TestValue1* and *TestValue2*, captures the primary certification content extracted from the regulatory text. These properties are string-valued fields that can store numerical values, formulas, and other encoded expressions as they appear in the source paragraph or subparagraph. The parsing routines provide the mechanisms to interpret these strings and convert them into executable elements used by the certification calculation suite.

On the corresponding Profile diagrams, enumerations provide the allowable values for the properties. Following a similar approach, the *ASTM* stereotype includes the following additional properties:

Designation: This property is a unique string identifier for the applicable consensus-standard document. When useful, the identifier can be structured to reflect the document hierarchy (e.g., standard number, Section, paragraph, and subparagraph), thereby preserving the depth of the referenced content.

Certification Level: This property encodes the regulatory classification used for general aviation aircraft and distinguishes the applicable requirements as a function of the maximum approved seating configuration (i.e., the maximum number of occupants) for the proposed design.

Performance Level: This property is an enumeration, discriminating between high-speed and low-speed design architectures.

This SysML modelization is sufficient to capture the minimum set of certification and substantiation information to enable a proper MBSE digital certification toolchain.

Subsection 2.3.1 describes the properties used to encode the certification documents. Figure 3.3 presents an example of a SysML model organized according to a hierarchical structure, imitating the certification rules paragraphs and subparagraphs. The textual content of certification document paragraphs and subparagraphs is enhanced for clarity.

A notable aspect of this modeling effort is the inclusion of SysML Requirement Diagrams in the overall model structure. SysML supports both requirement tables and requirement diagrams, which can relate *Requirement* elements through several standard relationships. In this research project, only the *refine* relationship was used. This relationship indicates that one requirement provides a more detailed, concrete expression of another: the refining requirement elaborates and substantiates the more abstract requirement to which it is linked. In diagram form, the dashed dependency arrow conventionally points toward the more concrete requirement that qualifies the more abstract one.

This modeling activity yielded an important insight, which is discussed in related work completed with colleagues at POLIMI and POLITO. In particular, EASA CS-23, Amendment 6, does not always provide sufficient detail for applicants to trace high-level requirements to a complete set of relevant ASTM International consensus standards. More specifically, several CS-23 paragraphs list an incomplete set of references for applicants seeking acceptable means of compliance. This limitation can have the following consequences:

- Incomplete *Compliance Demonstration Items*, which may require additional review of consensus standards to identify acceptable methods for completing compliance activities and associated verification and validation tasks.

- Increased risk of omissions in the compliance evidence base, potentially leading to late redesign activities or additional substantiation effort to address gaps identified during certification.

- Reduced traceability and correctness in the structure of the certification database.

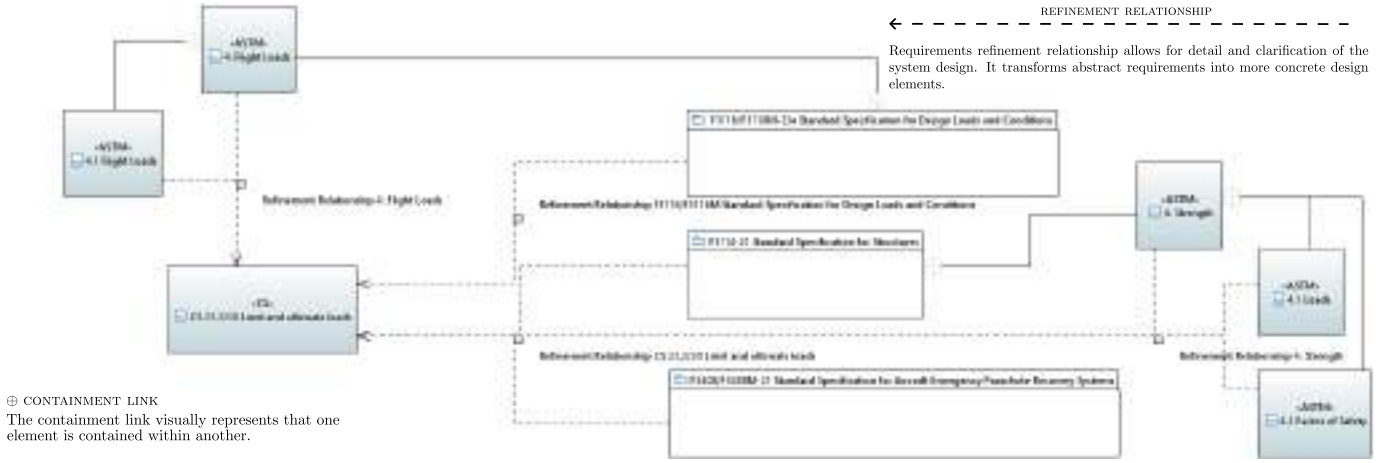


Figure 3.2: A SysML requirement diagram referring to a specific paragraph of the EASA CS-23 high-level requirements. The diagram shown is part of a larger set of diagrams drafted in the Eclipse Papyrus modeling environment. Requirement diagram for CS 23.2230 in Eclipse Papyrus, showing the refine relationships to the referenced consensus standards. Adapted from [67] for illustrative purposes.

Figure 3.2 shows the requirement diagram for CS 23.2230, Limit and ultimate loads. The *Refine* relationships link the high-level requirement to the consensus standards listed as its means of compliance. Building such diagrams for Subpart C exposed a recurring traceability gap. For CS 23.2230, AMC1 refers to the following:

> **ASTM F44 F3264-21 STANDARD SPECIFICATION FOR NORMAL CATEGORY AEROPLANES CERTIFICATION**
> *6.7 Limit and Ultimate loads*
>
> - **F3114-21** Standard Specification for Structures
> - **F3408/F3408M-21** Standard Specification for Aircraft Emergency Parachute Recovery Systems

Neither standard defines the limit load factors; these are specified in F3116/F3116M-23. An applicant relying only on the listed references would therefore assemble an incomplete compliance basis. Of all the fifteen Subpart C paragraphs modeled, six showed a comparable gap. Appendix B includes Table B.4 that summarizes the Subpart C modeled Requirement Diagrams; column AMC1 identifies ASTM documents included by EASA, while the analysis of every requirement and the comparison with the AMC2 led to the Requirement Diagram additional ASTM documents, listed in red in the last column. This thesis does not include Requirement Diagrams from Subpart B, although they were modeled by UNINA, who led most of the development efforts associated with building the SysML infrastructure. Requirement diagrams thus serve not only to record traceability but also to verify the completeness of the published means of compliance.

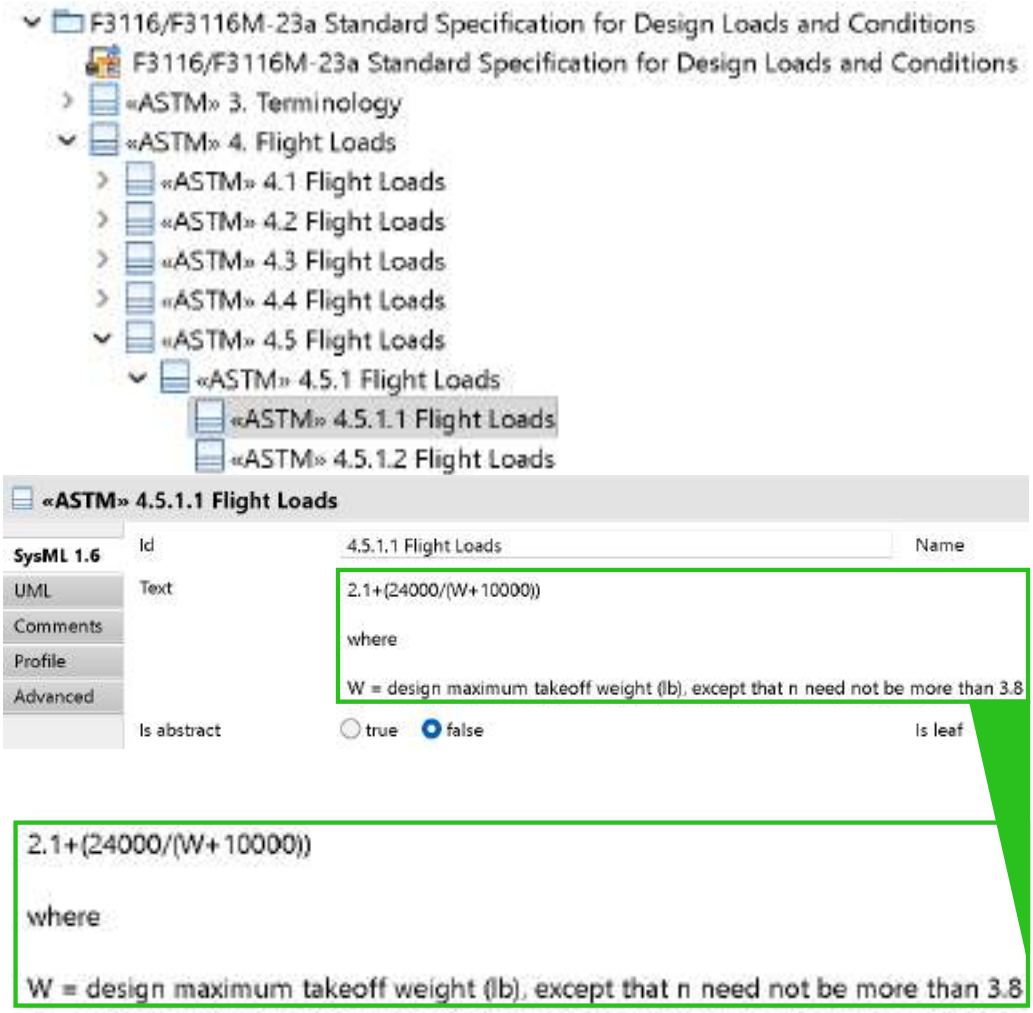


Figure 3.3: A general example of SysML hierarchical structure, as shown in the Papyrus containment tree. The textual content of the certification rule corresponds with the reference certification document. Adapted from [53] for illustrative purposes.

The incomplete correspondence between high-level CS-23 airworthiness requirements and the relevant consensus standards represents a substantive limitation that can hinder correct and efficient requirement application. Although discussions between applicants and the authority during certification-basis definition can mitigate this issue and often lead to a workable and reasonably complete compliance pathway, the underlying gap remains a practical concern for many stakeholders. In commercial aviation programs, similar shortcomings in document-centric traceability can impose additional coordination effort and can affect planning, evidence management, and the overall efficiency of certification activities. In this context, the MBSE approaches discussed in this thesis can support earlier identification of inconsistencies in the certification basis and can improve traceability between high-level requirements, applicable consensus standards, and acceptable means of compliance.

3.2.2 The MATLAB/Simulink substantiation routines

This Section describes the parsing routines and substantiation methods organized into separate libraries. The MATLAB-based certification toolchain developed within this research effort follows a modular design, with self-contained libraries that address specific flight-load certification tasks. A single MATLAB script, `RunFlightLoads.m`, orchestrates the end-to-end flight-loads workflow, including the generation of a complete certification report. This script initializes the execution environment and then invokes the individual certification routines in sequence. Users can execute the workflow from a single directory by running the main script, which calls the standalone components as needed. Throughout the execution of the code, a structured variable called *Aircraft* is created and progressively populated with new data, creating an effective digital data container that can be used for generating the certification report.

The execution of the MATLAB-implemented certification framework relies on two primary data containers:

- The *ASTM* data structure is created at the beginning of each run and contains the certification-related information used by the downstream routines. It represents a subset of the SysML content, including the data and rules required to evaluate the substantiation methods and generate the certification artifacts typically included in the *Flight Loads* report. To keep the scope of this implementation manageable, only ASTM International F3116/F3116M is considered. This choice does not constitute a fundamental limitation, because the approach can be extended to additional standards with limited conceptual changes. The structure includes function handles, numerical parameters, limits, constraints, and other certification-relevant information.

- The *Aircraft* data structure acts as a central container for aircraft-architecture data (geometry, mass properties, and aerodynamic inputs, just to name a few) and for the outputs of the Subpart C flight-load calculations. It provides a shared interface through which routines access the inputs required for execution and store intermediate and final results. At the end of the workflow, it supplies the report-generation routine with the information needed to assemble a complete *Flight Loads* certification report.

Additional auxiliary data structures support execution, but these are typically instantiated within lower-level scopes (e.g., inside functions or subsidiary M-files) and are not part of the main data interface.

The structure of the MATLAB library, presented in the order of execution, is as follows:

1. The *ExecMode* script serves as a container for execution flags and allows the user to run the certification routines either in test mode or in standard execution mode.

2. The *ImportModel* routine imports the SysML model for subsequent processing. In this workflow, the SysML model is exported as a UML interchange file, which is a

text-based representation that can be accessed using standard parsing methods. The current implementation relies on *regular expressions*, which are straightforward to implement and effective for extracting well-defined patterns. However, this approach is sensitive to changes in document structure and wording and is therefore insufficiently robust for industry-grade applications without additional controls and validation safeguards.

3. The *AircraftData* facility loads and evaluates the data required to initiate the certification calculations. It also performs supporting computations, such as estimating the maximum horizontal airspeed V_H at maximum continuous power. This routine includes powerplant data and other supplementary inputs.

4. The *AeroData* routines generate a coherent, simplified aerodynamic model of the wing-body configuration. This module provides the inputs required for subsequent evaluations, including balancing-load calculations at the boundaries of the maneuvering and gust flight envelope.

5. The *CalcAndStoreMassAndCG* routine performs preliminary operations required to initialize the certification calculations. These operations include evaluation of allowable center-of-gravity (CG) and mass combinations, derivation of the corresponding weight configurations, and computation of supporting quantities such as wing loading and empennage moment arms relative to the aircraft CG. The routine also assembles additional prerequisite data needed to define a consistent sequence of subsequent certification calculations.

6. The *GenerateFlightEnvelope* routine constructs the maneuvering and gust flight envelope as prescribed by ASTM International F3116/F3116M. It adapts the calculations automatically to the aircraft geometry and certification characteristics, applying the appropriate coefficients, limits, and boundary conditions.

7. The *BalancingLoadsData* routine computes wing-body and full-vehicle lift and the corresponding balancing loads acting on the horizontal tailplane, and plots these quantities as functions of airspeed.

8. The *CalcLoadDistributions* routine calls an external panel-method solver to compute the spanwise load distribution. In this work, the selected external package is *Open VSP*, which can be invoked from MATLAB through its API. This workflow supports multiple aircraft configurations (e.g., wing-body and full-vehicle) and enables assessment of propulsion-installation effects when the powerplant is mounted along the main-wing span. The powerplant and the propellers are supposed to work at their maximum continuous power setting.

9. A set of routines, orchestrated by the M-file `ShearBendingTorsion.m`, computes internal forces and moments for the most relevant points of the maneuvering and gust flight envelope. The outputs include spanwise distributions of shear forces, torsional moments, and bending moments.

10. After this step, two routines are executed: *LimitControlForces* establishes the applicable limits for control forces in accordance with the certification requirements, while *HingeMomentData* evaluates hinge moments based on the inputs defined earlier in the workflow.

11. The *HorizontalTailLoads*, *VerticalTailLoads*, and *PowerplantLoads* routines generate the corresponding databases of applicable and allowable loads for these three certification items.

12. Finally, the *ReportGenerator* routine executes the report-generation workflow and produces the technical *Flight Loads* certification report, exporting it in Word or PDF format.

3.2.3 The Flight Loads certification report

In this Section, a brief description of a typical *Flight Loads* certification report is provided. Among the implementations presented in this thesis, the Eclipse Papyrus/MATLAB MBSE digital certification toolchain is the only one that includes the automated generation of a technical certification report. Although this use-case scenario cannot yet be considered a fully deployable industrial solution and should be regarded as a research-oriented implementation, the insights gained during its development, together with this distinctive capability, represent a significant milestone. These outcomes support further experimentation in industrial environments and demonstrate the feasibility of applying digital certification approaches to practical design and manufacturing scenarios.

The report-generation template adopted in this work is intentionally general and therefore cannot be considered universally applicable to the diversity of certification cases encountered in the aviation industry. Nevertheless, at the time of writing, it represents a meaningful level of automation within the certification process. This process is inherently complex and involves numerous interdependent calculations and verification steps, which are often prone to human error when conducted using traditional, document-based methods.

The following provides an overview of the general structure of a *Flight Loads* certification report:

- The document typically begins with a *preamble*, which includes administrative information such as a revision history and a list of reference documents (e.g., previously issued certification reports already submitted to the regulator). The scope of the report is usually defined concisely at the outset.

- Subsequent Sections describe the principal characteristics of the aircraft design, including geometry, mass properties, and aerodynamic features.

- Several Sections address the flight loads and the data required for their determination, such as the design airspeeds. These Sections generally provide a detailed account of the calculation of maneuvering and gust limit load factors, the construction

of the flight envelope, and the evaluation of both symmetrical and unsymmetrical loading conditions. They also include the assessment of balancing airloads on the horizontal tailplane, fuel loads, and other relevant load cases (e.g., taxi or water loads, when applicable).

- Additional Sections focus on loads acting on the empennage, including those associated with control surfaces and maneuvering devices, in accordance with the applicable means of compliance defined in the substantiation documents.
- The loads associated with flap-deployed configurations are typically addressed in a dedicated Section.
- Auxiliary control devices, such as tabs or other movable surfaces, are examined either in dedicated Sections or within a single Section, depending on the level of detail required.
- A Section on powerplant loads is commonly included, particularly for conventional propulsion systems. This set of substantiation items applies to standard configurations and may require additional treatment when nonconventional systems (e.g., electric or hybrid propulsion) are considered under special conditions, which are beyond the scope of this work.
- The report concludes with a summary of the examined load cases and often includes appendices containing supplementary material that supports the main compliance evidence.

In practice, certification reports are typically subject to non-disclosure agreements. As a result, academic analyses can only provide generalized descriptions of their structure and content. Consequently, the analysis and results presented in this thesis should be interpreted as an approximation of an industrial scenario, developed within a research context to resemble the processes, data flows, and outputs of a typical certification industrial application. Nevertheless, the rationale underlying the development of the *Flight Loads* certification report was carefully verified against the relevant regulatory material governing load determination. Both established and non-standard substantiation procedures were considered to construct the compliance database required to demonstrate adherence to the high-level requirements of Subpart C. Where appropriate, additional guidance material applicable to transport-category aircraft was also consulted. The resulting set of operations implemented in the Eclipse Papyrus/MATLAB MBSE framework can therefore be regarded as technically sound and consistent with consensus-standard approaches widely adopted in the aviation industry [196, 194].

3.2.4 Results and conclusions

This Section describes an interesting use-case scenario where the MBSE digital certification framework established in the MATLAB programming environment is applied to several conventional architectures and to a non-conventional design architecture.

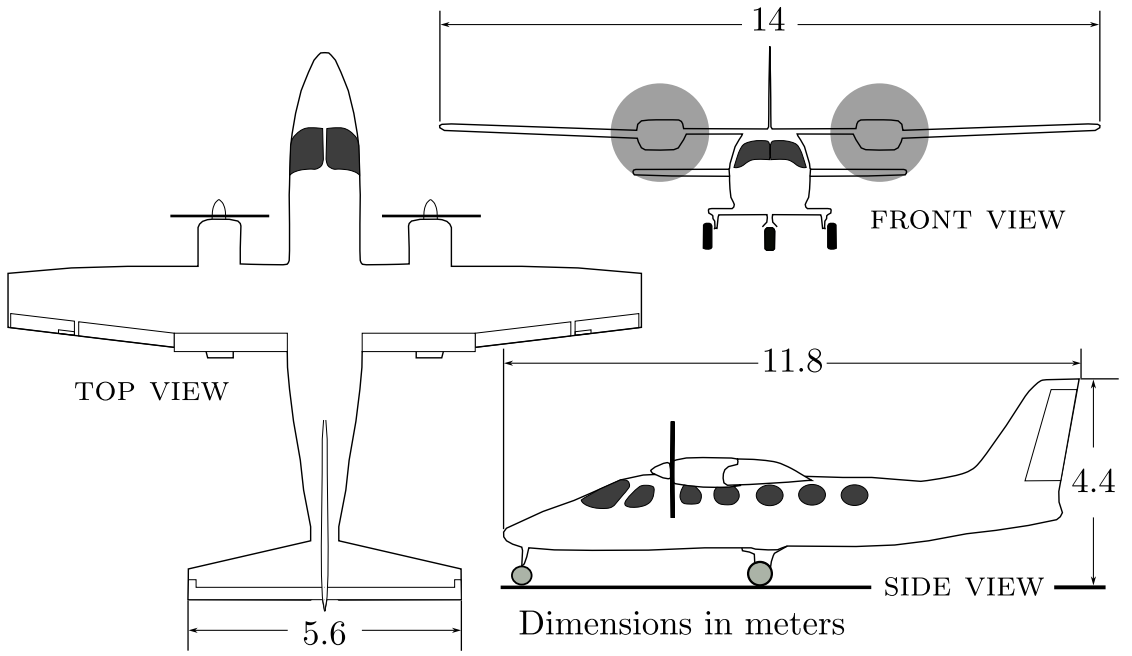


Figure 3.4: The Certification Level 4, twin-engine reference aircraft design architecture.

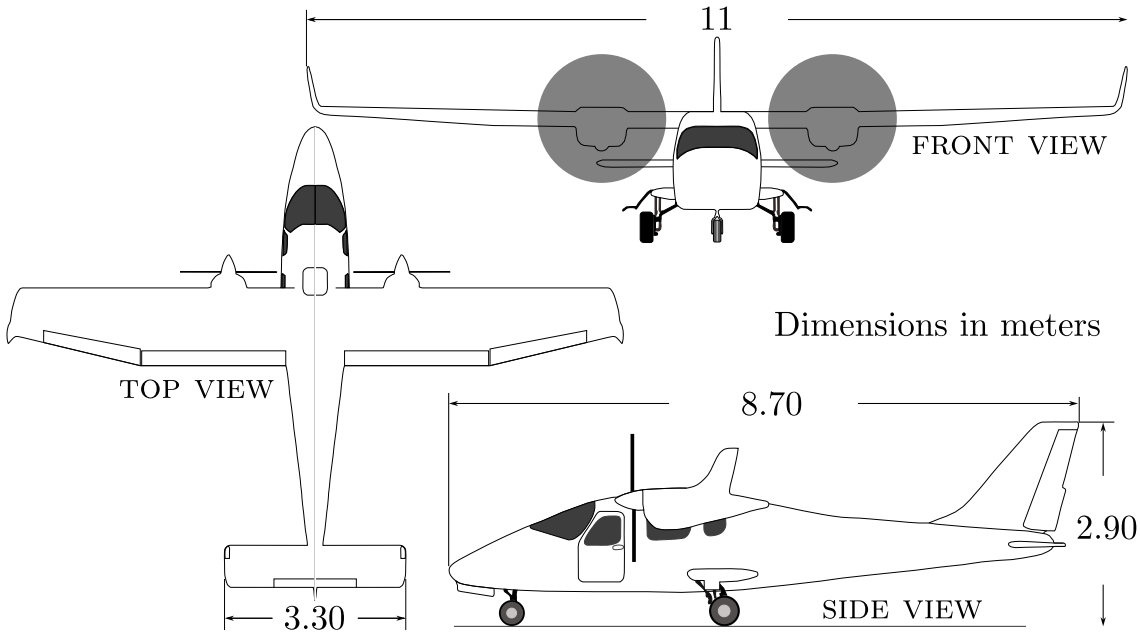


Figure 3.5: The Certification Level 2, twin-engine reference aircraft design architecture.

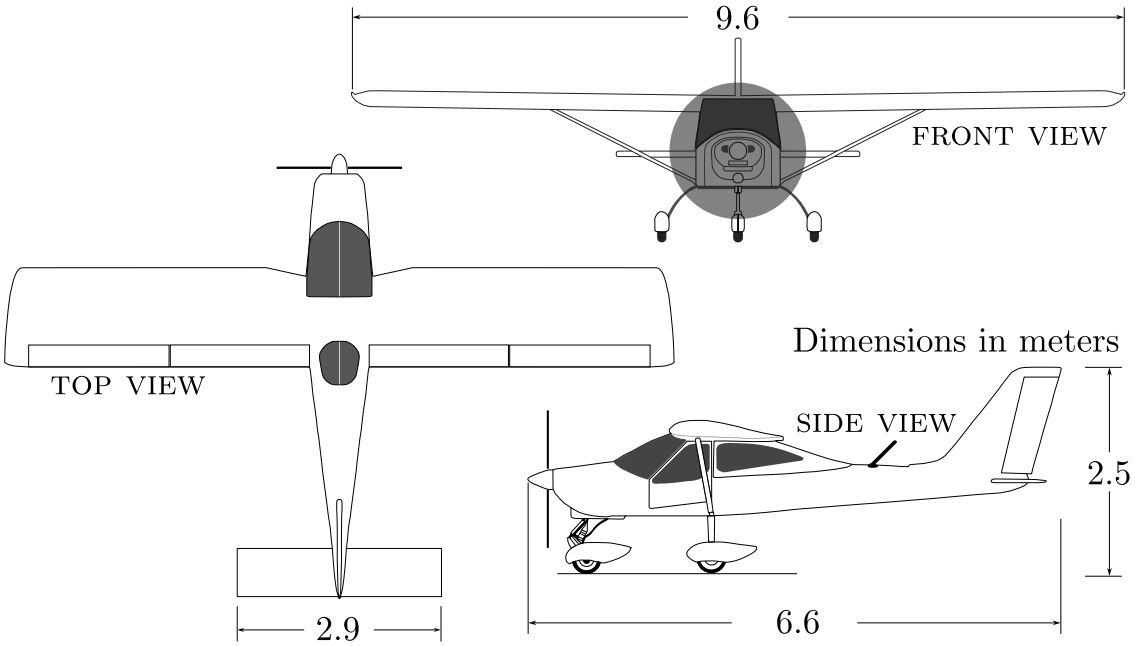


Figure 3.6: The Certification Level 2, single-engine reference aircraft design architecture.

Figures 3.4, 3.5, and 3.6 show the three conventional aircraft configurations used to test the MBSE digital certification framework. Their geometric characteristics are loosely based on three Tecnam aircraft and are intended to represent three different certification levels:

1. A low-speed, Level 4, twin-engine aircraft configured for short-haul commercial transport, with a maximum seating capacity of 11 passengers. Under EASA CS-23, Amendment 4, aircraft of this type were classified in the commuter category. Because these designs are typically subject to more demanding airworthiness requirements, this configuration is used as the reference case for activating requirements that do not apply at lower certification levels.

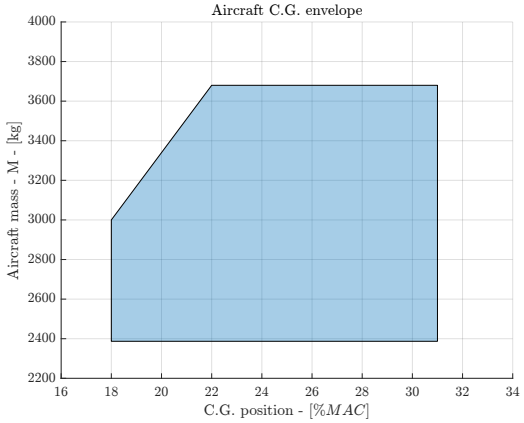
2. A low-speed, Level 2, twin-engine general aviation aircraft with a relatively low power-to-weight ratio, representative of a low-cost advanced trainer for pilots pursuing a commercial license. Although it is a complex aircraft with retractable landing gear, its characteristics are broadly representative of a common class of aircraft within the scope of EASA CS-23.

3. A low-speed, Level 2, single-engine general aviation aircraft that can be regarded as an intermediate case between a Very Light Aeroplane (VLA) and a small CS-23 aircraft. Its low mass and compact dimensions make it suitable for examining cases in which gust loads exceed maneuvering loads by a substantial margin.

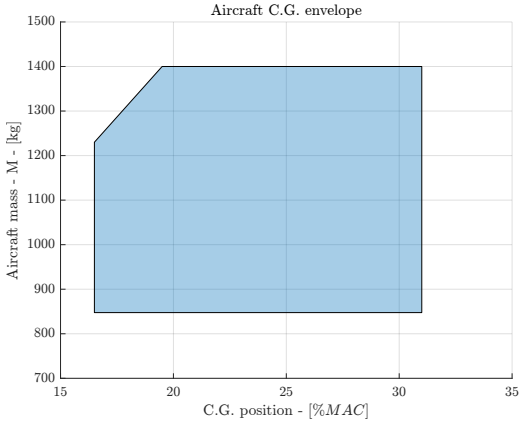
Most of the certification-related calculations presented in this thesis refer to these three

conventional configurations. Where additional aircraft are considered, this will be stated explicitly.

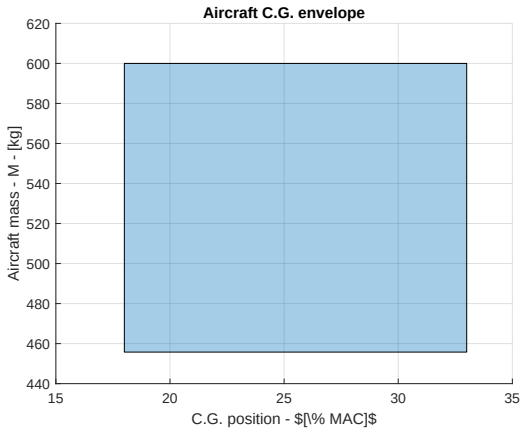
Conventional aircraft configurations also provide a useful basis for studying powerplant loads, evaluated in accordance with the applicable substantiation requirements. Powerplant loads associated with unconventional configurations are not considered in this work, because the scope of the MBSE digital certification framework does not include the application of special conditions.



(a) Level 4, twin-engine aircraft.



(b) Level 2, twin-engine aircraft.

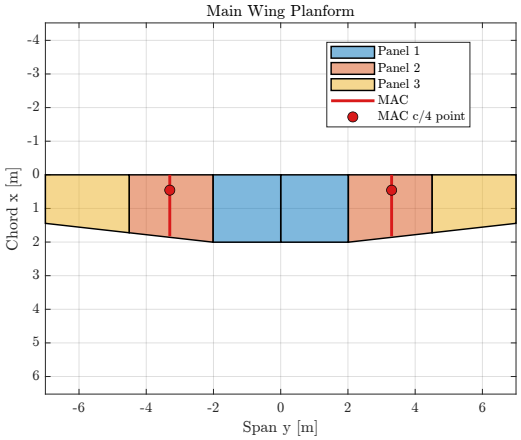


(c) Level 2, single-engine aircraft.

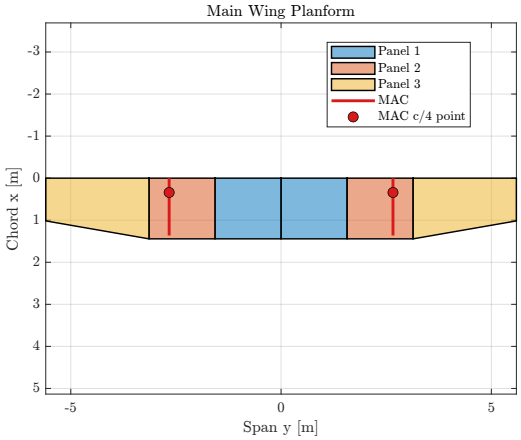
Figure 3.7: Position of the center of gravity, expressed in percentage of the mean aerodynamic chord, with respect to the aircraft mass. The mass boundaries are the operational empty weight and the maximum takeoff mass. The boundaries for the abscissa are the allowable most forward and most aft positions of the center of gravity.

Figure 3.7 illustrates the preliminary calculations performed by the MATLAB substantiation code, including evaluation of the selected mass (or weight) and center-of-gravity (CG) position. Once the user specifies the relevant boundaries for this analysis—namely, the operating empty mass, the maximum takeoff mass, and the most forward and aft allowable CG positions—the code determines whether the selected mass–CG combination

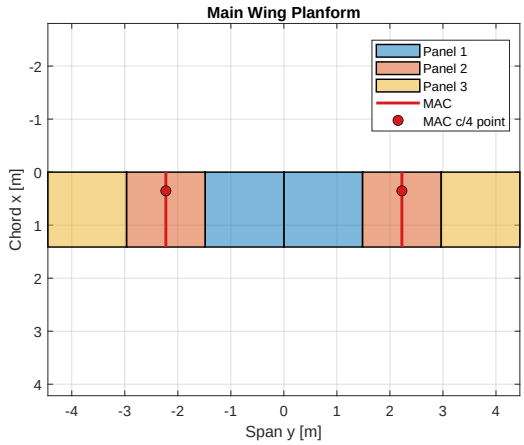
is admissible, that is, whether the point lies within the allowable CG–mass envelope. Figure 3.7a shows the allowable CG–mass envelope for the Level 4 aircraft, while Figures 3.7b and 3.7c present the corresponding envelopes for the twin-engine and single-engine Level 2 aircraft, respectively.



(a) Level 4, twin-engine aircraft.



(b) Level 2, twin-engine aircraft.



(c) Level 2, single-engine aircraft.

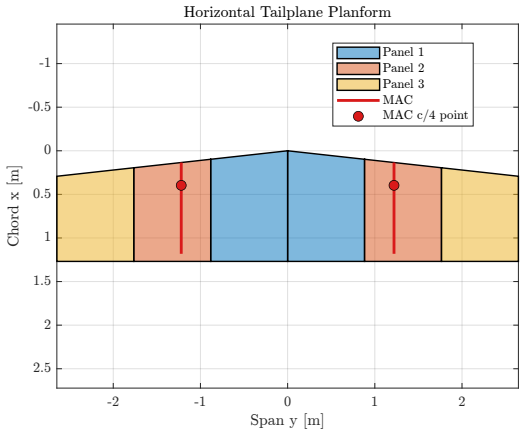
Figure 3.8: Wing planforms for all the conventional aircraft design architectures. All dimensions are expressed in meters.

Other preliminary calculations performed by the code ensure that the geometric characteristics of the aircraft's primary aerodynamic surfaces are represented consistently and used correctly in the subsequent certification analyses. Figures 3.8, 3.9, and 3.10 show the wing, horizontal tailplane, and vertical tailplane planforms for the aircraft configurations considered in this thesis. The procedures used to evaluate and represent these surfaces are generalized and can accommodate different levels of input data availability. The resulting object—whether for the main wing or for the empennage—stores the principal geometric characteristics of the surface, including sweep angle, dihedral angle, angle of incidence, washout, and related parameters. The graphical representation also includes the mean

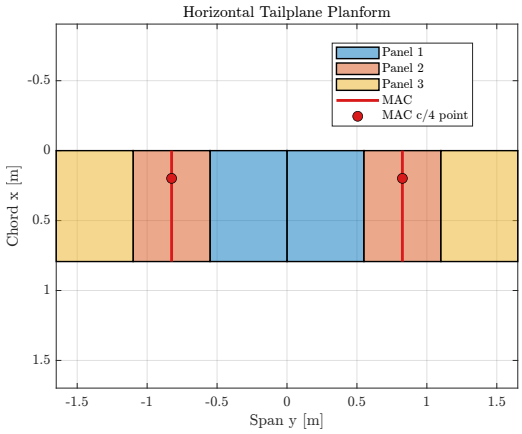
aerodynamic chord, positioned at its actual spanwise station y_{MAC}. As shown in the figures, the mean aerodynamic chord can differ substantially from the local chord at the same spanwise location. A red marker identifies the quarter-chord point of the mean aerodynamic chord.

These graphical representations serve two purposes. First, they allow the user to verify the consistency of the aircraft input data by inspecting the main aerodynamic surfaces. Second, they provide clear illustrative material that can assist the regulator during review of the certification report.

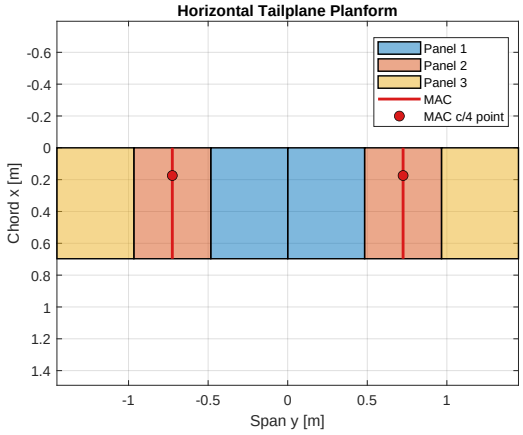
During execution, the code also evaluates the distance between the aircraft center of gravity and the aerodynamic center of the horizontal or vertical empennage. Together with the aircraft inertia properties, this quantity supports several calculations that are essential for demonstrating compliance with the applicable airworthiness requirements, particularly those governing maneuvering and gust loads acting on the empennage.



(a) Level 4, twin-engine aircraft.



(b) Level 2, twin-engine aircraft.



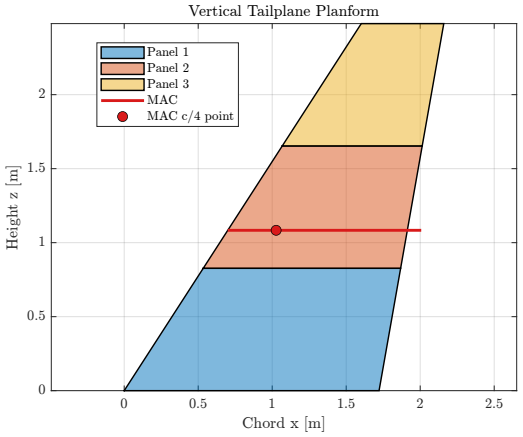
(c) Level 2, single-engine aircraft.

Figure 3.9: Horizontal tailplane planforms for all the conventional aircraft design architectures. All dimensions are expressed in meters.

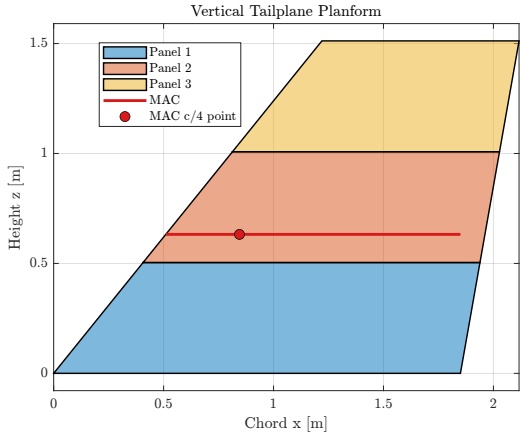
The MATLAB code can generate input instructions for external analysis tools through

a convenient application programming interface (API), such as Open *Vehicle Sketch Pad* (VSP). *Open VSP* allows users to create detailed three-dimensional models of aircraft components, including fuselages, wings, tail surfaces, nacelles, and propellers, which can then be exported for aerodynamic, structural, and multidisciplinary analyses. Once the MATLAB code generates the aircraft model, the VSPAERO solver—which is designed to operate on *Open VSP* geometries and thin-surface representations—can be executed to perform the aerodynamic analysis. The results can then be visualized in several forms.

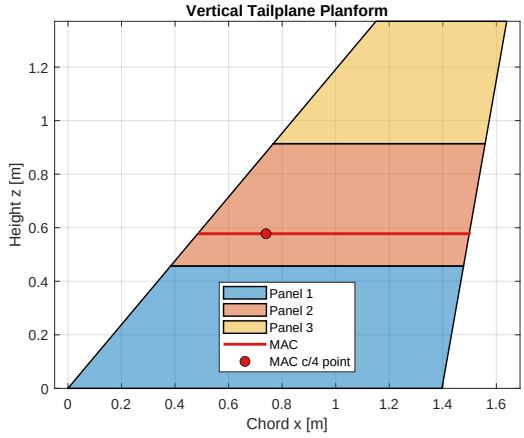
Figures 3.11 and 3.12 show the three-dimensional representation of the Level 4 twin-engine aircraft in the *Open VSP* environment, together with selected results from an analysis conducted over a configuration angle-of-attack range from −16° to 14°. The plots shown in the figures illustrate only a subset of the outputs available from the software, but they are sufficient for the purposes of a *Flight Loads* certification report.



(a) Level 4, twin-engine aircraft.



(b) Level 2, twin-engine aircraft.



(c) Level 2, single-engine aircraft.

Figure 3.10: Vertical tailplane planforms for all the conventional aircraft design architectures. All dimensions are expressed in meters.

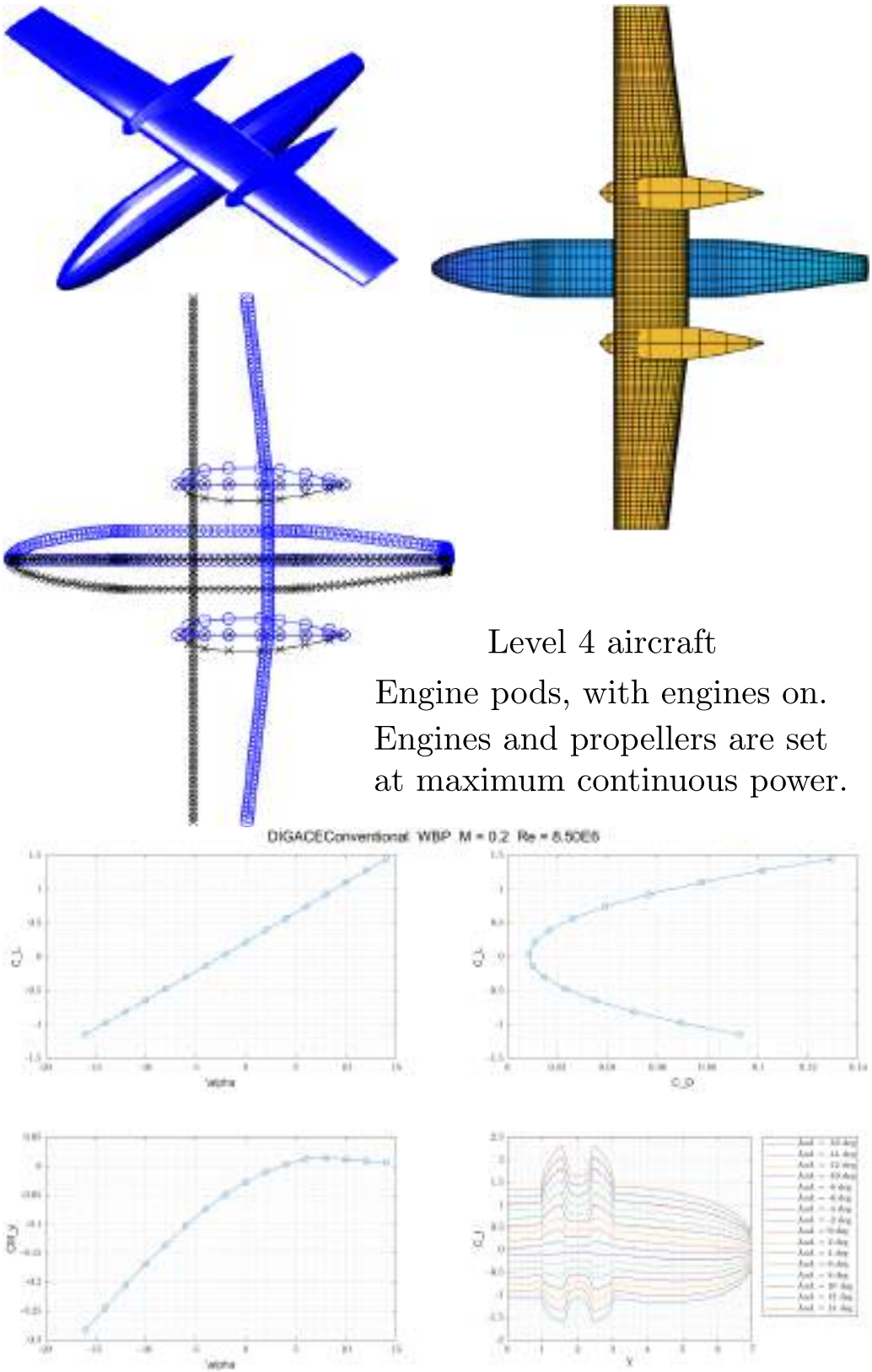


Figure 3.11: Level 4 aircraft design architecture with its engines working at maximum continuous power, loaded into the *Open VSP* analysis environment.

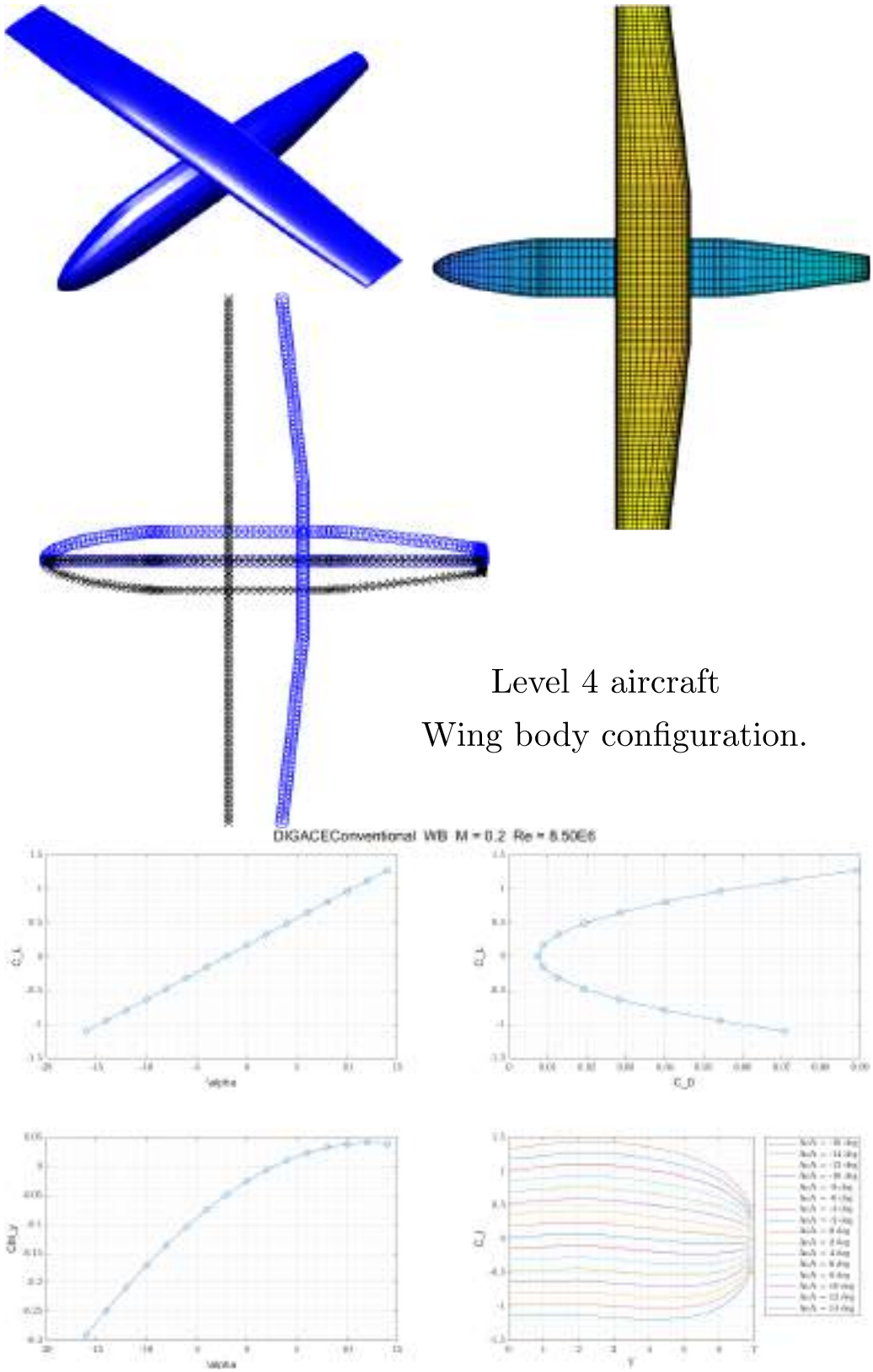


Figure 3.12: Level 4 aircraft design architecture. Its wing-body configuration as loaded into the *Open VSP* analysis environment.

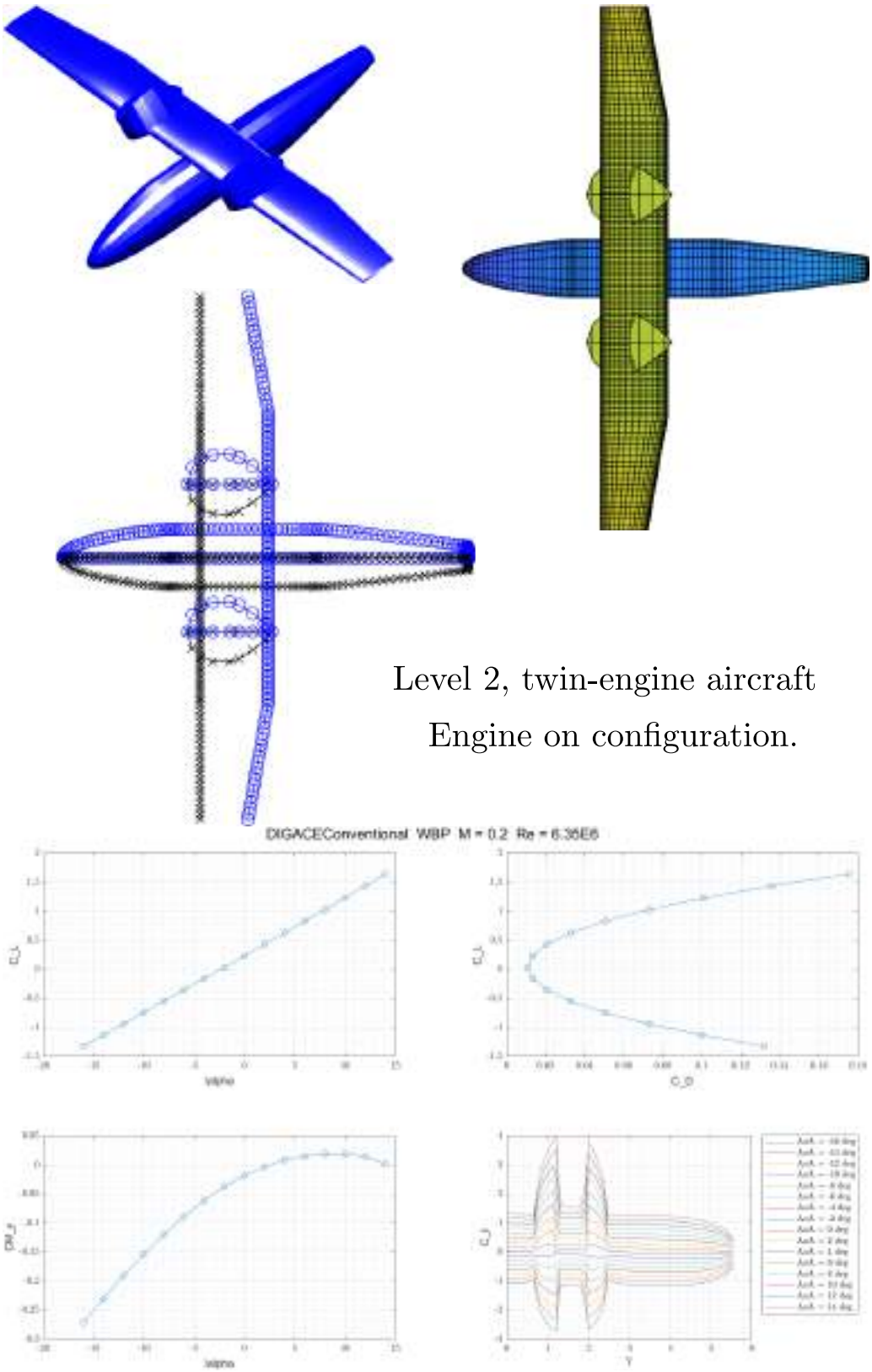


Figure 3.13: Level 2, twin-engine aircraft design architecture. Its engine-on configuration as loaded into the *Open VSP* analysis environment.

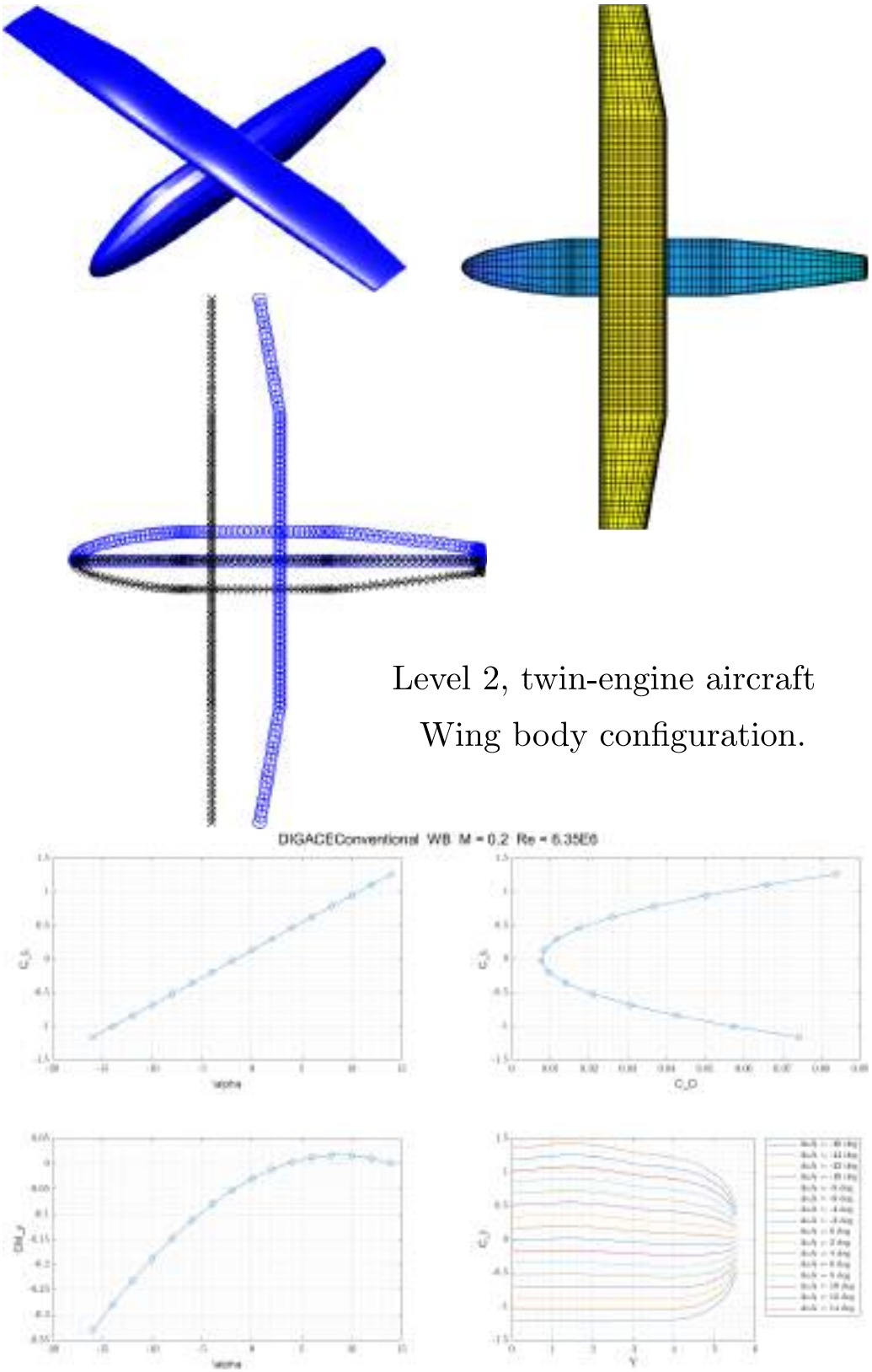


Figure 3.14: Level 2, twin-engine aircraft design architecture. Its wing-body configuration as loaded into the *Open VSP* analysis environment.

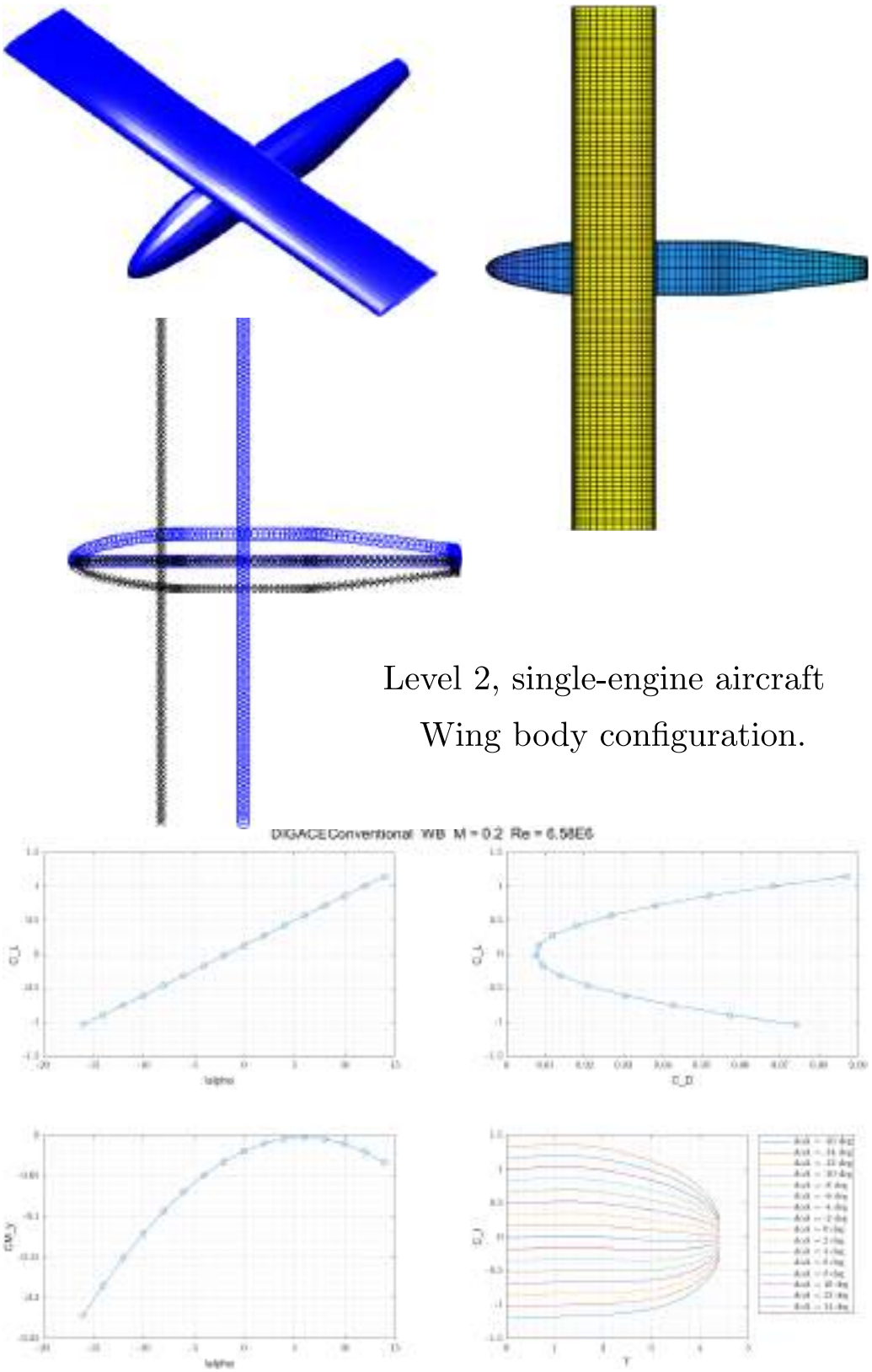
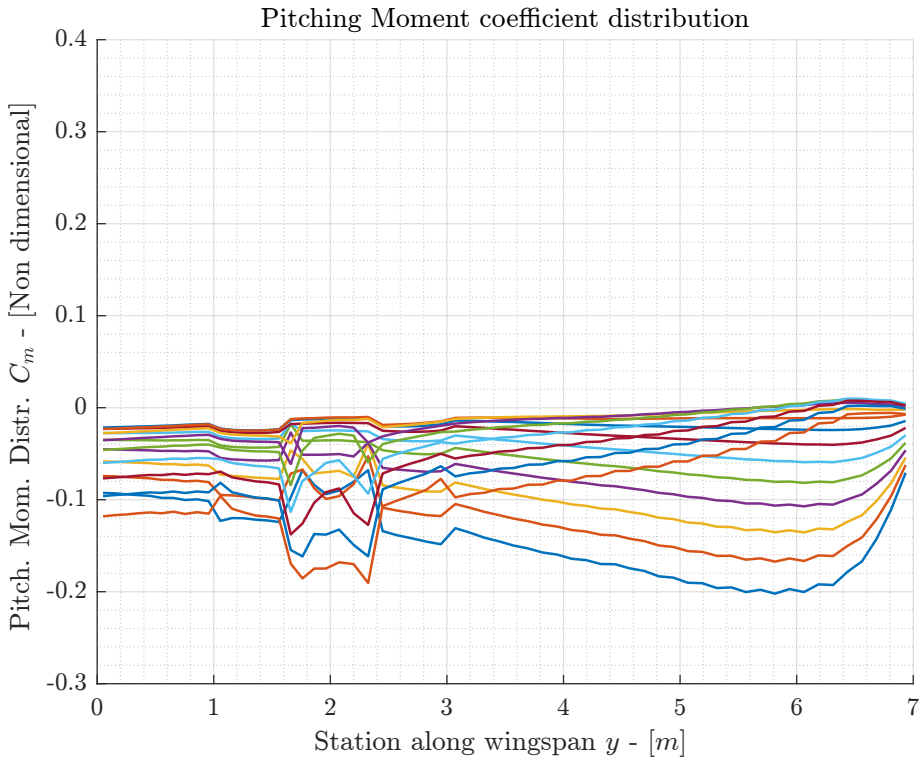
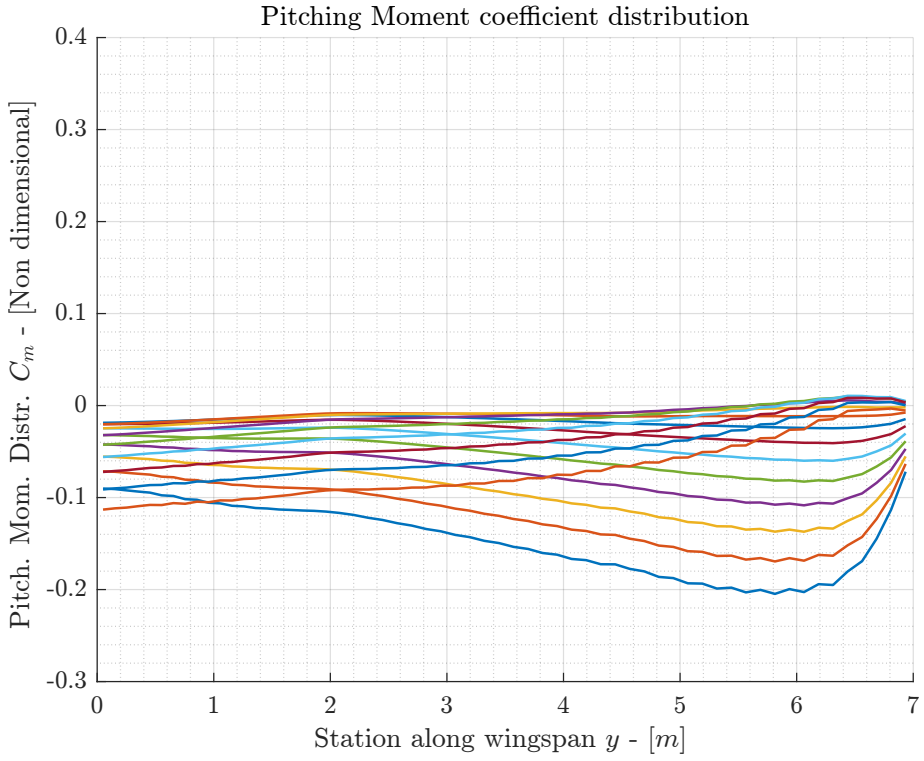


Figure 3.15: Level 2, single-engine aircraft design architecture. Its wing-body configuration as loaded into the *Open VSP* analysis environment.

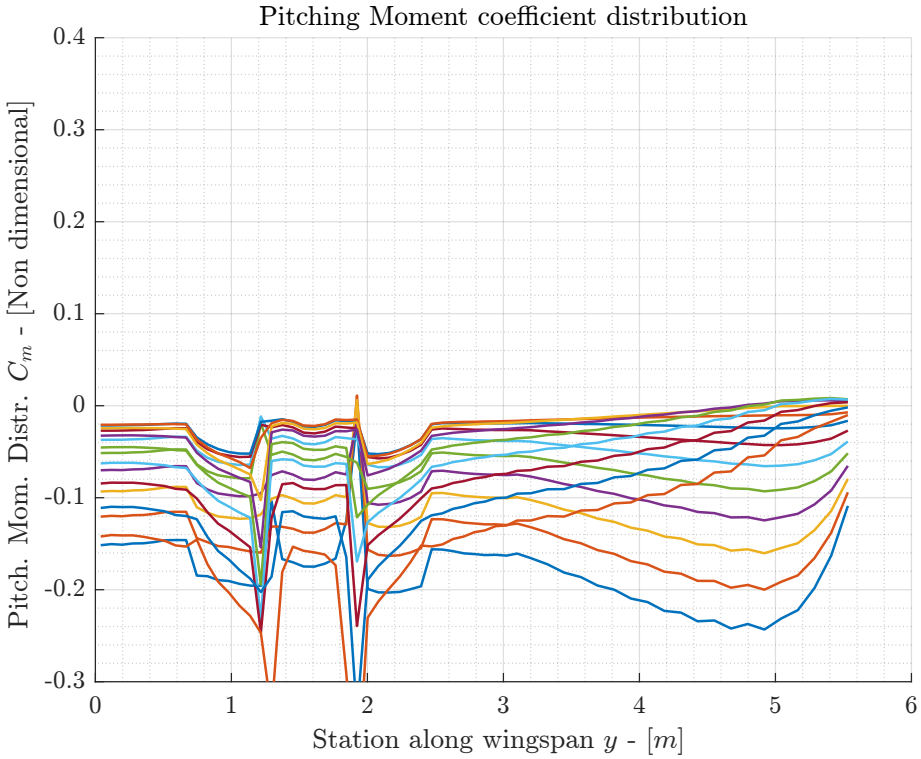


(a) Spanwise pitching moment distribution. Engine on.

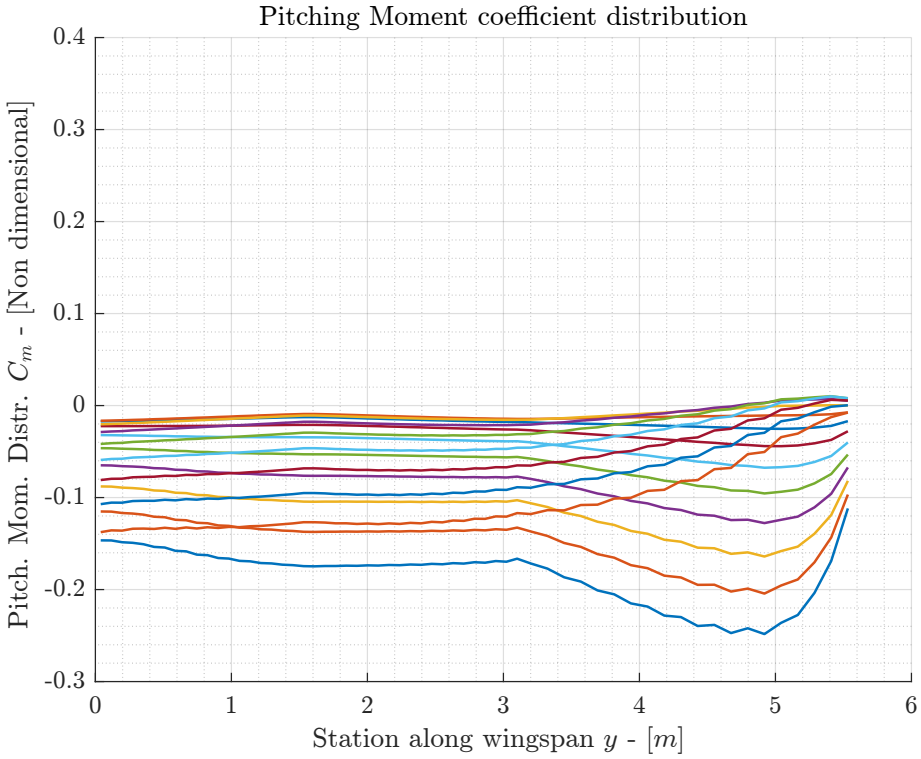


(b) Spanwise pitching moment distribution. Engines off.

Figure 3.16: Level 4 aircraft design architecture spanwise pitching moment coefficient $C_M(y)$ distribution, parameterized with respect to the angle of attack α.



(a) Spanwise pitching moment distribution. Engine on.



(b) Spanwise pitching moment distribution. Engines off.

Figure 3.17: Level 2, twin-engine aircraft design architecture spanwise pitching moment coefficient $C_M(y)$ distribution, parameterized with respect to the angle of attack α.

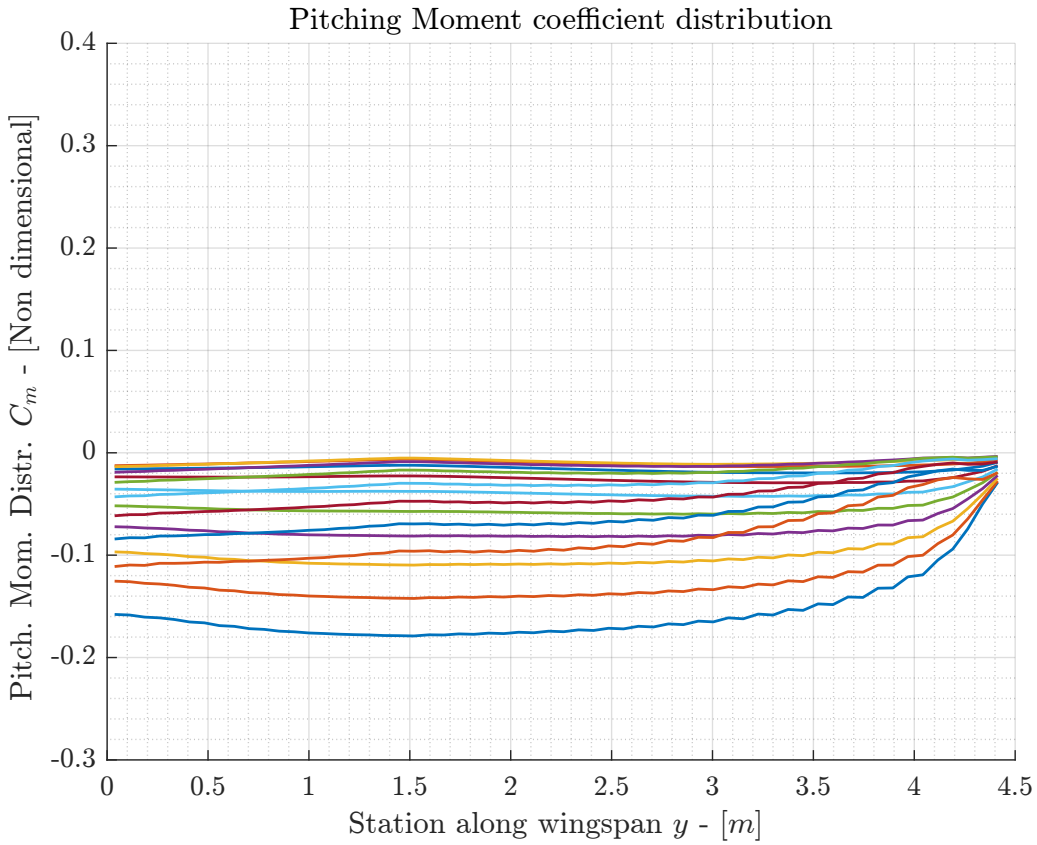
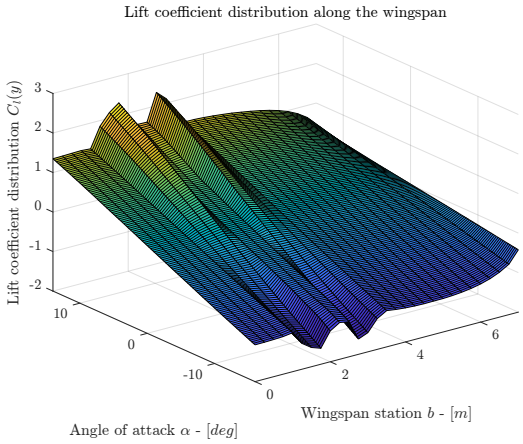


Figure 3.18: Level 2, single-engine aircraft design architecture spanwise pitching moment coefficient $C_M(y)$ distribution, parameterized with respect to the angle of attack α.

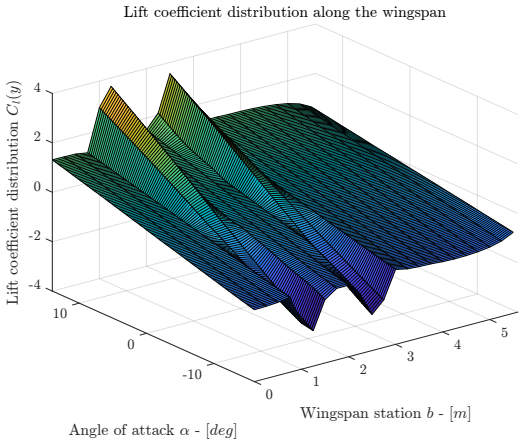
The reported quantities include the configuration lift coefficient $C_{L_{\text{config}}}$ as a function of α, the drag polar, and the configuration pitching-moment coefficient $C_{M_{\text{config}}}$ as a function of α. Of particular relevance for the evaluation of internal wing loads is the spanwise lift-coefficient distribution $C_l(y)$, where y denotes the spanwise coordinate and the distribution is parameterized by α. Comparison of the engine-on and engine-off cases highlights the aerodynamic effects associated with the propellers and engine nacelles.

Figure 3.16 illustrates in more detail the spanwise pitching moment coefficient $C_m(y)$ distribution, parameterized with respect to the angle of attack α. Figure 3.16a reports the engines-on case, while Figure 3.16b shows the engines-off case. The two scenarios offer the reader a clear picture of the potential of a digital certification tool such as the one designed in the context of this research project, which aims to provide the final user with a practical, immediately deployable digital certification tool.

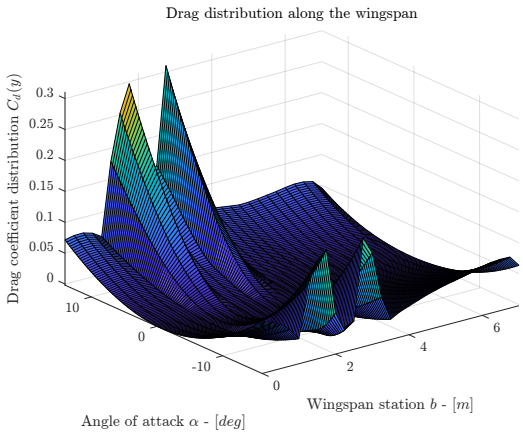
Figures 3.13, 3.14, and 3.17 are included for completeness and reinforce the idea of a generally applicable digital certification framework that can be reused with different aircraft design architectures.



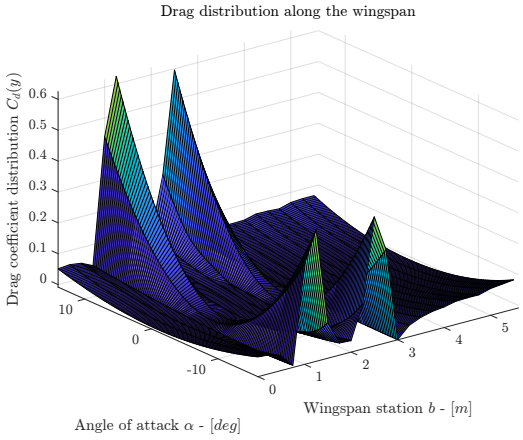
(a) Level 4, twin-engine aircraft. $C_l = C_l(y)$



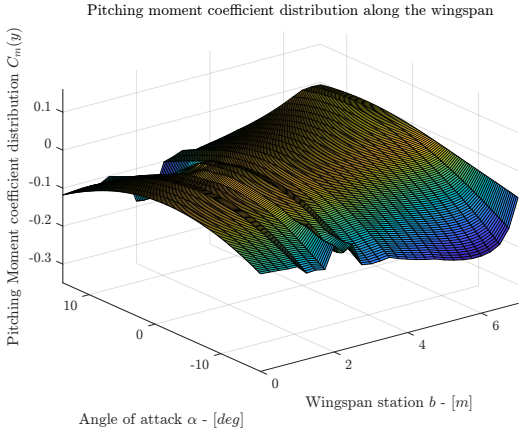
(b) Level 2, twin-engine aircraft. $C_l = C_l(y)$



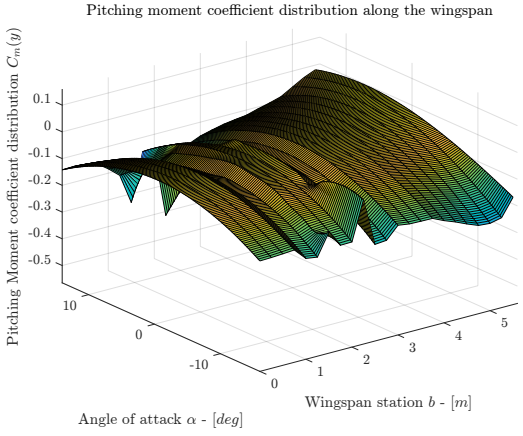
(c) Level 4, twin-engine aircraft. $C_d = C_d(y)$



(d) Level 2, twin-engine aircraft. $C_d = C_d(y)$

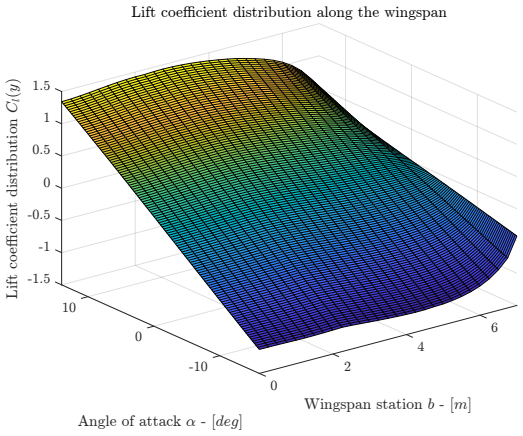


(e) Level 4, twin-engine aircraft. $C_m = C_m(y)$

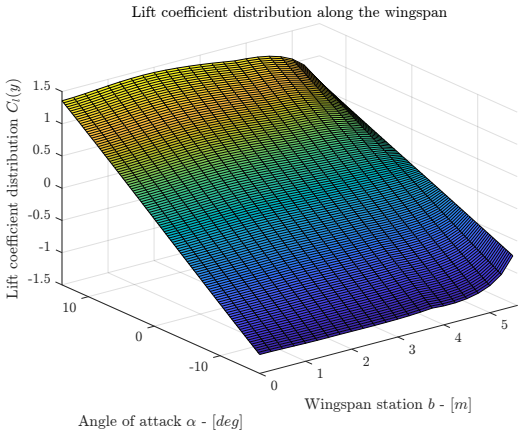


(f) Level 2, twin-engine aircraft. $C_m = C_m(y)$

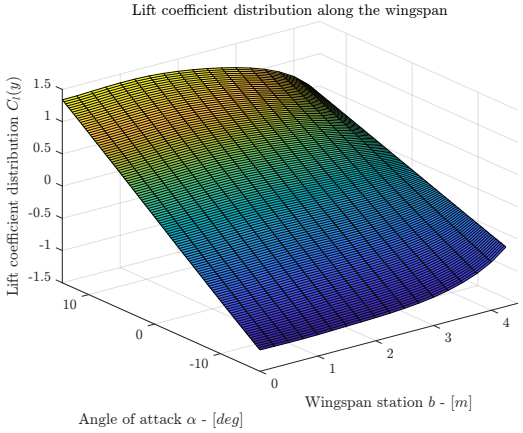
Figure 3.19: Three-dimensional representation of the spanwise distribution over the main wings of the lift ($C_l(y)$), drag ($C_d(y)$), and pitching moment coefficients ($C_m(y)$) for the engine on case. The aircraft design architectures considered are the Level 4 and Level 2, twin-engine aircraft.



(a) Level 4, twin-engine aircraft.



(b) Level 2, twin-engine aircraft.



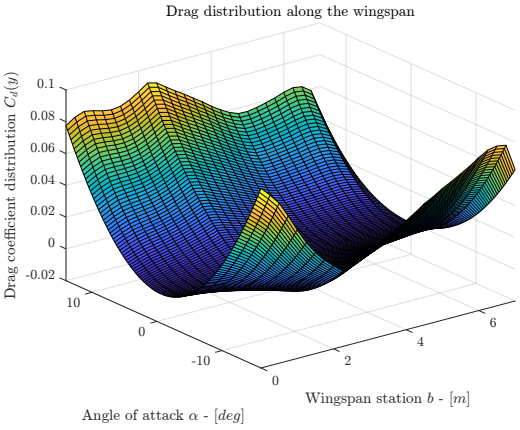
(c) Level 2, single-engine aircraft.

Figure 3.20: Engine-off spanwise lift coefficient distributions $C_l = C_l(y)$ for the Level 4 and Level 2 twin-engine aircraft and the Level 2 single-engine aircraft.

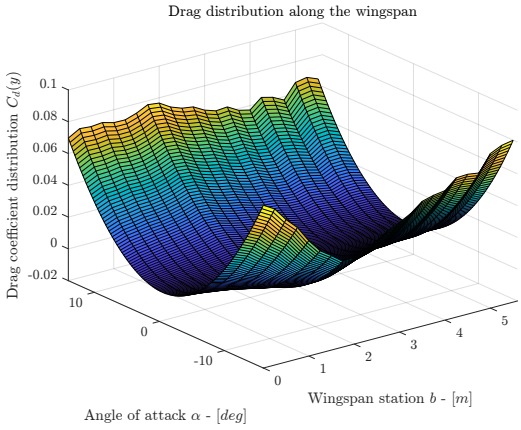
The Level 2 twin-engine aircraft 3D representation, aerodynamic results, and the spanwise pitching moment coefficient distribution are calculated without changing either the SysML model or the MATLAB code, yielding correct, ready-to-use results. Figures 3.15 and 3.18 complete the cases considered in this thesis, providing a smaller, Level 2, single-engine aircraft design architecture for comparison; this example enforces once more the inherent flexibility of the tools developed within this specific application of the MBSE methodologies to the problem of aeronautical certification.

Figure 3.19 collects the three-dimensional interpolation created using the powerful facilities available within the MATLAB/Simulink programming environment, for the Level 4 and Level 2, twin-engine aircraft with their engines working at maximum continuous power. The digital certification MATLAB code accesses the 3D interpolation for evaluating the spanwise lift, drag, and pitching moment coefficients at the maneuvering and gust flight envelope design points—such as the design maneuvering, cruising, and dive speeds with their corresponding design limit load factors—allowing calculation of the shear,

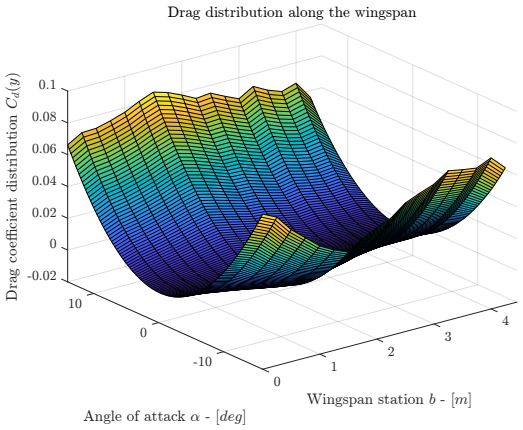
bending moment, and torsion distributions applied on the main wing structures when the aircraft is flying at that specific design point.



(a) Level 4, twin-engine aircraft.



(b) Level 2, twin-engine aircraft.



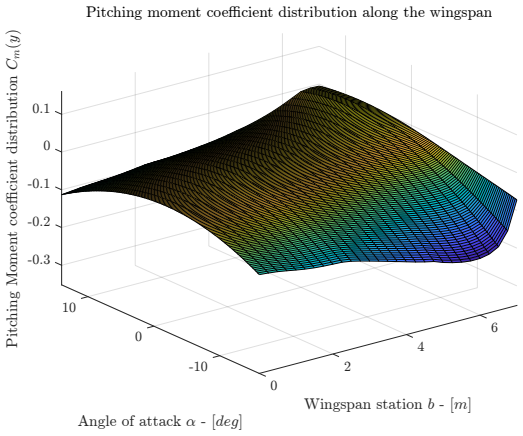
(c) Level 2, single-engine aircraft.

Figure 3.21: Engine off spanwise drag coefficient distributions $C_d = C_d(y)$ for the Level 4 and Level 2, twin-engine aircraft and the Level 2, single-engine aircraft.

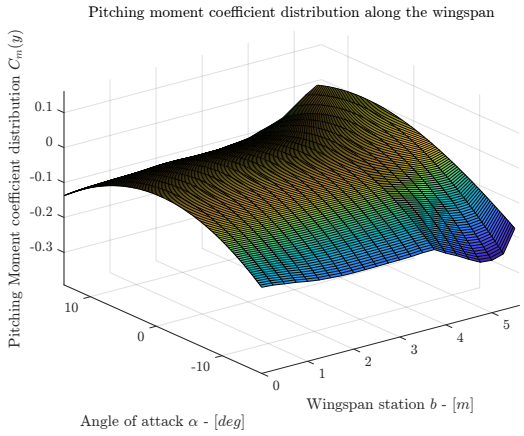
Figure 3.20 reports the spanwise lift coefficient distributions for all three conventional design architectures considered in this thesis, with their engines off. Figures 3.21 and 3.22 complete the collection of results obtained from a single run of the MATLAB digital certification code.

The spanwise load distributions provide a useful representation of how aerodynamic forces are distributed over the principal lifting surfaces of the aircraft configurations considered in this thesis. Although these distributions should be interpreted as analytical approximations, the resulting calculations are generally accepted by regulators and can therefore be used to generate certification artifacts to demonstrate compliance within the context of Subpart C (Structure).

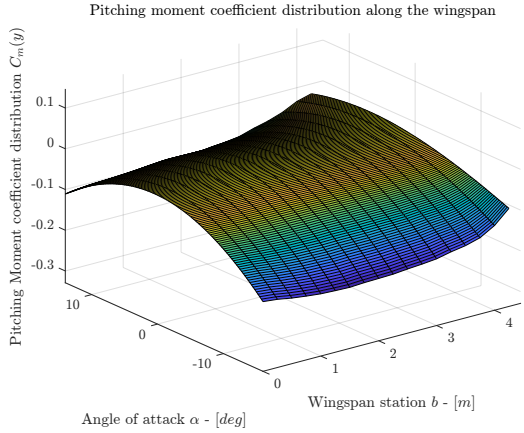
Another important preliminary evaluation concerns the wing-body aerodynamics of the aircraft.



(a) Level 4, twin-engine aircraft.



(b) Level 2, twin-engine aircraft.



(c) Level 2, single-engine aircraft.

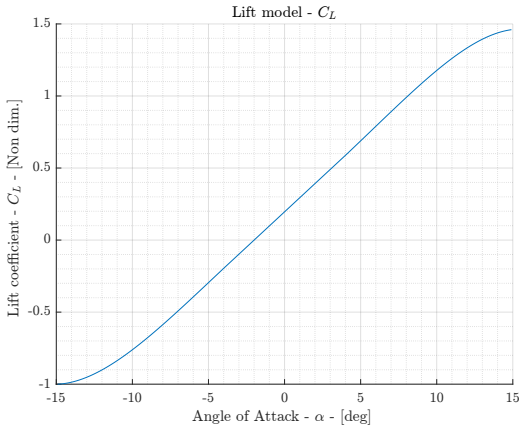
Figure 3.22: Engine-off spanwise pitching moment coefficient distributions $C_m = C_m(y)$ for the Level 4 and Level 2, twin-engine aircraft and the Level 2, single-engine aircraft.

The MATLAB/Simulink environment provides the flexibility to handle different input datasets while applying a consistent sequence of operations to evaluate the lift, drag, and pitching-moment coefficients of the wing–body configuration. These aerodynamic data are then used to construct the maneuvering and gust flight envelope and to evaluate the balancing-load diagrams, including the wing–body and full-vehicle lift along the boundaries of the flight envelope.

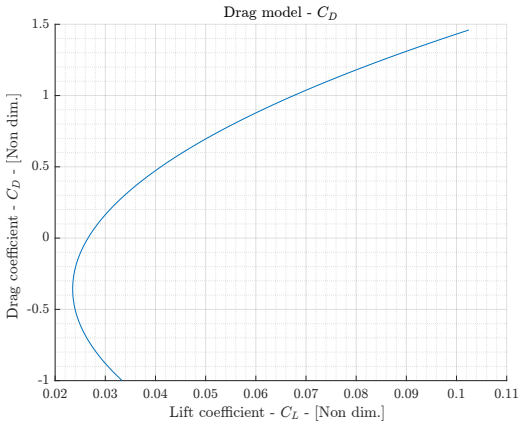
Figure 3.23 presents a selected set of wing-body aerodynamic data for the Level 2 single-engine reference aircraft configuration. Figures 3.23a and 3.23b show the wing-body lift coefficient and drag coefficient as functions of angle of attack α and lift coefficient C_L, respectively. Figures 3.23c and 3.23d present the corresponding pitching-moment characteristics, expressed as functions of C_L and α. The aerodynamic quantities reported in these figures refer consistently to the wing-body configuration.

The standard methodology for the evaluation of air loads acting on the horizontal tailplane is based on the following definition:

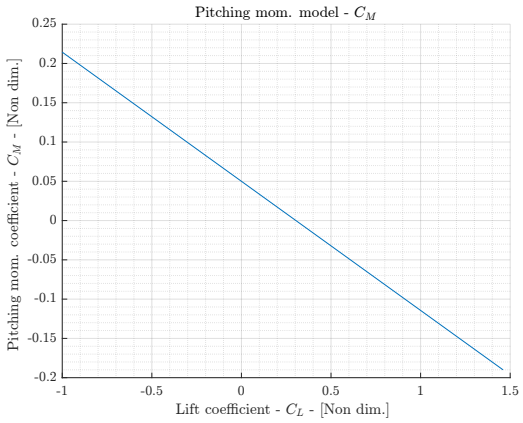
Definition 1 (of Balancing Loads)**.** A horizontal tailplane balancing load is the load necessary to maintain equilibrium in any condition with no pitching acceleration. The horizontal tailplane must be designed for the balancing loads occurring at any point on the limit maneuvering and gust flight envelope.



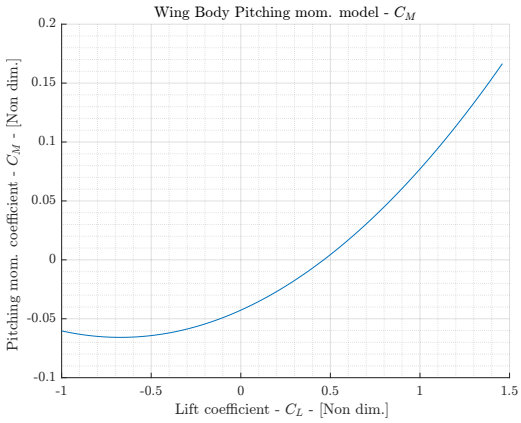
(a) Level 2, single-engine aircraft wing body lift coefficient.



(b) Level 2, single-engine aircraft wing body drag coefficient.



(c) Level 2, single-engine aircraft wing body pitching moment coefficient as a function of the lift coefficient.



(d) Level 2, single-engine aircraft wing body pitching moment coefficient as a function of the angle of attack.

Figure 3.23: A selected output from the wing-body aerodynamic model used for the Level 2, single-engine aircraft.

Definition 1 may be found in [230, p. 5, sec. 4.16] and paraphrases EASA CS-23, Amendment 4, paragraphs §23.421(a) and §23.421(b) [231]. Throughout this thesis, the application of this definition is based on the following assumptions:

1. The reference frame is defined with an x-axis, positive aft and parallel to the thrust vector, passing through the aircraft center of gravity in its most forward position. This choice yields the highest balancing loads. The z-axis is perpendicular to the

x-axis and also passes through the center of gravity. The y-axis completes the reference frame and is positive along the right wing.

2. The normal forces generated by the propellers are neglected.

3. The aerodynamic force acting on the horizontal tailplane is assumed to be equal to the lift and to act perpendicular to the x-axis.

In nondimensional form, the equilibrium equations are the following:

$$C_{L_{\text{wb}}} \cdot \cos\alpha_{\text{wb}} + C_{D_{\text{wb}}} \cdot \sin\alpha_{\text{wb}} + (C_L)_{\text{tail}} \cdot \frac{S_{\text{tail}}}{S_W} - \frac{n \cdot W \cdot \cos\alpha_{\text{wb}}}{q \cdot S_W} = 0 \tag{3.1a}$$

$$\begin{aligned} & C_{L_{\text{wb}}} \cdot \left[\cos\alpha_{\text{wb}} \cdot (\overline{x}_{\text{ac}} - \overline{x}_{\text{cg}}) - \frac{z_{\text{cg}}}{\overline{c}} \cdot \sin\alpha_{\text{wb}} \right] + \\ & + C_{D_{\text{wb}}} \cdot \left[\frac{z_{\text{cg}}}{\overline{c}} \cdot \cos\alpha_{\text{wb}} - \sin\alpha_{\text{wb}} \cdot (\overline{x}_{\text{ac}} - \overline{x}_{\text{cg}}) \right] + \\ & + \left[(C_L)_{\text{tail}} \cdot \left(\frac{l_{\text{tail}}}{\overline{c}} + (\overline{x}_{\text{ac}} - \overline{x}_{\text{cg}}) \right) \right] + \left[\frac{C_{D_{\text{wb}}}}{q \cdot S_W \cdot \overline{c}} \cdot \frac{h}{\overline{c}} \right] + C_{M_{\text{wb}}} = 0 \end{aligned} \tag{3.1b}$$

where

$$n = \text{Applicable limit load factor (nondimensional)} \tag{3.2}$$

$$W = \text{Aircraft analysis weight in N} \tag{3.3}$$

$$(\overline{x}_{\text{ac}} - \overline{x}_{\text{cg}}) = x\text{-axis distance between the a.c. and the CG (percentage of MAC)} \tag{3.4}$$

$$z_{\text{cg}} = z\text{-axis CG coordinate in m} \tag{3.5}$$

$$l_{\text{tail}} = x\text{-axis distance between the aircraft CG and the horiz. tail a.c. in m} \tag{3.6}$$

$$h = z\text{-axis distance between the thrust axis and the } x\text{-axis in m} \tag{3.7}$$

$$C_{L_{\text{tail}}} = \text{Horizontal tailplane lift coefficient} \tag{3.8}$$

$$S_{\text{tail}} = \text{Horizontal tailplane surface in m}^2 \tag{3.9}$$

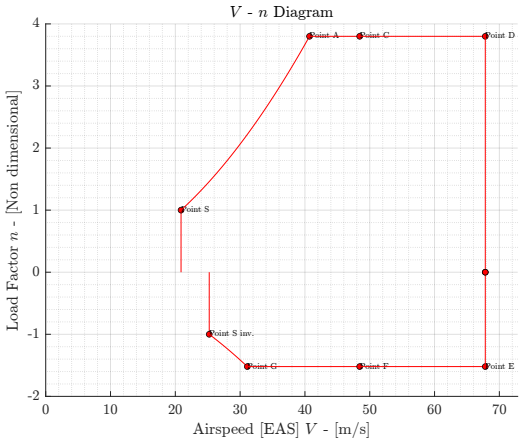
$$S_W = \text{Main wing surface in m}^2 \tag{3.10}$$

$$\overline{c} = \text{Main wing mean aerodynamic chord in m} \tag{3.11}$$

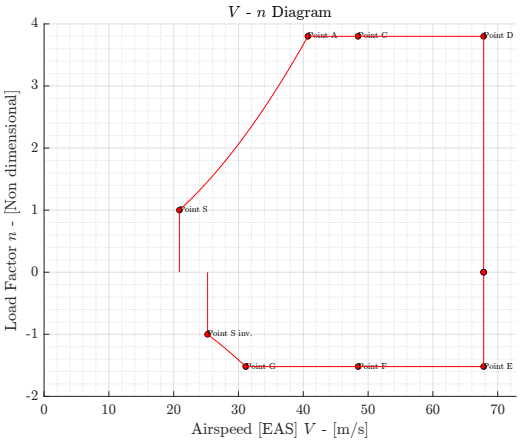
$$q = \text{Dynamic pressure in Pa} \tag{3.12}$$

Equation (3.1a) expresses force equilibrium along the z-axis, whereas Equation (3.1b) expresses rotational equilibrium about the y-axis, that is, the pitching-moment equilibrium. Taken together, the system of equations in (3.1) can be solved using standard numerical tools available in the MATLAB/Simulink environment.

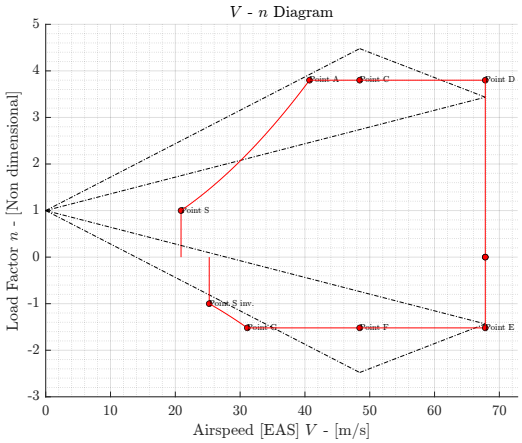
The maneuvering and gust flight envelope, therefore, represents the fundamental certification object for the evaluation of the structural soundness of the new design. The applicable limit load factors n, whether they come from maneuvering the aircraft or conducting it through gusts, and the corresponding airspeeds are points on the flight envelope boundaries. The detailed methods used to determine each design point are not discussed here.



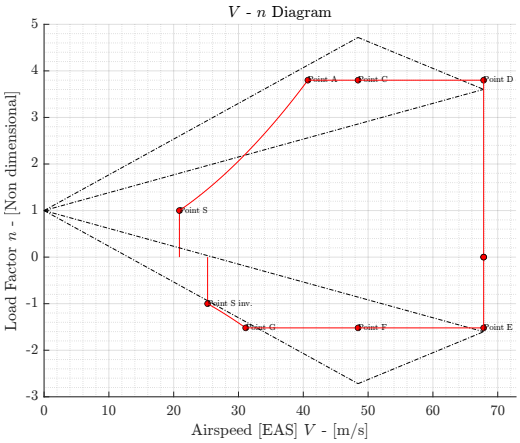
(a) Level 2, single-engine aircraft maneuvering flight envelope at sea level.



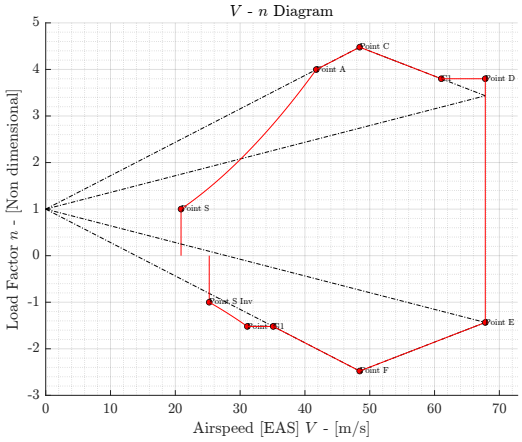
(b) Level 2, single-engine aircraft maneuvering flight envelope at $h = 6000$ ft.



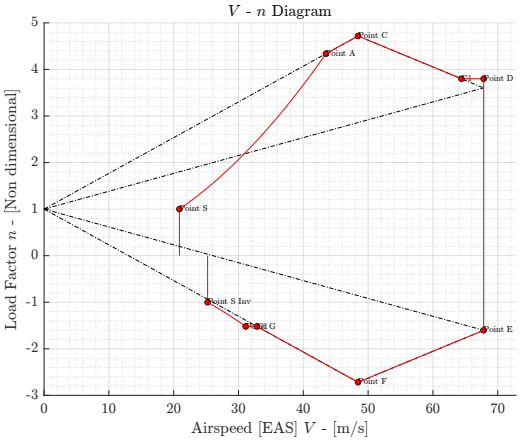
(c) Level 2, single-engine aircraft maneuvering flight envelope with gust lines at sea level.



(d) Level 2, single-engine aircraft maneuvering flight envelope with gust lines at $h = 6000$ ft.

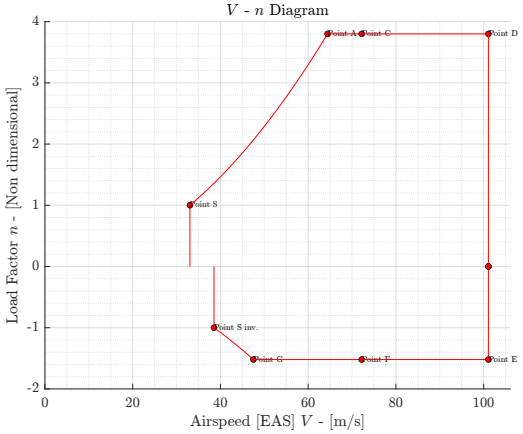


(e) Level 2, single-engine aircraft maneuvering and gust flight envelope at sea level.

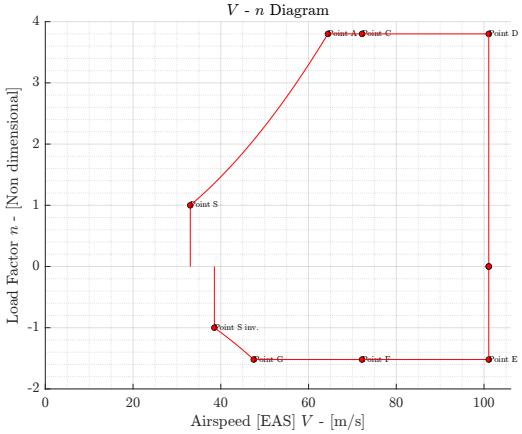


(f) Level 2, single-engine aircraft maneuvering and gust flight envelope at $h = 6000$ ft.

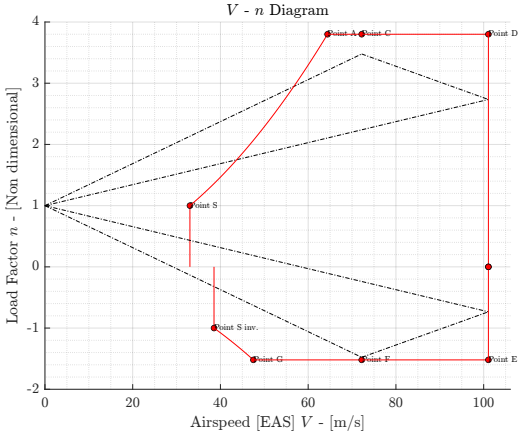
Figure 3.24: The maneuvering and gust flight envelope for the Level 2, single-engine aircraft evaluated at sea level and at $h = 6000$ ft.



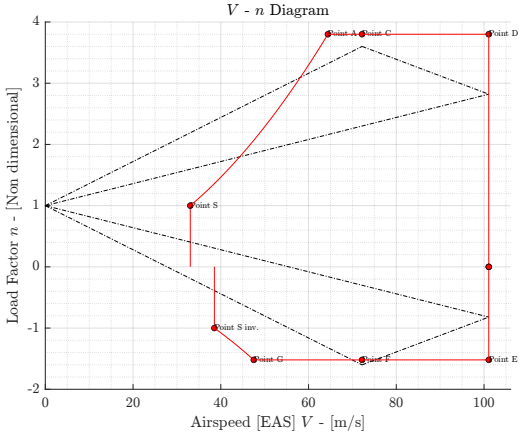
(a) Level 2, twin-engine aircraft maneuvering flight envelope at sea level.



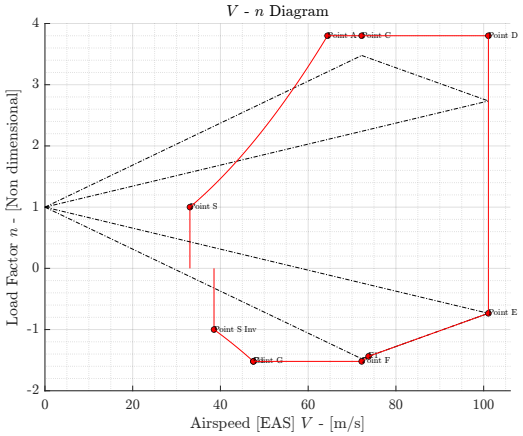
(b) Level 2, twin-engine aircraft maneuvering flight envelope at $h = 10\,000$ ft.



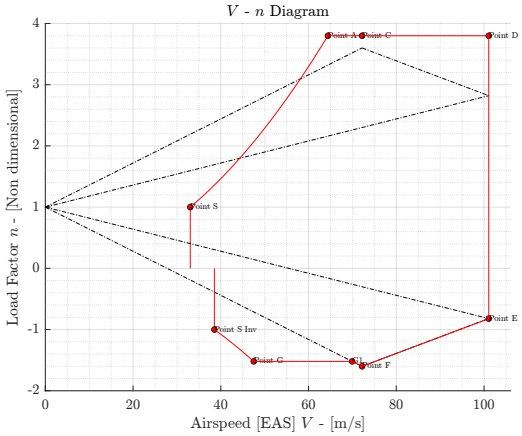
(c) Level 2, twin-engine aircraft maneuvering flight envelope with gust lines at sea level.



(d) Level 2, twin-engine aircraft maneuvering flight envelope with gust lines at $h = 10\,000$ ft.

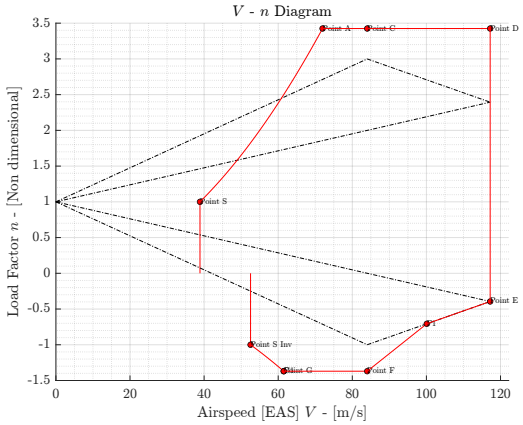


(e) Level 2, twin-engine aircraft maneuvering and gust flight envelope at sea level.

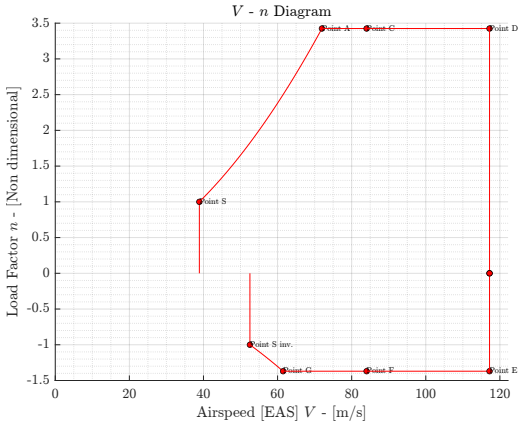


(f) Level 2, twin-engine aircraft maneuvering and gust flight envelope at $h = 10\,000$ ft.

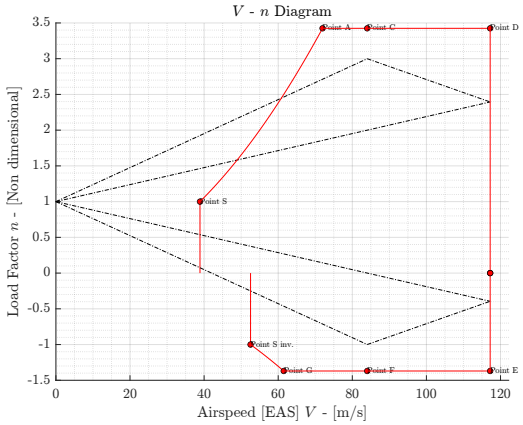
Figure 3.25: The maneuvering and gust flight envelope for the Level 2, twin-engine aircraft evaluated at sea level and at $h = 10\,000$ ft.



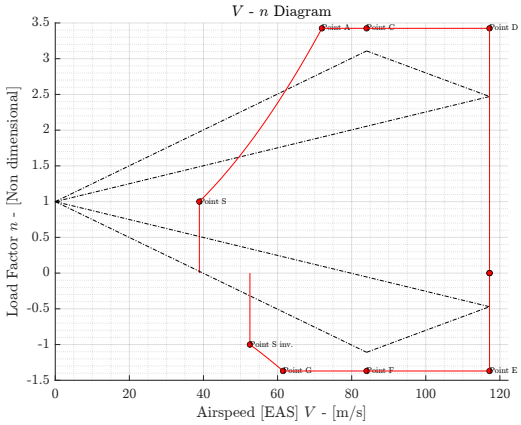
(a) Level 4, twin-engine aircraft maneuvering flight envelope at sea level.



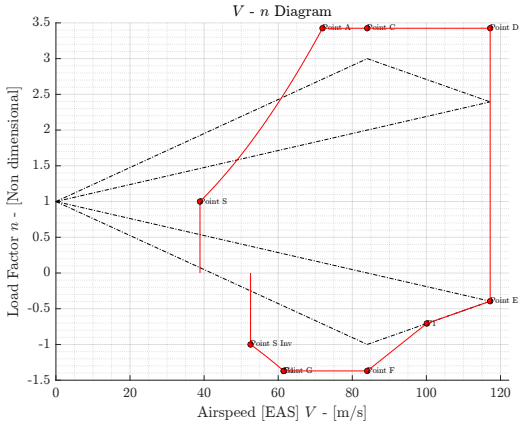
(b) Level 4, twin-engine aircraft maneuvering flight envelope at $h = 12\,000$ ft.



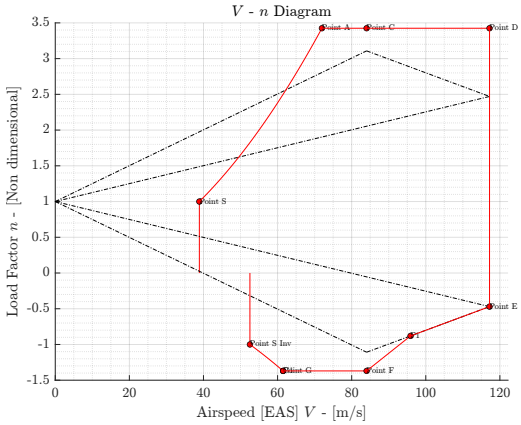
(c) Level 4, twin-engine aircraft maneuvering flight envelope with gust lines at sea level.



(d) Level 4, twin-engine aircraft maneuvering flight envelope with gust lines at $h = 12\,000$ ft.

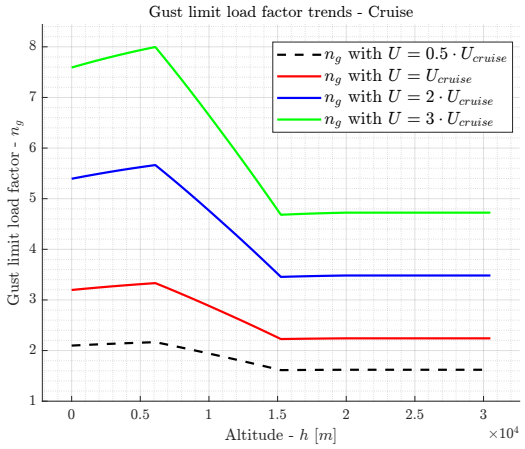


(e) Level 4, twin-engine aircraft maneuvering and gust flight envelope at sea level.

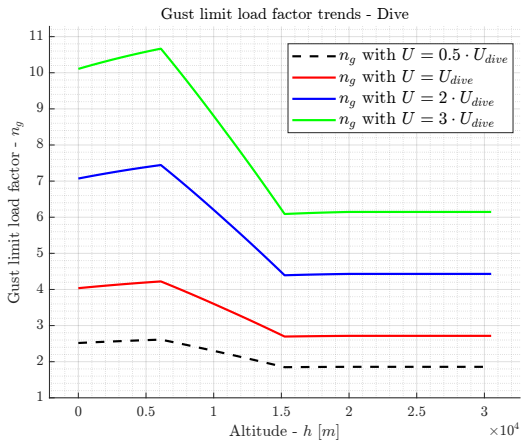


(f) Level 4, twin-engine aircraft maneuvering and gust flight envelope at $h = 12\,000$ ft.

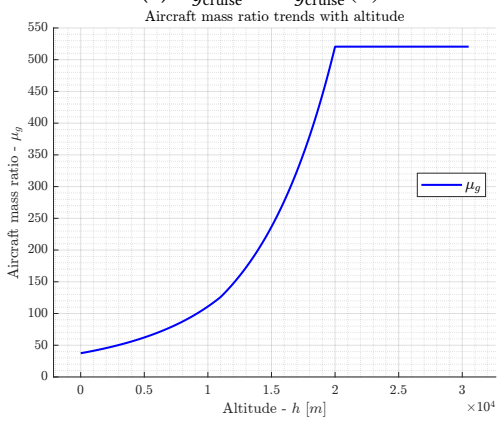
Figure 3.26: The maneuvering and gust flight envelope for the Level 4, twin-engine aircraft evaluated at sea level and at $h = 12\,000$ ft.



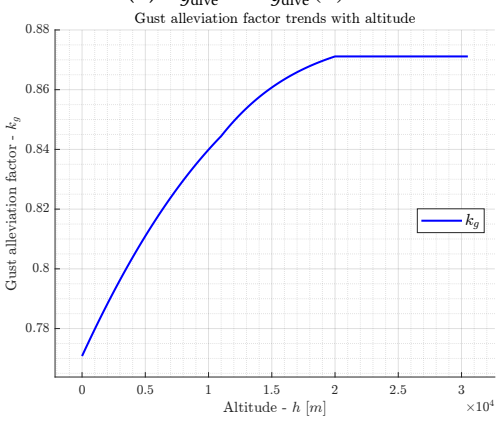
(a) $n_{g_{\text{cruise}}} = n_{g_{\text{cruise}}}(h)$.



(b) $n_{g_{\text{dive}}} = n_{g_{\text{dive}}}(h)$.



(c) $\mu_g = \mu_g(h)$



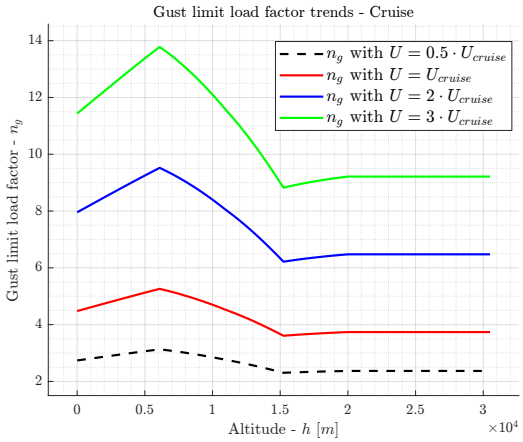
(d) $k_g = k_g(h)$

Figure 3.27: n_g evaluated using the EASA CS-23 formula with eq. (A.9), at both the design cruising and dive speeds; μ_g and k_g are included. The calculation is performed using the non-conventional design architecture (refer to Section 3.2.5).

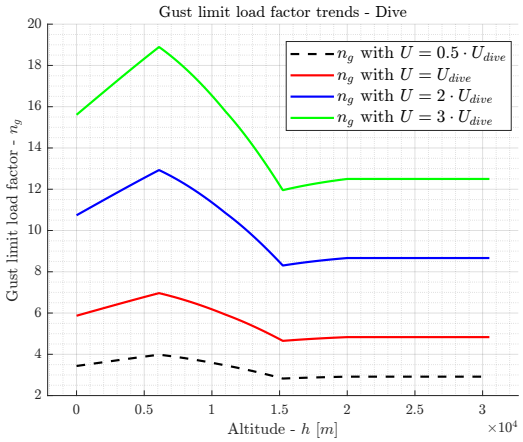
Figures 3.24, 3.25, and 3.26 present a set of graphs that illustrate the certification artifacts associated with constructing the maneuvering and gust flight envelope. The gust lines are always included to improve clarity and to support the interpretation of the plots. Engineers use the maneuvering and gust flight envelope to identify the design-critical loads, that is, the loads that must be considered when sizing the aircraft structure. For example, Figures 3.24e and 3.24f show the sensitivity of the Level 2 single-engine configuration to gust effects, particularly at higher altitude. By contrast, Figures 3.26e and 3.26f, which refer to the Level 4 aircraft, show a different pattern: the maneuvering loads govern the highest design limit load factors even at altitude. Figure 3.25f presents an intermediate case, in which the gust line associated with negative limit loads exceeds the maneuvering loads at the design cruise speed, although by a modest margin.

Figure 3.27 presents a set of useful graphs representing trends of quantities involved in the evaluation of the gust limit loads. The calculations are performed using the non-

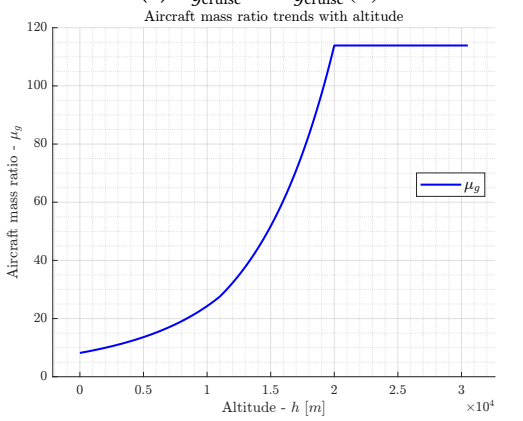
conventional design architecture.



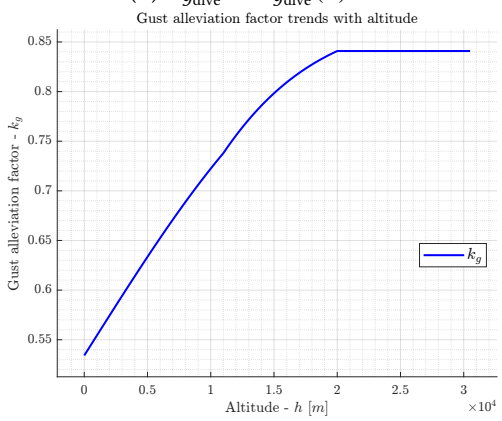
(a) $n_{g_{\text{cruise}}} = n_{g_{\text{cruise}}}(h)$.



(b) $n_{g_{\text{dive}}} = n_{g_{\text{dive}}}(h)$.



(c) $\mu_g = \mu_g(h)$



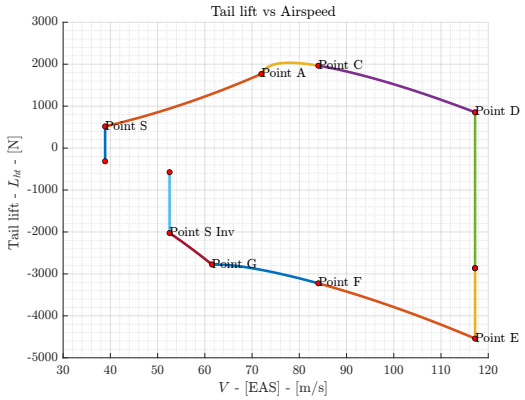
(d) $k_g = k_g(h)$

Figure 3.28: n_g evaluated using the EASA CS-23 formula with eq. (A.9), at both the design cruising and dive speeds; μ_g and k_g are included. The calculation is performed using the Level 2 single-engine architecture.

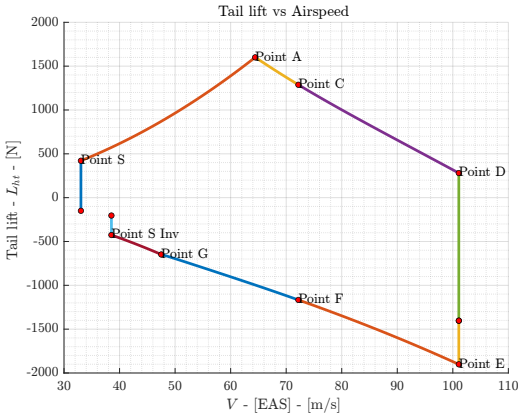
Figure 3.28 presents the same trends for the Level 2 single-engine design architecture.

Figures 3.27a and 3.27b show the variation in gust limit load factor with altitude at the design cruise speed V_C and the design dive speed V_D, respectively. Under the EASA CS-23 provisions applied in this analysis, the initial effect of increasing altitude is governed primarily by the increase in the gust alleviation factor k_g, because the design gust velocity U_{de} remains constant up to an altitude of 6096 m. Above this altitude, the prescribed linear reduction in U_{de} becomes dominant relative to the gradual increase in k_g, resulting in a substantial reduction in the gust limit load factor n_g. The figures also compare the same relationship for different values of U_{de}, illustrating the combined influence of gust velocity and the gust alleviation factor. Under the conditions considered, n_g reaches its maximum at $h = 6096\,\text{m}$. By contrast, EASA CS-25 specifies different design gust velocities and altitude thresholds. Within that regulatory framework, the prescribed reduction in U_{de}

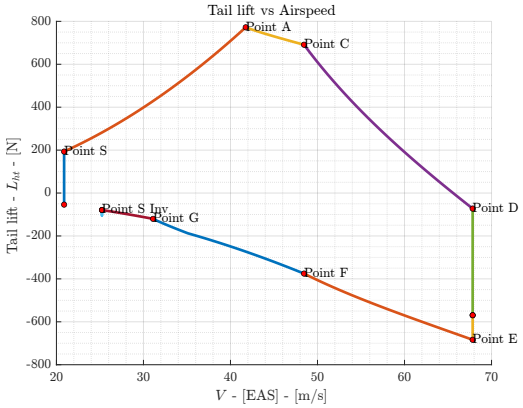
has a greater influence on n_g, while the contribution of variations in k_g is comparatively limited. Interestingly, μ_g and k_g have an upper bound with altitudes higher than 20 000 m. Thus, the gust limit load factors n_g have an upper bound too at these altitudes.



(a) Level 4, twin-engine aircraft balancing loads.



(b) Level 2, twin-engine aircraft balancing loads.



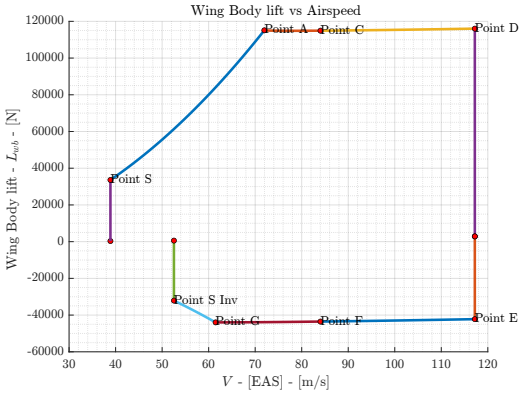
(c) Level 2, single-engine aircraft balancing loads.

Figure 3.29: Balancing loads diagrams for all the aircraft design architectures considered in this thesis.

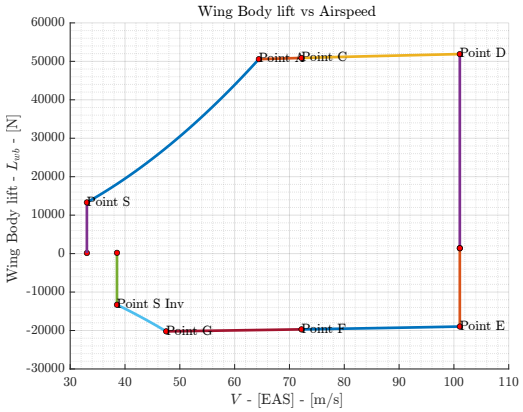
Figures 3.29, 3.30, and 3.31 show the results obtained by solving the system of equations (3.1), including the wing-body lift, and the full-vehicle lift corresponding to flight along the boundaries of the flight envelope. As expected, the overall shape of these diagrams closely resembles that of the maneuvering and gust flight envelope.

The influence of gust loads is most pronounced for the Level 2 single-engine configuration, as shown by Figures 3.29c, 3.30c, and 3.31c. For a typical general aviation aircraft, Point E is usually associated with the highest horizontal-tail load in absolute value. In the present Level 2 single-engine case, however, the aircraft is substantially lighter than a typical general aviation configuration. The highest load occurs at Point A, where the

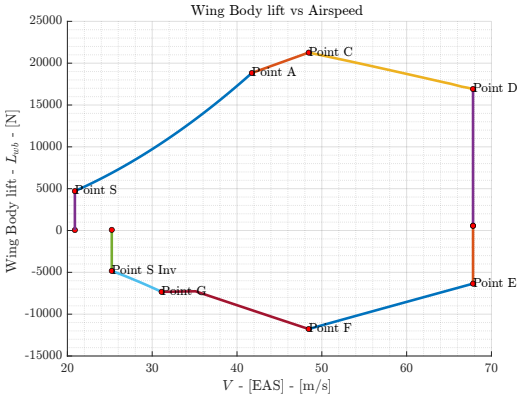
lift generated by the horizontal tailplane is approximately equal to 770 N. However, the difference with Point E is relatively small ($L_{\text{ht, E}} \approx -680$ N) in absolute value.



(a) Level 4, twin-engine aircraft wing-body lift.



(b) Level 2, twin-engine aircraft wing-body lift.



(c) Level 2, single-engine aircraft wing-body lift.

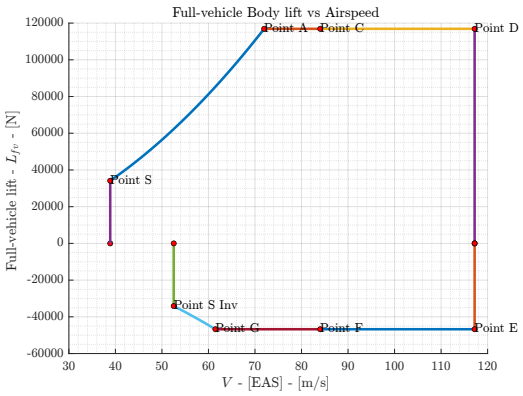
Figure 3.30: Wing-body lift diagrams for all the aircraft design architectures considered in this thesis.

Interestingly, the main-wing lift does not reach its maximum at the same design points. Figures 3.30c and 3.31c illustrate this clearly, showing that the maximum wing-body lift and full vehicle lift occur at the design cruise speed.

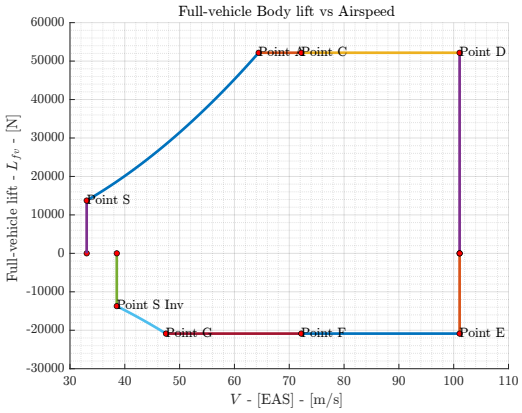
The important point, regardless of the numerical results, is that the code correctly drafts the maneuvering and gust flight envelope for radically different aircraft design architectures, adapting to their characteristics without requiring human intervention and automatically applying the airworthiness and substantiation prescriptions.

Figures 3.32, 3.33, 3.34, and 3.35 present representative compliance-demonstration items used to assess the critical internal loads in the main-wing structure; the spanwise stations, measured in m, are reported on the x-axis, while the forces and moments are on the y-axis, measured in N and N m, respectively. For comparison, the engine-on and

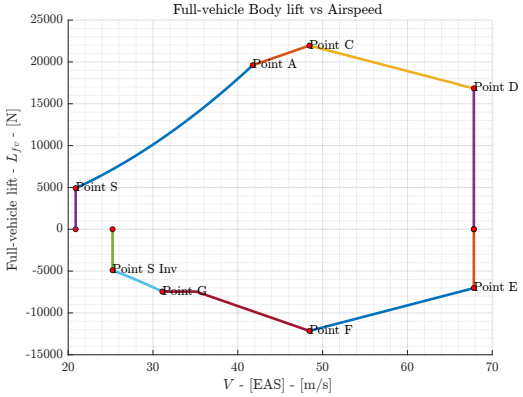
engine-off cases are shown side by side. In all engine-on cases, the propulsion system is assumed to operate at the maximum continuous power setting.



(a) Level 4, twin-engine aircraft full vehicle lift.



(b) Level 2, twin-engine aircraft full vehicle lift.



(c) Level 2, single-engine aircraft full vehicle lift.

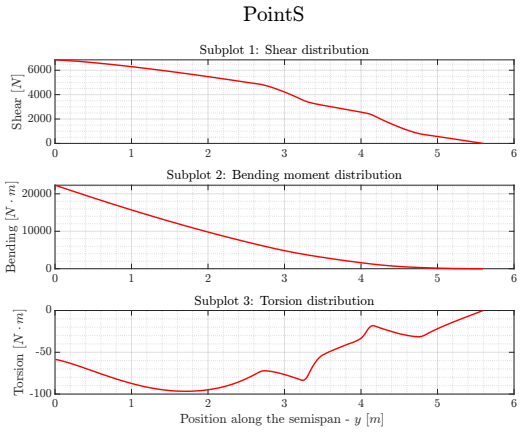
Figure 3.31: Full vehicle lift diagrams for all the aircraft design architectures considered in this thesis.

As shown in the diagrams, the effect of operating the propulsion system is most evident in the torsional-load distributions, particularly at the points on the positive side of the maneuvering and gust flight envelope, and especially for the Level 2 twin-engine configuration.

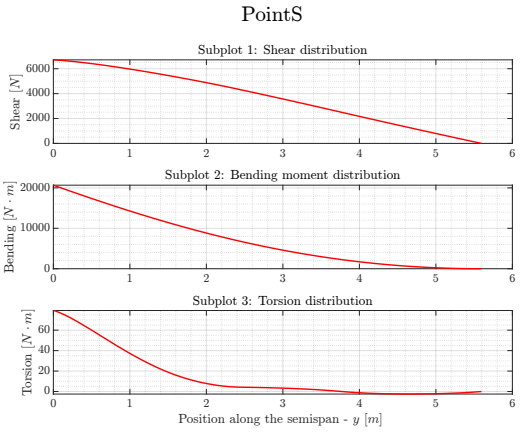
Figures 3.32a, 3.32c, and 3.32e illustrate this effect, while Figures 3.33a and 3.33e show a similar spanwise torsion pattern for the Level 4 twin-engine aircraft.

Figures 3.34, 3.36, 3.37, 3.38, and 3.39 complete the series of these certification artifacts. The maneuvering and gust flight-envelope points at the design dive speed—specifically Points D and E—are typically examined in detail because they can govern critical structural loads.

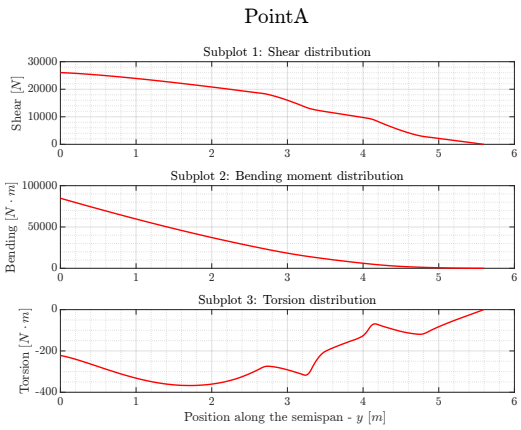
Careful evaluation of the corresponding internal loads is therefore essential for identifying the most demanding conditions acting on the primary aircraft structure.



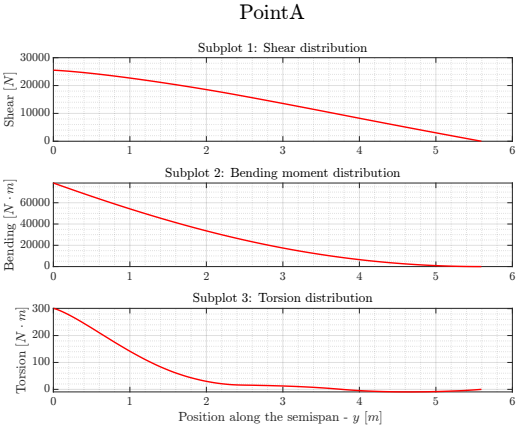
(a) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point S. Engine on.



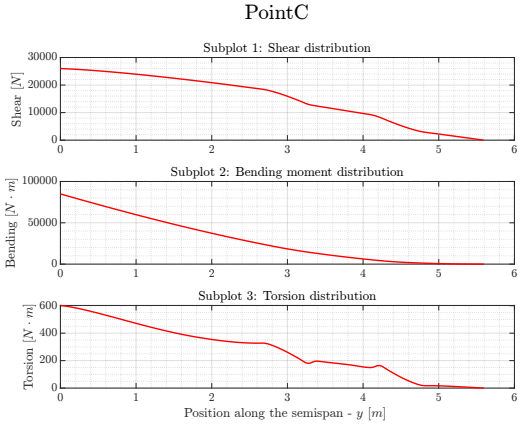
(b) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point S. Engine off.



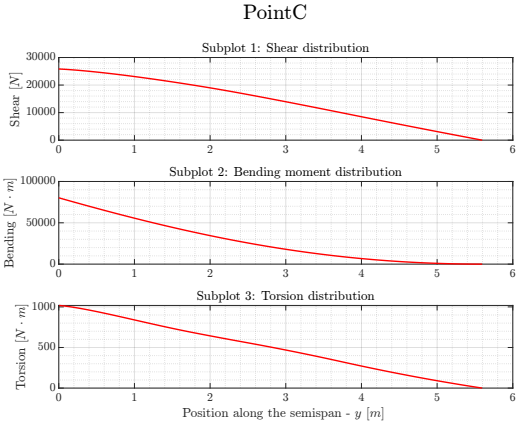
(c) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point A. Engine on.



(d) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point A. Engine off.

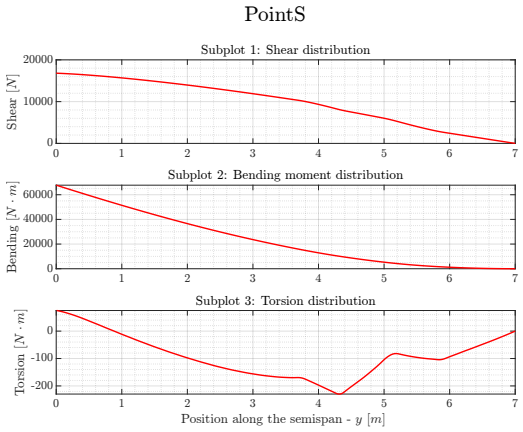


(e) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point C. Engine on.

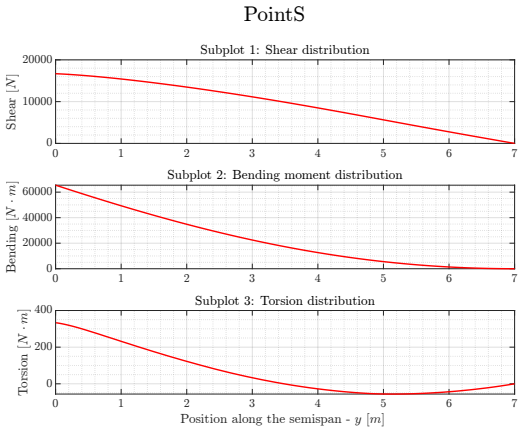


(f) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point C. Engine off.

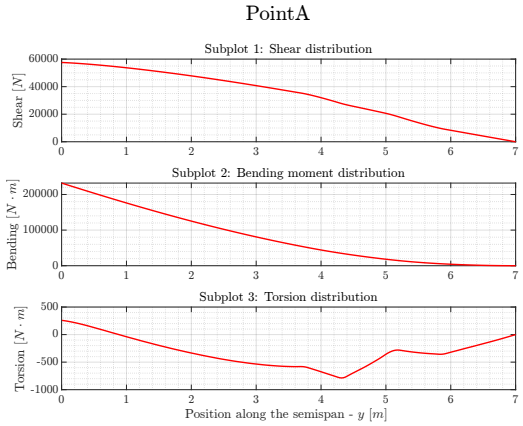
Figure 3.32: Shear, bending moment, and torsion for the Level 2, twin-engine aircraft. Points S, A, and C of the maneuvering and gust flight envelope.



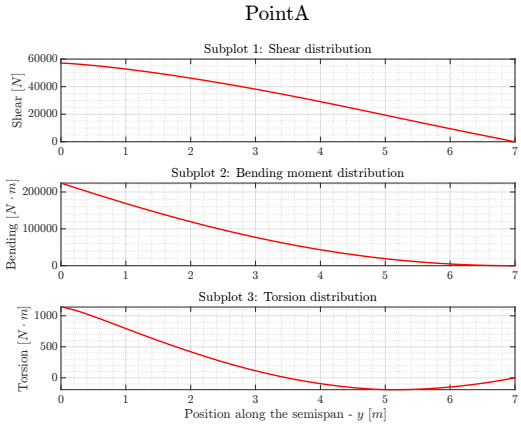
(a) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point S. Engine on.



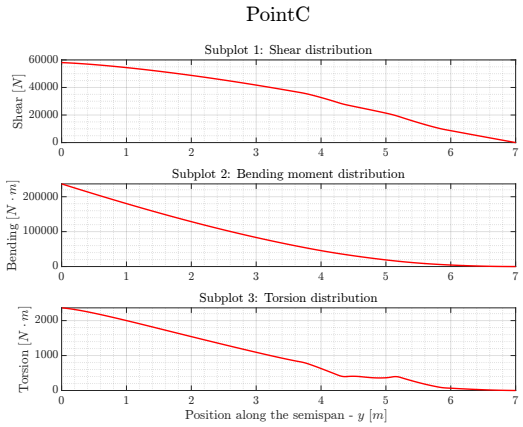
(b) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point S. Engine off.



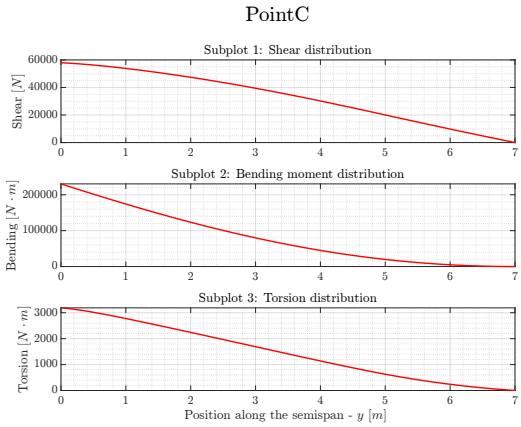
(c) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point A. Engine on.



(d) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point A. Engine off.

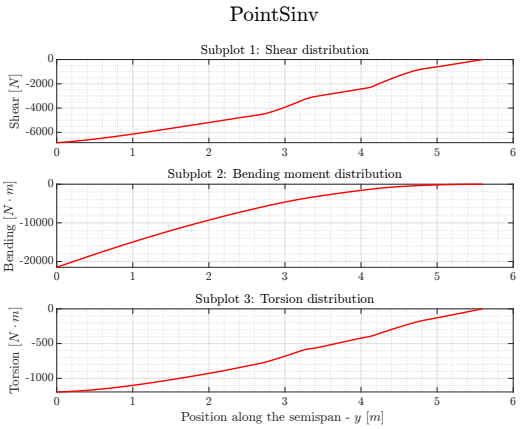


(e) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point C. Engine on.

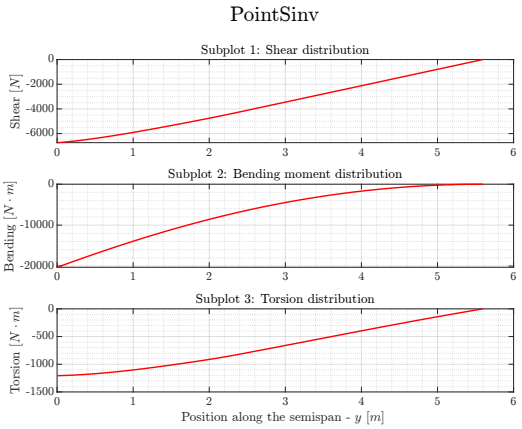


(f) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point C. Engine off.

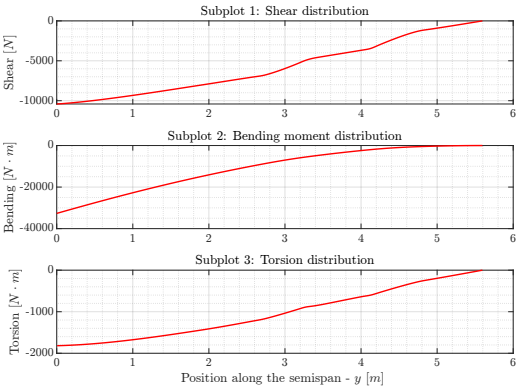
Figure 3.33: Shear, bending moment, and torsion for the Level 4, twin-engine aircraft. Points S, A, and C of the maneuvering and gust flight envelope.



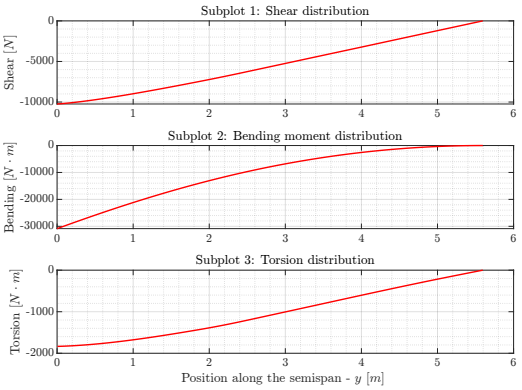
(a) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point S_{neg}. Engine on.



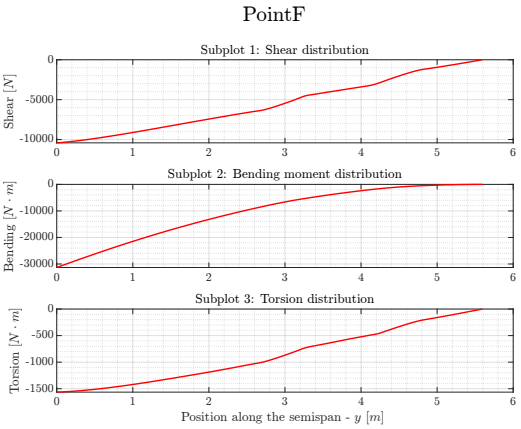
(b) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point S_{neg}. Engine off.



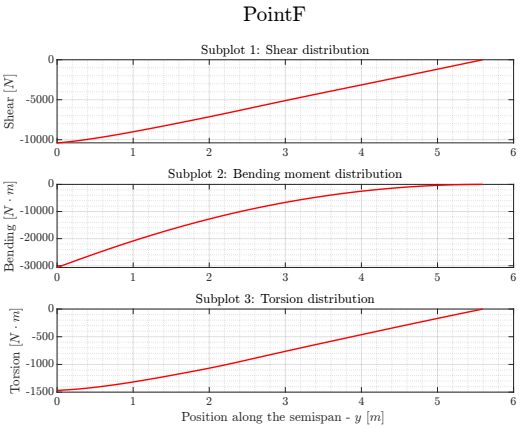
(c) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point G. Engine on.



(d) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point G. Engine off.

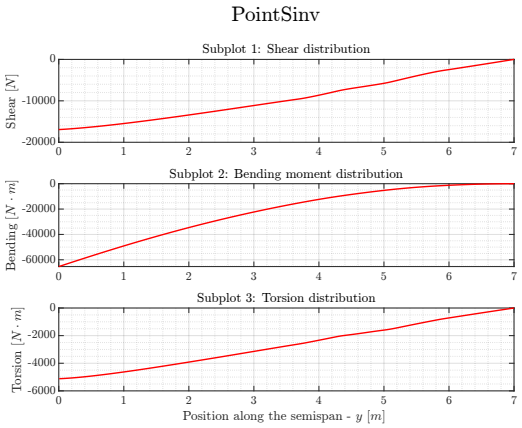


(e) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point F. Engine on.

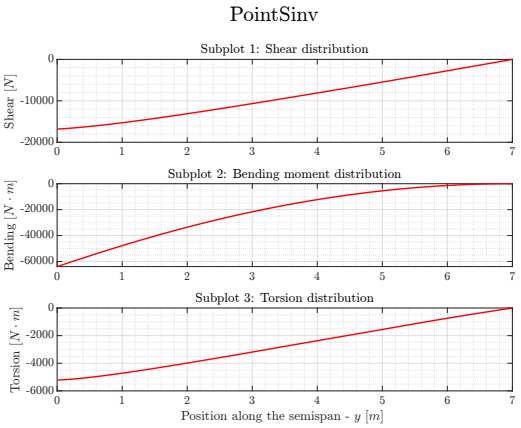


(f) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point F. Engine off.

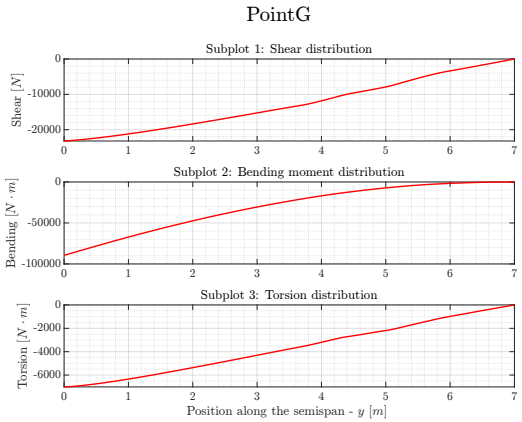
Figure 3.34: Shear, bending moment, and torsion for the Level 2, twin-engine aircraft. Points S_{neg}, G, and F of the maneuvering and gust flight envelope.



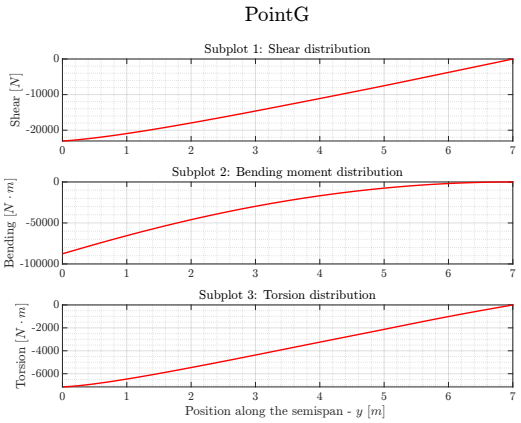
(a) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point S_{neg}. Engine on.



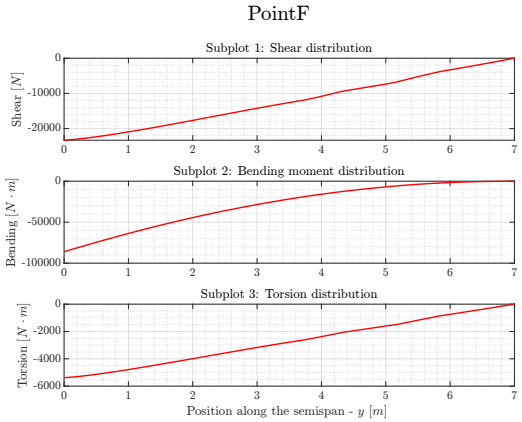
(b) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point S_{neg}. Engine off.



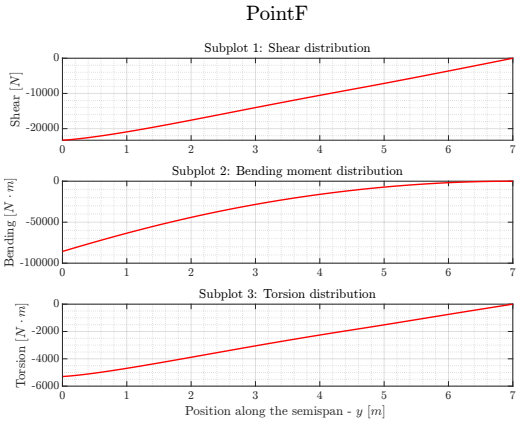
(c) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point G. Engine on.



(d) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point G. Engine off.

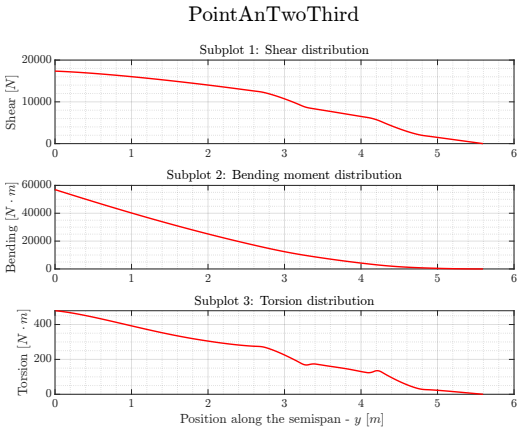


(e) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point F. Engine on.

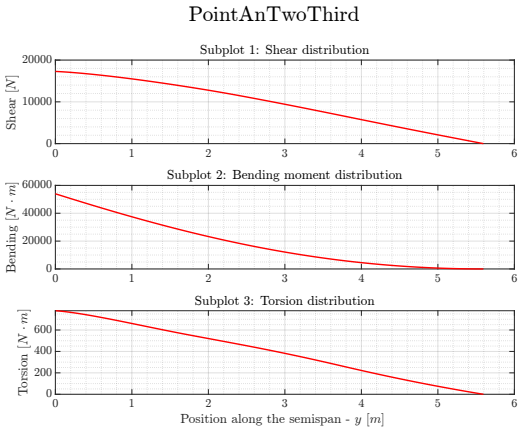


(f) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point F. Engine off.

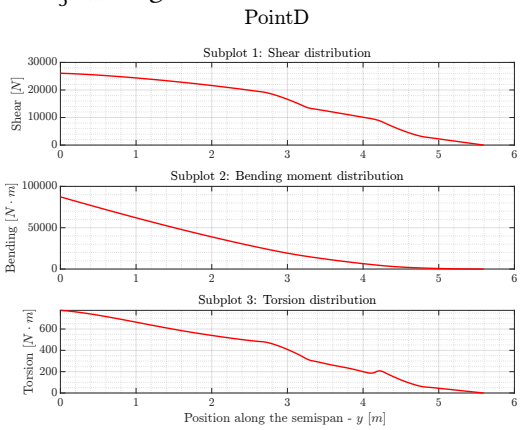
Figure 3.35: Shear, bending moment, and torsion for the Level 4, twin-engine aircraft. Points S_{neg}, G, and F of the maneuvering and gust flight envelope.



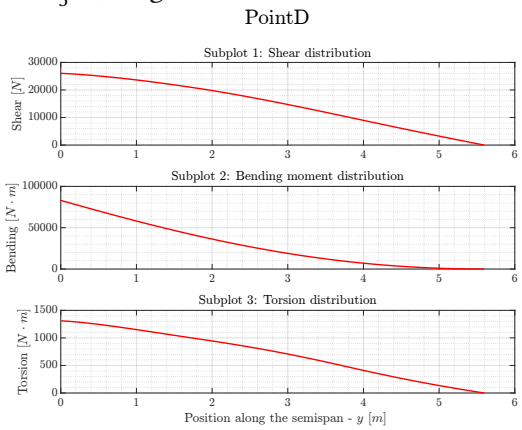
(a) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point A with $n = \frac{2}{3}n_{\mathrm{A}}$. Engine on.



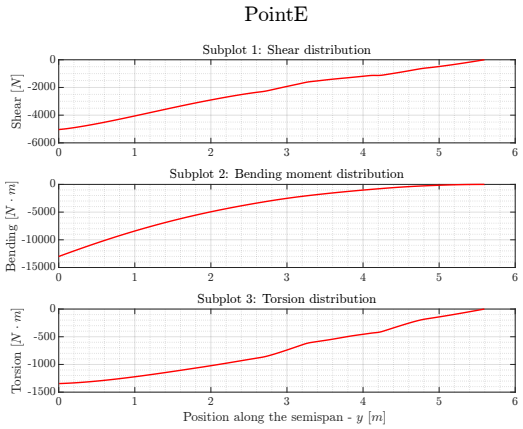
(b) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point A with $n = \frac{2}{3}n_{\mathrm{A}}$. Engine off.



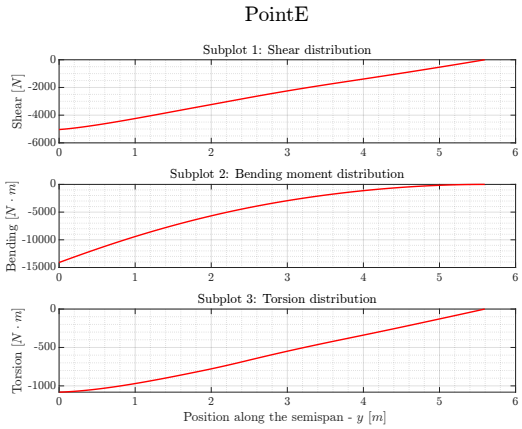
(c) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point D. Engine on.



(d) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point D. Engine off.

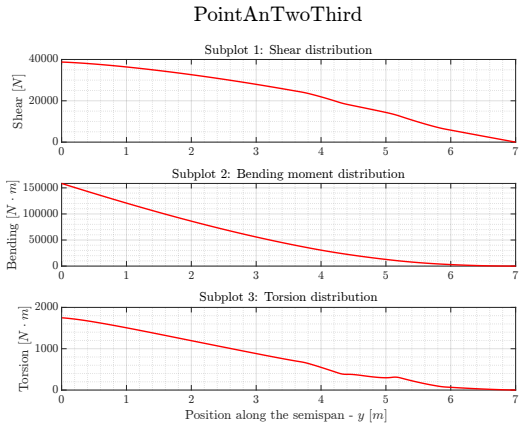


(e) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point E. Engine on.

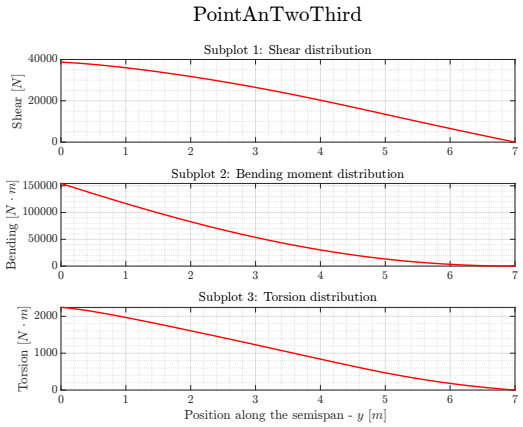


(f) Level 2, twin-engine aircraft. Shear, bending moment, and torsion at Point E. Engine off.

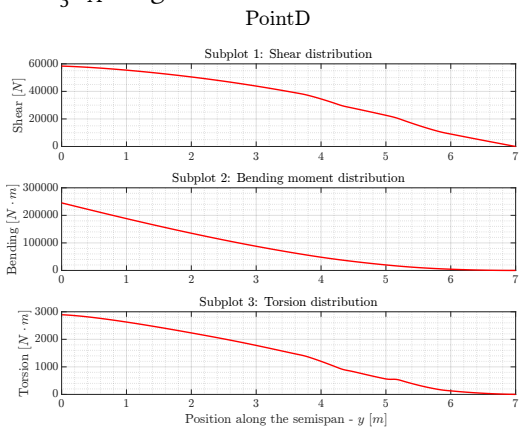
Figure 3.36: Shear, bending moment, and torsion for the Level 2, twin-engine aircraft. Points A with $n = \frac{2}{3}n_{\mathrm{A}}$, D, and E of the maneuvering and gust flight envelope.



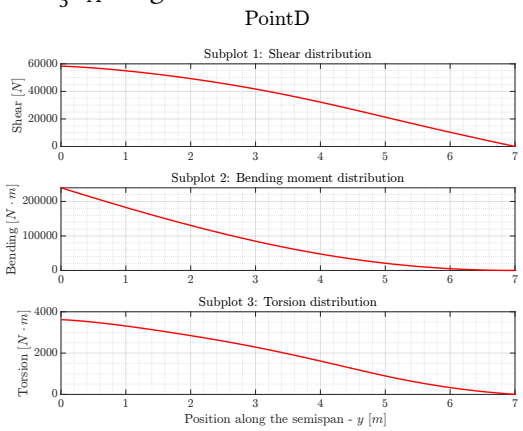
(a) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point A with $n = \frac{2}{3}n_{\mathrm{A}}$. Engine on.



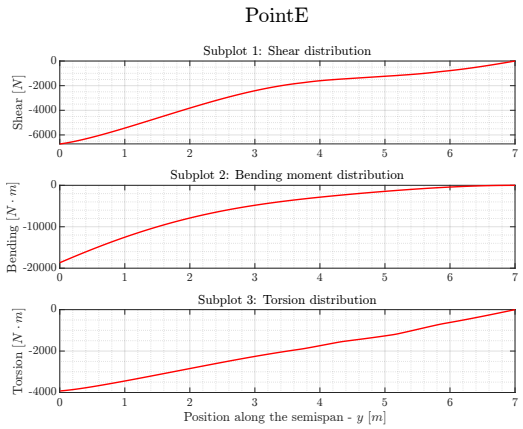
(b) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point A with $n = \frac{2}{3}n_{\mathrm{A}}$. Engine off.



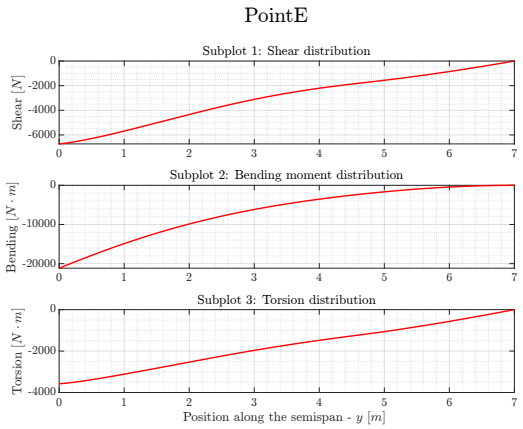
(c) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point D. Engine on.



(d) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point D. Engine off.

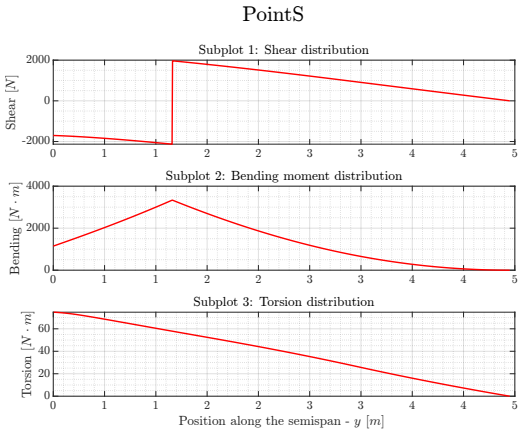


(e) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point E. Engine on.

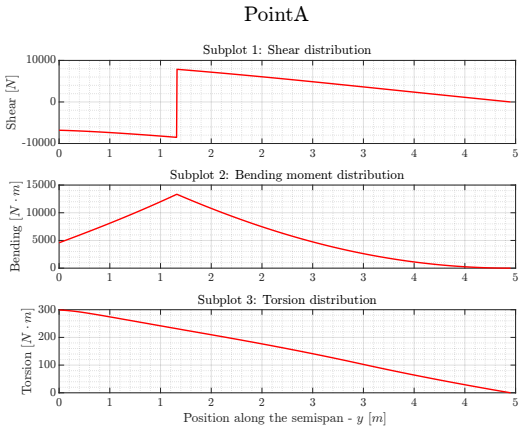


(f) Level 4, twin-engine aircraft. Shear, bending moment, and torsion at Point E. Engine off.

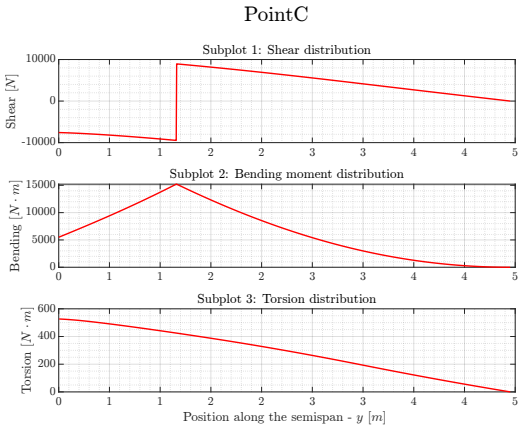
Figure 3.37: Shear, bending moment, and torsion for the Level 4, twin-engine aircraft. Points A with $n = \frac{2}{3}n_{\mathrm{A}}$, D, and E of the maneuvering and gust flight envelope.



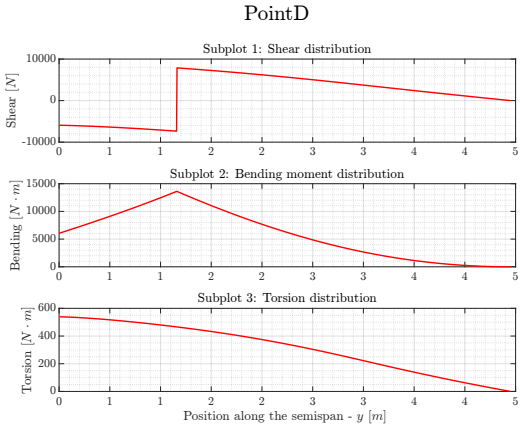
(a) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point S. Engine off.



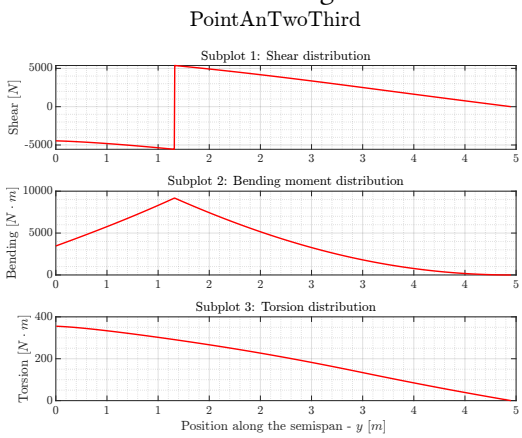
(b) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point A. Engine off.



(c) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point C. Engine off.

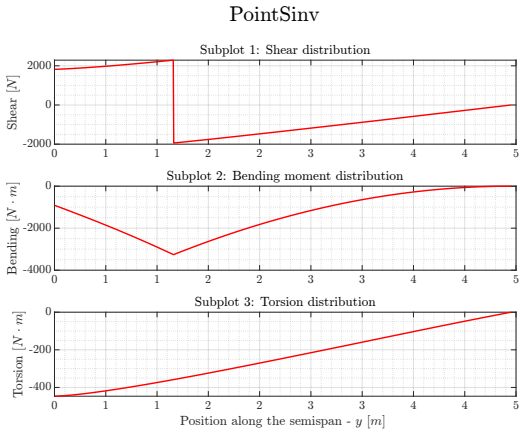


(d) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point D. Engine off.

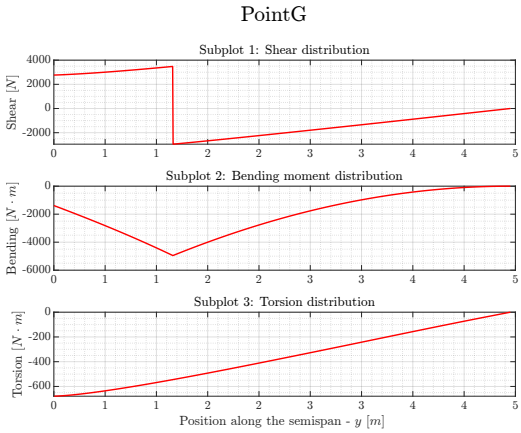


(e) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point A with $n = \frac{2}{3}n_{\mathrm{A}}$. Engine off.

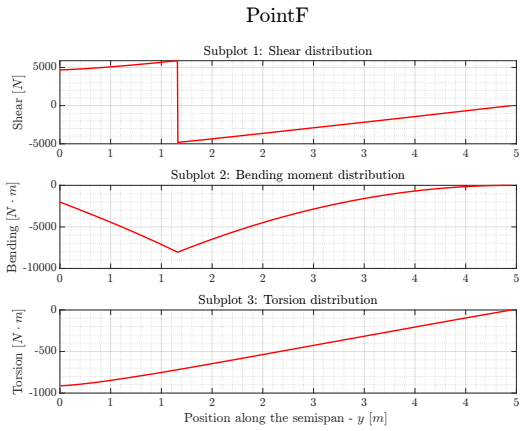
Figure 3.38: Shear, bending moment, and torsion for the Level 2, single-engine aircraft. Points S, A, A with $n = \frac{2}{3}n_{\mathrm{A}}$, C, and D of the maneuvering and gust flight envelope.



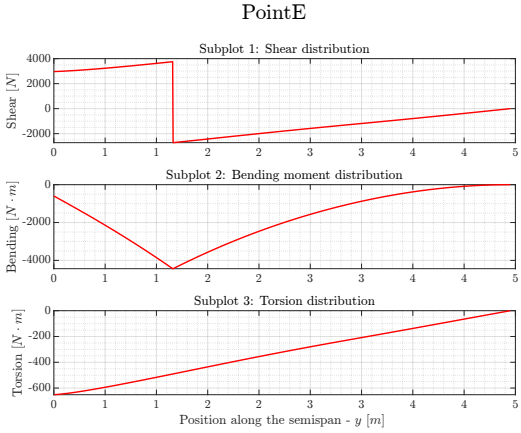
(a) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point S_{neg}. Engine off.



(b) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point G. Engine off.



(c) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point F. Engine off.



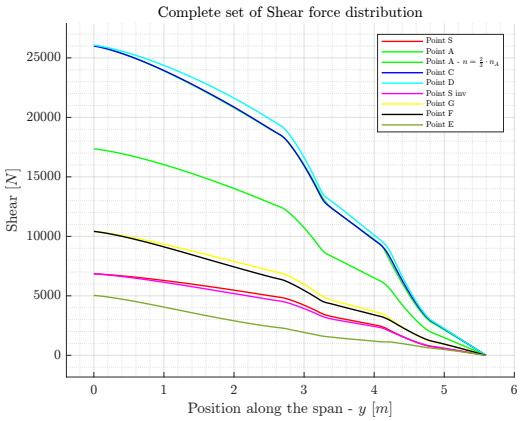
(d) Level 2, single-engine aircraft. Shear, bending moment, and torsion at Point E. Engine off.

Figure 3.39: Shear, bending moment, and torsion for the Level 2, single-engine aircraft. Points S_{neg}, G, F, and E of the maneuvering and gust flight envelope.

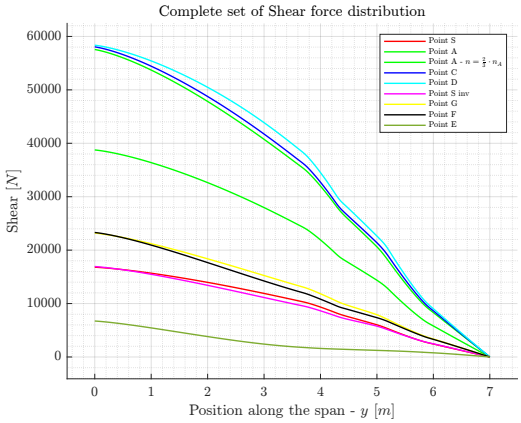
Figures 3.40, 3.41, and 3.42 provide an overall summary of the spanwise distributions of internal forces and moments for the three aircraft configurations considered in this thesis. These plots offer a compact reference for regulatory review by presenting a large amount of structural-load information in a clear and accessible form.

Figure 3.42 illustrates the flexibility of the code, which applies to different dispositions of the aircraft's main structural elements; in this particular case, this Level 2, single-engine aircraft has a strut-braced wing, and this is reflected by its shear and bending moment spanwise distributions.

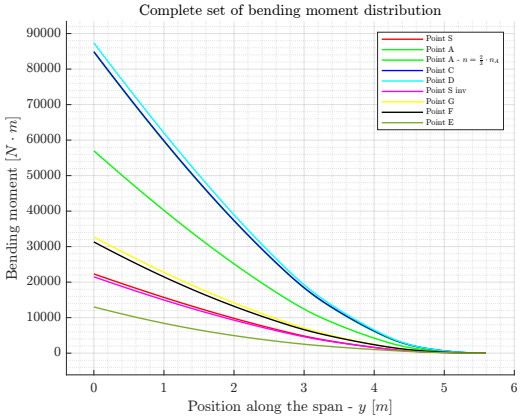
In particular, they allow the critical spanwise force or moment distribution, expressed in absolute value, to be identified readily and used as a practical reference for the sizing and design of the primary aircraft structures.



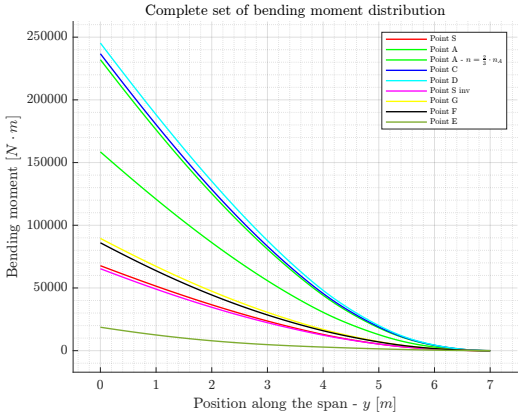
(a) Level 2, twin-engine aircraft. Summary of spanwise shear force distributions. Engine on.



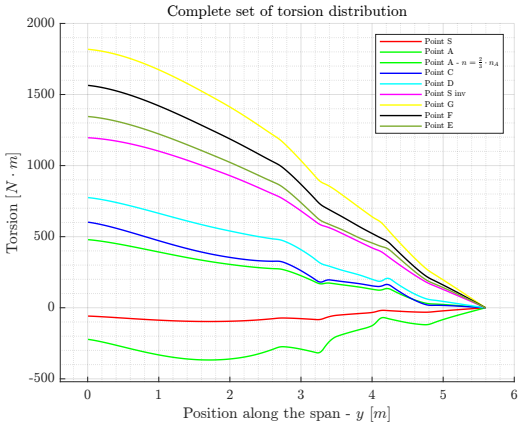
(b) Level 4, twin-engine aircraft. Summary of spanwise shear force distributions. Engine on.



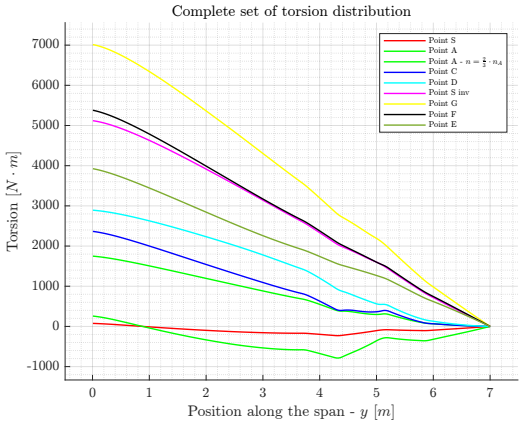
(c) Level 2, twin-engine aircraft. Summary of spanwise bending moment distributions. Engine on.



(d) Level 4, twin-engine aircraft. Summary of spanwise bending moment distributions. Engine on.

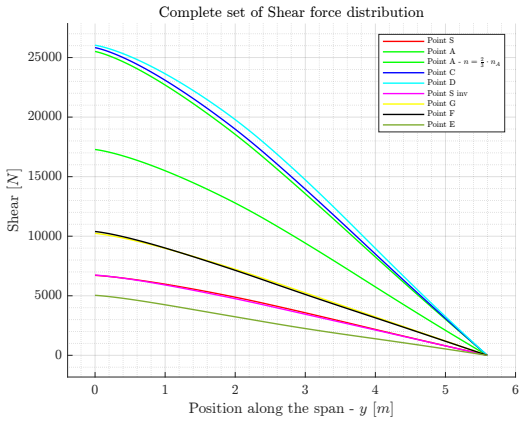


(e) Level 2, twin-engine aircraft. Summary of spanwise torsion distributions. Engine on.

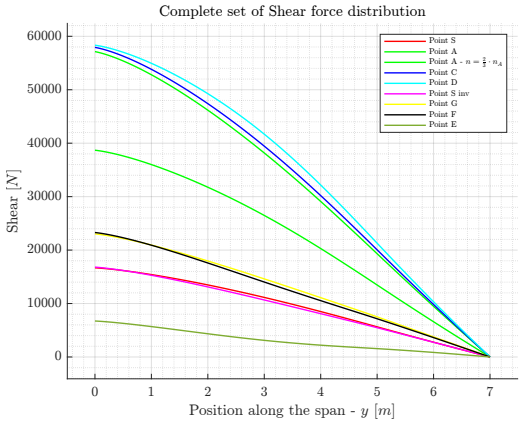


(f) Level 4, twin-engine aircraft. Summary of spanwise torsion distributions. Engine on.

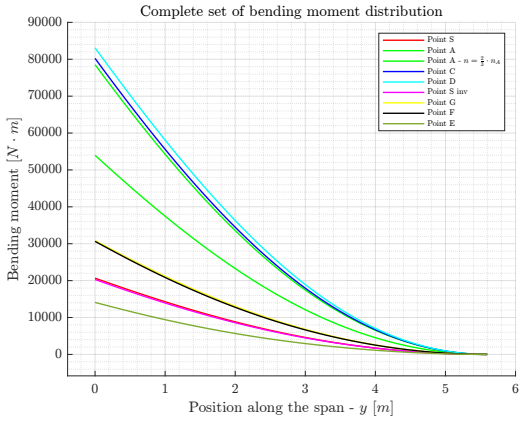
Figure 3.40: Shear, bending moment, and torsion summary graphs for the Level 4 and Level 2, twin-engine aircraft. Engine on.



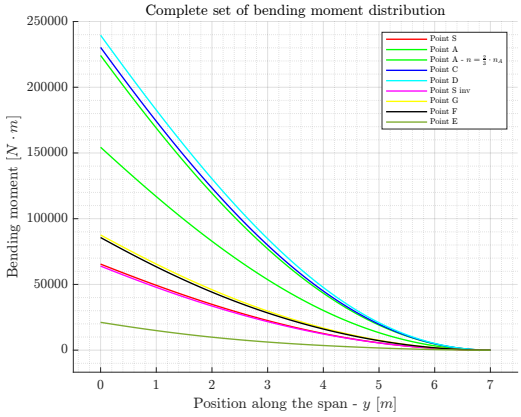
(a) Level 2, twin-engine aircraft. Summary of spanwise shear force distributions. Engine off.



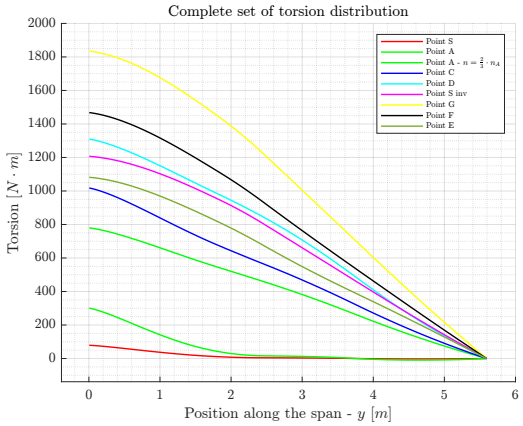
(b) Level 4, twin-engine aircraft. Summary of spanwise shear force distributions. Engine off.



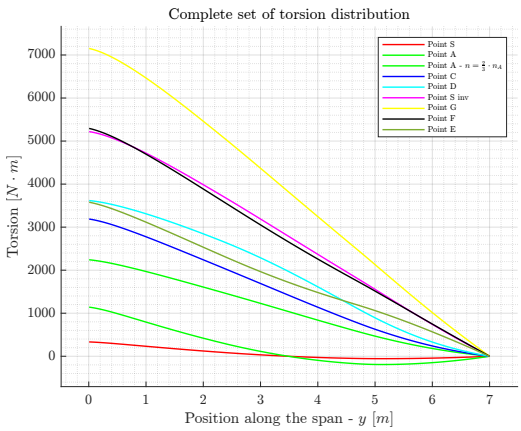
(c) Level 2, twin-engine aircraft. Summary of spanwise bending moment distributions. Engine off.



(d) Level 4, twin-engine aircraft. Summary of spanwise bending moment distributions. Engine off.

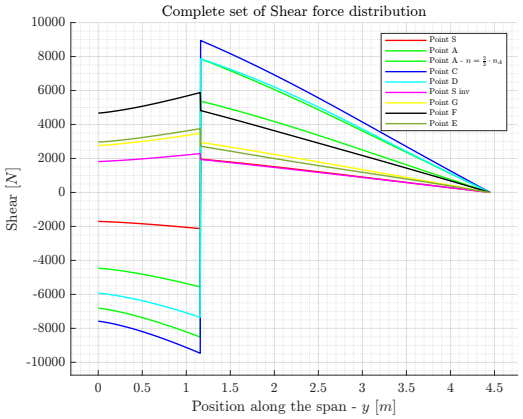


(e) Level 2, twin-engine aircraft. Summary of spanwise torsion distributions. Engine off.

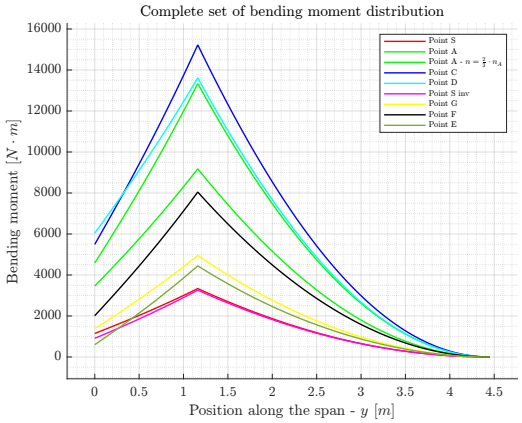


(f) Level 4, twin-engine aircraft. Summary of spanwise torsion distributions. Engine off.

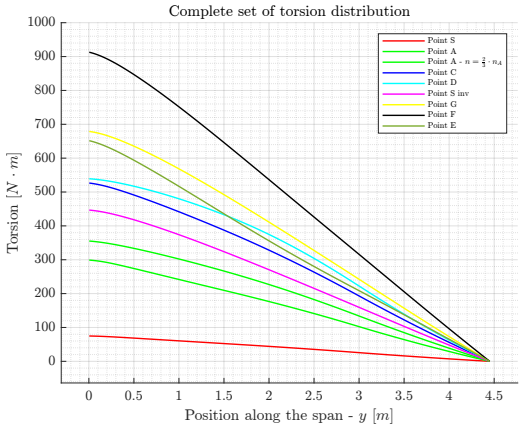
Figure 3.41: Shear, bending moment, and torsion summary graphs for the Level 4 and Level 2, twin-engine aircraft. Engine off.



(a) Level 2, single-engine aircraft. Summary of spanwise shear force distributions. Engine off.



(b) Level 2, single-engine aircraft. Summary of spanwise bending moment distributions. Engine off.

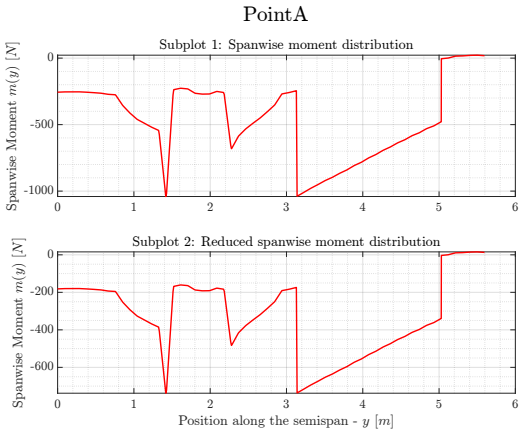


(c) Level 2, single-engine aircraft. Summary of spanwise torsion distributions. Engine off.

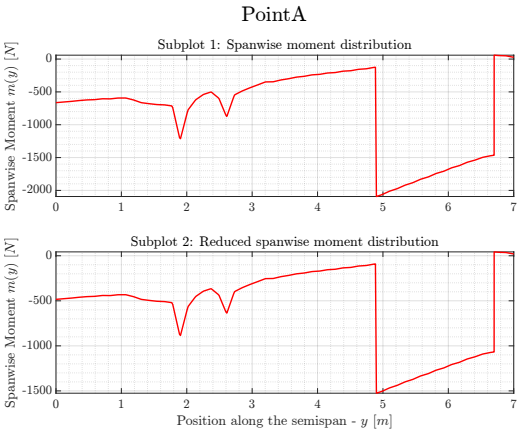
Figure 3.42: Shear, bending moment, and torsion summary graphs for the Level 2, single-engine aircraft. Engine off.

For the Level 2 single-engine aircraft, one notable result is that, among the flight conditions examined, the critical torsion distribution occurs at the design cruise speed under negative limit-load conditions, corresponding to Point F of the maneuvering and gust flight envelope.

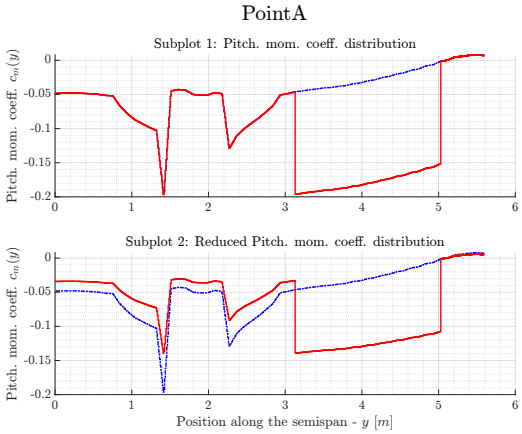
Figures 3.43, 3.44, and 3.45 illustrate another relevant compliance-demonstration item, addressing requirements applicable to the main-wing structure in the presence of aileron deflection, in accordance with paragraphs §23.347(a) and §23.347(b) (unsymmetrical flight conditions) and their corresponding acceptable means of compliance [7].



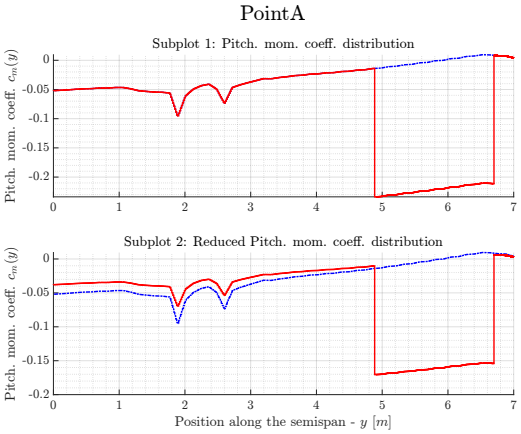
(a) Level 2, twin-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point A. Engine on.



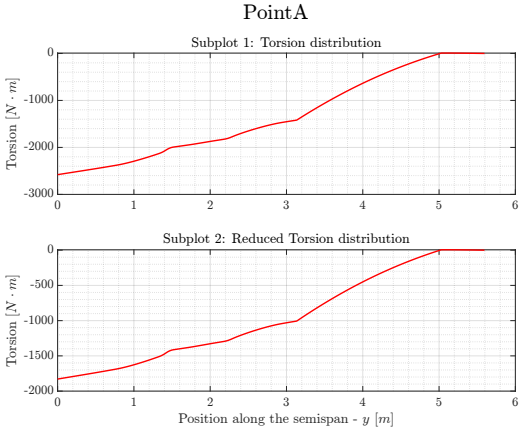
(b) Level 4, twin-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point A. Engine on.



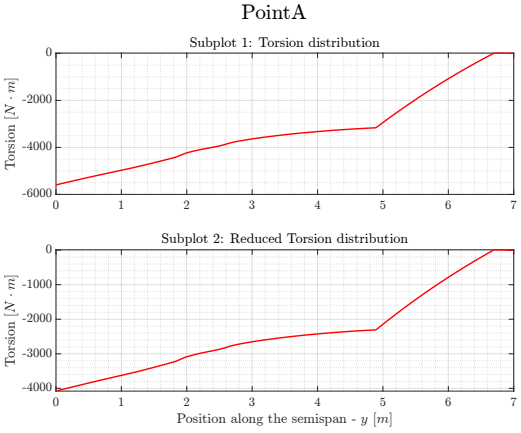
(c) Level 2, twin-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point A. Engine on.



(d) Level 4, twin-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point A. Engine on.

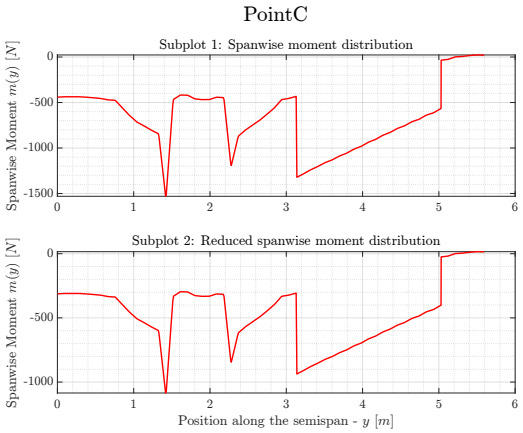


(e) Level 2, twin-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point A. Engine on.

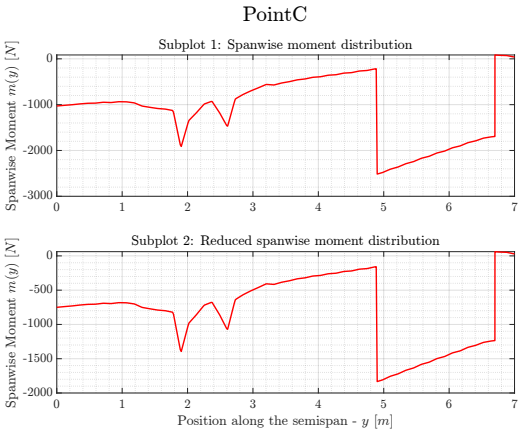


(f) Level 4, twin-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point A. Engine on.

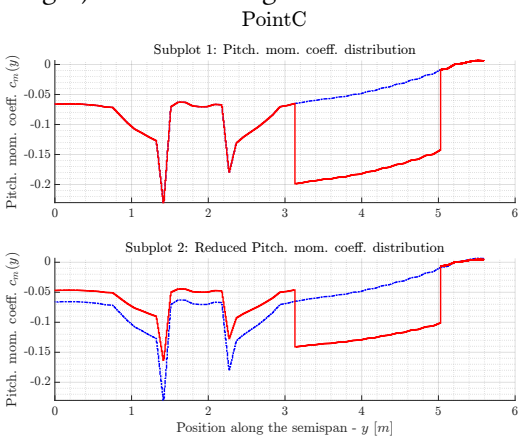
Figure 3.43: Unsymmetrical pitching moment, pitching moment coefficient, and torsion distribution for the Level 4 and Level 2, twin-engine aircraft at Point A of the maneuvering and gust flight envelope. Engine on.



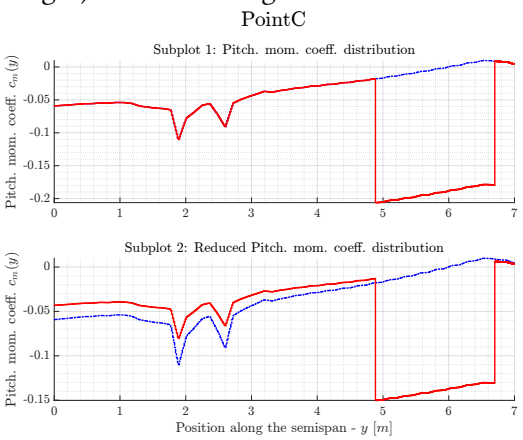
(a) Level 2, twin-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point C. Engine on.



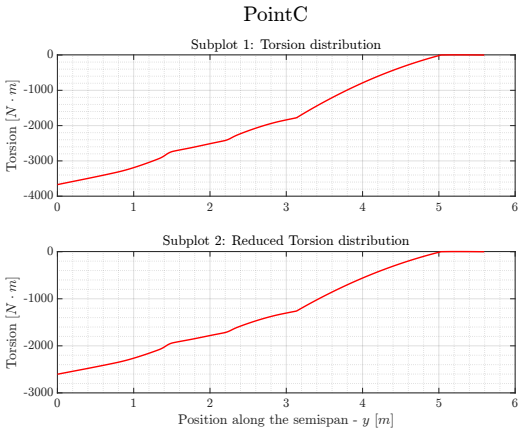
(b) Level 4, twin-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point C. Engine on.



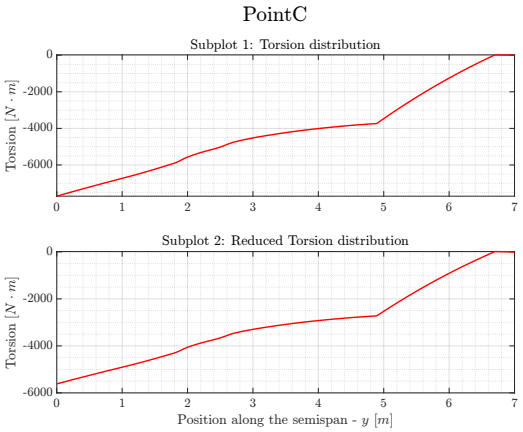
(c) Level 2, twin-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point C. Engine on.



(d) Level 4, twin-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point C. Engine on.

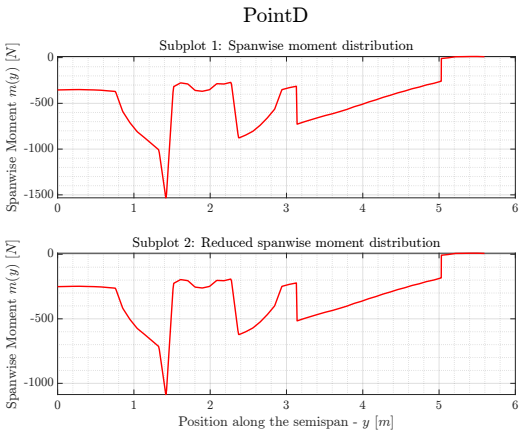


(e) Level 2, twin-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point C. Engine on.

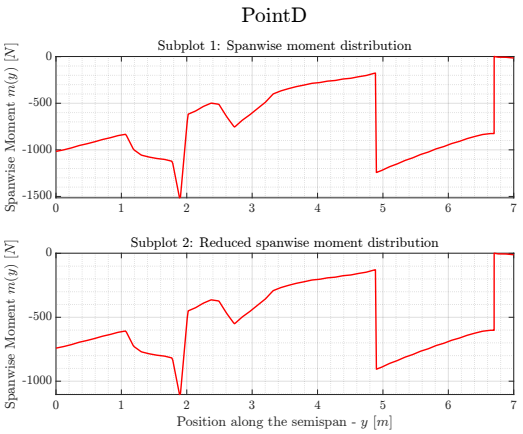


(f) Level 4, twin-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point C. Engine on.

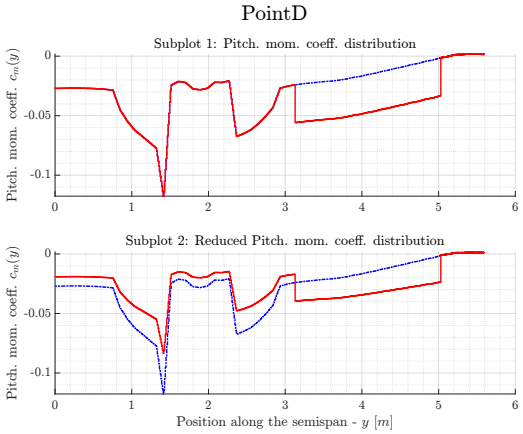
Figure 3.44: Unsymmetrical pitching moment, pitching moment coefficient, and torsion distribution for the Level 4 and Level 2, twin-engine aircraft at Point C of the maneuvering and gust flight envelope. Engine on.



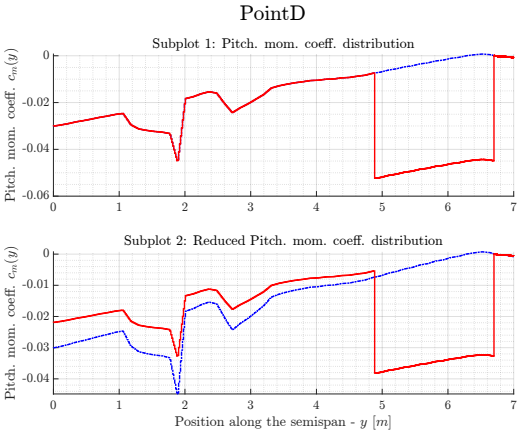
(a) Level 2, twin-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point D. Engine on.



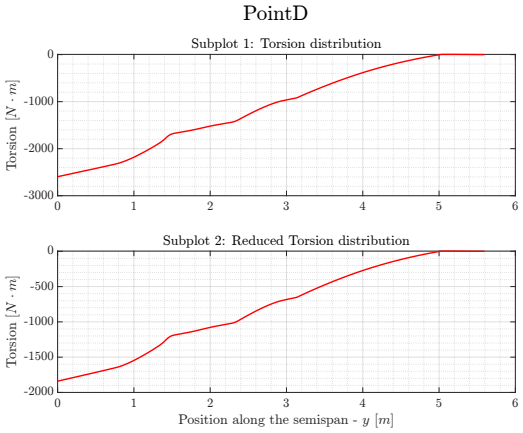
(b) Level 4, twin-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point D. Engine on.



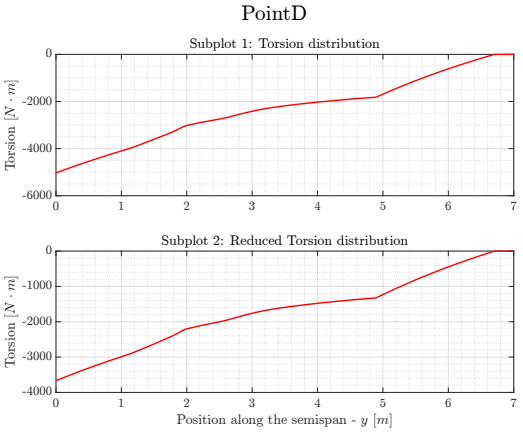
(c) Level 2, twin-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point D. Engine on.



(d) Level 4, twin-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point D. Engine on.

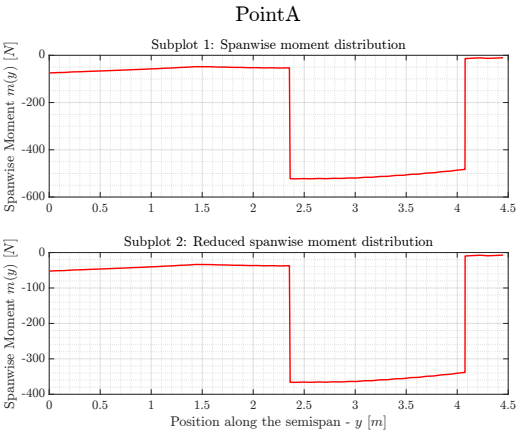


(e) Level 2, twin-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point D. Engine on.

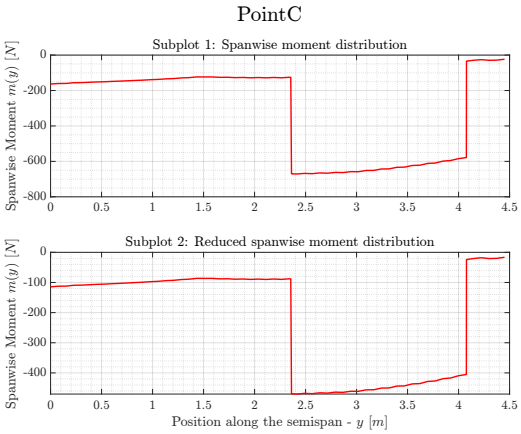


(f) Level 4, twin-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point D. Engine on.

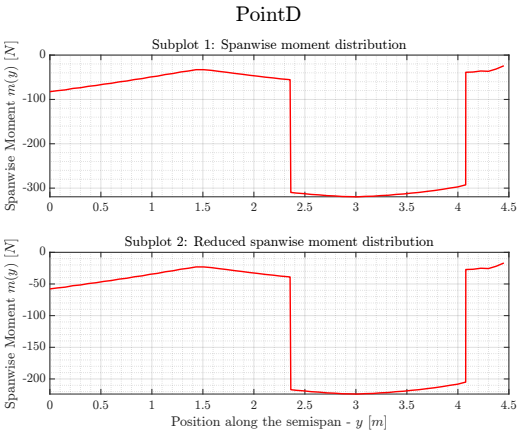
Figure 3.45: Unsymmetrical pitching moment, pitching moment coefficient, and torsion distribution for the Level 4 and Level 2, twin-engine aircraft at Point D of the maneuvering and gust flight envelope. Engine on.



(a) Level 2, single-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point A. Engine off.



(b) Level 2, single-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point C. Engine off.



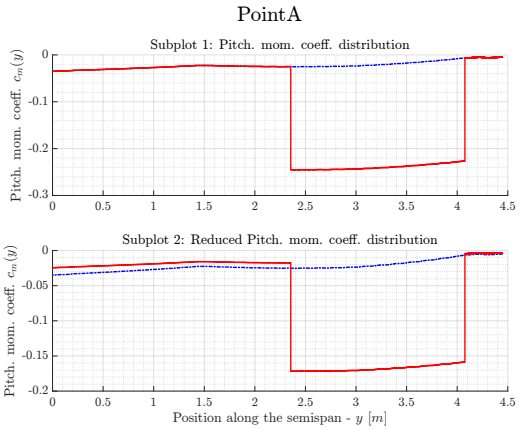
(c) Level 2, single-engine aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point D. Engine off.

Figure 3.46: Spanwise pitching moment distribution $m(y)$ (moment per unit length) at Points A, C, and D of the maneuvering and gust flight envelope for the Level 2, single-engine aircraft. Engine off.

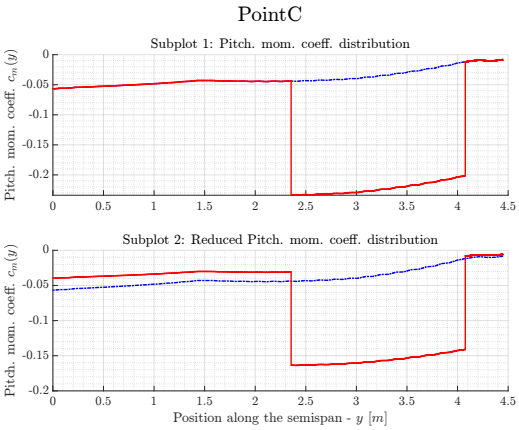
Each figure presents the full and reduced spanwise distributions of pitching moment $m(y)$, pitching-moment coefficient $C_m(y)$, and torsion $t(y)$ at the maneuvering design condition (Point A), the design cruise condition (Point C), and the design dive condition (Point D) with a power setting corresponding to the maximum continuous power.

Figures 3.46, 3.47, and 3.48 present the same analysis for the Level 2 single-engine configuration with the engine off. The figures compare the same quantities evaluated at different points of the maneuvering and gust flight envelope.

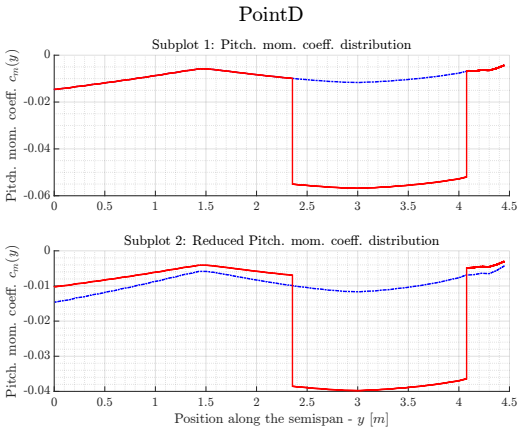
The methodology used to calculate unsymmetrical loads due to aileron deflection may be discussed with the certification authority.



(a) Level 2, single-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point A. Engine off.



(b) Level 2, single-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point C. Engine off.



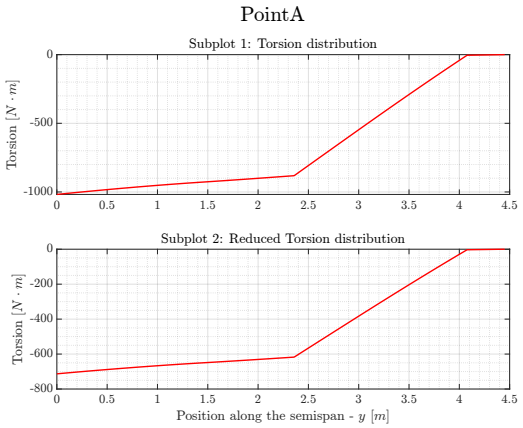
(c) Level 2, single-engine aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point D. Engine off.

Figure 3.47: Spanwise pitching moment coefficient distribution $C_m(y)$ (moment per unit length) at Points A, C, and D of the maneuvering and gust flight envelope for the Level 2, single-engine aircraft. Engine off.

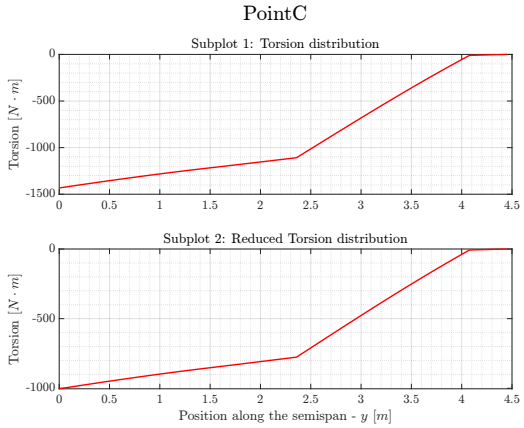
However, the applicable means of compliance and the supporting material contained in the consensus standards provide guidance and calculation methods for evaluating these loads conservatively using established procedures.

Interested readers may refer to ASTM F3116 [230, p. 4, sec. 4.9] (*Unsymmetrical Flight Conditions*) and to Pearson [232] for a theoretical overview based on investigations conducted during the 1930s.

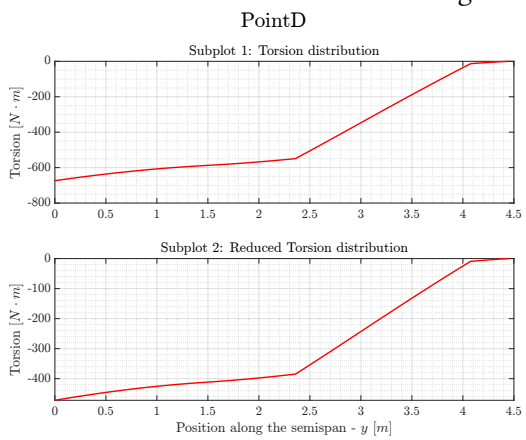
Although this thesis reports calculations performed with well-established methods implemented in MATLAB and Python, its primary focus is the integration of those methods into a single, self-contained digital certification workflow capable of generating a complete technical report for EASA CS-23, Subpart C (Structure).



(a) Level 2, single-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point A. Engine off.



(b) Level 2, single-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point C. Engine off.



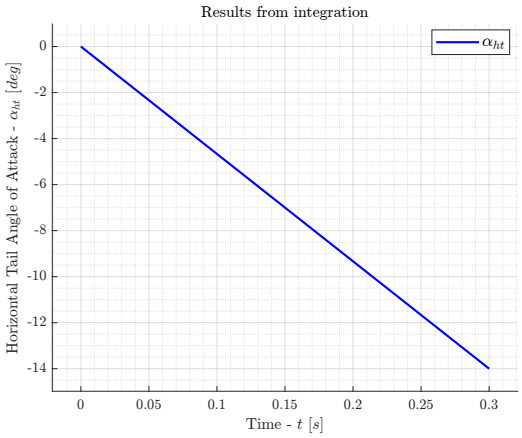
(c) Level 2, single-engine aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point D. Engine off.

Figure 3.48: Spanwise unsymmetrical torsion distribution $t(y)$ (moment per unit length) at Points A, C, and D of the maneuvering and gust flight envelope for the Level 2, single-engine aircraft. Engine off.

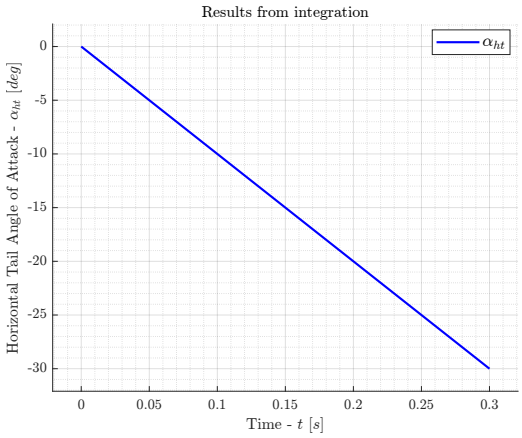
The certification demonstration items produced so far show that the workflow can be executed efficiently while allowing the user to vary key parameters, such as altitude, aircraft weight, center-of-gravity position, and engine power setting. Two additional applications remain to be presented in this thesis:

- the maneuvering (checked and unchecked) and gust loads acting on the empennage;
- the powerplant loads associated with engine torque, side loads, gyroscopic effects, and aerodynamic moments.

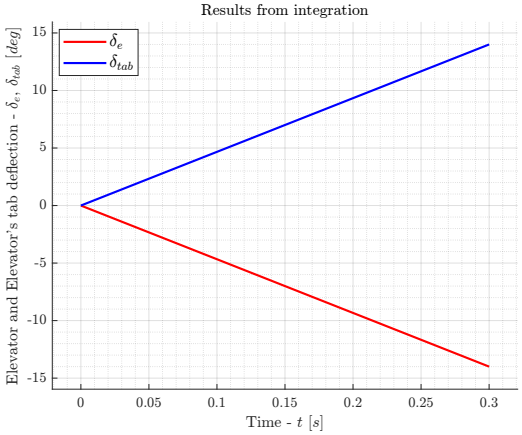
The first set of results is discussed in this Section, whereas the powerplant loads substantiation procedures are described in Appendix A.



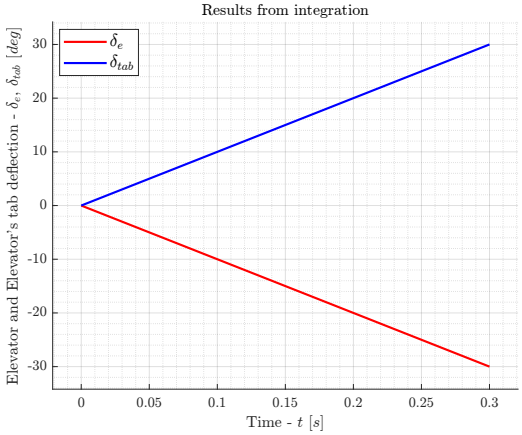
(a) Level 2, twin-engine aircraft. Horizontal tailplane angle of attack $\alpha_{\text{ht}}(t)$ during the unchecked maneuver. Engine off.



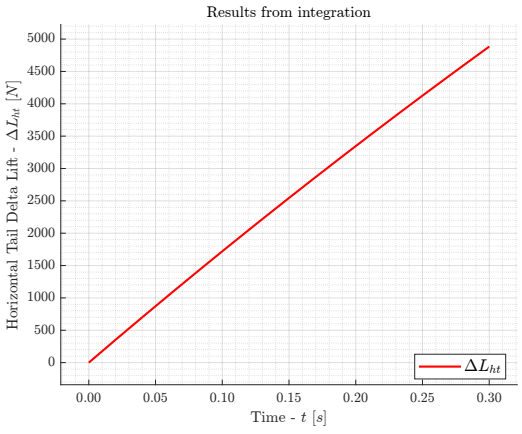
(b) Level 4, twin-engine aircraft. Horizontal tailplane angle of attack $\alpha_{\text{ht}}(t)$ during the unchecked maneuver. Engine off.



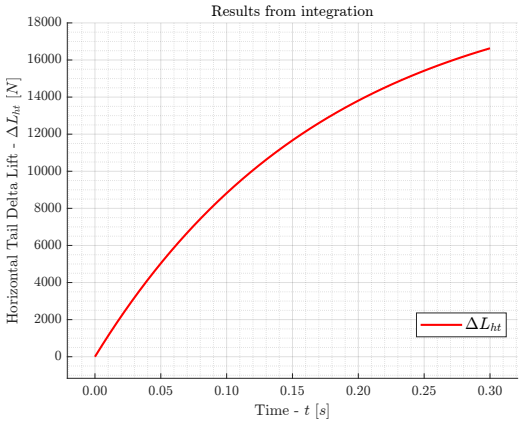
(c) Level 2, twin-engine aircraft. Elevator and elevator's deflection angles, δ_e and δ_{e_t}. Engine off.



(d) Level 4, twin-engine aircraft. Elevator and elevator's deflection angles, δ_e and δ_{e_t}. Engine off.

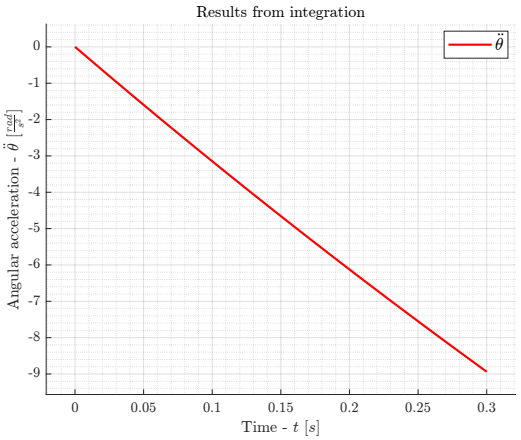


(e) Level 2, twin-engine aircraft. Horizontal tailplane lift-increase during the unchecked maneuver, ΔL_{ht}. Engine off.

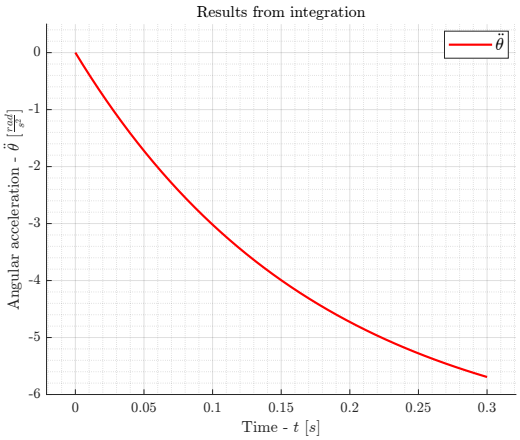


(f) Level 4, twin-engine aircraft. Horizontal tailplane lift-increase during the unchecked maneuver, ΔL_{ht}. Engine off.

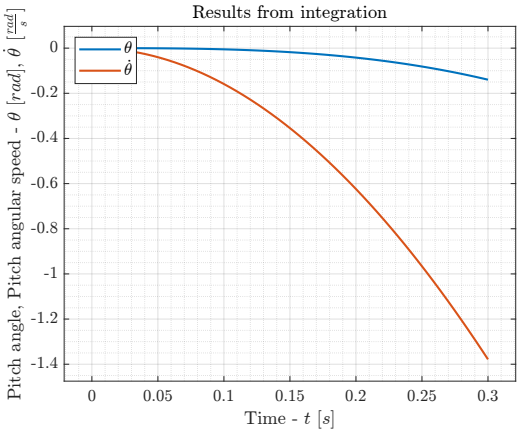
Figure 3.49: Horizontal tailplane angle of attack, elevator and elevator tab deflection angle, and lift increase for the Level 4 and Level 2, twin-engine aircraft. Engine off.



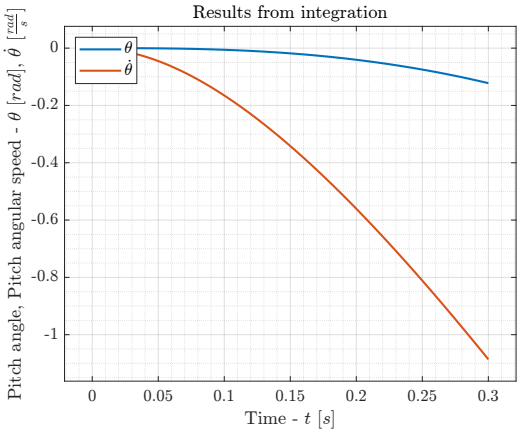
(a) Level 2, twin-engine aircraft. Pitching angular acceleration $\ddot{\theta}(t)$ during the unchecked maneuver. Engine off.



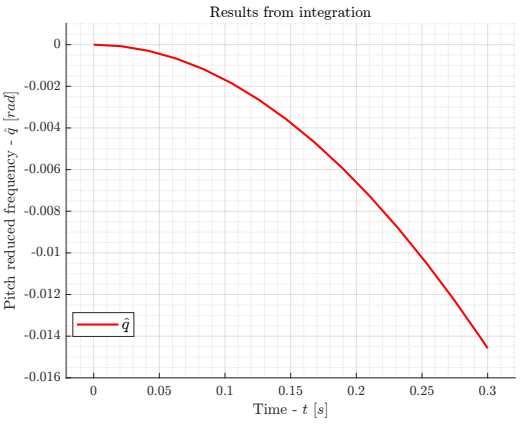
(b) Level 4, twin-engine aircraft. Pitching angular acceleration $\ddot{\theta}(t)$ during the unchecked maneuver. Engine off.



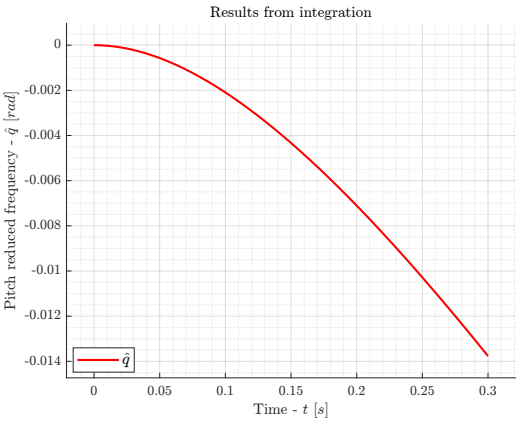
(c) Level 2, twin-engine aircraft. Pitching angle and angular speed, $\theta(t)$ and $\dot{\theta}(t)$. Engine off.



(d) Level 4, twin-engine aircraft. Pitching angle and angular speed, $\theta(t)$ and $\dot{\theta}(t)$. Engine off.

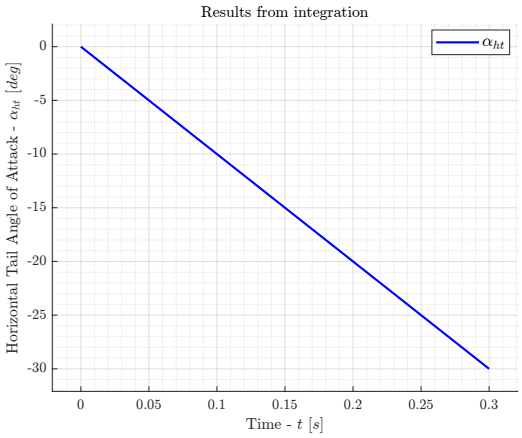


(e) Level 2, twin-engine aircraft. Pitching reduced frequency, $\hat{q}(t)$. Engine off.

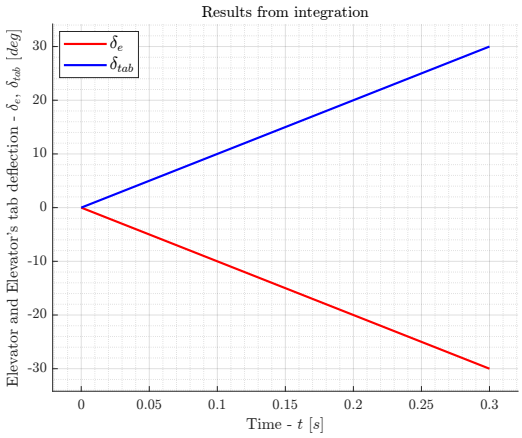


(f) Level 4, twin-engine aircraft. Pitching reduced frequency, $\hat{q}(t)$. Engine off.

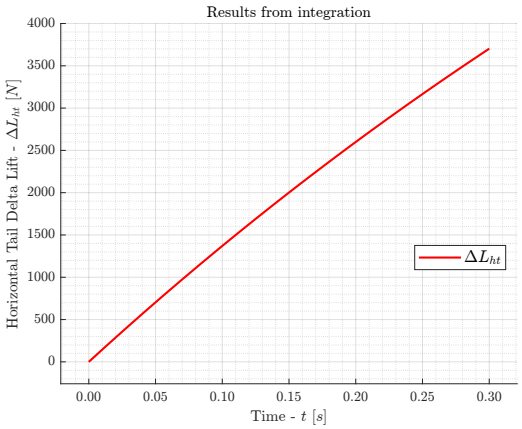
Figure 3.50: Horizontal tailplane pitching angle, angular speed, angular acceleration, and reduced frequency for the Level 4 and Level 2 twin-engine aircraft. Engine off.



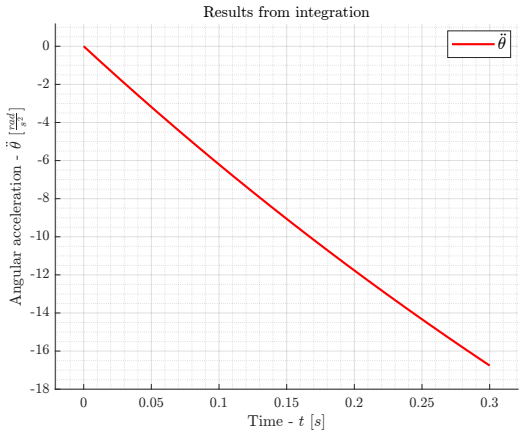
(a) Level 2, single-engine aircraft. Horizontal tailplane angle of attack $\alpha_{\text{ht}}(t)$ during the unchecked maneuver. Engine off.



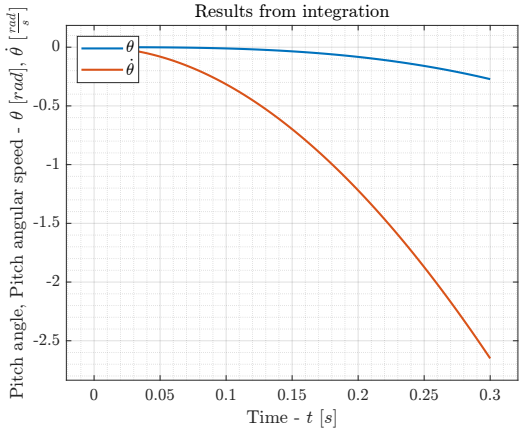
(b) Level 2, single-engine aircraft. Elevator and elevator's tab deflection angles, $\delta_e(t)$ and $\delta_{e_{\text{tab}}}(t)$, during the unchecked maneuver. Engine off.



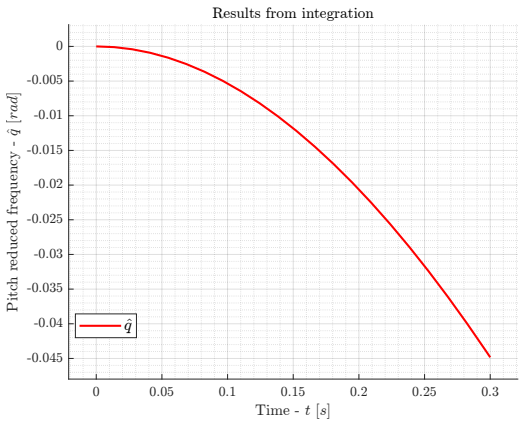
(c) Level 2, single-engine aircraft. Horizontal tailplane lift increase during the unchecked maneuver, ΔL_{ht}. Engine off.



(d) Level 2, single-engine aircraft. Pitching angular acceleration $\ddot{\theta}(t)$ during the unchecked maneuver. Engine off.

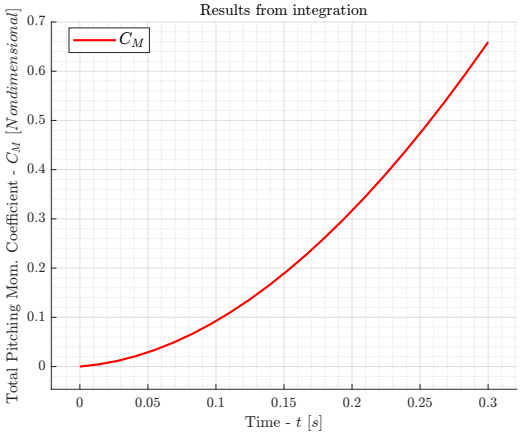


(e) Level 2, single-engine aircraft. Pitching angle and angular speed, $\theta(t)$ and $\dot{\theta}(t)$. Engine off.

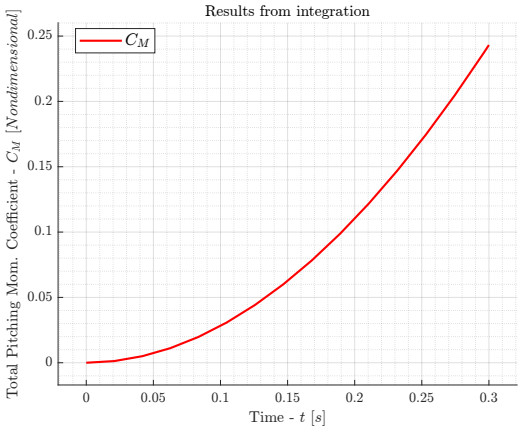


(f) Level 2, single-engine aircraft. Pitching reduced frequency, $\hat{q}(t)$. Engine off.

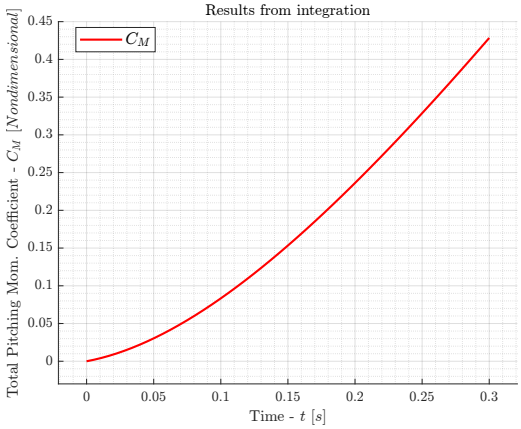
Figure 3.51: Horizontal tailplane unchecked maneuver for the Level 2, single-engine aircraft. Engine off.



(a) Level 2, single-engine aircraft. Total pitching moment coefficient $C_M(t)$. Engine off.



(b) Level 2, twin-engine aircraft. Total pitching moment coefficient $C_M(t)$. Engine off.



(c) Level 4, twin-engine aircraft. Total pitching moment coefficient $C_M(t)$. Engine off.

Figure 3.52: Total pitching moment coefficient for the Level 4 and Level 2 twin-engine and the Level 2 single-engine aircraft. Engine off.

Figures 3.49 and 3.50 present results obtained by integrating a one-dimensional dynamic model of the unchecked maneuver for the Level 2 and Level 4 twin-engine aircraft. The differential equation governing this maneuver is reported in a dedicated appendix. In the present context, it serves as a simplified but useful representation for including this certification demonstration item in the *Flight Loads* certification report.

A range of quantities can be derived from the solution and displayed graphically; in this thesis, the following are reported:

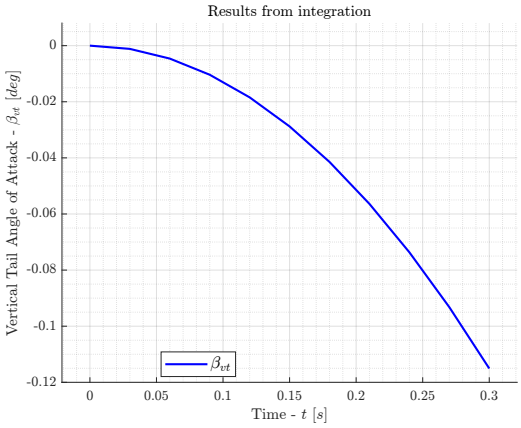
- the horizontal-tail angle of attack, α_{ht};
- the deflection angles of the elevator and elevator tab, δ_{e} and $\delta_{e_{\text{tab}}}$;
- the increment in horizontal-tail lift during the unchecked maneuver, ΔL_{ht};

- the pitching angular acceleration, $\ddot{\theta}$;
- the pitch angle and pitch rate, θ and $\dot{\theta}$; and
- the reduced pitch rate, $\hat{q}$.

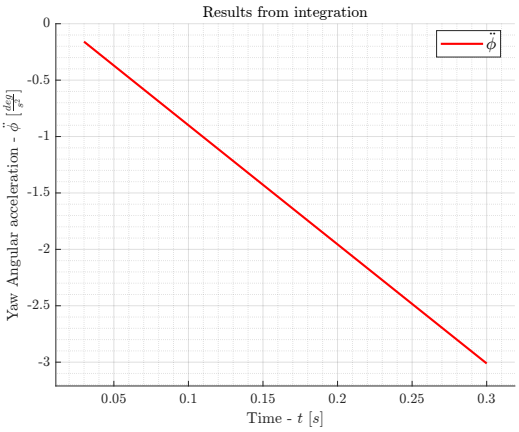
The approach used to obtain these results is practical, widely accepted, and technically sound. It combines simplicity, low computational cost, and rapid execution, thereby allowing the user to examine a wider set of unchecked maneuvers with different numerical parameters. For more complex high-speed, Level 4 aircraft, however, a more detailed six-degree-of-freedom simulation would be advisable. Such simulations are beyond the scope of this thesis. Future work will consider these higher-fidelity analyses for larger transport-category aircraft certified under EASA CS-25. The model adopted for the CS-23 unchecked maneuver is likely conservative for aircraft configurations with high pitch damping. Nevertheless, conservative results are generally acceptable in certification analyses, particularly for larger Level 4 aircraft within the CS-23 category. Figure 3.51 completes the series of results concerning the unchecked maneuver, illustrating the same calculations for the Level 2, single-engine aircraft. Figure 3.52 reports the total pitching moment coefficient during the unchecked maneuver for all the aircraft design architectures analyzed in this thesis. Figures 3.53, 3.54, and 3.55 present the unchecked maneuver for the vertical tailplane. The quantities reported in this thesis are:

- the sideslip angle, β;
- the deflection angles of the rudder and, where applicable, the rudder tab, δ_{r} and $\delta_{r_{\mathrm{tab}}}$;
- the increment in vertical-tail lift during the unchecked maneuver, ΔL_{vt};
- the yaw angular acceleration, $\ddot{\phi}$;
- the yaw angle and yaw rate, ϕ and $\dot{\phi}$; and
- the reduced yaw rate, $\hat{r}$.

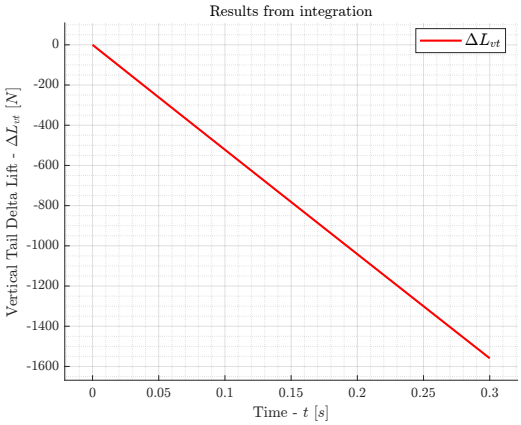
Figure 3.56 shows the total yawing-moment coefficient during the unchecked maneuver for all the aircraft configurations considered. As in the case of the pitching unchecked maneuver, these calculations can be repeated efficiently with different input parameters to examine multiple load cases in accordance with the applicable airworthiness requirements. The method is based on a simplified differential equation reported in the appendices of this thesis. For cases in which nonlinear or other complex lateral-directional effects become important, the certification authority may require a more rigorous modeling approach.



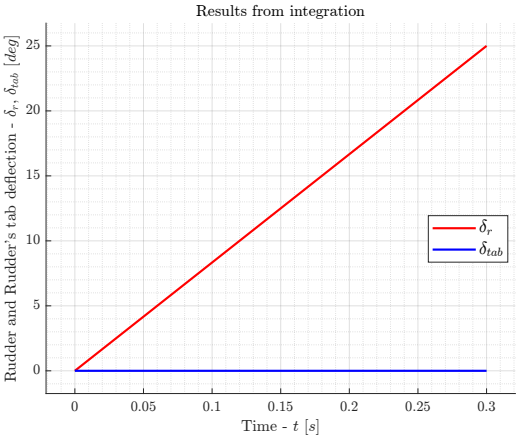
(a) Level 2, single-engine aircraft. Vertical tailplane sideslip angle $\beta(t)$ during the unchecked maneuver. Engine off.



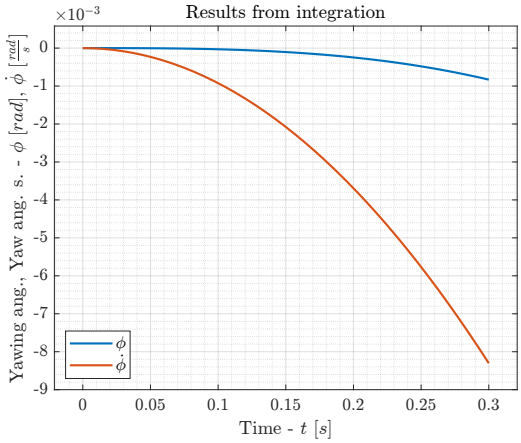
(b) Level 2, single-engine aircraft. Yawing angular acceleration $\ddot{\phi}(t)$ during the unchecked maneuver. Engine off.



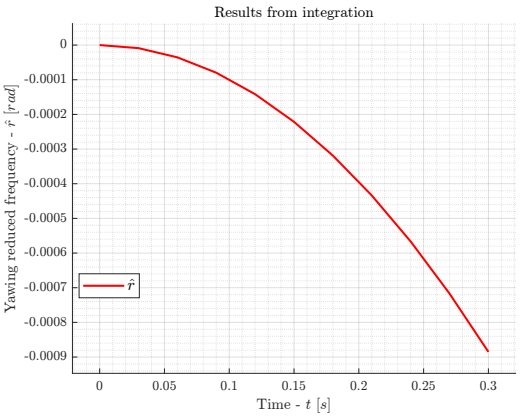
(c) Level 2, single-engine aircraft. Vertical tailplane lift-increase during the unchecked maneuver, ΔL_{vt}. Engine off.



(d) Level 2, single-engine aircraft. Rudder and rudder's tab (if present) deflection angles, $\delta_r(t)$ and $\delta_{v_{\text{tab}}}(t)$ during the unchecked maneuver. Engine off.

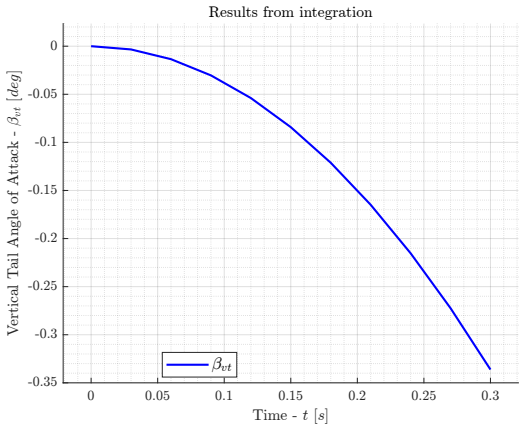


(e) Level 2, single-engine aircraft. Yawing angle and angular speed, $\phi(t)$ and $\dot{\phi}(t)$. Engine off.

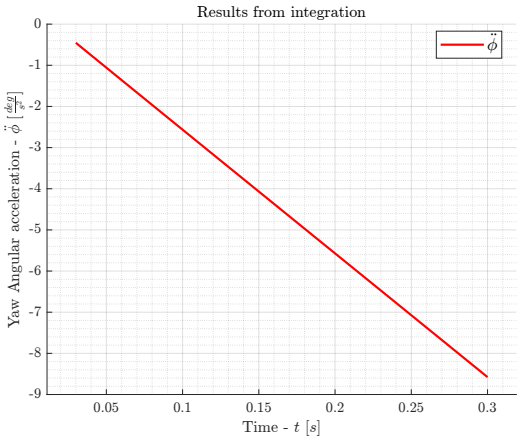


(f) Level 2, single-engine aircraft. Yawing reduced frequency, $\hat{r}(t)$. Engine off.

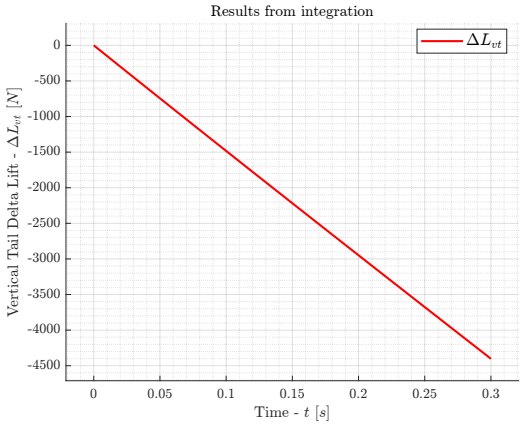
Figure 3.53: Vertical tailplane unchecked maneuver for the Level 2, single-engine aircraft. Engine off.



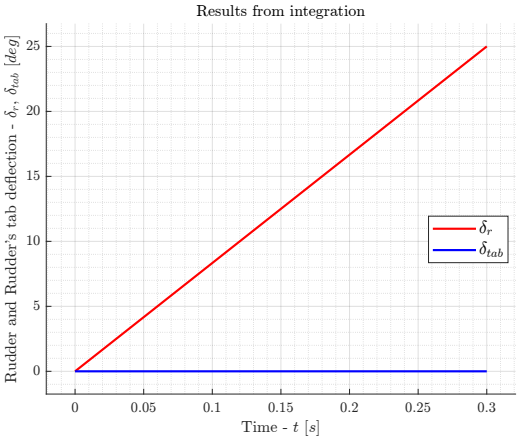
(a) Level 2, twin-engine aircraft. Vertical tailplane sideslip angle $\beta(t)$ during the unchecked maneuver. Engine off.



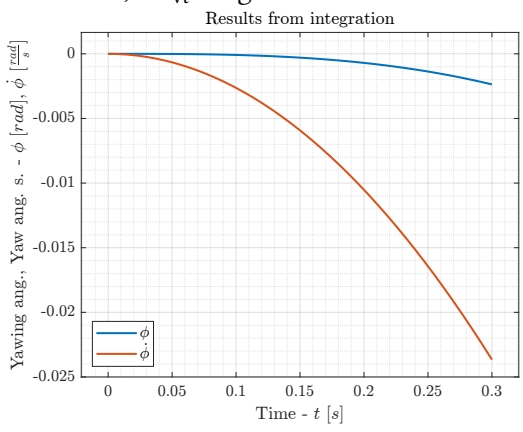
(b) Level 2, twin-engine aircraft. Yawing angular acceleration $\ddot{\phi}(t)$ during the unchecked maneuver. Engine off.



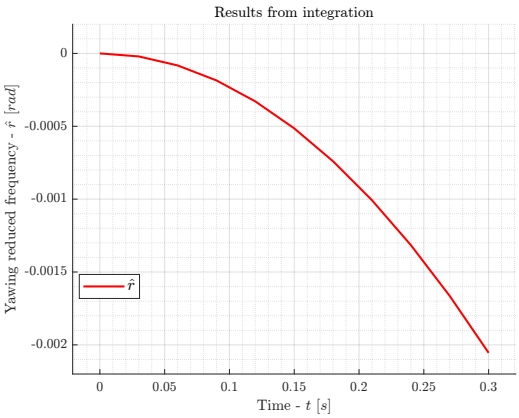
(c) Level 2, twin-engine aircraft. Vertical tailplane lift-increase during the unchecked maneuver, ΔL_{vt}. Engine off.



(d) Level 2, twin-engine aircraft. Rudder and rudder's tab (if present) deflection angles, $\delta_r(t)$ and $\delta_{v_{\text{tab}}}(t)$ during the unchecked maneuver. Engine off.

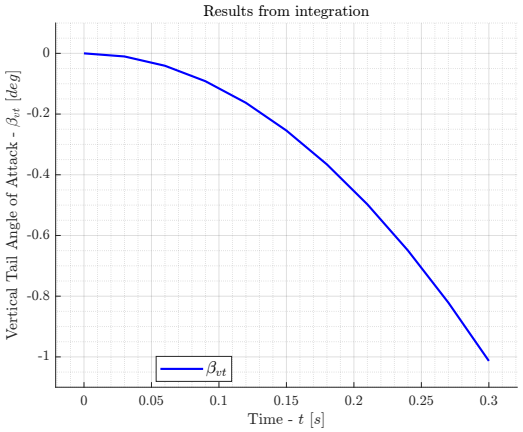


(e) Level 2, twin-engine aircraft. Yawing angle and angular speed, $\phi(t)$ and $\dot{\phi}(t)$. Engine off.

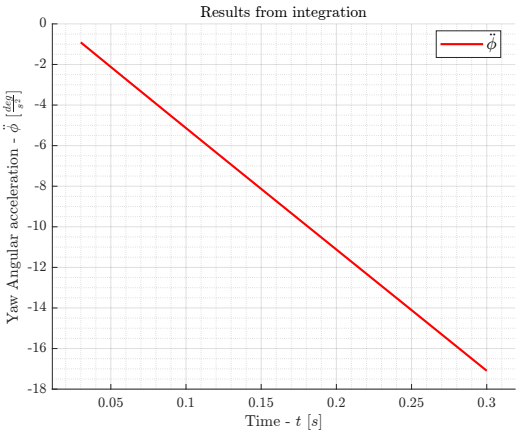


(f) Level 2, twin-engine aircraft. Yawing reduced frequency, $\hat{r}(t)$. Engine off.

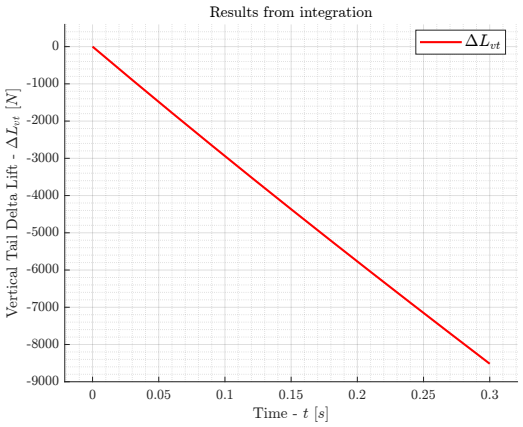
Figure 3.54: Vertical tailplane unchecked maneuver for the Level 2, twin-engine aircraft. Engine off.



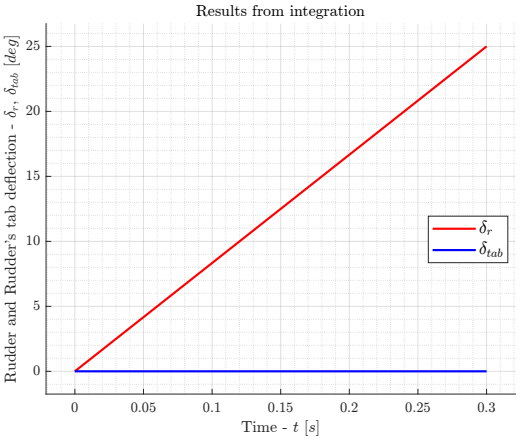
(a) Level 4, twin-engine aircraft. Vertical tailplane sideslip angle $\beta(t)$ during the unchecked maneuver. Engine off.



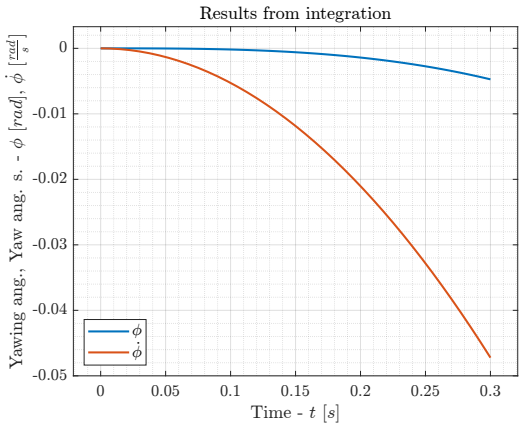
(b) Level 4, twin-engine aircraft. Yawing angular acceleration $\ddot{\phi}(t)$ during the unchecked maneuver. Engine off.



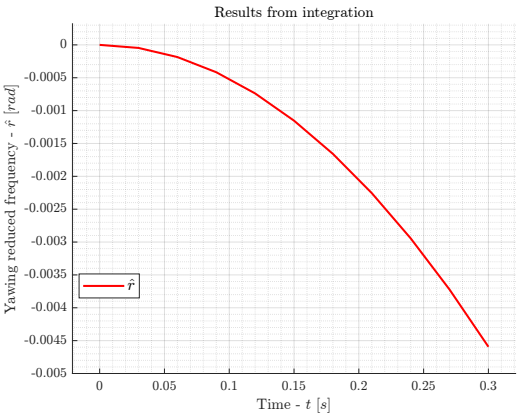
(c) Level 4, twin-engine aircraft. Vertical tailplane lift-increase during the unchecked maneuver, ΔL_{vt}. Engine off.



(d) Level 4, twin-engine aircraft. Rudder and rudder's tab (if present) deflection angles, $\delta_r(t)$ and $\delta_{v_{\text{tab}}}(t)$ during the unchecked maneuver. Engine off.

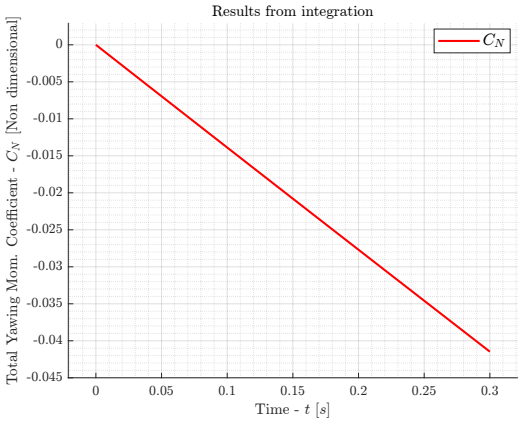


(e) Level 4, twin-engine aircraft. Yawing angle and angular speed, $\phi(t)$ and $\dot{\phi}(t)$. Engine off.

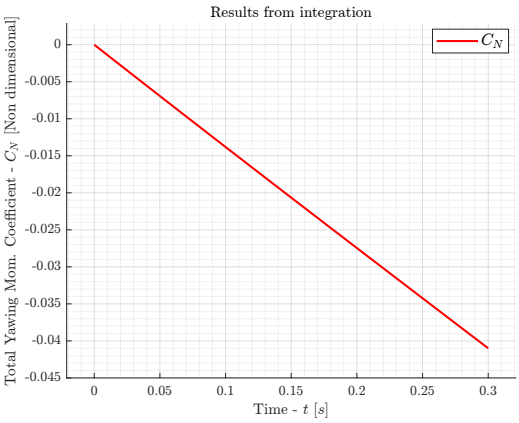


(f) Level 4, twin-engine aircraft. Yawing reduced frequency, $\hat{r}(t)$. Engine off.

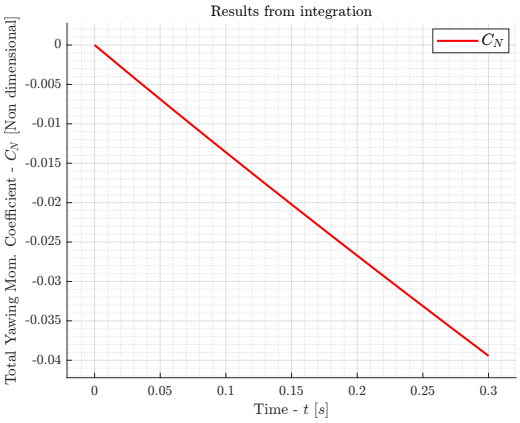
Figure 3.55: Vertical tailplane unchecked maneuver for the Level 4, twin-engine aircraft. Engine off.



(a) Level 2, single-engine aircraft. Total pitching moment coefficient $C_N(t)$. Engine off.



(b) Level 2, twin-engine aircraft. Total pitching moment coefficient $C_N(t)$. Engine off.



(c) Level 4, twin-engine aircraft. Total pitching moment coefficient $C_N(t)$. Engine off.

Figure 3.56: Total yawing moment coefficient for the Level 4 and Level 2 twin-engine and the Level 2 single-engine aircraft. Engine off.

Evaluating empennage flight loads is essential to the preparation of a complete *Flight Loads* certification report.

These calculations support the demonstration of compliance with the applicable substantiation requirements and provide the structural airworthiness evidence needed to identify the most adverse load combinations that may occur during normal and abnormal flight operations.

In summary, the application presented in this Section demonstrates the feasibility of a digitally managed certification process based on the MBSE paradigm.

This is achieved by defining an appropriate set of encoding *rules* within the SysML model and by deploying suitable tools and methodologies to import and utilize the airworthiness requirements embedded in the model.

A limitation of the approach, also addressed in Section 3.4, lies in the adopted parsing

strategy. The current implementation relies on regular expressions to extract relevant information from the textual content of the regulatory documents.

While effective for structured inputs, this approach is sensitive to changes in the underlying text. Modifications to the document database, such as those introduced by regulatory amendments, can disrupt the parsing routines, which are not inherently capable of adapting autonomously to such variations.

The most important certification artifact generated by the MBSE certification framework is the *Flight Loads* report. Report generation is managed programmatically in the MATLAB/Simulink environment through the *RepGen* API, which is available within this programming environment.

The initial implementation requires substantial effort, because the report template must define tables, include figures, provide explanatory comments and supporting text, and reference the applicable regulations and supporting documents.

Once implemented, however, the routine is sufficiently flexible to accommodate different aircraft configurations and can therefore be used to generate and compare flight-loads reports efficiently. For a single aircraft configuration, framework execution requires less than ten minutes, and the resulting report can also support verification of the numerical results.

The execution of the report generator routine is agnostic with respect to the considered aircraft architecture. In the case of an unconventional design—one for which the powerplant loads cannot be determined using conventional procedures and compliance methods—the execution automatically adjusts to the aircraft characteristics, avoiding critical Sections such as the powerplant loads.

Special conditions applicable to hybrid or full-electric powerplants are not included, and this Section will be completely excluded from the automatic report drafting routine.

Consistent with the applicable means of compliance and with the structure of the regulations, the report follows a regulation-oriented organization. It provides an overview of the applicable requirements and presents the results obtained by applying the corresponding formulas and substantiation methods described in the source material.

This organization is rational for the purposes of this thesis. However, because the detailed structure of industrial flight-loads reports is typically confidential, the exact format of reports accepted by EASA is not publicly available.

Excerpts from the *Flight Loads* report generated by the digital certification framework are included and briefly discussed in Appendix C. These excerpts provide examples of relevant reporting mechanisms, including tables, figures, and supporting textual content.

Section 3.2.5 below completes the set of results obtained with the Eclipse Papyrus/-MATLAB framework with the non-conventional design architecture use-case scenario, including the output from executing the MATLAB/Papyrus MBSE digital certification framework.

Appendix B includes Table B.3, which compares the numerical results generated by the Eclipse Papyrus/MATLAB MBSE digital certification framework with those produced by the other implementations described in this thesis.

3.2.5 Non-conventional design architecture

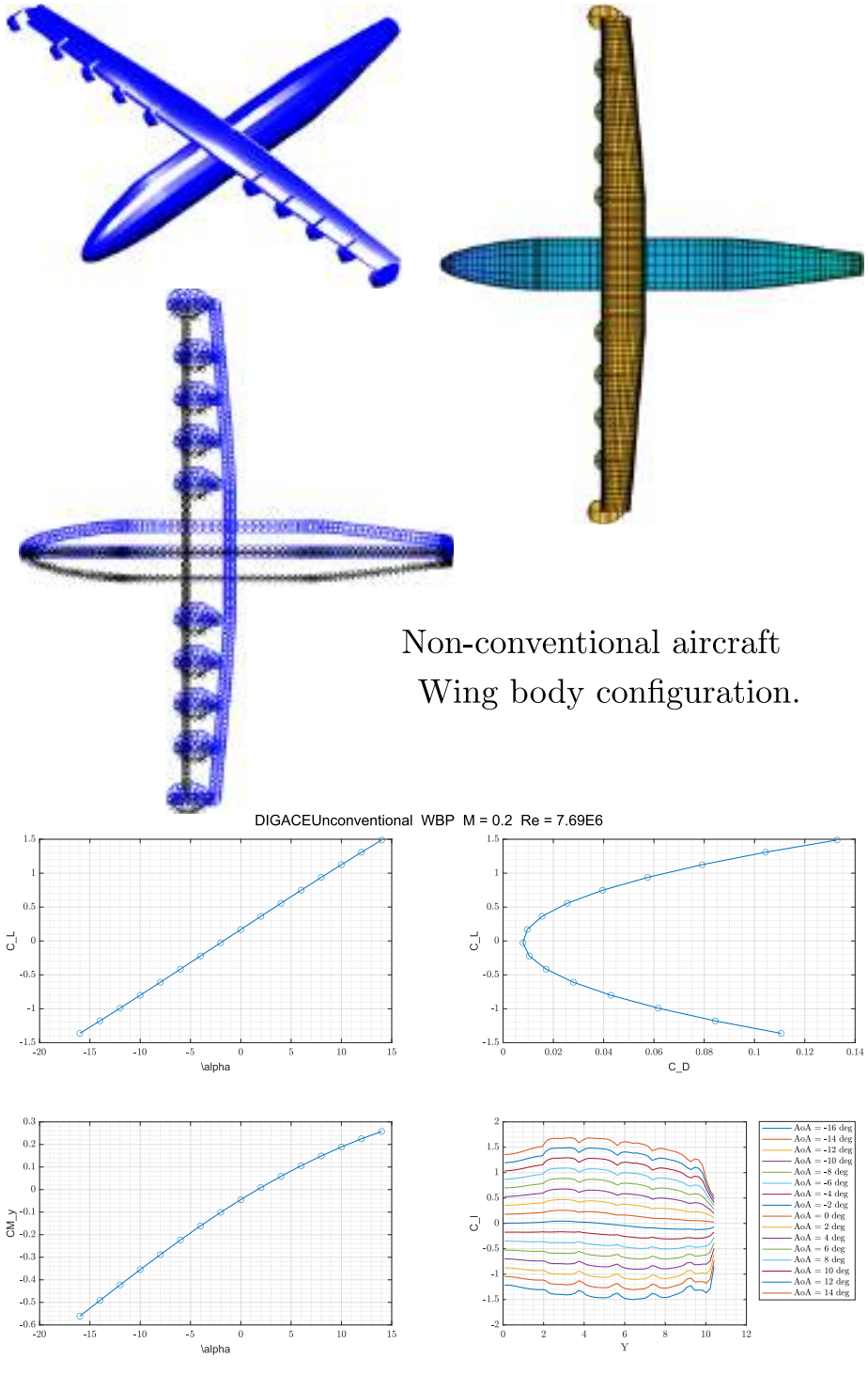
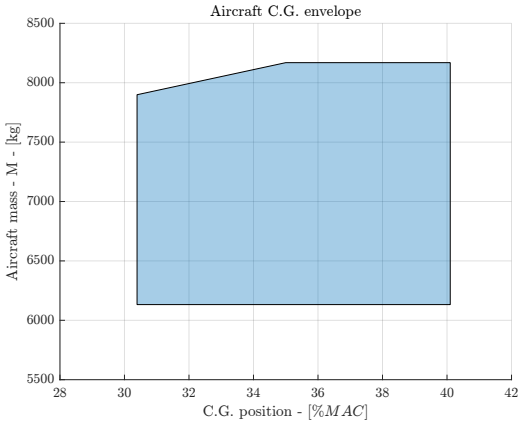
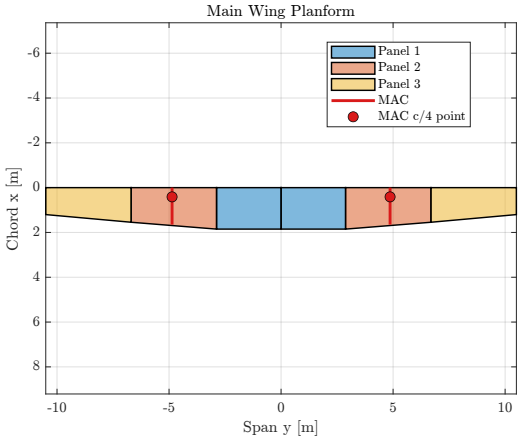


Figure 3.57: Non-conventional aircraft design architecture. Its wing-body configuration as loaded into the *Open VSP* analysis environment.

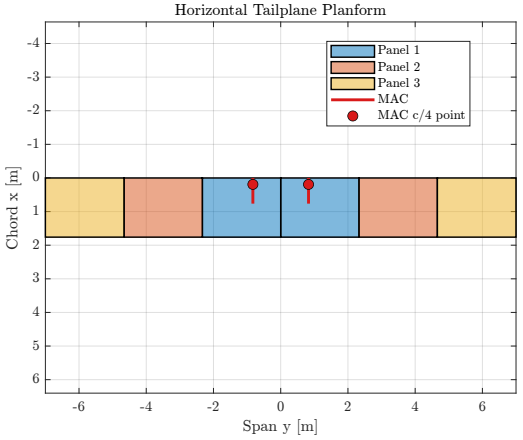
This Section briefly illustrates results from the application of the MBSE digital certification framework established in the MATLAB programming environment to a non-conventional aircraft embodying a distributed electric propulsion (DEP) mechanism. For this application, only the engine-on case will be considered. Figure 3.57 presents the non-conventional architecture when loaded into the *Open VSP* analysis environment. The main wings present eight smaller engines with their corresponding propellers, providing additional lift during takeoff and landing operations, and two main engines located at the wing tips, providing thrust during the cruising phase.



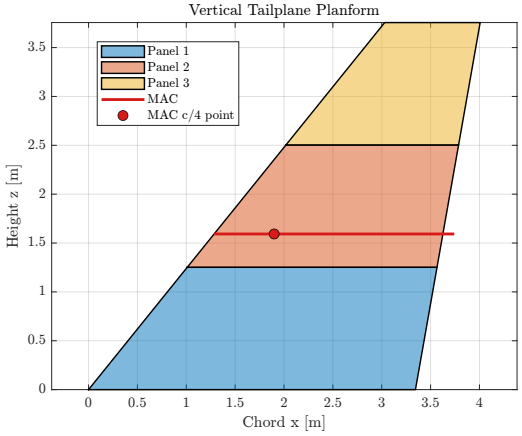
(a) Non-conventional design aircraft mass envelope.



(b) Non-conventional design aircraft wing planform.

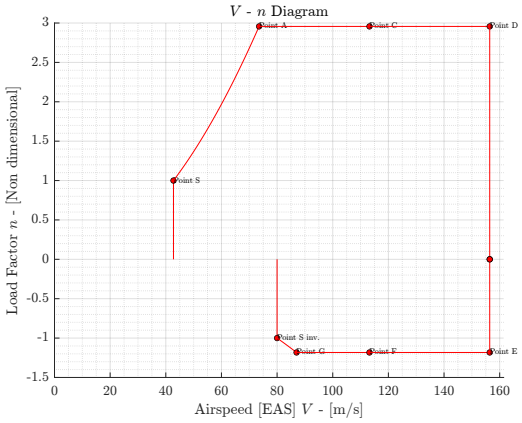


(c) Non-conventional design aircraft h. t. planform.

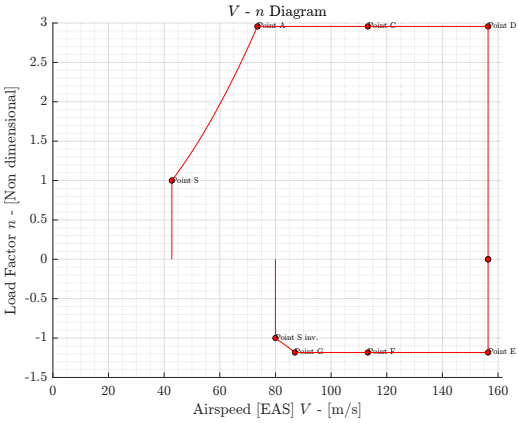


(d) Non-conventional design aircraft v. t. planform.

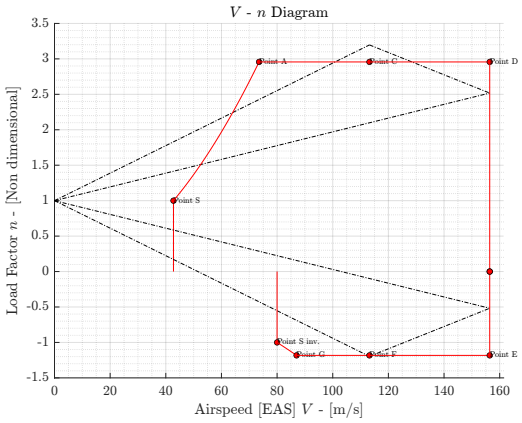
Figure 3.58: Position of the center of gravity, expressed as a percentage of the mean aerodynamic chord, with respect to the aircraft mass and main wing and empennage planforms. The mass boundaries are the operational empty weight and the maximum takeoff mass. The boundaries for the abscissa are the allowable most forward and most aft positions of the center of gravity.



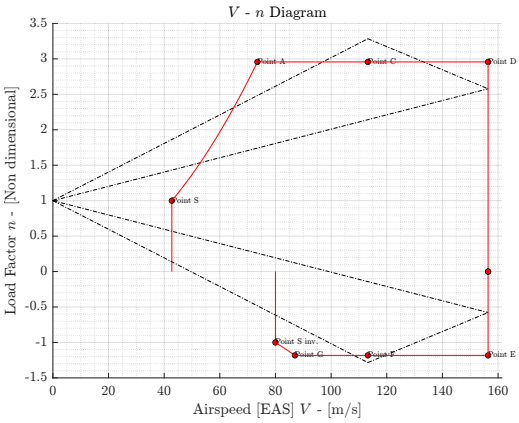
(a) Non-conventional design aircraft maneuvering flight envelope at sea level.



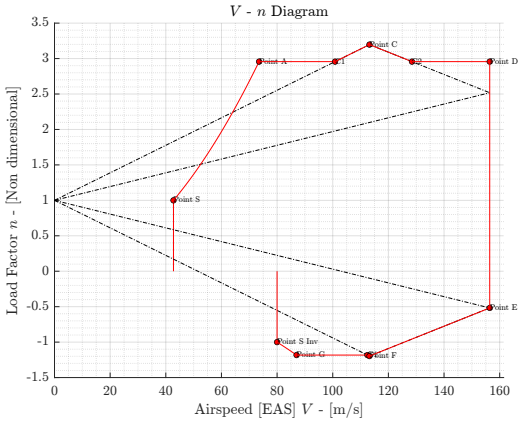
(b) Non-conventional design aircraft maneuvering flight envelope at $h = 12\,000$ ft.



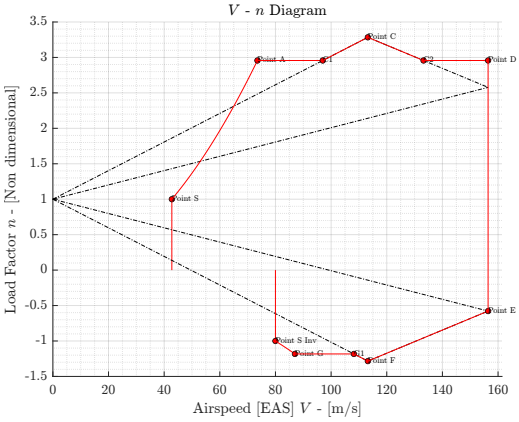
(c) Non-conventional design aircraft maneuvering flight envelope with gust lines at sea level.



(d) Non-conventional design aircraft maneuvering flight envelope with gust lines at $h = 12\,000$ ft.

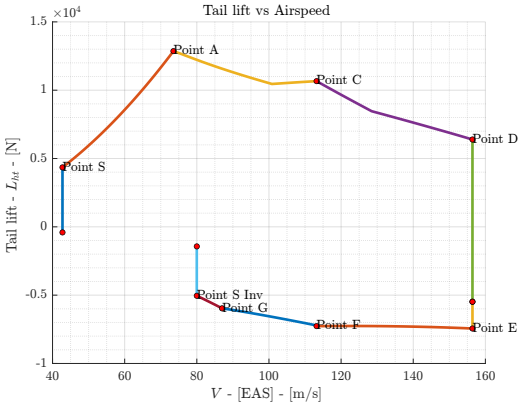


(e) Non-conventional design aircraft maneuvering and gust flight envelope at sea level.

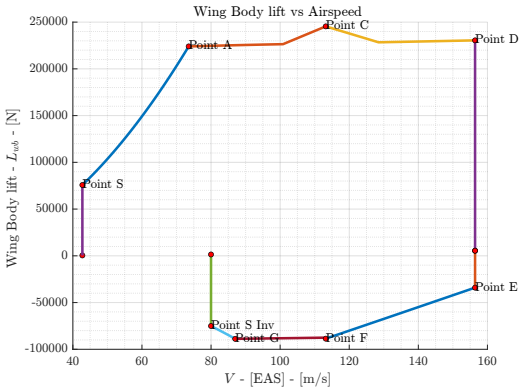


(f) Non-conventional design aircraft maneuvering and gust flight envelope at $h = 12\,000$ ft.

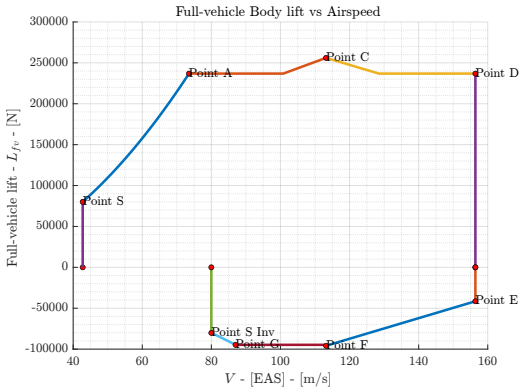
Figure 3.59: The maneuvering and gust flight envelope for the non-conventional design aircraft evaluated at sea level and at $h = 12\,000$ ft.



(a) Non-conventional aircraft balancing loads.



(b) Non-conventional aircraft wing-body lift.



(c) Non-conventional aircraft full-vehicle lift.

Figure 3.60: Balancing loads, wing-body, and full-vehicle lift diagrams for the non-conventional design architecture considered in this thesis.

Figure 3.58 shows the CG-mass envelope and the planform surface of the main wings, the horizontal and vertical tailplanes.

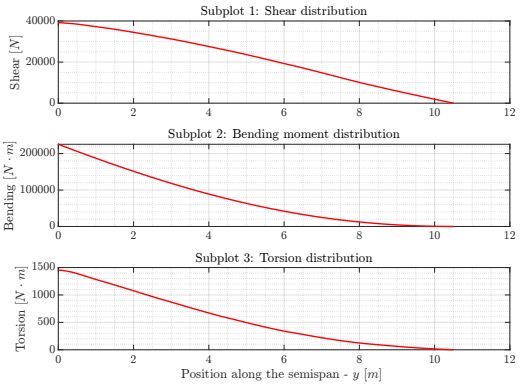
Figure 3.59 illustrates the construction of the maneuvering and gust flight envelope for the non-conventional aircraft, at sea level and at 12 000 ft.

Figure 3.60 shows the resulting balancing loads acting on the horizontal tailplane, the wing-body, and full-vehicle lift forces. It is important to note the influence of the gust limit load factors, especially at Point C, on the maneuvering and gust flight envelope diagram.

Figures 3.61 and 3.62 collect the corresponding spanwise distribution of shear, bending moment, and torsion evaluated at various points of the maneuvering and gust flight envelope.

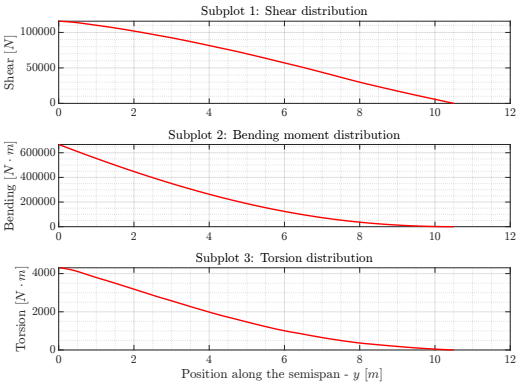
Figure 3.63 presents the spanwise distributions, parameterized with respect to the angle of attack α, of the lift, drag, and pitching moment coefficients. In particular, Figure 3.63a shows only the curves interpolated at the relevant points of the maneuvering and gust flight envelope diagram.

PointS



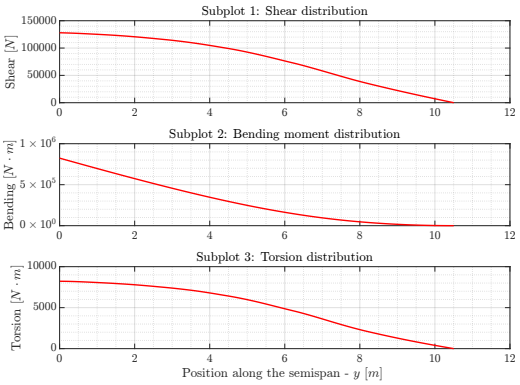
(a) Non-conventional aircraft. Shear, bending moment, and torsion at Point S. Engine on.

PointA



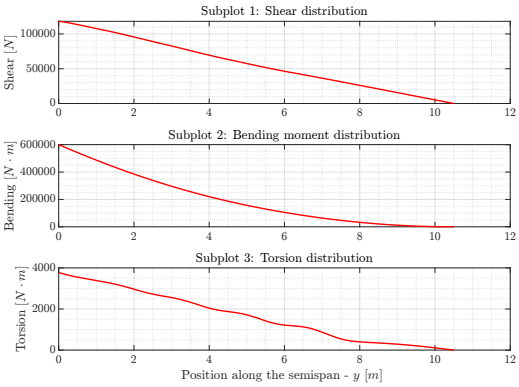
(b) Non-conventional aircraft. Shear, bending moment, and torsion at Point A. Engine on.

PointC



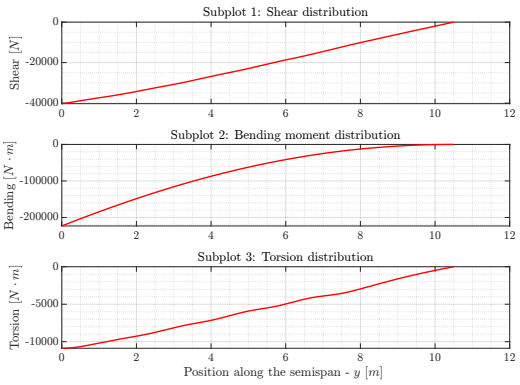
(c) Non-conventional aircraft. Shear, bending moment, and torsion at Point C. Engine on.

PointD



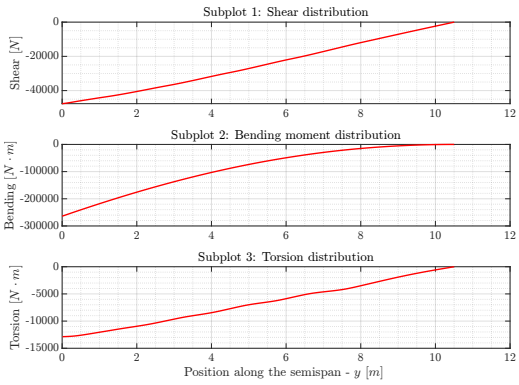
(d) Non-conventional aircraft. Shear, bending moment, and torsion at Point D. Engine on.

PointSinv



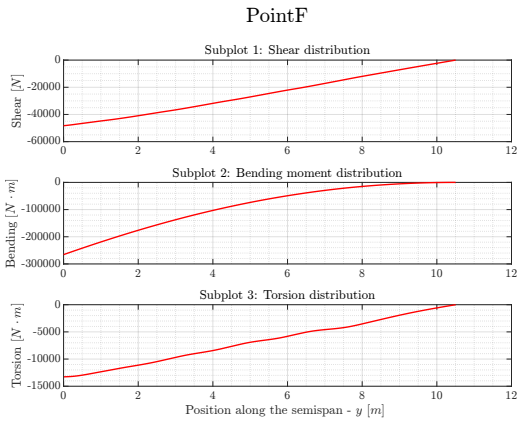
(e) Non-conventional aircraft. Shear, bending moment, and torsion at Point S_{inv}. Engine on.

PointG

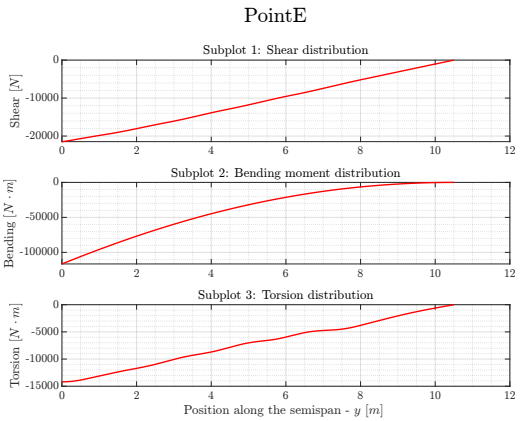


(f) Non-conventional aircraft. Shear, bending moment, and torsion at Point G. Engine on.

Figure 3.61: Shear, bending moment, and torsion for the non-conventional aircraft. Points S, A, C, S_{inv}, and G of the maneuvering and gust flight envelope. Engines on.

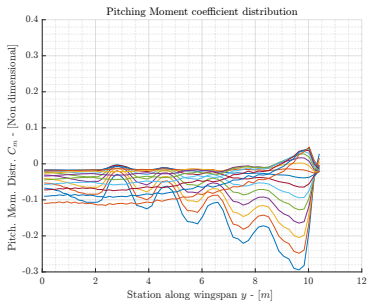


(a) Non-conventional aircraft. Shear, bending moment, and torsion at Point F. Engine on.

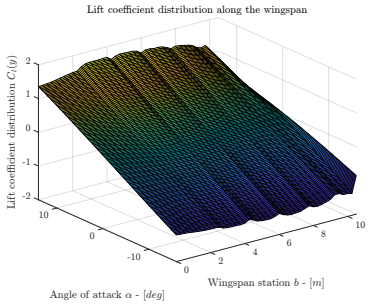


(b) Non-conventional aircraft. Shear, bending moment, and torsion at Point E. Engine on.

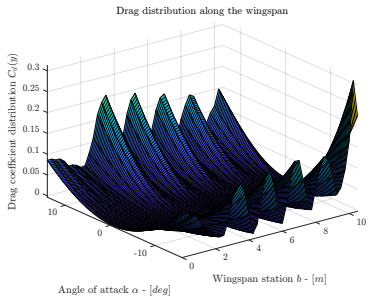
Figure 3.62: Shear, bending moment, and torsion for the non-conventional aircraft. Points F and E of the maneuvering and gust flight envelope. Engines on.



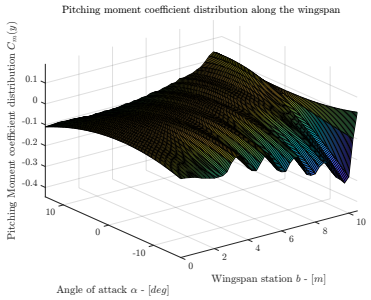
(a) Spanwise pitching moment distribution. Engine on.



(b) Non-conventional $C_l = C_l(y, \alpha)$. Engines on.

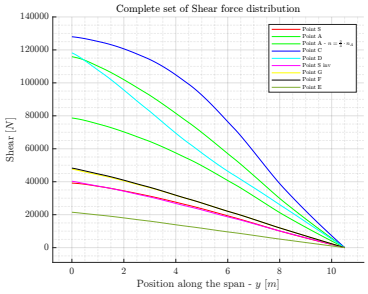


(c) Non-conventional $C_d = C_d(y, \alpha)$. Engines on.

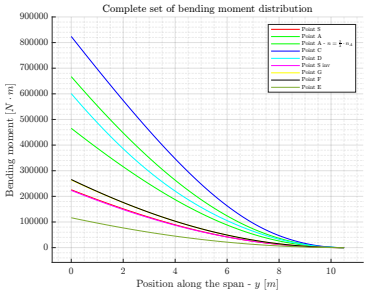


(d) Non-conventional $C_m = C_m(y, \alpha)$. Engines on.

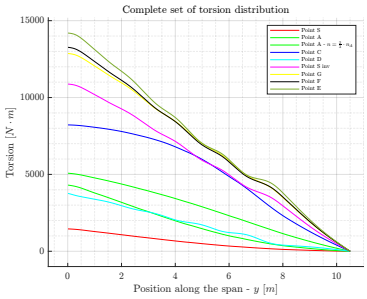
Figure 3.63: Non-conventional aircraft design architecture spanwise pitching moment coefficient $C_M(y)$ distribution, parameterized with respect to the angle of attack α, and three-dimensional representation of the C_l, C_d, and C_m. Engine on.



(a) Non-conventional aircraft. Summary of spanwise shear force distributions. Engine on.

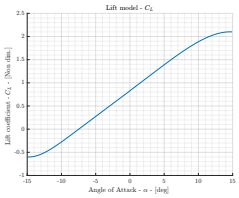


(b) Non-conventional aircraft. Summary of spanwise bending moment distributions. Engine on.

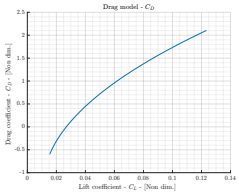


(c) Non-conventional aircraft. Summary of spanwise torsion distributions. Engine on.

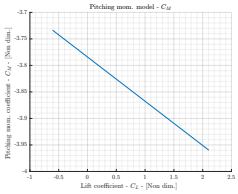
Figure 3.64: Shear, bending moment, and torsion summary graphs for the non-conventional aircraft. Engine on.



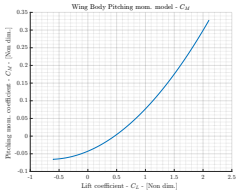
(a) Non-conventional aircraft wing-body lift coefficient.



(b) Non-conventional aircraft wing-body drag coefficient.

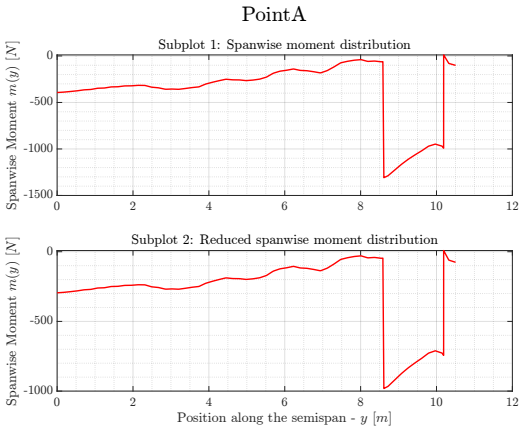


(c) Non-conventional aircraft wing-body pitching moment coefficient as a function of the lift coefficient.

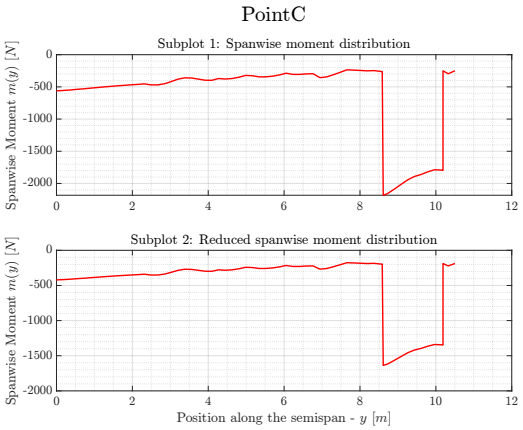


(d) Non-conventional aircraft wing-body pitching moment coefficient as a function of the angle of attack.

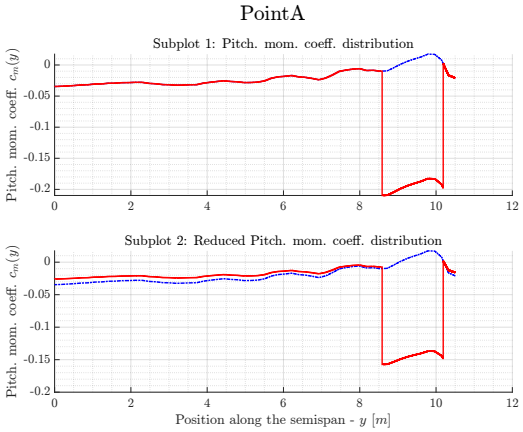
Figure 3.65: A selected output from the wing-body aerodynamic model used for the non-conventional aircraft.



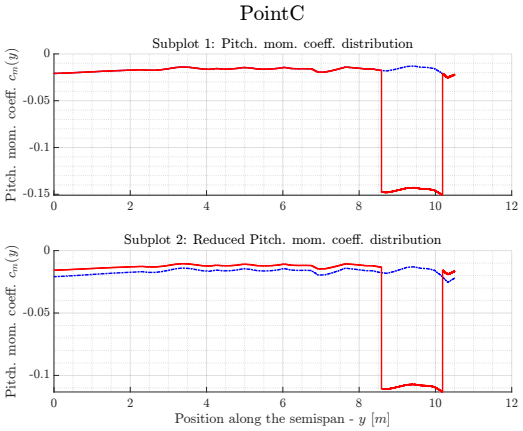
(a) Non-conventional aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point A. Engine on.



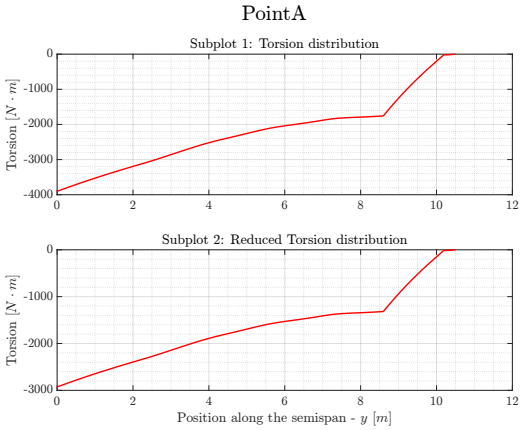
(b) Non-conventional aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point C. Engine on.



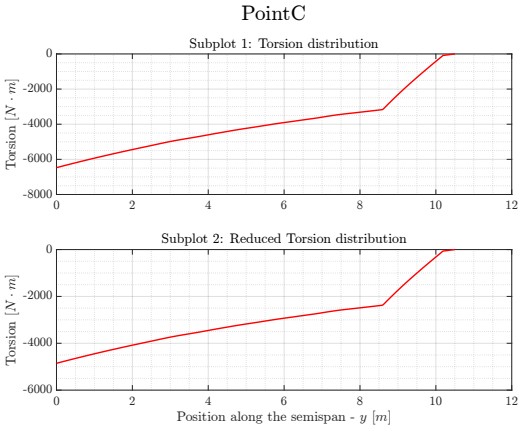
(c) Non-conventional aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point A. Engine on.



(d) Non-conventional aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point C. Engine on.

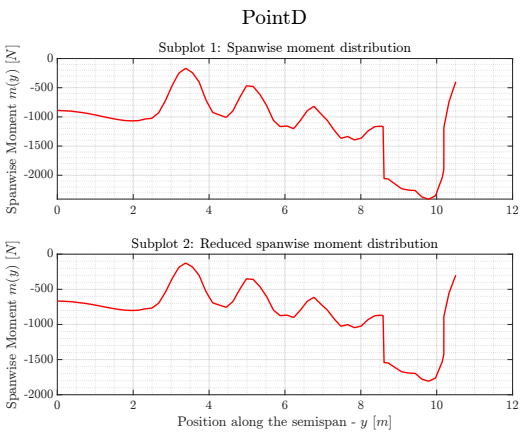


(e) Non-conventional aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point A. Engine on.

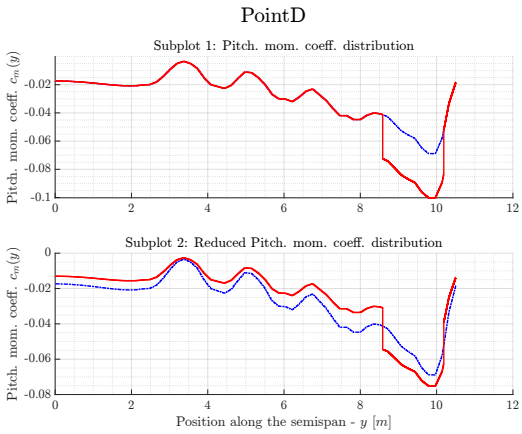


(f) Non-conventional aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point C. Engine on.

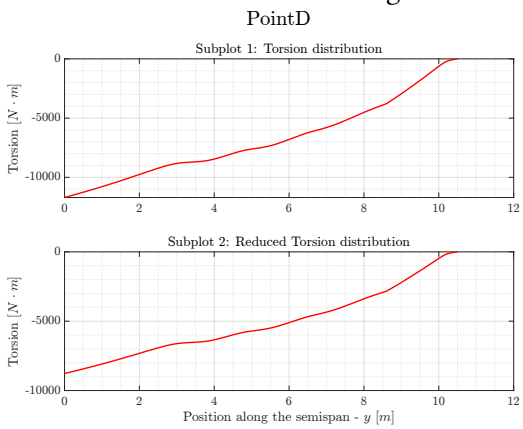
Figure 3.66: Unsymmetrical pitching moment, pitching moment coefficient, and torsion distribution for the non-conventional aircraft at points A and C of the maneuvering and gust flight envelope. Engine on.



(a) Non-conventional aircraft. Spanwise pitching moment $m_t(y)$ (moment per unit length) at Point D. Engine on.



(b) Non-conventional aircraft. Spanwise pitching moment coefficient $C_m(y)$ at Point D. Engine on.



(c) Non-conventional aircraft. Spanwise torsion $t(y)$ (pure couple per unit length) at Point D. Engine on.

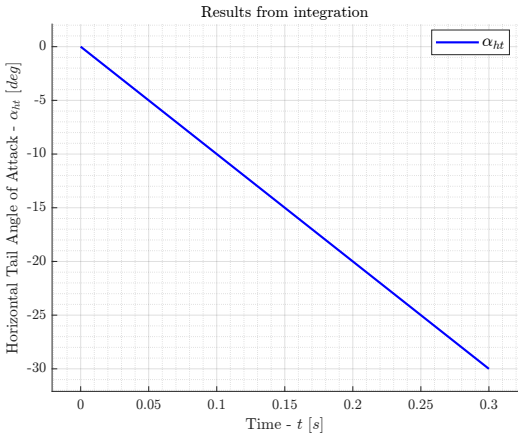
Figure 3.67: Unsymmetrical pitching moment, pitching moment coefficient, and torsion distribution for the non-conventional aircraft at Point D of the maneuvering and gust flight envelope. Engine on.

Figure 3.64 summarizes all the results concerning the spanwise distributions of Shear, Bending moment, and Torsion.

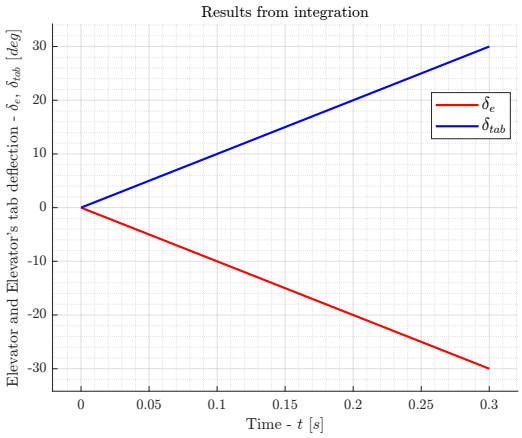
Figure 3.65 provides a synthesis of the wing-body reference aerodynamic model used for the non-conventional design architecture.

Figures 3.66 and 3.67 illustrate the unsymmetrical spanwise pitching moment coefficient, with the resulting torsion distribution, at points A, C, and D of the maneuvering and gust flight envelope diagram, as prescribed by the substantiation requirements for aileron deflection loads.

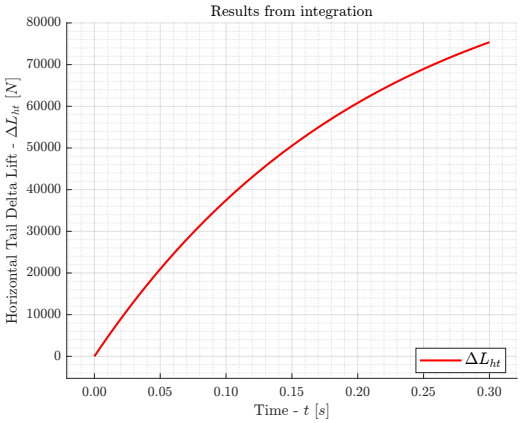
Figures 3.70a and 3.68 collect results obtained from integrating (A.17). Figures 3.70b and 3.69 collect results obtained from integrating (A.20).



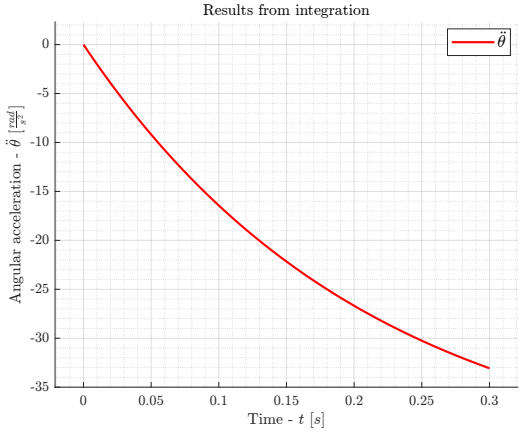
(a) Non-conventional aircraft. Horizontal tailplane angle of attack $\alpha_{\text{ht}}(t)$ during the unchecked maneuver. Engine on.



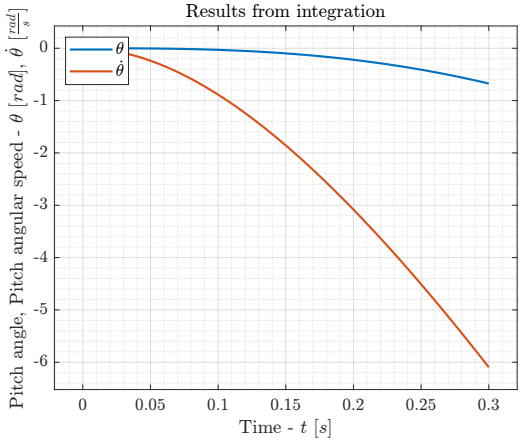
(b) Non-conventional aircraft. Elevator and elevator's tab deflection angles, $\delta_e(t)$ and $\delta_{e_{\text{tab}}}(t)$, during the unchecked maneuver. Engine on.



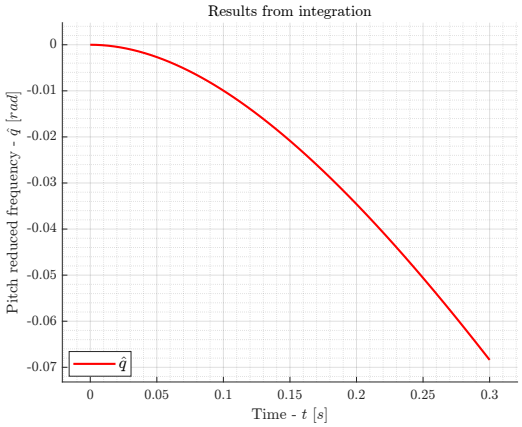
(c) Non-conventional aircraft. Horizontal tailplane lift increase during the unchecked maneuver, ΔL_{ht}. Engine on.



(d) Non-conventional aircraft. Pitching angular acceleration $\ddot{\theta}(t)$ during the unchecked maneuver. Engine on.

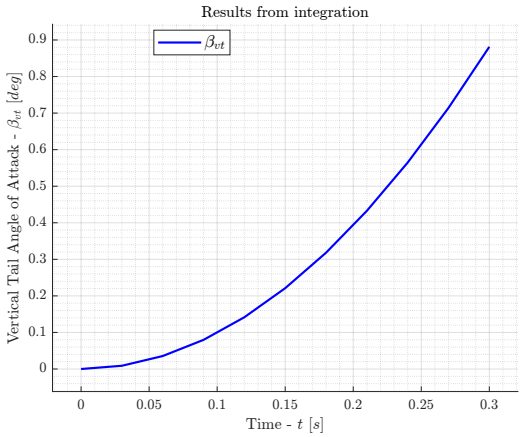


(e) Non-conventional aircraft. Pitching angle and angular speed, $\theta(t)$ and $\dot{\theta}(t)$. Engine on.

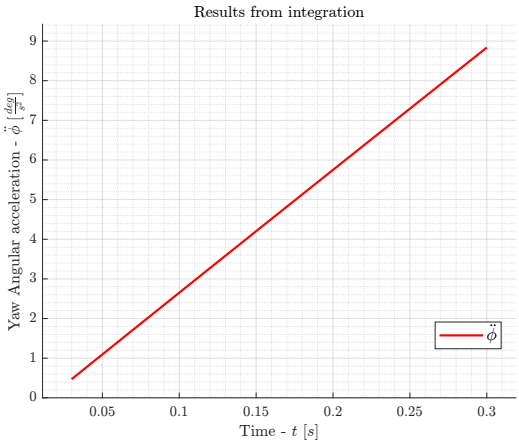


(f) Non-conventional aircraft. Pitching reduced frequency, $\hat{q}(t)$. Engine on.

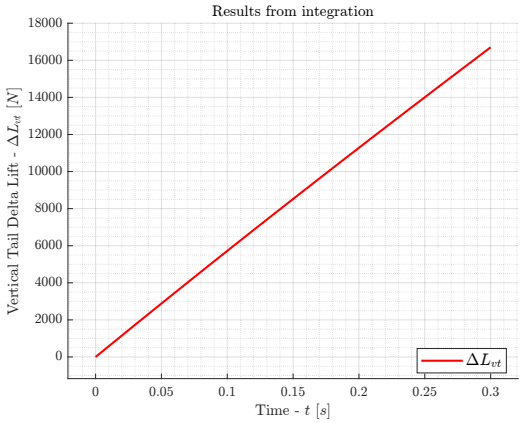
Figure 3.68: Horizontal tailplane unchecked maneuver for the non-conventional aircraft. Engine on.



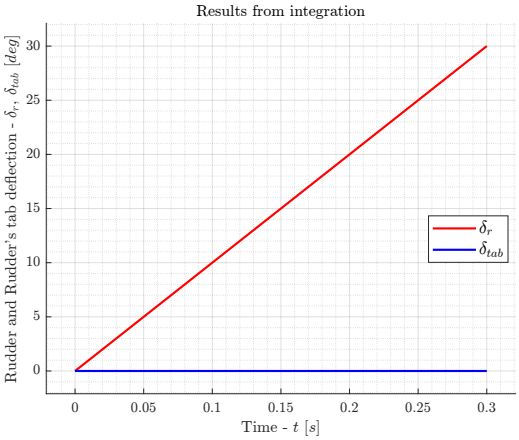
(a) Non-conventional aircraft. Vertical tailplane sideslip angle $\beta(t)$ during the unchecked maneuver. Engine on.



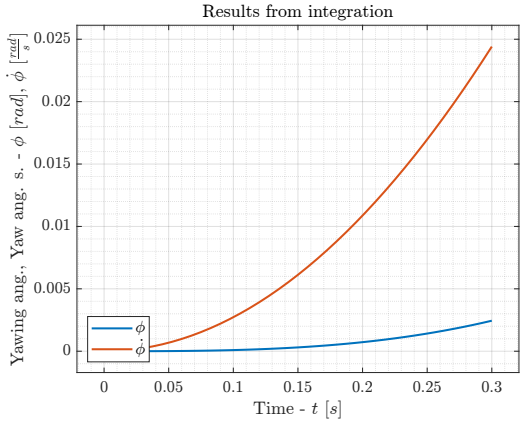
(b) Non-conventional aircraft. Yawing angular acceleration $\ddot{\phi}(t)$ during the unchecked maneuver. Engine on.



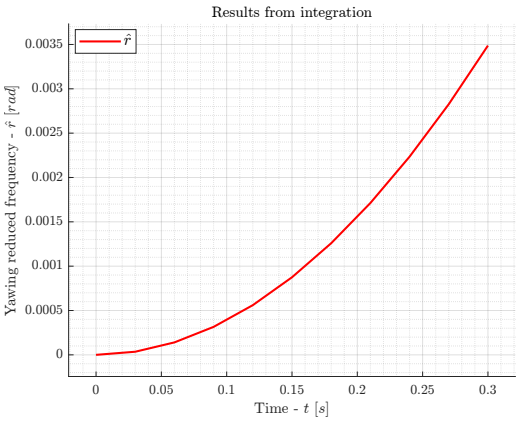
(c) Non-conventional aircraft. Vertical tailplane lift increase during the unchecked maneuver, ΔL_{vt}. Engine on.



(d) Non-conventional aircraft. Rudder and rudder tab (if present) deflection angles, $\delta_r(t)$ and $\delta_{v_{\text{tab}}}(t)$, during the unchecked maneuver. Engine on.

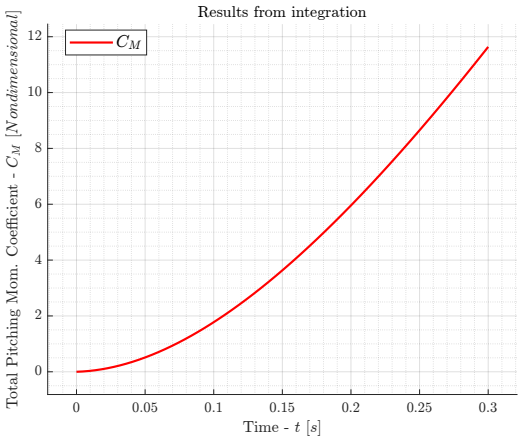


(e) Non-conventional aircraft. Yawing angle and angular speed, $\phi(t)$ and $\dot{\phi}(t)$. Engine on.

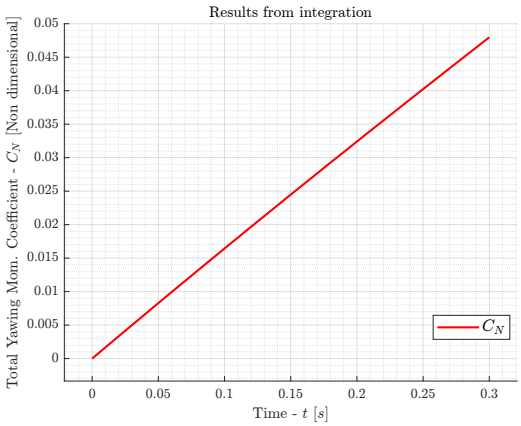


(f) Non-conventional aircraft. Yawing reduced frequency, $\hat{r}(t)$. Engine on.

Figure 3.69: Vertical tailplane unchecked maneuver for the non-conventional aircraft. Engine on.



(a) Non-conventional aircraft. Total pitching moment coefficient $C_M(t)$. Engine on.



(b) Non-conventional aircraft. Total yawing moment coefficient $C_N(t)$. Engine on.

Figure 3.70: Non-conventional total pitching and yawing moment coefficients. Engine on.

This Section further demonstrates the flexibility of the MBSE digital certification framework described in Section 3.2. Its design is focused on flexibility, aiming for general applicability and deployability of the framework to designs acceptable within CS-23, Amendment 5 or later, constraints and limitations.

The non-conventional design application includes the automatic generation of the Flight Loads certification reports, including most of the Subpart C certification artifacts and data. The framework always applies the same set of requirements, reading the aircraft data and the corresponding applicable substantiation requirements, with predictable output.

This scenario can be executed quickly to generate a family of similar designs along with their corresponding reports, allowing engineers to evaluate slight variations of weight, wing surfaces, altitude, and other design parameters.

In conclusion, the discussion presented here demonstrates that, conceptually speaking, the certification burden can be significantly alleviated once engineers and certification experts have access to a sufficiently complete and accessible SysML model which must include certification data and metadata.

It must be emphasized that the SysML model used in this digital certification framework presents requirements and substantiation data mostly in textual format, i.e., as strings. While this digitalization may not be the most convenient for engineering applications, it has been demonstrated that using regular expressions can transform textual information into data suitable for mathematical operations and certification processes. Of course, this process has the limitations previously discussed and will not be repeated here.

3.3 Activity Diagrams MBSE implementation

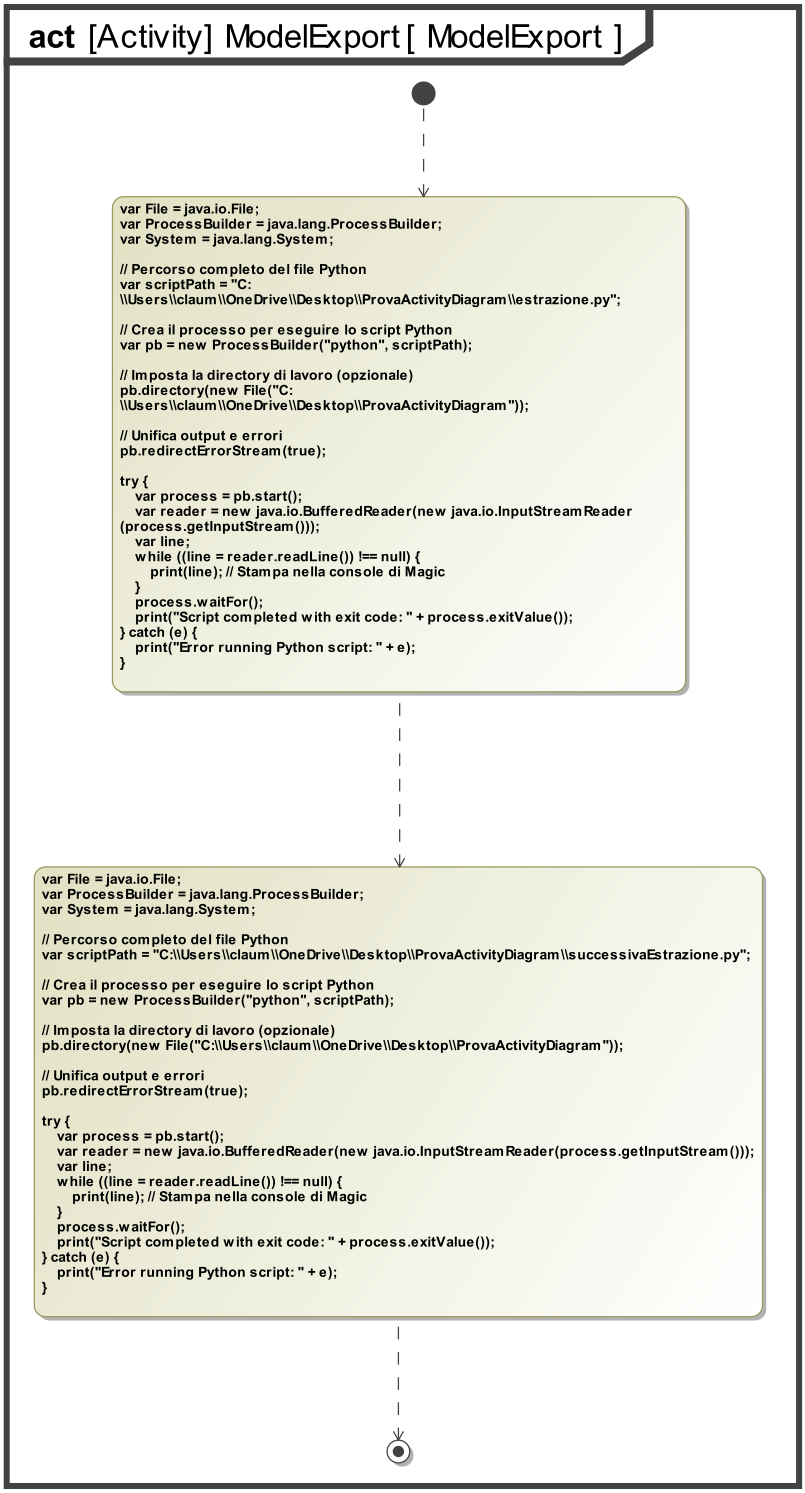


Figure 3.71: A simple example of an activity diagram, as it appears in the CATIA MSoSA modeling environment.

This Section describes an initial implementation of the MBSE approach using the CATIA MSoSA modeling environment. The application has a deliberately limited scope, as its primary objective is to demonstrate the automation capabilities of the Dassault Systèmes SysML environment. In this context, certification and substantiation operations are orchestrated through Activity Diagrams embedded in the SysML model. This implementation is ideally closer to a proper answer to the research question stated in RQ 2 and can be interpreted as an attempt to enrich the CATIA MSoSA environment with a new plugin capable of performing certification tasks in a digital environment.

Figure 3.71 shows one of the Activity Diagrams developed for this application. Although the execution workflow remains relatively cumbersome, the functionality already demonstrates the practical potential of this implementation for developing automated digital certification tools. The CATIA modeling environment provides access to several programming languages, including Python and Java. Activity Diagrams can invoke code fragments defined both within the environment and in external libraries.

These Activity Diagrams provide mechanisms for invoking external tools directly from the modeling environment, thereby eliminating the need to export model data and to rely on auxiliary interchange files. In the finalized version of the model, these diagrams can be encapsulated using *opaque actions*, which allow the execution of predefined operations without requiring explicit user interaction. This approach supports a more streamlined and user-transparent workflow for implementing automated certification procedures.

In this implementation, the digital certification framework automatically generates a JSON file that serves as the input for an external C++ code suite. These routines compute the flight envelope by combining aircraft design data with the applicable certification and substantiation information, including formulas, constraints, and boundary conditions defined within the model.

Subsection 3.3.1 presents an initial implementation of an MBSE-based digital certification framework within the CATIA MSoSA modeling environment, including the automation capabilities enabled through Activity Diagrams. Subsection 3.3.2 provides an overview of the external C++ routines used to execute certification and substantiation operations. Finally, Subsection 3.3.3 summarizes the outcomes of this MBSE certification framework.

3.3.1 The CATIA MSoSA SysML UNINA implementation

The SysML model implemented and illustrated in this thesis, using the CATIA MSoSA facilities, is deliberately simple and does not rely on complex SysML infrastructures, such as elaborate stereotypes or complex definition diagrams. The main purposes of this modeling effort were to demonstrate the soundness of the conceptual approach and to become familiar with the CATIA MSoSA modeling environment, gaining an understanding of the software before engaging in more complex implementations, such as the one developed by the Georgia Institute of Technology.

At the core of the model are two conceptually similar stereotypes: *CS* for the EASA CS-23 requirements and *ASTM* for the ASTM International paragraphs and subparagraphs.

For brevity, their definitions are not repeated here; the interested reader is referred to Section 3.2.1 for a detailed description of the modeling approach used to encode the airworthiness and substantiation rules. In this application, the SysML model is restricted to the requirements directly relevant to the construction of the maneuvering and gust flight envelope.

A distinctive feature of this SysML implementation is the inclusion of multiple Activity Diagrams. These diagrams can orchestrate *opaque actions*, that is, automated operations that do not require direct user intervention. In this application, the workflow comprises the following steps:

1. First, an Activity Diagram parses the model by invoking an internally defined Python routine. CATIA MSoSA provides access to scripting capabilities, including Python, which can be used to extract regulatory text and associated metadata by leveraging the user-defined stereotype properties. The output of this Activity Diagram is a JSON file structured to initiate the certification and substantiation operations. Once generated, the Python routine transfers the file to the appropriate directory for the external C++ tool suite.

2. A second Activity Diagram triggers execution of the external C++ library, initiating the certification calculations and generating certification artifacts, such as maneuvering and gust flight-envelope tables and figures. The PDF figures are produced through LaTeXstandalone files in a straightforward workflow. The resulting artifacts can then be imported into the SysML model and used as inputs to the CATIA MSoSA report wizard.

Developing a report-wizard template was outside the scope of this application and is not addressed in this thesis. However, this option will be examined in subsequent work.

In all other respects, this SysML model is logically consistent with the Eclipse Papyrus model described earlier; therefore, additional explanation is not required.

3.3.2 The C++ external library

This Section describes the external C++ library, which comprises multiple classes that address distinct functional needs. As noted in earlier Chapters, the maneuvering and gust flight-envelope calculations rely on established, widely used methods and do not require advanced theoretical developments. Accordingly, the underlying derivations are not repeated in detail in this thesis. Readers seeking a deeper treatment of the relevant formulations are referred to the specialized literature.

The following provides a high-level description of the features included in the C++ library. C++ is widely used in aerospace and other safety-critical domains because it supports efficient execution, explicit control of resources, and broad portability. In this work, an object-oriented design organizes the substantiation routines into self-contained classes that support the generation of the maneuvering and gust flight-envelope diagram.

A practical drawback of adopting C++ is the increased development effort, which can raise implementation time and cost.

Individual class files are not described in detail; instead, the main actions performed by the C++ substantiation library are summarized below:

- A set of class files performs the preliminary calculations using established numerical methods and standard implementation practices. The routines include solvers for nonlinear systems of equations, root-finding procedures, and JSON parsing based on predefined rules. These capabilities are used to compute, for example, the maximum horizontal speed V_H, evaluate key points on the maneuvering and gust flight envelope, and instantiate the objects required by the main execution workflow.

- The core portion of the library constructs the flight envelope. The methods combine certification information with aircraft input data to generate the maneuvering and gust boundaries and to produce tabular output files. These outputs are then used to generate PDF plots of the flight envelope, including gust lines and labeled points.

- The implementation includes safeguards intended to improve robustness and maintainability. Dedicated utilities support data-type validation, memory-safe handling of inputs and outputs, and structured error reporting. The overall design follows established practices for developing durable C++ libraries.

- Finally, a set of auxiliary routines produces output files in a convenient and readable format that can be processed by a LaTeXscript designed to generate high-quality figures. In this way, the user can assess the results efficiently through graphical outputs without needing to inspect the raw data directly.

The C++ code libraries execute within seconds and can be associated with different parts of the SysML model, embedding them in its structure. This will be studied in subsequent works.

An additional integration challenge can arise when the MBSE digital certification framework is extended with external analysis components, such as a panel-method tool for evaluating spanwise wing-load distributions. To harmonize data exchange and execution across heterogeneous tools, the framework may require wrappers and dedicated interface libraries (e.g., Python-based adapters or custom C++ bindings). These integration layers can increase both development effort and long-term maintenance costs, particularly as external tools evolve and interfaces must be updated to preserve compatibility.

The preceding discussion is consistent with the qualitative assessment of this strategy for introducing MBSE methods and tools into digital certification. Despite its complexity and limitations, the workflow based on *opaque actions* provides the most streamlined user experience during oversight activities. It also allows certification data to remain within a single integrated environment, in accordance with the concept of a *Single Source of Truth* (SSoT) for managing certification tasks. Integrating a report-generation phase through the CATIA MSoSA report wizard would further strengthen the framework by enabling automated preparation of certification documents. This capability could enhance the

framework's practical value and facilitate its adoption by domain experts and certification practitioners.

3.3.3 Results and conclusions

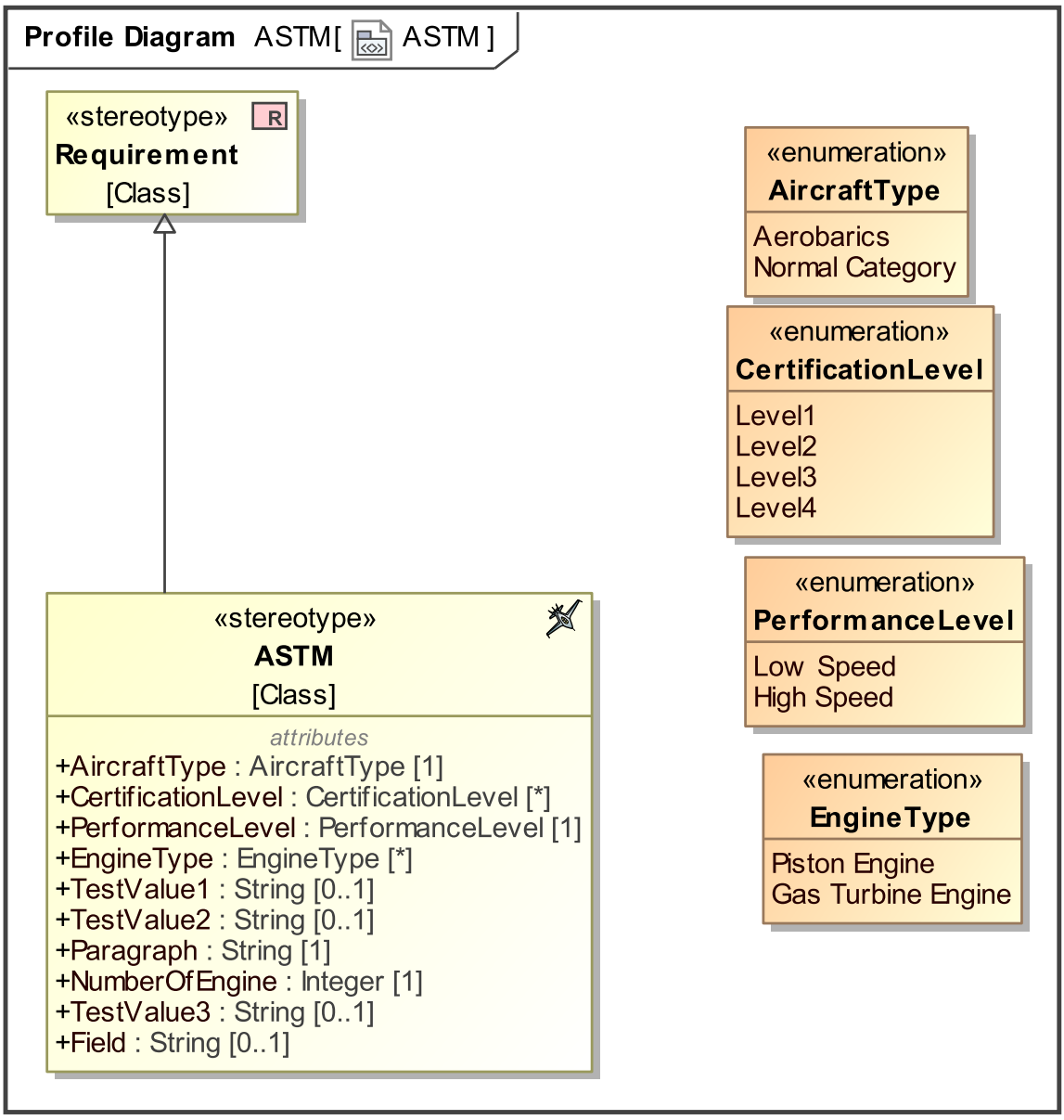
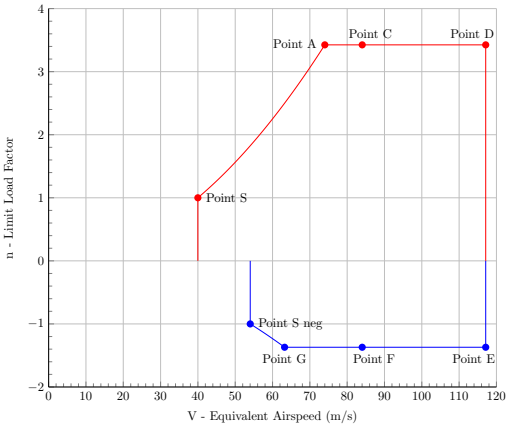


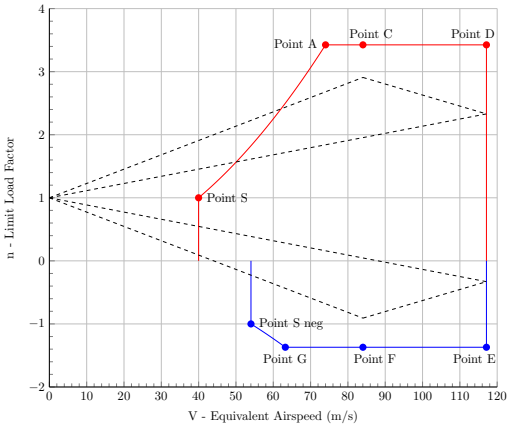
Figure 3.72: The ASTM Profile Diagram as it appears in the CATIA MSoSA modeling environment.

Figure 3.72 shows the ASTM Profile Diagram, which defines the stereotypes used in this application. The stereotype is conceptually similar to that developed for the Eclipse

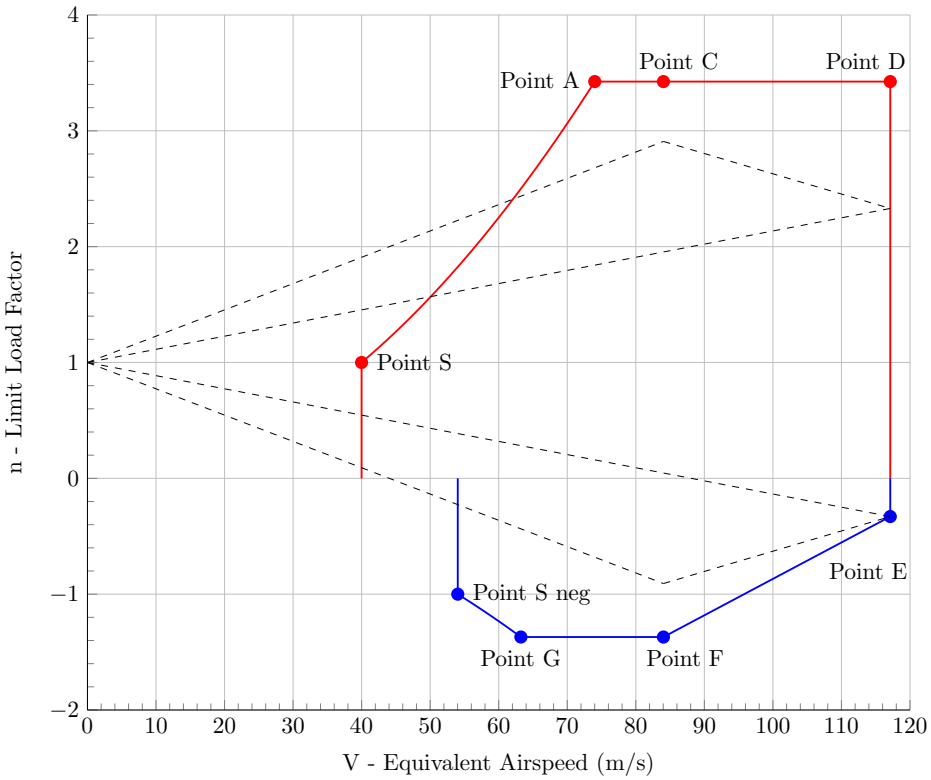
Papyrus SysML model. It is based on the SysML *Requirement* element, from which it inherits the standard requirement properties and relationships, and it extends that element through enumerations and additional attributes used to represent the relevant characteristics of paragraphs and subparagraphs in the substantiation documents.



(a) Maneuvering flight envelope.



(b) Flight envelope with gust lines.



(c) Maneuvering and gust flight envelope.

Figure 3.73: The maneuvering and gust flight envelope obtained by launching the C++ calculation routines. A LaTeXcode draws the diagrams using the output data.

Figure 3.73 shows the output generated by this MBSE-based digital certification tool. The resulting diagram can be included directly in a technical certification report and is produced from output data formatted in a report-ready form. All external routines involved in this workflow, including those written in C++ and LaTeX, are executed through user-defined Activity Diagrams. Appendix B includes Table B.3 that offers a comparison between numerical results from the various tools developed within the context of this research effort and presented in this thesis.

Figure 3.74 shows the containment tree as displayed in the CATIA MSoSA modeling environment. The SysML model adopts a hierarchical structure, and a dedicated icon marks elements to which the ASTM stereotype has been applied. This visual cue helps the user identify the context and properties of the modeled elements more clearly.

Figure 3.75 presents the outcome of the modeling effort for this application. The CATIA MSoSA environment allows the user to explore the model and inspect its properties, features, and fields.

Numerical data and equations are stored directly in the model elements, specifically in the *TestValues* and *TestExpressions* fields. In addition, CATIA MSoSA includes an internal glossary that assists in the interpretation of requirement statements expressed through ordered relationships—or, more generally, by using any other natural language expression.

Expressions such as *more than* and *less than* can be recognized automatically by the software, associated with glossary entries, and translated into metadata that may be useful in specific use-case scenarios or during information retrieval after model export. Figure 3.75 presents an example of this—the expression *more than 3.8* that appears underlined.

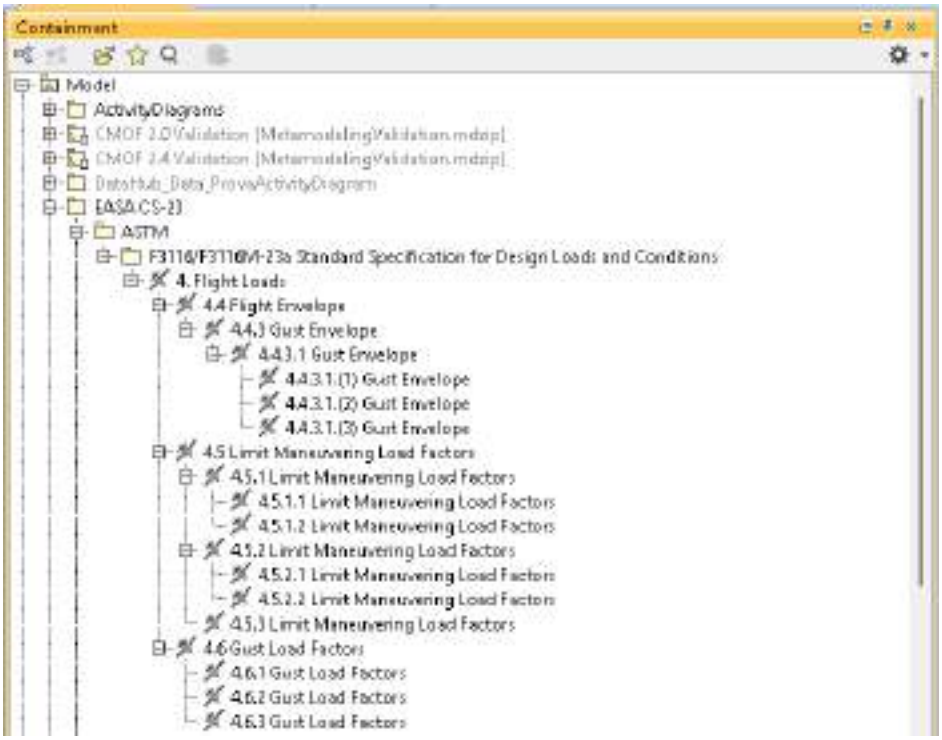


Figure 3.74: The containment tree of the CATIA MSoSA modeling environment. The hierarchical structure of the SysML model is clearly visible.

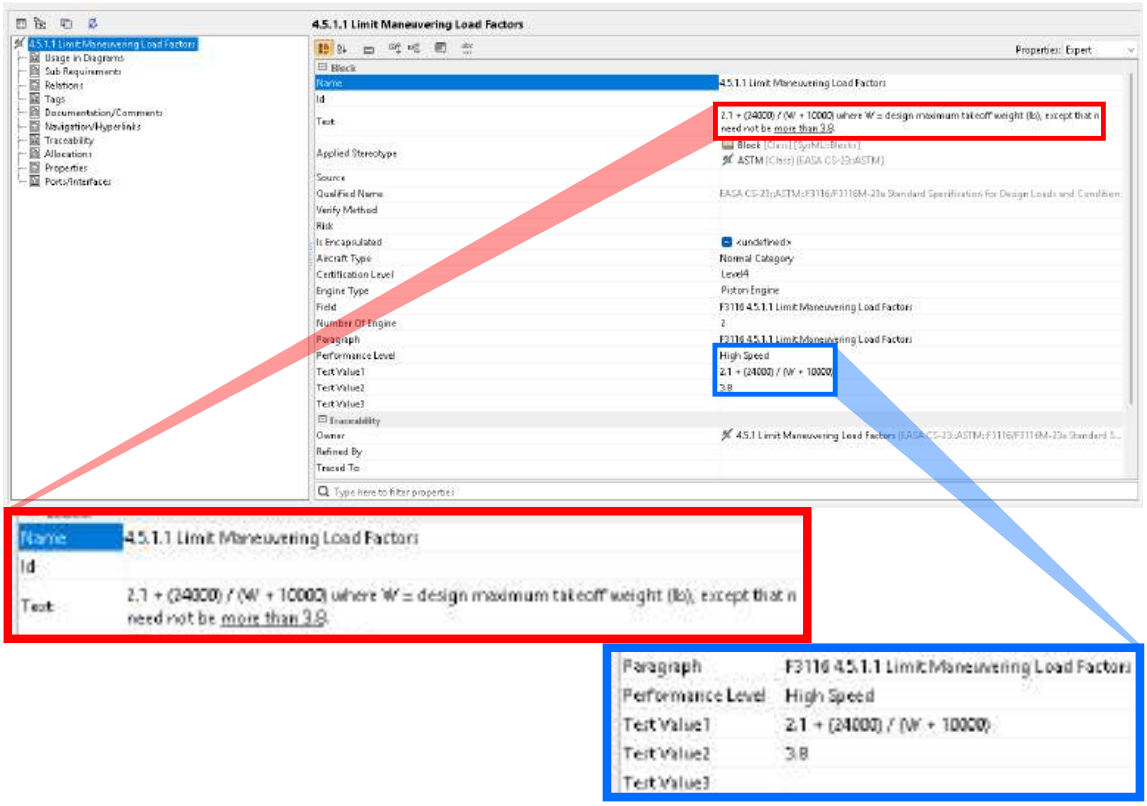


Figure 3.75: The CATIA MSoSA modeling environment allows the user to explore the model in all its features. In this case, the ASTM F3116 4.5.1.1 subparagraphs is shown, as it appears in the modeling environment.

This application demonstrates, at least at the research level, the feasibility of establishing an MBSE-based digital certification framework whose execution is automated through SysML Activity Diagrams, as supported in the CATIA MSoSA modeling environment. When applied carefully to a certification workflow leading to the issuance of a new type certificate, the use of *opaque actions* can make execution more user-transparent and repeatable. In turn, this can reduce avoidable inconsistencies and improve the coherence and reliability of the certification baseline, while preserving the need for domain expertise in interpreting requirements and substantiating compliance.

These capabilities can be established by implementing verification and validation frameworks, either to support the selection of appropriate substantiation pathways for a given aircraft configuration or to assess the soundness of computed results against the high-level, performance-based requirements.

3.4 Georgia Tech digital certification pipeline

During a six-month collaborative research effort with the Georgia Institute of Technology, the author of this thesis designed and implemented a use-case scenario from a larger

SysML infrastructure that modeled the FAA FAR Part 23 airworthiness rules, designed and developed by the Aerospace Systems Design Laboratory (ASDL) researchers. This application leveraged the experience gathered with the MBSE techniques, with the main purpose of improving the MBSE integration level and performance. The stereotypes available within the context of this model have been expanded with additional stereotypes, representing the ASTM International consensus standard documents. The model has also been expanded, adding the majority of the ASTM International material published by the general aviation committee, the F44 committee, and its subcommittees. The successful integration of the new use case into an already complex SysML infrastructure indicates that the proposed approach can support additional certification applications without requiring a substantial redesign of the underlying model. These results therefore provide an initial, research-based answer to the research question stated in RQ 2.

Figures 3.76 and 3.77 illustrate the addition of the *Consensus Standard Paragraph* stereotype to the Profile Diagram of the Georgia Tech FAR Part 23 SysML model and the connections between the ASTM consensus standards and the FAR Part 23 paragraphs through Advisory Circular 23-20-10-1, *FAA Accepted Means of Compliance Process for 14 CFR Part 23*.

The block diagram is a representative example extracted from the model and shows how templates for assessing the certification basis can be managed through the digital SysML model of the FAR Part 23 airworthiness requirements. By combining standard and user-defined SysML relationships, the model can establish direct traceability between FAR Part 23 paragraphs and subparagraphs and the corresponding consensus-standard material.

This traceability can be extended to the paragraph and subparagraph level within the standards, thereby linking high-level airworthiness requirements to the specific substantiation provisions used to demonstrate compliance.

The block diagram presented in figure 3.77 also shows the compliance mechanism that the user should apply, following the Part 21 notation thoroughly; the only hypothesis introduced in this modelization approach is the name of the corresponding compliance document produced at the end of the process.

Figure 3.78 presents a further and slightly more complex example of a block diagram used to derive not only the appropriate certification basis but also the corresponding compliance and substantiation activities, together with the compliance documentation to be submitted to the airworthiness authority.

Subsection 3.4.1 describes the use-case scenario developed based on the existing SysML airworthiness model created by researchers at ASDL. Subsection 3.4.2 presents the Python libraries used to execute the certification and substantiation tasks for this application. Finally, Subsection 3.4.3 concludes the Section by summarizing the certification artifacts generated during execution and by providing final observations on the results.

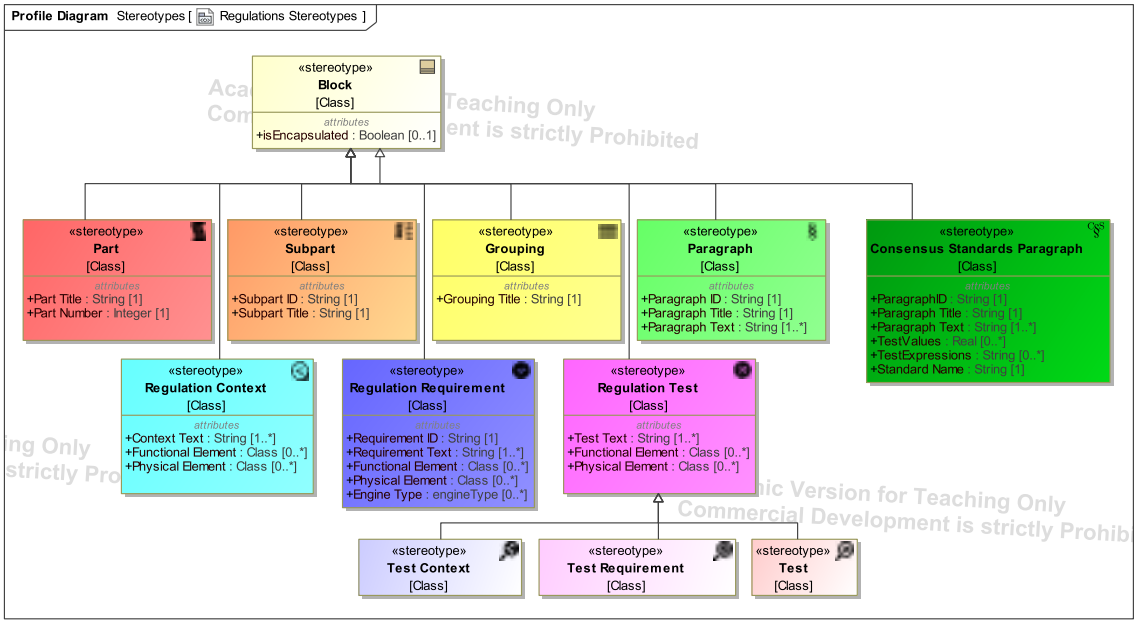


Figure 3.76: The CATIA MSoSA Profile Diagram used for the definition of FAR 23 stereotypes.

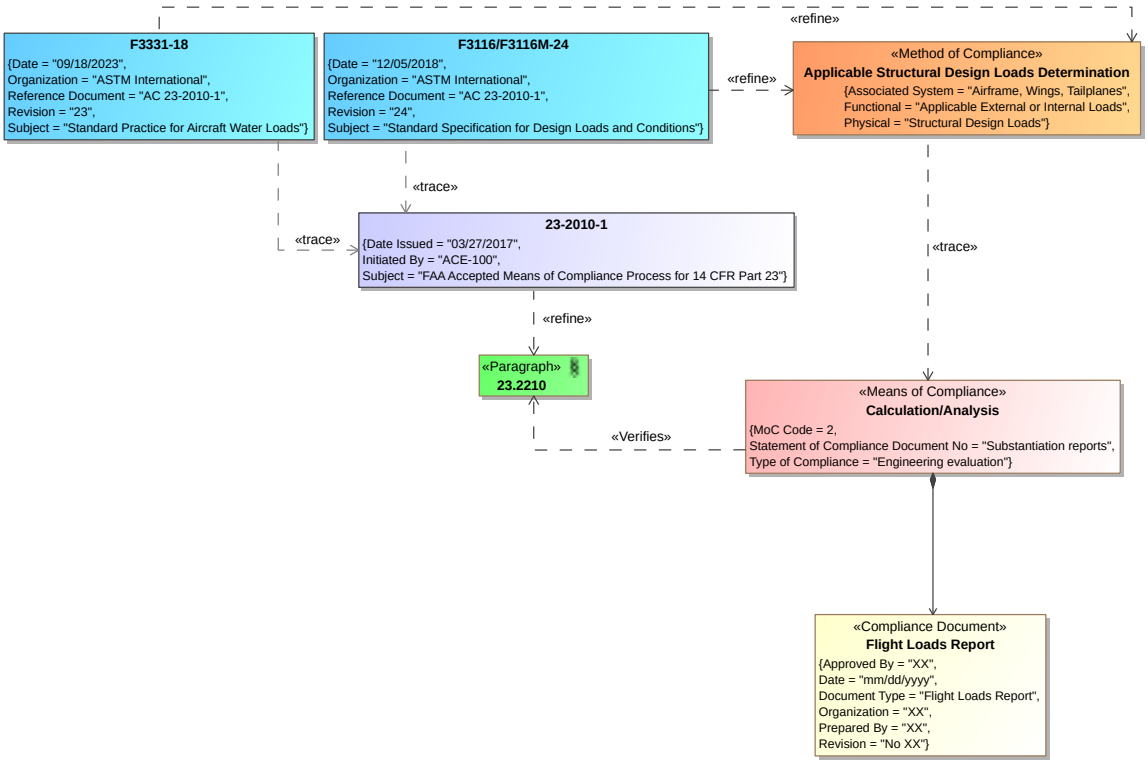
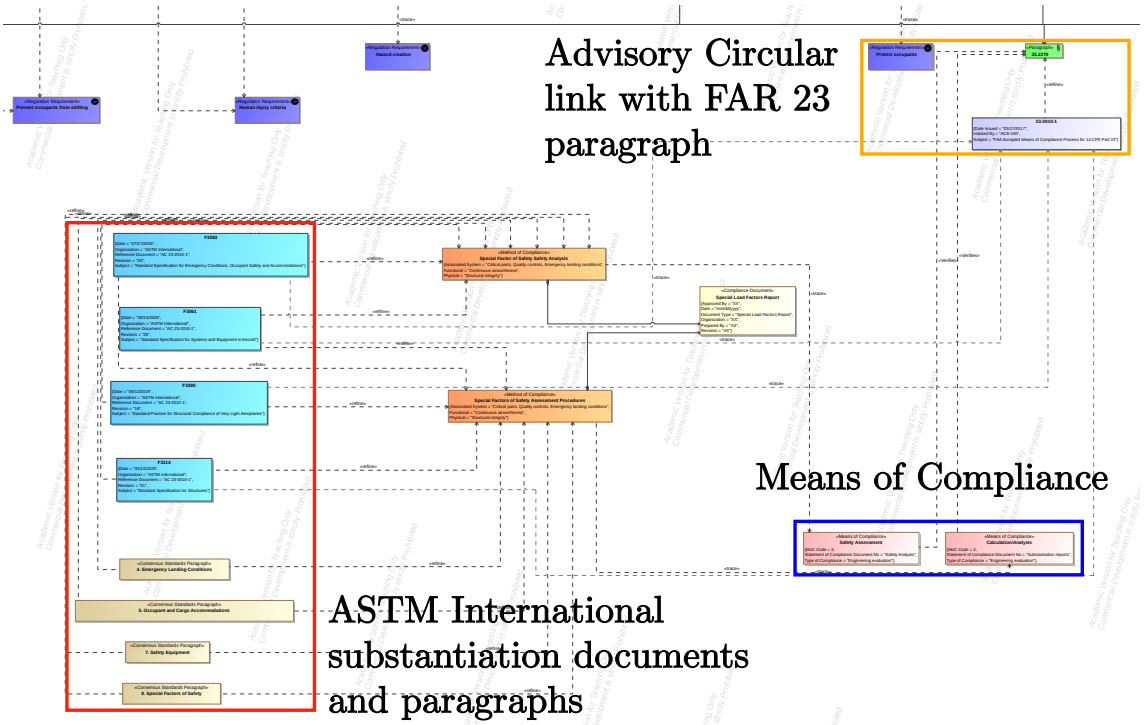


Figure 3.77: An example of block diagrams from the FAR 23 Georgia Tech SysML airworthiness rules model. The FAR §23.2210 paragraph is shown.



Some Block Definition Diagram associated with FAR 23 paragraphs can have more than one or two ASTM International standards.
In this instance, the substantiation procedures requires at least four different standards and two different Means of Compliance (Safety Assessment and Calculation/Analysis).

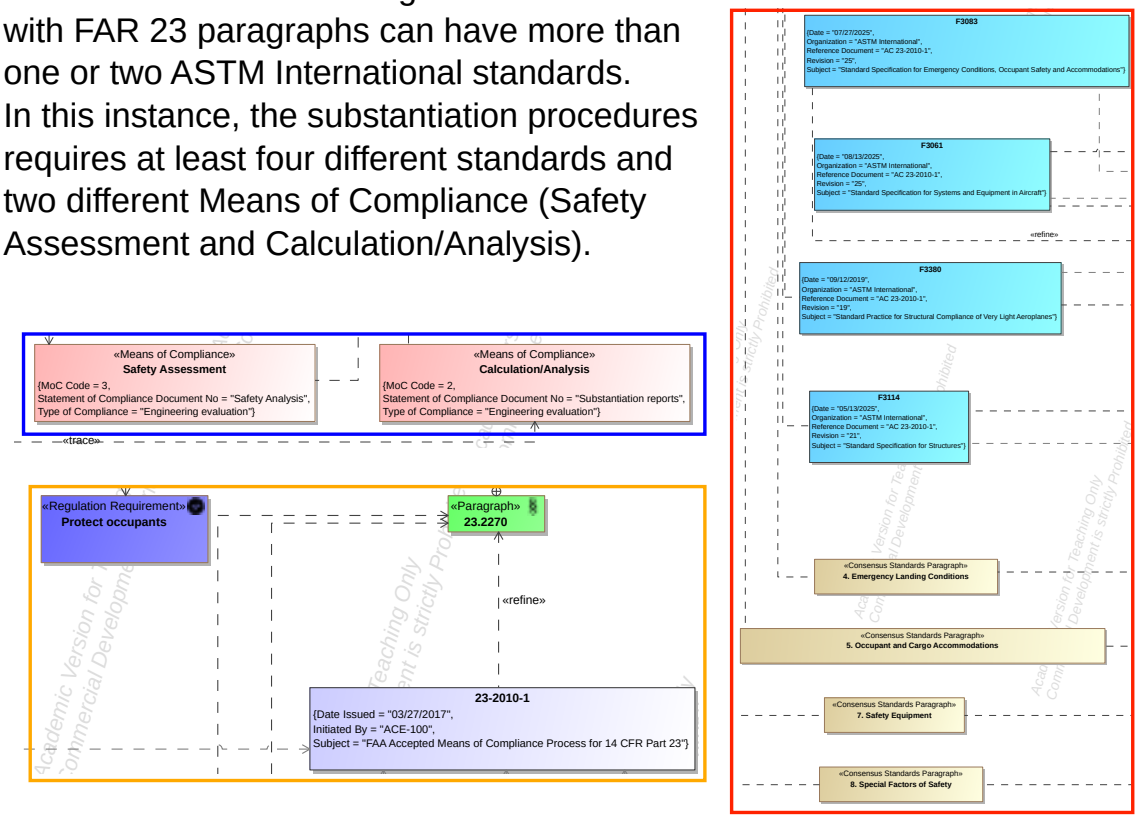
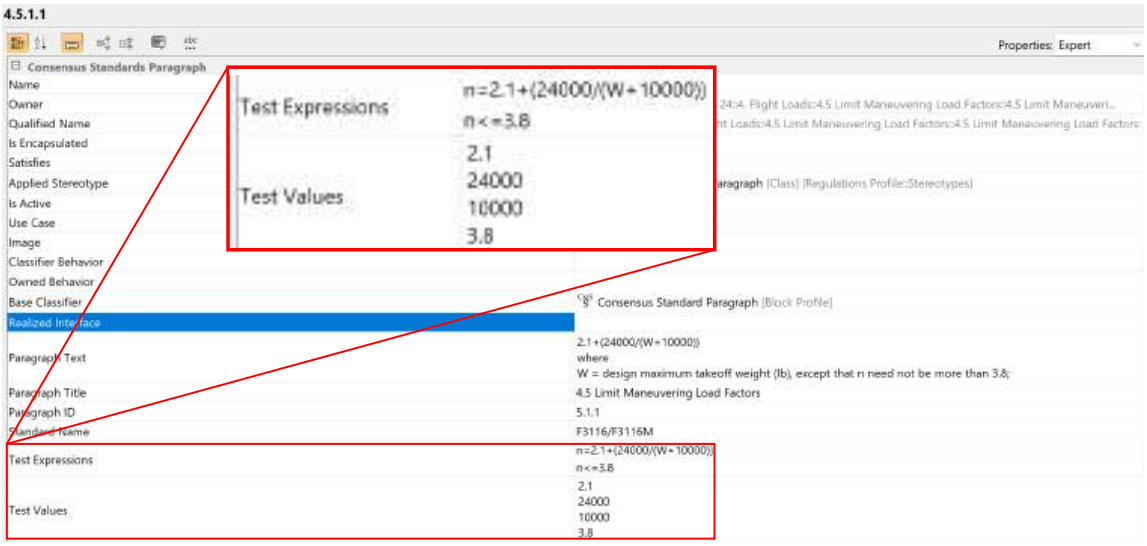
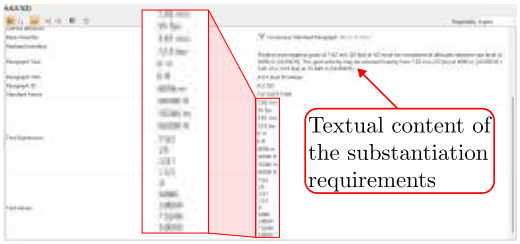


Figure 3.78: An additional example of block diagrams from the FAR 23 Georgia Tech SysML airworthiness rules model. The FAR §23.2270 paragraph is shown.

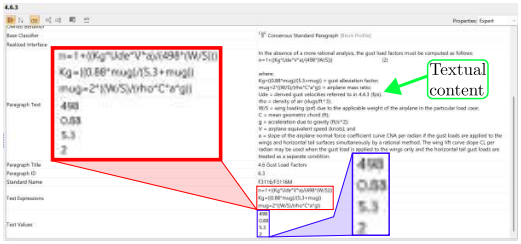
3.4.1 Developing a use-case scenario from the model



(a) ASTM F3116, paragraph 4.5.1.1.



(b) ASTM F3116, paragraph 4.4.3.1(2).



(c) ASTM F3116, paragraph 4.6.3.

Figure 3.79: Some examples of ASTM consensus standard documents modeled in the CATIA MSOSA modeling environment. Some features of the model's elements are evidenced.

In this application, the modeling effort was limited because the SysML infrastructure was already available and substantially broader in scope than the certification activities considered here. The model was nonetheless extended to include the relevant ASTM International consensus standards, using a newly introduced, application-oriented stereotype to preserve consistency with the existing model structure and extraction workflow. By using the SysML *Package* element to encapsulate the consensus-standard content, the model extension remained modular and could be implemented in a straightforward manner.

Figure 3.79 presents a selection of consensus-standard paragraphs modeled in the CATIA MSoSA environment, all drawn from ASTM F3116/F3116M, *Standard Specification for Design Loads and Conditions*. In particular, Figure 3.79a shows how the formulas and numerical values contained in Section 4, *Flight Loads*, paragraph 4.5.1.1, are encoded in the fields of a single *Consensus Standard Paragraph* block element. Figures 3.79b and 3.79c provide two additional examples used in this thesis.

Consequently, the primary objective of the modeling phase was to implement reliable extraction capabilities through an efficient export strategy based on stereotype properties. Under this approach, extraction of certification information is amendment-agnostic: it requires only that model elements representing ASTM paragraphs and subparagraphs carry the relevant stereotype and that the associated property fields are non-empty.

The key properties can be summarized as follows:

- A property named *TestExpressions*, defined as a zero-to-many string vector. This property collects formulas, mathematical expressions, ordered relationships between quantities, and other constraints expressed in the requirement text.

- A property named *TestValues*, defined as a zero-to-many vector of floating-point values. This property collects numerical data contained in the corresponding requirement text.

The remaining features of the *ASTM* stereotype were inherited from the previously established stereotype infrastructure. This design choice preserves consistency across the model and reduces the risk of introducing incoherence or inconsistency through a divergent modeling approach.

To keep the model as simple as possible, and given the time constraints of this research effort, no Activity Diagrams or other SysML automation mechanisms were implemented. Instead, the extraction phase follows the same approach adopted in the Eclipse Papyrus/-MATLAB framework described in Section 3.2. The user exports the SysML model manually as a UML interchange file and provides this file to the external parsing tools by placing it in the designated working directory. Standard Python packages, such as `pandas` or similar ones, provide adequate facilities for establishing the parsing infrastructure.

Again, the selected interchange data format is the standard JSON file, arranging the information using a standard nomenclature used among the entire Python code suites.

3.4.2 The Python supporting code

The execution component of this version of the MBSE-based digital certification toolchain consists of a set of Python libraries that support substantiation operations and the generation of certification artifacts. As in the previous use cases, the underlying analytical methods are well established and are not repeated here. Python was selected for the following reasons:

- The ASDL research engineers routinely use Python-based tools. Using Python also improves compatibility with the existing software ecosystem within the laboratory.

- Python is widely adopted in engineering practice and provides access to a mature scientific computing stack. Version 3.9 was selected to maintain compatibility with existing scripts and external tool interfaces, including workflows that invoke *Open VSP*.

- Python supports rapid development under time constraints while still allowing implementation of robust engineering software practices, including structured error handling, safeguards, and disciplined resource management.

A major feature of this routine set is the ability to generate executable functions directly from the string-based expressions stored in the certification input JSON file. The implementation leverages well-established scientific and symbolic-computation libraries, including `sympy` and `scipy`, and in particular uses *sympify* and *lambdify* to transform textual expressions into callable functions. The procedure is simple, repeatable, and transparent:

1. Using standard Python string-handling operations, the code tokenizes and separates certification formulas and mathematical expressions encoded in the JSON file. For example, when processing an equation, the routine splits the expression using the equal sign.

2. The relevant side of each expression is converted into a symbolic object within the Python namespace using *sympify*.

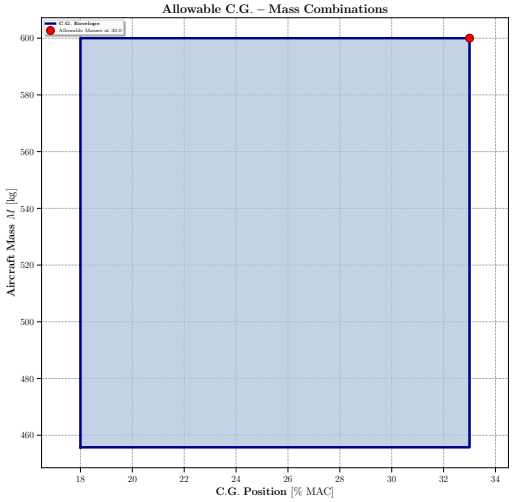
3. The symbolic expression is then converted into an executable function using *lambdify*, enabling direct evaluation within the certification-calculation workflow. The name of this built-in method reflects the use of inline *lambda functions* in Python.

The preliminary and certification activities performed by the Python routines are broadly analogous to those implemented in the corresponding MATLAB toolchain. After defining the certification scenario and completing the required preliminary calculations, the Python workflow constructs the maneuvering and gust flight envelope in accordance with the applicable certification instructions. It then evaluates the balancing loads acting on the horizontal tailplane and computes the associated wing-body and full-vehicle lift acting on the main wing.

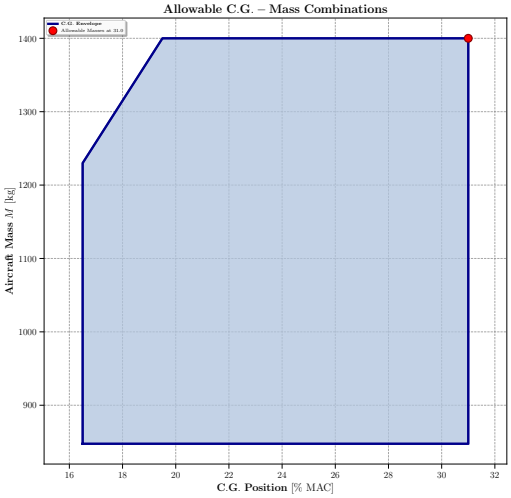
The resulting database of certification data and artifacts can be imported into the CATIA MSoSA modeling environment and used to support the preparation of a technical *Flight Loads* certification report using its built-in report wizard. At the moment, this feature is not implemented; however, the main purpose of using the CATIA MSoSA report wizard is to create a self-contained, digital certification toolchain that can avoid human intervention for generating complex, technical reports by applying user-defined templates. Each template can accept only a limited subset of output data or certification demonstration items, creating a structured document, ready for submission to the regulators.

3.4.3 Results and conclusions

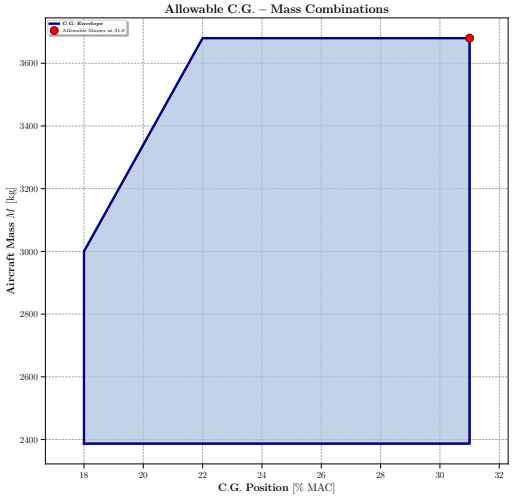
This Section presents a representative subset of the results generated by the Python companion code developed during this joint research effort. As in the MATLAB-based implementation discussed in Section 3.2, the scope of the Python code is limited to FAR Part 23, Subpart C (Structure).



(a) Level 2, single-engine aircraft. Allowable CG-mass combination, obtained from the Python code.

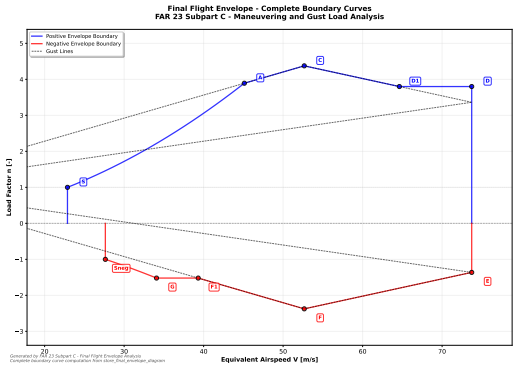


(b) Level 2, twin-engine aircraft. Allowable CG-mass combination, obtained from the Python code.

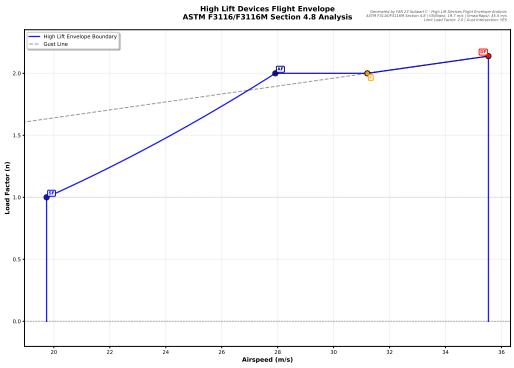


(c) Level 4, twin-engine aircraft. Allowable CG-mass combination, obtained from the Python code.

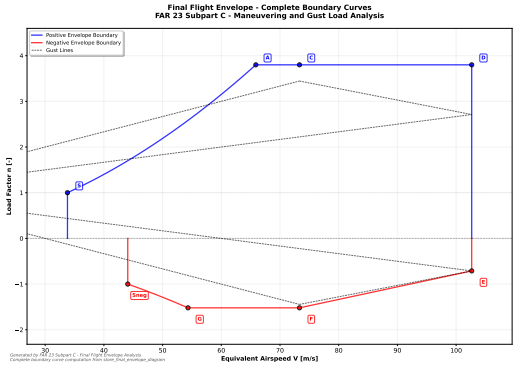
Figure 3.80: Allowable CG-mass combination for the Level 4 and Level 2, twin-engine aircraft, and for the Level 2 single-engine aircraft, obtained with the Python companion code.



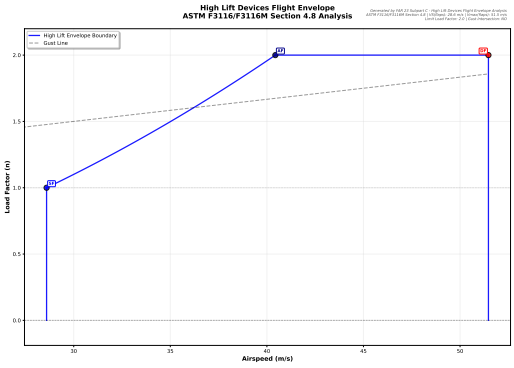
(a) Level 2, single-engine aircraft. Maneuvering and gust flight envelope, obtained from the Python code.



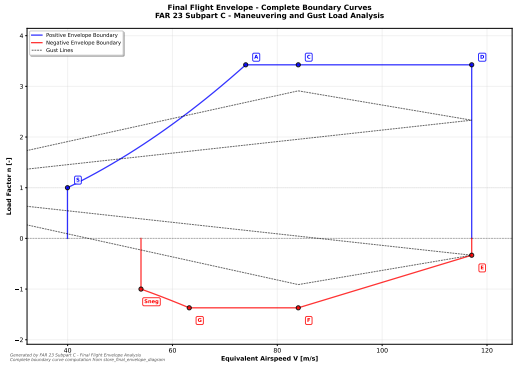
(b) Level 2, single-engine aircraft. High-lift flight envelope, obtained from the Python code.



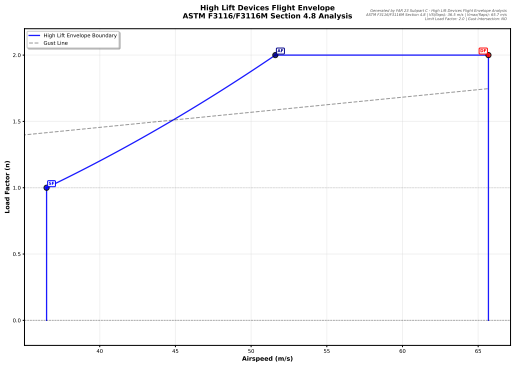
(c) Level 2, twin-engine aircraft. Maneuvering and gust flight envelope, obtained from the Python code.



(d) Level 2, twin-engine aircraft. High-lift flight envelope, obtained from the Python code.

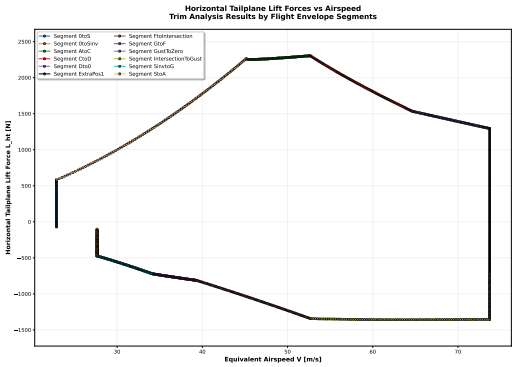


(e) Level 4, twin-engine aircraft. Maneuvering and gust flight envelope, obtained from the Python code.

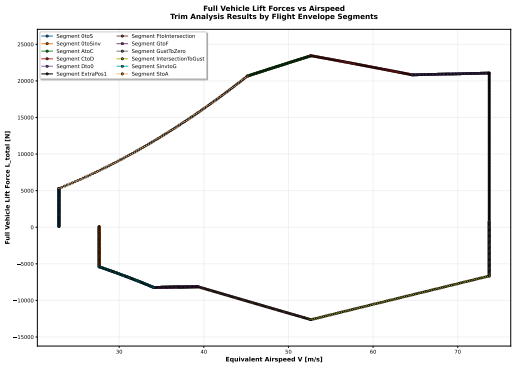


(f) Level 4, twin-engine aircraft. High-lift flight envelope, obtained from the Python code.

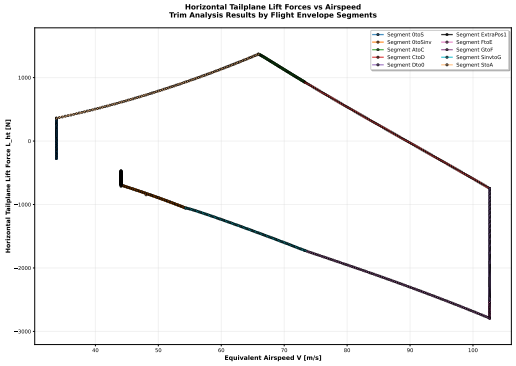
Figure 3.81: Maneuvering and gust flight envelope and high-lift flight envelope for the Level 4 and Level 2, twin-engine aircraft, and for the Level 2 single-engine aircraft, obtained with the Python companion code.



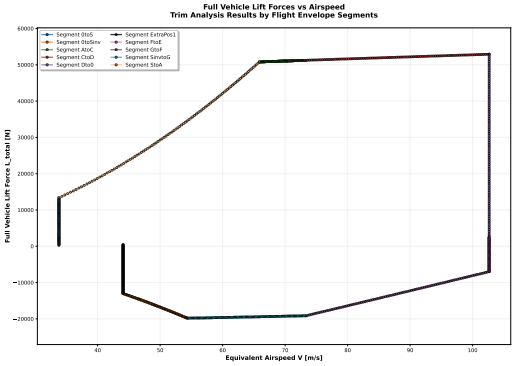
(a) Level 2, single-engine aircraft. Balancing loads, obtained from the Python code.



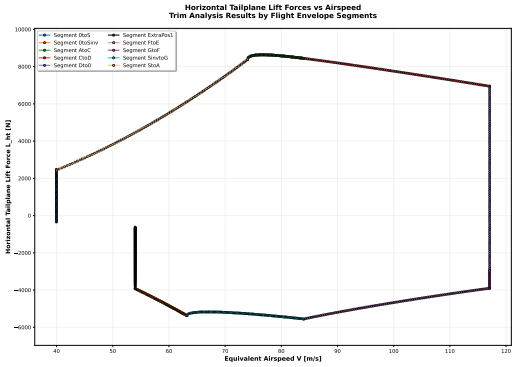
(b) Level 2, single-engine aircraft. Full vehicle lift, obtained from the Python code.



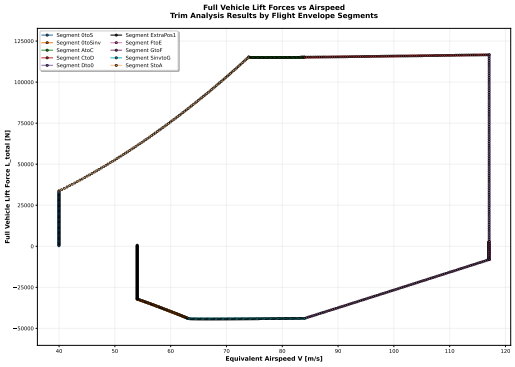
(c) Level 2, twin-engine aircraft. Balancing loads, obtained from the Python code.



(d) Level 2, twin-engine aircraft. Full vehicle lift, obtained from the Python code.



(e) Level 4, twin-engine aircraft. Balancing loads, obtained from the Python code.



(f) Level 4, twin-engine aircraft. Full vehicle lift, obtained from the Python code.

Figure 3.82: Balancing Loads and full vehicle lift as a function of airspeed for the Level 4 and Level 2, twin-engine aircraft, and for the Level 2 single-engine aircraft, obtained with the Python companion code.

All the results presented in this Section refer to the aircraft configurations shown in Figures 3.4, 3.5, and 3.6. Figure 3.80 is analogous to Figure 3.7 and presents the corresponding allowable center-of-gravity-mass combinations. A small red marker identifies

the configuration selected for the certification calculations.

Figure 3.81 presents the maneuvering, gust, and high-lift flight envelopes. This result illustrates the usefulness of an automated certification tool that can be executed for each aircraft configuration relevant to the demonstration of compliance with the airworthiness requirements governing flight loads. Standard Python libraries, such as `matplotlib`, can be used to generate certification artifacts of publication quality.

Figure 3.82 presents the balancing-load and full-vehicle-load graphs for all the aircraft configurations considered. The equations used to generate these plots are the same as those described in Section 3.2.4. The MBSE-based digital certification framework presented in this Section represents a step toward an industry-grade implementation of digital certification. In particular, the export phase—which is critical for robust deployment in practical settings—is designed to remain resilient to changes in the regulatory baseline.

Appendix B contains a numerical comparison between results obtained with the MBSE digital certification framework described in this Section and the ones described in Sections 3.2 and 3.3, summarized in Table B.3.

When multiple airworthiness amendments must be considered within a given application, the export mechanism can operate without amendment-specific logic, provided that the required model elements and their properties are populated consistently.

3.5 Conclusions

This Chapter presented three distinct applications of the proposed approach to the digitalization of certification, substantiation, verification, and validation processes required for the issuance of a new type certificate (TC) for normal-category, privately operated, general aviation aircraft. Although the overarching MBSE paradigm was applied consistently across all three applications, their specific characteristics differ substantially.

- The Eclipse Papyrus/MATLAB MBSE digital certification framework features a relatively simple SysML implementation, with sufficient capabilities to capture and encode the textual content of the EASA CS-23 airworthiness requirements and ASTM International standards. The modeling phase is labor-intensive and requires iterative development, including validation against expected outcomes, identification and correction of errors, and progressive refinement to address gaps and inconsistencies. The SysML model is exported as a UML interchange file and interpreted by a MATLAB parsing routine based on regular expressions. The techniques developed and the experience gained during the construction of this framework were essential for understanding MBSE concepts and tools, identifying limitations, and introducing improvements in subsequent applications. Although designed to operate with a single amendment of the airworthiness rules, this framework remains a valuable research tool for experimentation and for exploring the capabilities and limitations of the MBSE-based digital certification approach. Despite its research-oriented nature, it also demonstrates several practical and transferable features, producing in the meantime a complete certification report.

- Building on the experience gained with the Eclipse Papyrus/MATLAB framework, the second application explores the capabilities of advanced industrial MBSE tools, specifically CATIA MSoSA. This environment provides a comprehensive, OMG-compliant SysML implementation. In this case, the SysML airworthiness model is intentionally limited in scope, including only a subset of the requirements necessary for compliance with EASA CS-23, Subpart C. Nevertheless, this application introduces automated execution capabilities through the use of SysML Activity Diagrams, enabling a workflow in which certification operations are triggered without manual data export. The model automatically generates structured input data (e.g., JSON files), which are used by external C++ libraries to perform certification calculations and produce the corresponding artifacts. This implementation represents a proof of concept for a more integrated and automated digital certification infrastructure, characterized by reduced manual intervention and reliance on robust, high-performance computational tools. With further development and expansion of its scope, this approach could support practical adoption in industrial environments and improve the management and generation of certification artifacts by both applicants and regulatory authorities.

- Finally, during the collaboration with the Aerospace Systems Design Laboratory (ASDL), the existing SysML infrastructure encoding FAA FAR Part 23 airworthiness requirements was extended and refined by the author of this thesis to include the relevant ASTM International consensus standards applicable to general aviation aircraft. A set of additional stereotypes was introduced, building on previously defined elements and extending them with properties suitable for encoding certification data, mathematical relations, and other substantiation-relevant information. Once exported as a UML interchange file, the model is processed by external Python routines that perform the certification and substantiation operations. This implementation, which constitutes a component of a broader SysML ecosystem, builds on the experience gained in the previous case studies. To complete the substantiation workflow for Part 23, Subpart C, further requirements must be incorporated into the Python routines. In addition, the generated certification artifacts should be reintegrated into the SysML model to enable the automated generation of a *Flight Loads* report using the built-in report-generation facilities.

–4–
Conclusions

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This Chapter reexamines the central research problem of the thesis in order to identify further insights and potential pathways toward practical, industrial application. Because SysML provides a rich and flexible syntax, airworthiness requirements can be modeled in more than one way. In this context, the decisive criterion is not strict formal uniformity in the use of the language, but the ability of the modeling strategy to support effective automation of the certification process in practice. This perspective underlies the applications developed in this thesis, each of which addresses a different aspect of this complex problem through a distinct implementation approach exploring different tools and a variety of programming languages.

4.1 Summary of the research

This thesis addressed the complex industrial process of aeronautical certification. The increasing complexity of certification documents, the extensive exchange of information between applicants and regulators during verification and validation, and the need to introduce new aircraft configurations and technologies without compromising safety or

quality are driving the development, controlled implementation, and eventual adoption of new methods for managing certification processes.

The research presented in this thesis applies Model-Based Systems Engineering (MBSE) methods and principles to develop strategies, paradigms, and tools for managing aeronautical certification in a digital environment. Several MBSE-based digital frameworks were developed and evaluated using both open-source and commercial software. These frameworks exhibit different levels of maturity and readiness for practical deployment in industrial settings. Section 3.2 presented a framework built with Eclipse Papyrus and MATLAB that used regular expressions to extract certification data. This is the only framework presented in this thesis capable of producing a complete certification *Flight Loads* technical report. Excerpts of the report are presented in Appendix C. Section 3.3 presented a more integrated tool developed with CATIA MSoSA and C++ that used *opaque actions* and Activity Diagrams to orchestrate certification calculations and generate certification artifacts. Finally, Section 3.4 presented a use-case scenario derived from a broader SysML infrastructure developed by researchers at Georgia Tech. This framework initiated substantiation activities by exporting the model and parsing it with companion Python code. Although further validation is required, they have the potential to reduce the certification burden and lower the barriers to the introduction of new technologies and aircraft configurations across general aviation manufacturers of different sizes.

Chapter 3 presents the results produced by the three MBSE-based certification pipelines for three conventional aircraft architectures and one unconventional aircraft architecture. The selected aircraft represent a sufficiently broad range of general aviation designs and show the flexibility of the conceptual approaches, methods, and tools presented in this thesis. The output of each framework has been presented and described to illustrate the certification database generated at the end of each execution.

The proposed digital certification approach is sufficiently general to be extended, in principle, to all EASA CS-23 subparts. However, the scope of this thesis is limited to Subpart C (Structure). This choice reflects the characteristics of its requirements, many of which can be substantiated through established analytical methods and consensus standards recognized as acceptable means of compliance. When the applicability conditions, input data, assumptions, and prescribed procedures are correctly implemented, these methods provide a structured and reproducible basis for generating compliance evidence. Nevertheless, the resulting calculations and certification artifacts remain subject to appropriate verification, validation, and regulatory review.

In many cases, applicants must also demonstrate compliance through testing, including ground tests on individual test articles and flight tests conducted under restricted permits to fly in support of new type certification. Even in these cases, the MBSE-based digital certification framework can support the organization and management of certification audit activities. In particular, it enables applicants to maintain traceability between test articles, applicable requirements, and the resulting compliance evidence, thereby promoting the consistent generation of certification artifacts. Although these topics are beyond the scope of this thesis, they have received considerable attention in the literature.

4.2 Contributions and limitations of this thesis

This Section briefly analyzes the applications presented in this thesis and discusses their main strengths, limitations, and areas for improvement.

4.2.1 Original contributions

A summary of the original contributions included in this thesis is listed below:

1. A SysML profile with CS and ASTM stereotypes that stores numerical values, expressions, and constraints in typed properties, making substantiation rules machine-executable (see Sections 2.3.1 and 3.2.1).

2. An end-to-end, model-driven workflow from encoded regulation to a *Flight Loads* report for CS-23 Subpart C (refer to Section 3.2 and to Appendix C). The report format had to be inferred from public sources. However, it should be sufficiently detailed for presenting compliance evidence to regulators.

3. A comparative assessment of three implementation architectures: regular-expression parsing, orchestration through Activity Diagrams, and property-based extraction with symbolic generation of executable functions (refer to Sections 3.2, 3.3, 3.4, and 2.5).

4. Automatic generation of executable calculation functions from encoded expressions (refer to Sections 3.2.1 and 3.4.2).

5. Evidence, obtained through requirement diagrams, that the CS-23 Amendment 6 means of compliance do not trace completely to the ASTM standards needed to show compliance (Section 3.2.1 and in particular Figure 3.2).

6. The extension of the Georgia Tech FAR Part 23 model with ASTM F44 consensus standards and new stereotypes (renewed Requirement Diagrams; refer to Section 3.4).

4.2.2 Answers to the research questions

RQ 1 Regulatory and consensus-standard paragraphs can be encoded as SysML requirement elements extended through two stereotypes, one for CS-23 paragraphs and one for ASTM paragraphs, whose typed properties store numerical values, expressions, and constraints. Once exported in UML/XML/XMI files, these properties are sufficient for a companion code suite to reconstruct and execute the substantiation methods of ASTM F3116/F3116M for Subpart C flight loads, as demonstrated for three conventional reference aircraft and one distributed-propulsion configuration (Sections 3.2 to 3.4). Report sections cite the ASTM paragraph from which each calculation is taken (see Appendix C). The answer holds for algebraic and tabulated

provisions; procedures that require engineering judgment, flight tests, or special conditions were not encoded and remain outside the demonstrated scope.

RQ 2 The three architectures trade automation against robustness and effort. Extraction by regular expressions (Eclipse Papyrus with MATLAB) supported the broadest automation, including complete report generation, but is affected by changes in the source text (the certification information). Extraction from stereotype properties (CATIA MSoSA with C++ or Python) is designed to be insensitive to wording changes, and it is expected to be robust against tests with different EASA CS-23 Amendments and substantiation documents, and orchestration through Activity Diagrams removed the manual export step. For the Level 4 reference aircraft, the three codes agree on load factors at all points except E, while design airspeeds differ negligibly; these differences are presented and illustrated in Table B.3. The comparison therefore shows that encoding rules once in a model does not by itself guarantee consistent results for all the spectrum of certification and substantiation tasks.

RQ 3 This research proposes five figures of merit: adoption feasibility, operational efficiency, robustness and correctness, maintainability, and compliance evidence (Section 2.5). These indicators suggest that recurring substantiation effort can be reduced once the model exists. However, the initial encoding effort and the absence of industrial trials prevent a quantitative estimate of the net reduction in certification burden. To answer this question fully requires a controlled industrial evaluation. Nonetheless, the MBSE frameworks present several advantages, such as execution times measured in less than ten minutes per configuration and robustness under different design architectures, enhanced insights and understandability of requirements' context through Requirement Diagrams (refer to Table B.4), and agreement between different companion code suites (refer to Table B.3).

4.2.3 Limitations

Limitations of the methodologies and applications proposed in this thesis are listed below:

- The most important limitation is the lack of industry experts' feedback. The MBSE digital certification frameworks presented in this thesis are research prototypes valuable for further investigations. However, without a thorough validation process in a practical setting, it is not possible to quantitatively evaluate the impacts of deploying MBSE certification tools. Section 2.5 illustrates a qualitative set of KPIs for evaluating the MBSE frameworks.

- Another limitation concerns the *Flight Loads* certification reports, which must be improved in collaboration with EASA domain experts. This would represent a valuable validation and verification for the output generated by executing the digital certification.

- The application presented in Section 3.2, the Eclipse Papyrus and MATLAB framework, adopts fragile exporting techniques based upon regular expressions. This is inherently unstable under changes in the certification information.
- The certification and substantiation material is considered *frozen* in this thesis. Moreover, only one version of the certification and consensus standards documents has been modeled. This represents a limitation that may be addressed through packaging and segregation techniques, well-established among the engineering procedures applied to collaborative coding efforts.
- This thesis addresses only standard certification and substantiation material, without considering special conditions applicable to non-conventional design architectures. This may be considered a limitation for the material presented in Section 3.2.5.
- The final relevant limitation concerns the partial coverage of Subpart C provided by the MBSE frameworks presented in Sections 3.3 and 3.4. The development of these infrastructures extended beyond the original scope of this PhD thesis. Their completion and expansion to achieve more comprehensive coverage of Subpart C are therefore left to future work.

4.2.4 Future work

1. Encode a second amendment of CS-23 and the corresponding standards, and measure extraction failures for each architecture, to test the robustness claimed in RQ 2.
2. Develop CATIA MSoSA report-wizard templates and re-import the generated artifacts into the model.
3. Extend the Python and C++ implementations to the remaining Subpart C load cases.
4. Conduct a pilot study with a design organization and EASA specialists, measuring the figures of merit of RQ 3, including the initial encoding effort.
5. Migrate the profile to SysML v2 and KerML, and evaluate AI-assisted model construction with a deterministic verification layer.
6. Apply the framework to Subparts B and F, and to a CS-25 subset.

4.3 Concluding remarks

The applications presented in the previous Chapters are intended to clarify the motivation for developing an industry-grade digital certification thread capable of managing the complex information required for certification activities in aeronautics in an integrated and coherent manner, while reducing the inefficiencies and fragilities associated with a

document-based process. The implementation of an MBSE approach to the digitalization of aeronautical certification has led to the development of several applications, using both open-source modeling environments coupled with MATLAB and licensed modeling platforms combined with Python or C++. In each case, the emphasis has been on streamlining certification information, reducing the burden on certification personnel, and automating the execution of substantiation tasks, the generation of compliance items, and the preparation of technical certification reports required to demonstrate compliance to the regulatory authorities.

Aeronautical certification is a process of demonstrating, requirement by requirement, that a design is safe, and today that demonstration is assembled by hand from documents. This thesis has shown that, for the flight-loads requirements of CS-23 and Part 23, Subpart C, the requirements and their means of compliance can be encoded in a SysML model and executed to produce traceable compliance evidence, including a complete Flight Loads report. The same modeling effort exposed incomplete references in the published means of compliance, suggesting that formal models can verify certification material and apply it. The work remains a research demonstration: industrial validation, robustness across amendments, and quantified effort savings are still to be established. Within those limits, it provides a concrete path for small design organizations to move from document-based to model-based substantiation.

Although the MBSE-based digital certification toolchain developed in this thesis was conceived primarily for design and manufacturing organizations involved in developing aeronautical products, the proposed methodologies may also be valuable for regulators. In particular, information exchange can be simplified when compliance evidence is represented as model-based certification data and accessed directly within a shared cloud-based modeling environment, as supported by several advanced commercial platforms. In such a setting, integrating demonstration and substantiation activities within a common digital environment may reduce the cost and administrative burden associated with the certification process, including tasks that support the issuance of a type certificate. Error corrections and feedback concerning the certification material would be performed in the same context where the details of that specific part or issue can be easily and consistently accessed.

Modern aviation programs require careful planning and effective resource management. Aeronautical products are becoming increasingly complex as new technologies, advanced features, and more demanding operational capabilities are introduced. In this context, MBSE techniques and digital tools can support a more integrated design and manufacturing approach while helping the aeronautical sector increase product value without compromising quality or safety. From this perspective, certification can be understood not only as a regulatory obligation but also as a structured driver of technical rigor, performance, and continuous improvement. In this sense, the material presented in this thesis describes how MBSE is increasingly promoted and accepted among certification experts in replacement of legacy, document-based methods; in future applications, SysML v2 will be adopted in combination with AI-based modeling assistants, which can expedite the interpretation and inclusion of airworthiness requirements in a structured

digital SysML model, introducing reusability, interoperability, and a higher level of clarity in sharing information and presenting the substantiation, verification, and validation material.

At the time of writing, certification remains predominantly document-based, with only limited parts of the process managed digitally. Even so, this thesis does not propose impractical or unrealistic scenarios. Rather, it shows that MBSE best practices can support the development of digital certification toolchains that are both deployable and practically useful, including verification and validation activities and the automated generation of the associated certification documents. The introduction of MBSE approaches may be gradual, including scenarios where regulators and other stakeholders in the aeronautical industrial landscape adopt these tools and methodologies in limited, restricted applications and procedures, or in a hybrid fashion without the elimination of documents from the process loop.

The first application is based on the creation of a SysML model in the Eclipse Papyrus modeling environment. The automation features of this MBSE-based digital certification framework are enabled by a set of MATLAB routines that perform the certification and substantiation calculations, together with the verification tasks required to demonstrate compliance. As described in Chapter 3, Section 3.2, the airworthiness and substantiation requirements are extracted through regular expressions by exploiting the unique identifiers of paragraphs and subparagraphs. The resulting data—including equations, numerical values, constraints, and other relevant information—are organized in a structured MATLAB object, which then supports execution of the certification workflow. The final results are assembled into a technical report by means of the MATLAB Report Generator API.

This implementation of the MBSE digital certification framework is capable of performing a broad range of certification operations and, by using the same conceptual principles, can be extended beyond EASA CS-23 Subpart C to other subparts, including verification and validation activities and the corresponding compliance evidence. However, the parsing procedure is sensitive to changes in the airworthiness and substantiation documents because it relies on regular expressions. As a result, requirement identification and extraction can become cumbersome when the source text changes. In addition, the need to implement automation in an external programming environment such as MATLAB/Simulink introduces a further practical barrier to adoption. In a practical context, the expansion of the scope of the framework could impair its maintenance over time, as INCOSE best practices and recommendations suggest.

Further investigation led to two alternative implementations of the MBSE-based digital certification approach. CATIA MSoSA, a commercial modeling environment, provided a more integrated set of capabilities for defining SysML elements and for structuring the certification information captured for subsequent use. The first implementation was designed to model the airworthiness and substantiation rules as simply as possible by extending the SysML *Requirement* element with additional properties that encode certification data. Execution of the certification tasks was then orchestrated through a set of Activity Diagrams, developed to launch the certification calculations, generate

the certification artifacts, and prepare the data needed to draft a compliance technical certification report, which can be assembled through the CATIA MSoSA report wizard. The certification calculations were performed by using external routines, written in C++. The main contributions of this application are as follows:

- It demonstrates the automation capabilities that can be implemented within the CATIA MSoSA modeling environment. Through the use of Activity Diagrams, the user can orchestrate a variety of certification tasks and store the resulting certification artifacts and compliance items in a structured way.
- It shows that, even within an MBSE-based workflow, technical documents and certification reports can still be generated, without direct user intervention, by combining the CATIA MSoSA report wizard with the data produced by the external routines. This capability remains consistent with established certification practices and reporting procedures.

The disadvantages of using licensed software to create the SysML model are primarily related to acquisition costs, the effort required to train certification personnel in the new tools and workflow logic, the time needed to develop and maintain the SysML model, and the specialized expertise required to monitor results and verify the soundness of the certification process. These factors can have a substantial impact on small aeronautical design and manufacturing organizations, which are often not in a position to sustain large investments in innovation, modernization, and digitalization. At the same time, regulators can strongly influence these conditions through the standards, tools, and processes they endorse. In such a context, changes in the regulatory environment could accelerate the adoption of new technologies, particularly where firms perceive that failure to modernize may reduce their long-term competitiveness.

The final MBSE application to the problem of aeronautical certification is particularly compelling in demonstrating the need for a digital certification tool. This research effort is the result of a collaboration between the University of Naples Federico II and the Georgia Institute of Technology. The SysML digital model of the FAA FAR Part 23 high-level requirements, created using CATIA MSoSA, was expanded to include ASTM International consensus standards containing the substantiation rules and other certification provisions relevant to demonstrating compliance with FAA requirements. New stereotypes and related features were added to the model, creating a use-case application of this broader SysML infrastructure. The enriched model was then exported and used to generate certification-demonstration items in the context of FAA FAR Part 23, Subpart C (Structure). In this application, the model was exported manually in a readable format, parsed using standard Python libraries to create structured variables, and then processed by Python certification routines that generated the artifacts required to support compliance demonstration.

The SysML model used in this application is highly detailed and includes a wide range of features, modeling techniques, and intended uses. At the same time, and as previously stated in this Chapter, the use of commercial software for building and maintaining the

model introduces costs and organizational burdens that may significantly affect small aeronautical firms. The complexity of the model itself is also an important factor and requires careful trade-off analysis to determine whether its implementation is justified in a given industrial context.

Appendix A
Theoretical background for the Eclipse Papyrus and MATLAB MBSE digital certification framework

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The MBSE digital certification framework is based on simple, well-established mathematical procedures for determining flight loads and creating the certification database.

This database includes numerical results, which may be collected in tables; certification artifacts, such as figures and diagrams; and the information required to generate the *Flight Loads* certification report. This appendix provides a detailed explanation of the mathematical tools and formulas that are commonly accepted by regulators, particularly within the international regulatory framework established by the Convention on International Civil Aviation (Chicago Convention) and the International Civil Aviation Organization (ICAO). The discussion also considers the historical development of these methods through the Joint Aviation Requirements (JAR) and their evolution into the current regulatory landscape.

A.1 Aircraft design architectures

Definition 2 (of Aircraft Design Architecture)**.** An Aircraft Design Architecture is the collection of aerostructural elements, systems, and equipment that must be collectively and singularly approved and certified for the issuance or the modification of a Type Certificate.

A.2 Characteristic airspeeds

Characteristic airspeeds or design airspeeds are defined in the context of the so-called *V-Speeds*. These design airspeeds are defined as equivalent airspeeds (EAS); the reference document adopted in the context of this thesis is the ASTM International F3116/F3116M *Standard Specification for Design Loads and Conditions*. In Chapter 5, *Design Airspeeds*, we can read the following definitions:

- **Design Maximum Speed in level flight**, $\mathbf{V_H}$, at maximum continuous power. The evaluation of this quantity is based on the application of standard MATLAB libraries. The reference formula used and solved is the following:

$$g(V) = \frac{1}{2} \cdot f \cdot \rho_0 \cdot V^3 + 2 \cdot \frac{(\text{MTOW} \cdot b_{\text{wing}})^2}{\pi \cdot \rho_0 \cdot e \cdot V} - N_{\text{engines}} \cdot \text{MCP} \tag{A.1}$$

where

ρ_0 = Atmospheric density expressed in $\mathrm{kg\,m^{-3}}$
$f = S_{\text{wing}} \cdot C_{D_0}$ = Equivalent parasite area expressed in $\mathrm{m^2}$
MTOW = Maximum Takeoff Weight expressed in N
b_{wing} = Main wingspan measured in m
e = Oswald efficiency factor for the complete aircraft, non-dimensional
N_{engines} = Number of engines, non-dimensional
MCP = Maximum Continuous Power expressed in W
$g(V)$ = Input mathematical function to the MATLAB `fzero`
V = Aircraft airspeed measured in $\mathrm{m\,s^{-1}}$

- **Design Cruising Speed** [230, p. 9, sec. 5.1.1], $\mathbf{V_C}$, expressed in kn which is calculated considering the wing loading $\frac{W}{S}$ at the design maximum takeoff weight expressed in $\mathrm{lb\,ft^{-2}}$:
  - $33 \cdot \sqrt{\frac{W}{S}}$ for airplanes not approved for aerobatics.
  - $36 \cdot \sqrt{\frac{W}{S}}$ for airplanes approved for aerobatics.

  For $\frac{W}{S} > 20\,\mathrm{lb\,ft^{-2}}$ the multiplying factors may be decreased linearly with $\frac{W}{S}$ to a value of $28.6\,\mathrm{kn\,ft/lb^{\frac{1}{2}}}$ where $\frac{W}{S} = 100\,\mathrm{lb\,ft^{-2}}$. The resulting equation is therefore:

$$\mathrm{MF} = \mathrm{MF}_0 - \left(\frac{\mathrm{MF}_0 - 28.6}{80}\right) \cdot \left(\frac{W}{S} - 20\right) \tag{A.2}$$

  The design cruising speed need not be more than $0.9 \cdot V_H$ at sea level. At altitudes where an M_D is established, a cruising M_C limited by compressibility effects may be selected.

- **Design Dive Speed** [230, p. 9, sec. 5.1.2], $\mathbf{V_D}$, expressed in kn may be evaluated as follow:
  - $V_D < 1.25 \cdot V_C$ or $M_D < 1.25 \cdot M_C$.
  - With $V_{C_{\min}}$, which is the required minimum design cruising speed, V_D measured in kn may not be less than:
    - $1.40 \cdot V_{C_{\min}}$ for airplanes not approved for aerobatics.
    - $1.55 \cdot V_{C_{\min}}$ for airplanes approved for aerobatics.

    For $\frac{W}{S} > 20\,\mathrm{lb\,ft^{-2}}$ the multiplying factors may be decreased linearly with $\frac{W}{S}$ to a value of 1.35 where $\frac{W}{S} = 100\,\mathrm{lb\,ft^{-2}}$. The resulting equation is therefore:

$$\mathrm{MF} = \mathrm{MF}_0 - \left(\frac{\mathrm{MF}_0 - 1.35}{80}\right) \cdot \left(\frac{W}{S} - 20\right) \tag{A.3}$$

Compliance with the previously defined requirements need not be shown if V_D or M_D is selected so that the minimum speed margin between V_C or M_C and V_D or M_D is the greater of the following:

1. The speed increase resulting when, from the initial condition of stabilized flight at V_C or M_C, the airplane is assumed to be upset, flown for 20 s along a flight path 7.5 deg below the initial path, and then pulled up with a load factor of 1.5 (0.5 g acceleration increment). At least 75 % maximum continuous power for reciprocating engines, and maximum cruising power for turbines, or, if less, the power required for V_C or M_C for both kinds of engines, must be assumed until the pull-up is initiated, at which point power reduction and pilot-controlled drag devices may be used.
2. Mach 0.05 at altitudes where M_D is established.
3. Mach 0.07 for level 4 airplanes at altitudes where M_D is established unless a rational analysis, including the effects of automatic systems, is used to determine a lower margin. If a rational analysis is used, the minimum speed margin must be enough to provide for atmospheric variations, such as horizontal gusts, and the penetration of jet streams or cold fronts, instrument errors, airframe production variations, and must not be less than Mach 0.05.

- **Design Maneuvering Speed** [230, p. 9, sec. 5.1.3], $\mathbf{V_A}$, the following applies:
  - $V_A < V_S \cdot \sqrt{n}$ where:
    1. V_S is a 1 g computed stalling speed with flaps retracted, normally based on the maximum airplane normal force coefficients, C_{N_A}, at either the particular weight under consideration or the design maximum takeoff weight.
    2. n is the limit maneuvering load factor used in design.
  - The value of V_A need not exceed the value of V_C used in design.
- **Design Speed for Maximum Gust Intensity** [230, p. 9, sec. 5.1.4], $\mathbf{V_B}$, the following applies:
  - V_B may not be less than the speed determined by the intersection of the line representing the maximum positive lift, $C_{N_{\max}}$, and the line representing the rough air gust velocity on the plane $V - n$, or $V_{S_1} \cdot \sqrt{n_{\text{gust}}}$, whichever is less, where:
    1. n_{gust} is the positive airplane gust load factor due to gust, at speed V_C, in accordance with Section 4.6 of the same document and at the particular weight under consideration, usually equal to the maximum takeoff weight.
    2. V_{S_1} is the 1 g stalling speed with the flaps retracted at the particular weight under consideration.

- V_B need not be greater than V_C.

Other ASTM standards that address high-level Subpart C requirements include:

- F3396/F3396M, *Standard Practice for Aircraft Simplified Loads Criteria* [233], which applies to small, simple, and very light airplanes with a stalling speed not exceeding 45 kn.
- F3331-18, *Standard Practice for Aircraft Water Loads* [234], which provides structural design-load provisions for seaplanes.

ASTM also publishes additional standards that include provisions relevant to structural design loads and conditions, such as F2245-23, *Standard Specification for Design and Performance of a Light Sport Airplane* [235]. Although F2245-23 also addresses other EASA subparts and remains applicable to level 1 aircraft, this study focuses on F3116/F3116M because it applies to all aircraft eligible for certification in the normal category. This standard provides procedures for constructing the maneuvering and gust flight envelope, including the relationship between airspeed V and the applicable limit load factors n. The applicant must demonstrate compliance with strength requirements for all relevant combinations of airspeed and limit load factor within the resulting envelope.

A.3 Gust velocities

Referring to paragraph 4.4.3.1 of the F3116 document, the airplane is assumed to be subjected to symmetrical vertical gusts in level flight. The resulting limit load factors must correspond to the conditions determined as follows:

- Positive (up) and negative (down) gusts of $15.24\,\mathrm{m\,s^{-1}}$ ($50\,\mathrm{ft\,s^{-1}}$) at V_C must be considered at altitudes between sea level and 6096 m (20 000 ft). The gust velocity may be reduced linearly from $15.24\,\mathrm{m\,s^{-1}}$ ($50\,\mathrm{ft\,s^{-1}}$) at 6096 m (20 000 ft) to $7.62\,\mathrm{m\,s^{-1}}$ ($25\,\mathrm{ft\,s^{-1}}$) at 15 240 m (50 000 ft) [230, p. 2, sec. 4.4.3.1(1)]. In more mathematical terms:

$$U_{\mathrm{de}} = \begin{cases} 15.24\,\mathrm{m\,s^{-1}} & \text{if } h < 6096\,\mathrm{m} \\ 15.24 - 0.000833 \cdot (h - 6096) & \text{if } 6096\,\mathrm{m} \leq h < 15\,240\,\mathrm{m} \\ 7.62\,\mathrm{m\,s^{-1}} & \text{if } h \geq 15\,240\,\mathrm{m} \end{cases} \tag{A.4}$$

- Positive (up) and negative (down) gusts of $7.62\,\mathrm{m\,s^{-1}}$ ($25\,\mathrm{ft\,s^{-1}}$) at V_D must be considered at altitudes between sea level and 6096 m (20 000 ft). The gust velocity may be reduced linearly from $7.62\,\mathrm{m\,s^{-1}}$ ($25\,\mathrm{ft\,s^{-1}}$) at 6096 m (20 000 ft) to $3.81\,\mathrm{m\,s^{-1}}$ ($12.5\,\mathrm{ft\,s^{-1}}$) at 15 240 m (50 000 ft) [230, p. 2, sec. 4.4.3.1(2)]. In more mathematical terms:

$$U_{\mathrm{de}} = \begin{cases} 7.62\,\mathrm{m\,s^{-1}} & \text{if } h < 6096\,\mathrm{m} \\ 7.62 - 0.000417 \cdot (h - 6096) & \text{if } 6096\,\mathrm{m} \leq h < 15\,240\,\mathrm{m} \\ 3.81\,\mathrm{m\,s^{-1}} & \text{if } h \geq 15\,240\,\mathrm{m} \end{cases} \tag{A.5}$$

- In addition, for level 4 airplanes, positive (up) and negative (down) rough air gusts of $20.12\,\mathrm{m\,s^{-1}}$ ($66\,\mathrm{ft\,s^{-1}}$) at V_B must be considered at altitudes between sea level and 6096 m (20 000 ft). The gust velocity may be reduced linearly from $20.12\,\mathrm{m\,s^{-1}}$ ($66\,\mathrm{ft\,s^{-1}}$) at 6096 m (20 000 ft) to $11.58\,\mathrm{m\,s^{-1}}$ ($38\,\mathrm{ft\,s^{-1}}$) at 15 240 m (50 000 ft) [230, p. 2, sec. 4.4.3.1(3)]. In more mathematical terms:

$$U_{\mathrm{de}} = \begin{cases} 20.12\,\mathrm{m\,s^{-1}} & \text{if } h < 6096\,\mathrm{m} \\ 20.12 - 0.000934 \cdot (h - 6096) & \text{if } 6096\,\mathrm{m} \leq h < 15\,240\,\mathrm{m} \\ 11.58\,\mathrm{m\,s^{-1}} & \text{if } h \geq 15\,240\,\mathrm{m} \end{cases} \tag{A.6}$$

Referring to paragraph 4.4.3.2, the F3116 document gives us a formula for the functional evaluation of the gust shape [230, p. 2, sec. 4.4.3.2(1)]:

$$U = \frac{U_{\mathrm{de}}}{2} \cdot \left(1 - \frac{2 \cdot \pi \cdot s}{25 \cdot \overline{c}}\right) \tag{A.7}$$

where

s = Distance penetrated into gust measured in m or ft
$\overline{c}$ = Mean geometric or aerodynamic chord of the main wing measured in m or ft
U_{de} = Derived gust velocity previously evaluated in $\mathrm{m\,s^{-1}}$ or $\mathrm{ft\,s^{-1}}$

Gust load factors vary linearly with speed between V_C and V_D [230, p. 2, sec. 4.4.3.2(2)].

A.4 Maneuvering and gust flight envelope

The applicant should refer to the following mathematical formulation when drafting the maneuvering and gust flight envelope:

- For the evaluation of the positive limit maneuvering load factor n, the following applies:

$$n \leq 2.1 + \frac{24000}{W + 10000} \tag{A.8}$$

  where W is the maximum takeoff weight expressed in lb, except that n need not be more than 3.8 for airplanes not approved for aerobatics, and 6 for designs seeking approval for aerobatics [230, p. 3, sec. 4.5.1].

- The negative limit maneuvering load factor may not be less than [230, p. 3, sec. 4.5.2]:
  - $0.4 \cdot n$ for airplanes not approved for aerobatics.
  - $0.5 \cdot n$ for airplanes approved for aerobatics.

> Maneuvering load factors lower than those specified in this Section may be used if the airplane has design features that make it impossible to exceed these values in flight [230, p. 3, sec. 4.5.3].

Each airplane must be designed to withstand loads on each lifting surface resulting from gusts, associated with the following equation [230, p. 3, sec. 4.6.3]:

$$n_{\text{gust}} = 1 \pm \frac{k_g \cdot \rho_0 \cdot U_{\text{de}} \cdot V \cdot a}{2 \cdot \left(\frac{W}{S}\right)} \tag{A.9}$$

where

$$\begin{aligned}
k_g &= \frac{0.88 \cdot \mu_g}{5.3 + \mu_g} = \text{Gust alleviation factor} \\
\mu_g &= \frac{2 \cdot \frac{W}{S}}{\rho \cdot \overline{c} \cdot a \cdot g} = \text{Aeroplane mass ratio} \\
U_{\text{de}} &= \text{Derived gust velocities, defined in Section A.3} \\
\rho_0 &= \text{Air density at sea level measured in kg m}^{-3} \\
\rho &= \text{Air density at the reference altitude measured in kg m}^{-3} \\
\frac{W}{S} &= \text{Wing loading in the particular load case measured in N m}^{-2} \\
\overline{c} &= \text{Mean Aerodynamic or Geometric Chord measured in m} \\
g &= \text{Acceleration due to gravity measured in m s}^{-2} \\
a &= \text{Lift curve slope measured in rad}^{-1} \\
V &= \text{Airplane equivalent speed measured in m s}^{-1}
\end{aligned}$$

On the $V - n$ plane, the last formula appears as lines starting from the point at $V = 0$ and $n = 1$, and ending at points C and D of the diagram, with different values of U_{de}. The gust load factors for a canard or tandem wing configuration must be computed using a rational analysis, or may be computed in accordance with the above formula, provided that the resulting net loads are shown to be conservative with respect to the gust criteria list in Section A.3.

A.5 High-lift flight envelope

Paragraph 4.8 *High Lift Devices* from F3116 ASTM International standard [230, p. 4, sec. 4.8.1] provides guidelines for evaluating the loads when the aircraft is operating with its high lift devices in their deployed configuration. If wing flaps or similar high lift devices are installed for use in takeoff, approach, or landing, the airplane, with the flaps fully deflected at the maximum design speed with flaps deflected, V_F, is assumed to be subjected to symmetrical maneuvers and gusts resulting in limit load factors within the range determined by:

- Maneuvering to a positive limit load factor of 2 [230, p. 4, sec. 4.8.1.1].
- Positive (up) and negative (down) gusts of $7.62\,\mathrm{m\,s^{-1}}$ ($25\,\mathrm{ft\,s^{-1}}$) acting normal to the flight path in level flight [230, p. 4, sec. 4.8.1.2].
- V_F must be assumed to be no less than $1.4 \cdot V_S$ or $1.8 \cdot V_{S_F}$, whichever is greater where [230, p. 4, sec. 4.8.2]

  V_S = 1 g stall speed with flaps retracted at MTOW [230, p. 4, sec. 4.8.2.1]

  V_{S_F} = 1 g stall speed with flaps fully extended at MTOW [230, p. 4, sec. 4.8.2.2]

- If an automatic flap load limiting device is used, the airplane may be designed for the critical combinations of airspeed and flap position allowed by that device [230, p. 4, sec. 4.8.1.3].
- In determining external loads on the airplane as a whole, thrust, slipstream, and pitching acceleration may be assumed to be zero [230, p. 4, sec. 4.8.3].

The flaps, their operating mechanism, and their supporting structures must be designed for the conditions prescribed above. In addition, with the flaps fully extended at V_F, the following conditions, taken separately, must be accounted for [230, p. 4, sec. 4.8.4]:

- A head-on gust having a velocity of $7.62\,\mathrm{m\,s^{-1}}$ ($25\,\mathrm{ft\,s^{-1}}$) equivalent airspeed, combined with propeller slipstream corresponding to 75 % of maximum continuous power [230, p. 4, sec. 4.8.4.1].
- The effects of propeller slipstream corresponding to maximum takeoff power [230, p. 4, sec. 4.8.4.2].
- For the investigation of slipstream effects, the load factor may be assumed to be 1 [230, p. 4, sec. 4.8.4.3].

A.6 Balancing loads, wing-body and full vehicle lift

A simplified discussion will be included in this appendix concerning the balancing loads acting on an aircraft. The relationship between horizontal tail loads, pitching moment, stabilizer angle of attack, and elevator angle will be considered for a steady-state maneuver in which pitching acceleration is considered equal to zero. This is the logic applied in evaluating the balancing loads all across the spectrum of methods applied within the developed MBSE digital certification framework.

The equations for horizontal tail load and pitching moment are:

$$L_{\text{tail}} = L_{\alpha_S} \cdot \alpha_S + L_{\delta_e} \cdot \delta_e + L_{\text{camber}} \tag{A.10}$$

$$M_{\text{tail}} = M_{\alpha_S} \cdot \alpha_S + M_{\delta_e} \cdot \delta_e + M_{\text{camber}} \tag{A.11}$$

where

α_S = Stabilizer angle of attack measured in deg
δ_e = Elevator deflection angle measured in deg
L_{tail} = Horizontal tailplane load measured in N
M_{tail} = Horizontal tailplane pitching moment about $0.25 \cdot \bar{c}$ measured in N m
L_{α_S} = Horizontal tailplane load due to unit of α_S measured in N/deg
M_{α_S} = Horizontal tailplane pitch. mom. due to unit of α_S measured in N m/deg
L_{δ_e} = Horizontal tailplane load due to unit of δ_e measured in N/deg
M_{δ_e} = Horizontal tailplane pitch. mom. due to unit of δ_e measured in N m/deg
L_{camber} = Horizontal tailplane load due to unit of camber measured in N
M_{camber} = Horizontal tailplane pitch. mom. due to unit of camber measured in N m

Within the limit of this appendix, it can be found that the horizontal tailplane angle of attack corresponds to the stabilizer angle of attack α_S, which may be defined as a function of the main wing angle of attack, the curvilinear flight increment due to pitching velocity, and fuselage flexibility terms plus the initial trim setting of the stabilizer. The interested reader should refer to the specialized literature, reported in the bibliography of this thesis.

According to [230, p. 5, sec. 4.16], a horizontal tailplane balancing load is a load necessary to maintain equilibrium, in any specified flight condition with no pitching acceleration. Horizontal balancing surfaces must be designed for the balancing loads occurring at any point on the limit maneuvering and gust flight envelope and in the flap conditions specified in Section A.5.

A.7 Shear, bending moment, and torsion spanwise distributions

A.7.1 Shear spanwise distribution

For the cantilever wing configuration, the shear force is obtained by integrating the spanwise normal force distribution from the considered Section to the wing tip. Assuming that the shear force is zero at the free tip, the continuous expression is

$$\text{Shear}(y) = \int_{y}^{b/2} N(\eta)\, d\eta, \tag{A.12}$$

where $N(y)$ is the local normal force per unit span. In the numerical implementation, the integral is evaluated by means of the trapezoidal rule. Let y_i be the spanwise stations ordered from the wing tip to the wing root, such that $y_1 = b/2$ and $y_n = 0$. The shear force is initialized as

$$\text{Shear}_1 = 0,$$

and then computed recursively as

$$\text{Shear}_i = \text{Shear}_{i-1} + \frac{1}{2}\left(N_{i-1} + N_i\right)\left(y_{i-1} - y_i\right), \qquad i = 2, \ldots, n$$

Equivalently,

$$\text{Shear}_i = \sum_{k=2}^{i} \frac{1}{2}\left(N_{k-1} + N_k\right)\left(y_{k-1} - y_k\right) \tag{A.13}$$

For the strut-braced wing configuration, the distributed normal load $N(y)$ is first integrated along the semi-span to obtain the resultant

$$R_N = \int_0^{b/2} N(y)\, dy$$

The bending moment about the wing root due to the same distributed load is

$$M_N = \int_0^{b/2} y\, N(y)\, dy$$

The corresponding spanwise location of the load resultant is

$$\bar{y}_N = \frac{M_N}{R_N} = \frac{\int_0^{b/2} y\, N(y)\, dy}{\int_0^{b/2} N(y)\, dy}$$

If the strut is attached to the wing at the spanwise station y_a, the vertical component of the strut reaction is obtained by imposing moment equilibrium about the wing root:

$$S_v = \frac{M_N}{y_a} = \frac{1}{y_a}\int_0^{b/2} y\, N(y)\, dy$$

According to the adopted sign convention, the force introduced by the strut into the shear-force distribution is

$$S_{\text{strut}} = -S_v$$

The shear force due to the distributed load alone is

$$\text{Shear}_0(y) = \int_0^{y} N(\eta)\, d\eta.$$

Therefore, including the concentrated strut reaction, the shear-force distribution becomes

$$\text{Shear}(y) = \int_0^{y} N(\eta)\, d\eta + S_{\text{strut}} \cdot H(y - y_a),$$

or equivalently,

$$\text{Shear}(y) = \int_0^{y} N(\eta)\, d\eta - \frac{1}{y_a}\left(\int_0^{b/2} \eta\, N(\eta)\, d\eta\right) H(y - y_a), \tag{A.14}$$

where $H(y - y_a)$ is the Heaviside step function defined as

$$H(y - y_a) = \begin{cases} 0, & y < y_a, \\ 1, & y \geq y_a. \end{cases}$$

A.7.2 Bending moment spanwise distribution

Let the spanwise coordinate y be measured from the wing root towards the wing tip, with $0 \leq y \leq b/2$. The bending moment distribution is obtained from the shear force distribution as

$$M(y) = \int_{y}^{b/2} V(\eta)\, \mathrm{d}\eta, \qquad M\left(\frac{b}{2}\right) = 0$$

For a discretized semi-span with stations $y_1, \ldots, y_n$, where

$$y_1 = 0, \qquad y_n = \frac{b}{2},$$

the trapezoidal approximation gives

$$M_i = \sum_{k=i+1}^{n} \frac{1}{2} \left[\mathrm{Shear}(y_{k-1}) + \mathrm{Shear}(y_k)\right] (y_k - y_{k-1}), \qquad i = 1, \ldots, n-1, \tag{A.15}$$

with the boundary condition

$$M_n = 0$$

A.7.3 Torsion moment spanwise distribution

For the numerical implementation, the semi-span is discretized into n spanwise stations. If the points are ordered from the wing tip to the wing root, $y_1 = b/2$ and $y_n = 0$, the distance between two consecutive stations is

$$\Delta y_k = y_{k-1} - y_k, \qquad k = 2, \ldots, n$$

The local pitching moment per unit span is

$$m_{t,k} = q\, c_k^2\, c_{m,k}$$

Neglecting lift eccentricity, the torsional moment is evaluated through the cumulative trapezoidal integration

$$T_j = \sum_{k=2}^{j} \frac{1}{2} \left(m_{t,k-1} + m_{t,k}\right) \Delta y_k, \qquad j = 2, \ldots, n$$

Including the lift-induced torsion, the discrete formulation becomes

$$T_j = \sum_{k=2}^{j} \left[\frac{1}{2} \left(m_{t,k-1} + m_{t,k}\right) + e_k L'_k\right] \Delta y_k, \qquad j = 2, \ldots, n, \tag{A.16}$$

where

$$e_k = x_{\mathrm{sc},k} - x_{\mathrm{cp},k}$$

A.8 Horizontal tailplane flight loads

A.8.1 Maneuvering loads for the horizontal tailplane

The MBSE MATLAB/Simulink digital certification framework is capable of performing, at least on a preliminary basis, the study of the loads associated with the *unchecked maneuver*. The *unchecked maneuver* analysis simulates an abrupt aft or forward displacement of the pitching control to its full-stop position. This condition is typically most critical at the Design Maneuvering Flight Speed (V_A) and is assumed to occur over a time interval of $t_{\text{deflection}} = 0.3\,\text{s}$. This certification task revolves around the following differential equation:

$$\ddot{\theta} = \frac{q \cdot S_{\text{ht}} \cdot C_{L_{\alpha,\,\text{ht}}} \cdot l_{\text{ht}}}{I_{\text{yy}}} \cdot \left(\omega \cdot t - \frac{\dot{\theta} \cdot l_{\text{ht}}}{V} \right) \cdot \text{DF} \tag{A.17}$$

where

$$\begin{aligned}
\theta &= \text{Pitching angle measured in rad} \\
\dot{\theta} &= \text{Pitching angular speed measured in rad}\,\text{s}^{-1} \\
\ddot{\theta} &= \text{Pitching angular acceleration measured in rad}\,\text{s}^{-2} \\
q &= \text{Dynamic pressure measured in Pa} \\
C_{L_{\alpha,\,\text{ht}}} &= \text{Lift curve slope of the horizontal tailplane measured in rad}^{-1} \\
l_{\text{ht}} &= \text{Distance between aircraft CG and h.t. AC measured in m} \\
I_{\text{yy}} &= \text{Pitching moment of inertia measured in kg}\,\text{m}^2 \\
\omega = \frac{\delta_{e,\,\text{max}}}{t_{\text{deflection}}} &= \text{Control angular deflection speed measured in rad}\,\text{s}^{-1} \\
V &= \text{Airspeed measured in m}\,\text{s}^{-1} \\
\text{DF} &= \text{Damping factor, conservatively imposed equal to 0.35}
\end{aligned}$$

The rationale for determining the deflection time interval $t_{\text{deflection}} = [0, 0.3\,\text{s}]$ derives from a well-established literature. More specifically, for unpowered control surfaces, if a maneuver analysis is used to predict the maneuvering loads on the pitch control surface, the time for sudden deflection from neutral position to the stops or vice versa may be assumed as follows:

- For aerobatic category aircraft:
    1. 0.1 s for stick controlled surfaces.
    2. 0.2 s for wheel controlled surfaces.
- For aircraft not approved for aerobatics:
    1. 0.2 s for stick controlled surfaces.
    2. 0.3 s for wheel controlled surfaces.

It is safe to assume that the deflection time is equal to 0.3 s in any possible aircraft design architecture. For powered control surfaces, the deflection time should be measured experimentally.

The product $\dot{\theta} \cdot l_{\text{ht}}$ is collectively equal to the Δv measurable during the maneuver; the ratio $\frac{\Delta v}{V}$ is known as the damping angle. The integration of this equation is relatively easy and can be performed using standard MATLAB facilities. This particular modelization is generally applicable with limited data and provides conservative results that can be considered a sufficiently realistic simulated flight loads database for sizing the horizontal tailplane structures.

According to the F3116 ASTM International [230, p. 5, sec. 4.17], each horizontal surface and its supporting structure, and the main wing of a canard or tandem wing configuration, if that surface has pitch control, must be designed for the maneuvering loads imposed by conditions examined below:

- A sudden movement of the pitching control at the maneuvering design speed V_A [230, p. 5, sec. 4.17.1] to:
  - The maximum aft movement (upward deflection) [230, p. 5, sec. 4.17.1.1].
  - The maximum forward movement (downward deflection) [230, p. 5, sec. 4.17.1.2], as limited by the control stops, or pilot effort, whichever is critical.
- A sudden aft movement of the pitching control at speeds above V_A, followed by a forward movement of the pitching control resulting in the following combinations of normal and angular acceleration [230, p. 5, sec. 4.17.2]:
  - A nose-up pitching (down load) with $n = 1$ and $\dot{q} = \frac{39}{V} \cdot n_m \cdot (n_m - 1.5)$.
  - A nose-down pitching (up load) with $n = n_m$ and $\dot{q} = -\frac{39}{V} \cdot n_m \cdot (n_m - 1.5)$.

  where

  n_m = Pos. lim. man. load factor used in the design of the airplane.
  V = Initial speed measured in kn

The conditions in this Section involve loads corresponding to the loads that may occur in a *checked maneuver* (a maneuver in which the pitching control is suddenly displaced in one direction and then suddenly moved in the opposite direction). The deflections and timing of the *checked maneuver* must avoid exceeding the limit maneuvering load factor. The total horizontal surface load for both nose-up and nose-down pitching conditions is the sum of the balancing loads at V and the specified value of the normal load factor n, plus the maneuvering load increment due to the specified value of the angular acceleration [230, p. 6, sec. 4.17.2.1]. The following assumption must be made [230, p. 6, sec. 4.17.3.5]:

- The airplane is initially in level flight, and its attitude and airspeed remain constant.
- Inertia forces balance the air loads.

The applicant should analyze a sudden deflection of the elevator such as to cause the normal acceleration to change from an initial value to a final value; the cases considered are collected in Table A.1. Figure A.1 shows the location of the points listed in Table A.1 on the $V - n$ plane.

Table A.1: Reference conditions for the *checked maneuver*.

	Initial Cond.	**Final Cond.**	Δn
$\mathbf{V_A}$	A_1	A	$n_1 - 1$
	A	A_1	$1 - n_1$
	A_1	G	$n_4 - 1$
	G	A_1	$1 - n_4$
$\mathbf{V_D}$	D_1	D	$n_2 - 1$
	D	D_1	$1 - n_2$
	D_1	E	$n_3 - 1$
	E	D_1	$1 - n_3$

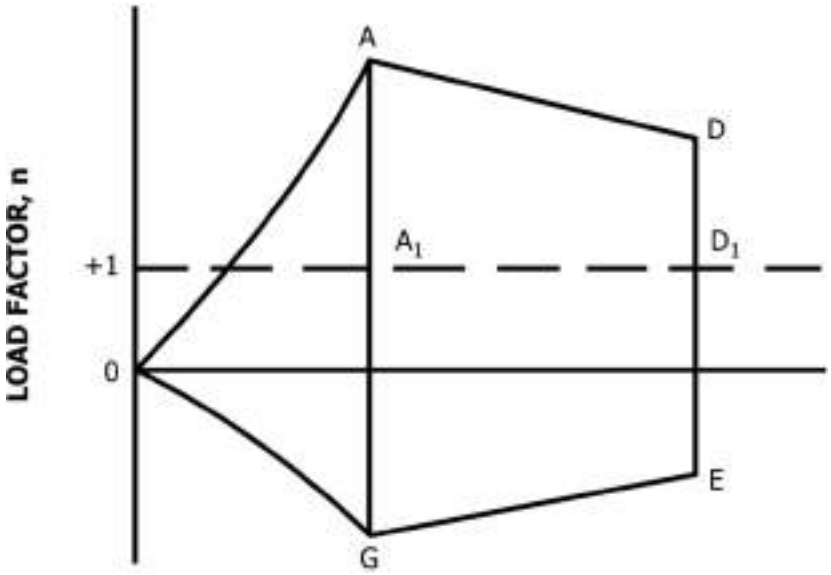

Figure A.1: Reference pitching maneuvers diagrams [230, p. 6].

A.8.2 Gust loads for the horizontal tailplane

According to ASTM International F3116 [230, p. 7, sec. 4.18.4], the incremental load resulting from the gusts must be added to the initial balancing load to obtain the total load. In the absence of a more rational analysis, the incremental horizontal tailplane load

due to the gust must be computed as follows:

$$\Delta L_{\text{ht}} = \frac{k_g \cdot U_{\text{de}} \cdot V \cdot C_{L_{\alpha,\,\text{ht}}} \cdot S_{\text{ht}}}{16.3} \cdot \left(1 - \frac{d\epsilon}{d\alpha}\right) \tag{A.18}$$

where

$$\begin{aligned}
\Delta L_{\text{ht}} &= \text{Incremental horizontal tailplane load measured in N} \\
k_g &= \text{Gust alleviation factor as already defined in Section A.4} \\
U_{\text{de}} &= \text{Gust intensity as defined in Section A.3} \\
V &= \text{Aircraft equivalent speed measured in m s}^{-1} \\
C_{L_{\alpha,\,\text{ht}}} &= \text{Horizontal tailplane lift curve slope measured in rad}^{-1} \\
S_{\text{ht}} &= \text{Horizontal tailplane surface measured in m}^2 \\
\left(1 - \frac{d\epsilon}{d\alpha}\right) &= \text{Downwash factor}
\end{aligned}$$

A.8.3 Unsymmetrical loads for the horizontal tailplane

In accordance with the F3116 ASTM International document [230, p. 7, sec. 4.19.2], in the absence of more reliable data for aircraft that are conventional in regard to location of engines, wings, horizontal surfaces other than the main wing, and fuselage shape:

$$\text{Load}_{\text{ht}} = \begin{cases} 1.0 \cdot \text{Load}_{\text{max}} \text{ on one side} \\ \left(1 - \frac{1}{10} \cdot (n-1)\right) \cdot \text{Load}_{\text{max}} \leq 0.8 \cdot \text{Load}_{\text{max}} \text{ on the opposite side} \end{cases} \tag{A.19}$$

where

$$\begin{aligned}
n &= \text{Specified positive maneuvering load factor} \\
\text{Load}_{\text{ht}} &= \text{Unsymmetrical load on the horizontal tailplane measured in N} \\
\text{Load}_{\text{max}} &= \text{Maximum reference symmetrical load on the horiz. tailplane in N}
\end{aligned}$$

For aircraft that are not conventional [230, p. 7, sec. 4.19.3], such as aircraft with horizontal surfaces other than the main wing having appreciable dihedral or supported by the vertical tail surfaces, the surfaces and supporting structures must be designed for combined vertical and horizontal surface loads resulting from each prescribed flight condition taken separately.

A.9 Vertical tailplane flight loads

A.9.1 Maneuvering loads for the vertical tailplane

The *unchecked maneuver* analysis, performing an abrupt left or right displacement of the rudder control surface to its full-stop position, is performed also for the vertical tailplane.

Again, this maneuver is usually performed at $V = V_A$ with the same deflection time. The following differential equation has been used:

$$\ddot{\phi} = \frac{q \cdot S_{\text{wing}} \cdot b_{\text{wing}}}{I_{zz} \cdot l_{\text{vt}}} \cdot \left[C_{N_{\delta_r}} \cdot \delta_r + C_{N_{\delta_{r,\,\text{tab}}}} \cdot \delta_{r,\,\text{tab}} \cdot \text{GR}_{\text{tab}} + \frac{\dot{\phi} \cdot b_{\text{wing}}}{2 \cdot V} \cdot C_{N_{\text{r}}} + \left(\phi - \frac{\dot{\phi} \cdot l_{\text{vt}}}{V} \cdot C_{N_\beta} \right) \right] \quad \text{(A.20)}$$

where

ϕ = Yawing angle measured in rad
$\dot{\phi}$ = Yawing angular speed measured in rad s^{-1}
$\ddot{\phi}$ = Yawing angular acceleration measured in rad s^{-2}
q = Dynamic pressure measured in Pa
$C_{N_{\delta_r}}$ = Yawing mom. coeff. per unit of δ_r measured in rad^{-1}
$C_{N_{\delta_{r,\,\text{tab}}}}$ = Yawing mom. coeff. per unit of $\delta_{r,\,\text{tab}}$ measured in rad^{-1}
$C_{N_{\text{r}}}$ = Yawing mom. coeff. per unit of yawing angular speed r measured in rad^{-1}
C_{N_β} = Yawing mom. coeff. per unit of sideslip angle β measured in rad^{-1}
S_{wing} = Main wing surface measured in m^2
b_{wing} = Main wing span measured in m
l_{vt} = Distance between aircraft CG and v.t. AC measured in m
I_{zz} = Yawing moment of inertia measured in kg m^2
δ_r = Rudder deflection angle measured in rad
$\delta_{r,\,\text{tab}}$ = Rudder tab deflection angle measured in rad
β = Sideslip angle measured in rad
V = Airspeed measured in m s^{-1}
GR_{tab} = Tab gear ratio

Standard MATLAB facilities can handle this problem easily and integrate the equation. The vertical tailplane loads are usually treated in a similar manner of the horizontal tailplane loads; for unpowered control surfaces, if a maneuver analysis is used to predict the maneuvering loads on the yaw control surfaces, the time for sudden deflection from neutral position to the stops or vice versa may be assumed as follows:

- For aerobatic category aircraft 0.2 s (pedal control surfaces).
- For aircraft non-approved for aerobatics 0.3 s (pedal control surfaces).

The following considerations are included in this appendix for the sake of completeness. For powered-controlled surfaces the deflection time should be measured. For aircraft

where the horizontal tailplane is supported by the vertical tailplane, the tail surfaces and their supporting structure including the rear portion of the fuselage should be designed to withstand the prescribed loadings on the vertical tailplane and the rolling moment induced by the horizontal tailplane acting in the same direction.

For T-tail designs, in the absence of a more rational analysis, the rolling moment induced by sideslip or deflection of the vertical rudder may be computed as follows:

$$M_r = 0.3 \cdot \frac{\rho_0}{2} \cdot \beta \cdot S_{\text{ht}} \cdot b_{\text{ht}} \cdot V^2 \tag{A.21}$$

where

M_r = Induced rolling moment at horizontal tailplane measured in N m
S_{ht} = Area of the horizontal tailplane measured in m^2
b_{ht} = Span of the horizontal tailplane measured in m
β = Effective sideslip angle of the vertical tailplane measured in rad
ρ_0 = Air density at sea level measured in kg m^{-3}

The ASTM International document F3116 [230, p. 7, sec. 4.20] describes the maneuvering loads on the vertical tailplane as well. At speeds up to the maneuvering design airspeed V_A, the vertical surfaces must be designed to withstand the following conditions [230, p. 7, sec. 4.20.1]:

- With the airplane in unaccelerated flight at zero yaw, it is assumed that the rudder control is suddenly displaced to the maximum deflection, as limited by the control stops or by limit pilot forces, whichever is critical [230, p. 7, sec. 4.20.1.1].
- With the rudder deflected as specified above, it is assumed that the airplane yaws to the overswing sideslip angle. In lieu of a rational analysis, an overswing angle may be assumed equal to $1.5 \cdot \beta_{\text{static}}$ [230, p. 7, sec. 4.20.1.2].
- A yaw angle of 15 deg with the rudder control maintained in the neutral position, except as limited by pilot strength [230, p. 7, sec. 4.20.1.3].

For level 4 aircraft, the loads imposed by the following additional maneuver must be substantiated at speeds from V_A to V_D or M_D [230, p. 7, sec. 4.20.3]. When computing the tail loads:

- The airplane must be yawed to the largest attainable steady state sideslip angle, with the rudder at maximum deflection caused by any one of the following [230, p. 7, sec. 4.20.3.1]:
    1. Control surface stops.
    2. Maximum available booster effort.
    3. Maximum pilot rudder force.

- The rudder must be suddenly displaced from the maximum deflection to the neutral position [230, p. 7, sec. 4.20.3.2].
- The yaw angles specified above may be reduced if the yaw angle chosen for a particular speed cannot be exceeded in [230, p. 7, sec. 4.20.4]:
    1. Steady slip conditions.
    2. Uncoordinated rolls from steep banks.
    3. For multi-engines aircraft, the sudden failure of the critical engine with delayed corrective action.

A.9.2 Gust loads for the vertical tailplane

The ASTM International F3116 document [230, p. 7, sec. 4.21] provides guidelines for evaluating the loads due to gusts. Vertical surfaces must be designed to withstand, in unaccelerated flight at speed V_C, lateral gusts or the values prescribed for V_C evaluated according to the methods explained in Section A.3. In addition, for level 4 aircraft, the aircraft is assumed to encounter derived gusts normal to the plane of symmetry while in unaccelerated flight at V_B, V_C, V_D, and V_F. The derived gusts and airplane speeds corresponding to these conditions must be investigated. The shape of the gust must be specified in Section A.3.

In the absence of a more rational analysis, the gust load must be computed as follows:

$$L_{\mathrm{vt}} = \frac{1}{2} \cdot \rho_0 \cdot k_{\mathrm{gt}} \cdot U_{\mathrm{de}} \cdot V \cdot C_{L_{\alpha,\,\mathrm{vt}}} \cdot S_{\mathrm{vt}} \tag{A.22}$$

where

$$k_{\mathrm{gt}} = \frac{0.88 \cdot \mu_{\mathrm{gt}}}{5.3 + \mu_{\mathrm{gt}}} = \text{Gust alleviation factor}$$

$$\mu_{\mathrm{gt}} = \frac{2 \cdot W}{\rho \cdot \bar{c}_{\mathrm{vt}} \cdot C_{L_{\alpha,\,\mathrm{vt}}} \cdot S_{\mathrm{vt}} \cdot g} \cdot \left(\frac{K}{l_{\mathrm{vt}}}\right)^2 = \text{Lateral mass ratio}$$

ρ_0 = Density of air at sea level measured in $\mathrm{kg\,m^{-3}}$

ρ = Density of air at the reference altitude measured in $\mathrm{kg\,m^{-3}}$

U_{de} = Derived gust velocity measured in $\mathrm{m\,s^{-1}}$

W = Applicable weight of the aircraft in the particular load case measured in N

S_{vt} = Area of the vertical tailplane measured in $\mathrm{m^2}$

l_{vt} = Distance from aircraft CG to lift centre of vertical surface measured in m

$\bar{c}_{\mathrm{vt}}$ = Mean aerodynamic or geometric chord of vertical tailplane measured in m

V = Aircraft equivalent airspeed measured in $\mathrm{m\,s^{-1}}$

g = Acceleration due to gravity measured in $\mathrm{m\,s^{-2}}$

K = Radius of gyration in yaw measured in m

For aircraft where the vertical tailplane supports the horizontal tailplane, the tailplane surfaces and their supporting structure, including the rear portion of the fuselage, should be designed to withstand the prescribed loading on the vertical tailplane and the rolling moment induced by the horizontal tailplane acting in the same direction.

For T-tail designs, in the absence of a more rational analysis, the rolling moment induced by the gust loads may be computed as follows:

$$M_r = 0.3 \cdot \frac{\rho_0}{2} \cdot S_{\text{ht}} \cdot b_{\text{ht}} \cdot V \cdot U_{\text{de}} \cdot k_{\text{gt}} \tag{A.23}$$

where

M_r = Induced rolling moment at vertical tailplane measured in N m
S_{ht} = Area of the horizontal tailplane measured in m^2
b_{ht} = Span of the horizontal tailplane measured in m
U_{de} = Gust intensity measured in m s^{-1}
ρ_0 = Air density at sea level measured in kg m^{-3}
k_{gt} = Gust alleviation factor

In this specific certification task, in computing the S_{ht} and b_{ht}, the horizontal tailplane root has to be assumed on a vertical plane through the centreline of the aircraft fuselage.

A.9.3 Other loads for the vertical tailplane

The scope of this appendix does not allow for including a thorough discussion of the other applicable paragraphs and subparagraphs for other vertical surfaces. The interested reader should refer to the ASTM International document F3116, paragraphs 4.22 *Outboard Fins or Winglets* [230, p. 8, sec. 4.22], 4.23 *Combined Loads on Tail Surfaces for airplanes meeting the limitations of Practice F3396/F3396M, Control Surface Loading (Level 1 Aeroplanes)* [230, p. 8, sec. 4.23], and 4.24 *Additional Loads Applicable to V-tails—for airplanes meeting the limitations of Practice F3396/F3396M, Control Surface Loading (Level 1 Aeroplanes)* [230, p. 8-9, sec. 4.23].

A.10 Ailerons flight loads

The ASTM International F3116 document provides guidelines and certification information for the structural design of the ailerons and their supporting structure [230, p. 9, sec. 4.25]. The ailerons must be designed for the loads to which they are subjected in the following conditions:

- In the neutral position during symmetrical flight conditions.

- By the following deflections, except as limited by pilot effort, during unsymmetrical flight conditions:

    1. Sudden maximum displacement of the aileron control at V_A. Suitable allowance may be made for control system deflections [230, p. 9, sec. 4.25.1.2(1)].
    2. Sufficient deflection at V_C, where V_C is more than V_A, to produce a rate of roll not less than the one obtained at the previous point [230, p. 9, sec. 4.25.1.2(2)].
    3. Sufficient deflection at V_D to produce a rate of roll not less than one-third of that obtained at point 1 [230, p. 9, sec. 4.25.1.2(3)].

The reference rolling conditions are included [230, p. 4, sec. 4.10]; the wing and wing bracing must be designed for the following loading conditions:

- First, the unsymmetrical wing loading conditions must be addressed. Unless the following values result in unrealistic loads, the rolling accelerations may be obtained by modifying the symmetrical flight conditions prescribed by the regulations as follows:

    1. At Point A, assume that 100 % of the semispan wing airload acts on one side of the airplane and 70 % of this load acts on the other side. For airplanes of more than 454 kg (1000 lb) design weight, the latter percentage may be increased linearly with weight up to 75 % at 5670 kg (12 500 lb).
    2. For airplanes approved for aerobatics, in Points A and F, assume that 100 % of the semispan wing airload acts on one side of the plane of symmetry and 60 % of this load acts on the other side.

- The loads resulting from the aileron deflections and speeds given above, in combination with an airplane load factor of at least two-thirds of the positive maneuvering load factor used for design. Unless the following values result in unrealistic loads, the effect of aileron displacement on wing torsion may be accounted for by adding the following increment to the basic airfoil moment coefficient over the aileron portion of the span in the critical condition determined for the symmetrical flight condition:

$$\Delta C_m = -0.01 \cdot \delta_a \tag{A.24}$$

  where

ΔC_m = Moment coefficient increment
δ_a = Down aileron deflection at the critical condition

A.11 Powerplant loads

The evaluation of powerplant loads starts with the determination of the mean torque at maximum takeoff power and the corresponding limit torque. The applied relations are the following:

$$\text{Mean Torque} = \text{MT} = P \cdot \frac{1000 \cdot 60}{2 \cdot \pi \cdot \text{RPM}_{\text{propeller}}} \qquad \text{measured in N m} \tag{A.25}$$

$$\text{Limit Torque} = \text{LT} = \text{MF}_{\text{ASTM}} \cdot \text{MT} \qquad \text{measured in N m} \tag{A.26}$$

where MF_{ASTM} is the assigned multiplied factor, in accordance with [230, p. 10, sec. 6.1.3]. This multiplying factor varies with the characteristics of the installed engine. This digital certification framework determines the value of MF_{ASTM} automatically. Another important load is determined in accordance with [230, p. 9, sec. 6.1.1.1], by using the following equation:

$$\text{Engine Mount Load, Point A} = L_{\text{Eng}_\text{A}} = n_A \cdot \text{ASTM}_{\text{coeff}} \cdot W_{\text{TO}} \qquad \text{measured in N} \tag{A.27}$$

$$\text{Balancing Load, Point A} = \text{BL}_{\text{Eng}_\text{A}} = \text{ASTM}_{\text{coeff}} \cdot \text{BL}_A \qquad \text{measured in N} \tag{A.28}$$

Inertia load will be evaluated following a well-established, conservative approach:

$$\dot{\omega} = \frac{W \cdot (n_{\text{gust}} - 1) \cdot (x_{\text{ac}} - x_{\text{cg}}) \cdot g}{I_{\text{yy}}} \tag{A.29}$$

where

$\dot{\omega}$ = Pitching angular acceleration measured in rad s^{-1}
W = Aircraft weight in N
n_{gust} = Limit load factor due to gust. Non dimensional.
g = Acceleration due to gravity in m s^{-1}
x_{ac} = Position of the main wing aerodynamic center in m
x_{cg} = Position of the center of gravity in m
I_{yy} = Aircraft pitching moment of inertia in kg m^{-2}

The pitching angular acceleration can be used for the evaluation of the rotational limit load factor during a pitching maneuver with the aircraft in that specific configuration:

$$n_{\text{rot}} = \frac{\dot{\omega} \cdot d}{g} \tag{A.30}$$

The total inertial limit load factor is, therefore, the summation of the gust and rotational limit load factors, $n_{\text{in}} = n_{\text{gust}} + n_{\text{rot}}$, so that the total inertial load can be evaluated with the following equation:

$$\text{Inertial load} = n_{\text{in}} \cdot W_{\text{powerplant}} \qquad \text{measured in N} \tag{A.31}$$

where $W_{\text{powerplant}}$ is the total powerplant weight measured in N. Another important powerplant load, the engine sideload, is evaluated in accordance with [230, p. 10, sec. 6.2.1]; first, the code evaluates the limit sideload factor:

$$n_{\text{sideload}} = \max\left(\frac{n_A}{3}, \text{MF}_{\text{ASTM}}\right) \tag{A.32}$$

where MF_{ASTM} is another multiplying factor extracted from the ASTM document. Then, it evaluates the limit sideload:

$$\text{Side load} = n_{\text{sideload}} \cdot W_{\text{powerplant}} \qquad \text{measured in N} \tag{A.33}$$

Lastly, the pitch and yaw gyroscopic couples have been evaluated using this simple relationship:

$$\text{Pitch and Yaw gyroscopic couple} = M_{\text{gyro}} = I_p \cdot \omega_1 \cdot \omega_2 \text{ measured in N m} \tag{A.34}$$

where I_p is the propeller polar moment of inertia measured in kg m^2, ω_1 is the propeller rotation speed, and ω_2 is the rate of pitch or yaw, both measured in rad s^{-1}. For completeness, the code includes the evaluation of the critical normal load caused by the presence of the normal force acting on the propeller:

$$N_{\text{critical}} = \text{MF}_{\text{ASTM}} \cdot F_{\text{normal}} \qquad \text{measured in N} \tag{A.35}$$

The code also includes the calculation of the maximum continuous static thrust acting on the powerplant structures; the following semi-empirical formula has been used:

$$T_{\text{critical static thrust}} = 1.35 \cdot (\text{MCP}_{\text{lb}})^{\frac{2}{3}} \cdot (2 \cdot \rho \cdot A_{\text{propeller}}) \cdot \left(1 - \frac{A_{\text{spinner}}}{A_{\text{propeller}}}\right) \tag{A.36}$$

where

$$\begin{aligned} T_{\text{critical static thrust}} &= \text{Critical static thrust in lb} \\ \text{MCP}_{\text{lb}} &= \text{Maximum continuous thrust in ft lb s}^{-1} \\ \rho &= \text{Air density in slug ft}^{-3} \\ A_{\text{propeller}} &= \text{Propeller disk area in ft}^2 \\ A_{\text{spinner}} &= \text{Propeller spinner area in ft}^2 \end{aligned}$$

Appendix B
Aircraft input data and other data

This Section presents a selected subset of the aircraft input data. The reference configurations are loosely based on conventional aircraft currently in service, including the Tecnam P92, P2010, P2006, and P2012. These aircraft were selected based on data availability, familiarity with their performance and operating characteristics, and direct flight experience; these design architectures represent almost the full spectrum of design architectures certifiable under the EASA CS-23 requirements. The unconventional configuration is loosely based on conceptual aircraft designs that incorporate technologies such as distributed electric propulsion (DEP) and were developed primarily for research purposes. Tables B.1 and B.2 present selected data describing the aircraft geometry, aerodynamic characteristics, powerplants, flight controls, mass and inertia properties, allowable center-of-gravity (CG) positions at different weights, and empennage configurations. Refer to the nomenclature included at the beginning of this thesis for the symbols used in the tables.

Table B.1: Aircraft input data. Mass, CG, and main wing.

	Data	P92	P2010	P2006	P2012	Non-conventional
Aircraft mass	AM	500 kg	1200 kg	1400 kg	3480 kg	8168.97 kg
	BEM	386.81 kg	720 kg	847.50 kg	2250 kg	5782.50 kg
	OEM	455.76 kg	893 kg	847.50 kg	2387 kg	6132.23 kg
	MZFM	455.76 kg	1160 kg	1180 kg	3480 kg	7898.97 kg
	MTOM	600 kg	1200 kg	1400 kg	3680 kg	8168.97 kg
	EM	386.81 kg	720 kg	847.50 kg	2286 kg	5782.50 kg
	UM	213.19 kg	445 kg	552.50 kg	1414 kg	1766.74 kg
	MidM	500 kg	950 kg	1230 kg	3000 kg	7898.97 kg
CG envelope	$CG_{\text{fwd, OEM}}$	18 % $\overline{c}$	19 % $\overline{c}$	16.5 % $\overline{c}$	18 % $\overline{c}$	30.39 % $\overline{c}$
	$CG_{\text{fwd, MidM}}$	18 % $\overline{c}$	19 % $\overline{c}$	16.5 % $\overline{c}$	18 % $\overline{c}$	30.39 % $\overline{c}$
	$CG_{\text{fwd, MTOM}}$	18 % $\overline{c}$	19 % $\overline{c}$	19.5 % $\overline{c}$	22 % $\overline{c}$	35 % $\overline{c}$
	$CG_{\text{fwd, MZFM}}$	18 % $\overline{c}$	19 % $\overline{c}$	19.5 % $\overline{c}$	21.20 % $\overline{c}$	30.39 % $\overline{c}$
	$CG_{\text{aft, MTOM}}$	33 % $\overline{c}$	32 % $\overline{c}$	31 % $\overline{c}$	31 % $\overline{c}$	40.10 % $\overline{c}$
	$CG_{\text{aft, OEM}}$	33 % $\overline{c}$	32 % $\overline{c}$	31 % $\overline{c}$	31 % $\overline{c}$	40.10 % $\overline{c}$
	$l_{\text{ht, fwd}}$	4.253 m	4.40 m	4.710 m	6.204 m	8.147 m
	$l_{\text{ht, aft}}$	4.041 m	4.39 m	4.520 m	5.965 m	8.148 m
	$l_{\text{vt, fwd}}$	3.390 m	4.10 m	4.227 m	5.431 m	6.805 m
	$l_{\text{vt, aft}}$	3.178 m	4.06 m	3.997 m	5.191 m	6.806 m
	$I_{xx, \text{fwd}}$	742.70 kg m^2	1862 kg m^2	2961.451 kg m^2	12 905 kg m^2	91 887.70 kg m^2
	$I_{xx, \text{aft}}$	735.27 kg m^2	1860 kg m^2	2879.883 kg m^2	12 894 kg m^2	91 887.71 kg m^2
	$I_{yy, \text{fwd}}$	915.13 kg m^2	2063 kg m^2	2884.924 kg m^2	17 598 kg m^2	28 111.00 kg m^2
	$I_{yy, \text{aft}}$	890.38 kg m^2	2060 kg m^2	2484.119 kg m^2	17 122 kg m^2	28 111.01 kg m^2
	$I_{zz, \text{fwd}}$	1423.08 kg m^2	3703 kg m^2	5537.347 kg m^2	28 262 kg m^2	115 024.87 kg m^2
	$I_{zz, \text{aft}}$	1399.61 kg m^2	3700 kg m^2	5037.156 kg m^2	27 796 kg m^2	115 024.96 kg m^2
Wing	b	8.9 m	10.3 m	11.19 m	14 m	21.006 m
	S	12.58 m^2	14.41 m^2	15.10 m^2	25.24 m^2	34.10 m^2
	c_{root}	1.413 m	1.457 m	1.443 m	2.004 m	1.851 m
	c_{tip}	1.413 m	1.133 m	1.020 m	1.444 m	1.203 m
	$\overline{c}$	1.413 m	1.406 m	1.362 m	1.824 m	1.650 m

Table B.2: Aircraft input data. Horizontal and vertical tailplanes.

	Data	P92	P2010	P2006	P2012	Non-conventional
Horiz. tailplane	b_{ht}	8.9 m	10.3 m	11.19 m	14 m	7 m
	S_{ht}	12.58 m^2	14.41 m^2	15.10 m^2	25.24 m^2	21.19 m^2
	$c_{\text{root, ht}}$	1.413 m	1.457 m	1.443 m	2.004 m	1.760 m
	$c_{\text{tip, ht}}$	1.413 m	1.133 m	1.020 m	1.444 m	1.268 m
	$\overline{c}_{\text{ht}}$	1.413 m	1.406 m	1.362 m	1.824 m	0.764 m
Vert. tailplane	b_{vt}	8.9 m	10.3 m	11.19 m	14 m	3.775 m
	S_{vt}	12.58 m^2	14.41 m^2	15.10 m^2	25.24 m^2	8.64 m^2
	$c_{\text{root, vt}}$	1.413 m	1.457 m	1.443 m	2.004 m	3.345 m
	$c_{\text{tip, vt}}$	1.413 m	1.133 m	1.020 m	1.444 m	1.254 m
	$\overline{c}_{\text{vt}}$	1.413 m	1.406 m	1.362 m	1.824 m	2.457 m
Aerodynamics	$C_{L_{\alpha,\text{fv}}}$	5.615 rad^{-1}	4.587 rad^{-1}	4.641 rad^{-1}	4.71 rad^{-1}	6.343 rad^{-1}
	$C_{L_{\alpha,\text{ht}}}$	4.1 rad^{-1}	4.1 rad^{-1}	3.266 rad^{-1}	3.873 rad^{-1}	3.873 rad^{-1}
	$C_{L_{\text{max}}}$	1.4	1.4	1.36	1.46	2.1
	$C_{L_{\text{min}}}$	−1	−1	−1	−0.8	−0.6
	C_{D_0}	0.0418	0.0411	0.0265	0.0410	0.0265
	e_{fv}	0.85	0.87	0.86	0.84	0.85
	$\alpha_{0_{\text{L}}}$	−2.2 deg	−2.2 deg	−2.2 deg	−2.2 deg	−7.47 deg
Powerplant	N_{Eng}	1	1	2	2	2 + 8
	P_{MC}	92.5 hp	180 hp	92.5 hp	375 hp	1206.92 hp
	N_{Cylinder}	4	4	4	6	0
	RPM_{MC}	2388 rpm	2700 rpm	2388 rpm	2575 rpm	5500 rpm
	M_{Engine}	93 kg	184 kg	93 kg	358.2 kg	213.94 kg
Controls	$y_{\text{a, in}}$	2.362 m	3.323 m	3.139 m	4.9 m	8.612 m
	$y_{\text{a, out}}$	4.0764 m	4.757 m	5.0285 m	6.7 m	10.188 m
	$\delta_{\text{a, max}}$	22 deg	25 deg	15 deg	22 deg	20 deg
	$\delta_{\text{e, max}}$	30 deg	19 deg	4 deg	30 deg	30 deg
	$\delta_{\text{r, max}}$	25 deg	25 deg	25 deg	25 deg	30 deg

Table B.3: Comparison between numerical results; only the Level 4, conventional twin-engine data are reported, since this design architecture was investigated extensively. The data concern the maneuvering and gust flight envelope evaluated at sea level.

Point	Data	MATLAB	Python	C++
S	V_S	$38.89\,\mathrm{m\,s^{-1}}$	$39.97\,\mathrm{m\,s^{-1}}$	$39.99\,\mathrm{m\,s^{-1}}$
	n_S	1	1	1
A	V_A	$71.97\,\mathrm{m\,s^{-1}}$	$73.97\,\mathrm{m\,s^{-1}}$	$74.01\,\mathrm{m\,s^{-1}}$
	n_A	3.42	3.42	3.42
C	V_C	$84.03\,\mathrm{m\,s^{-1}}$	$83.99\,\mathrm{m\,s^{-1}}$	$84.03\,\mathrm{m\,s^{-1}}$
	n_C	3.42	3.42	3.42
D	V_D	$117.21\,\mathrm{m\,s^{-1}}$	$117.10\,\mathrm{m\,s^{-1}}$	$117.12\,\mathrm{m\,s^{-1}}$
	n_D	3.42	3.42	3.42
S inv.	$V_{S_{inv}}$	$52.53\,\mathrm{m\,s^{-1}}$	$53.99\,\mathrm{m\,s^{-1}}$	$54.02\,\mathrm{m\,s^{-1}}$
	$n_{S_{inv}}$	-1	-1	-1
G	V_G	$61.49\,\mathrm{m\,s^{-1}}$	$63.20\,\mathrm{m\,s^{-1}}$	$63.23\,\mathrm{m\,s^{-1}}$
	n_G	-1.37	-1.37	-1.37
F	V_F	$84.03\,\mathrm{m\,s^{-1}}$	$83.99\,\mathrm{m\,s^{-1}}$	$84.03\,\mathrm{m\,s^{-1}}$
	n_F	-1.37	-1.37	-1.37
E	V_E	$117.21\,\mathrm{m\,s^{-1}}$	$117.10\,\mathrm{m\,s^{-1}}$	$117.56\,\mathrm{m\,s^{-1}}$
	n_E	-0.40	-0.33	-0.33
H	V_H	$93.37\,\mathrm{m\,s^{-1}}$	$93.33\,\mathrm{m\,s^{-1}}$	$93.37\,\mathrm{m\,s^{-1}}$

Table B.4: Comparison between the requirement diagrams drafted in the Eclipse Papyrus modeling environment and the EASA CS-23, Amendment 6. Standards referenced by AMC1 and additional standards identified through requirement diagrams.

CS paragraph	AMC1	Requirement Diagram
CS 23.2225	F3061/F3061M-20 F3232/F3232M-20 F3116/F3116M-18e2 F3396/F3396M-20	F3114-21
CS 23.2230	F3114-21 F3408/F3408M-20	F3116/F3116M-18e2
CS 23.2235	F3114/F114M-21 F3408/F3408M-20	F061/F3061M-20 F3232/F3232M-20
CS 23.2245	F061/F061M-20 F3093/F3093M-21	F3232/F3232M-20
CS 23.2250	F3061/F3061M-20 F3232/F3232M-20 F3083/F3083M-20a F3408/F3408M-21	F3227/F3227M-21 F3233/F3233M-21 F3066/F3066M-18 F3083/F3083M-20a
CS 23.2265	F3061/F3061M-20 F3114-21 F3380-19	F3232/F3232M-20 F3083/F3083M-20a

Appendix C
Excerpts from the Flight Loads Report

This appendix is included in this thesis to show excerpts from the most important certification artifacts—in this case, a document including other certification artifacts, such as figures and tables—that is produced by the MATLAB/Simulink digital certification framework, which is the flight loads certification report.

Figures C.1, C.2, and C.3 are the table of contents, the list of figures, and the list of tables, respectively, of the report; as the remaining material of the report, they are generated by the code through the API and the final user can edit them, using the Microsoft Word document editing environment. Figure C.4 presents the first actual page of the material, the Introduction, with text programmed into the code. Most of the textual content of the report has been taken from the ASTM documents, providing references to the paragraphs and subparagraphs when needed.

Figures C.5 and C.6 show how the design airspeeds are calculated and included directly in the report, with formulas and calculations reported for ease of reading the document and the results. To be more specific, the numbers included in the calculation formulas are not hard-written by hand, but instead come from the structured data variables created at the end of the execution phase. The calculations are shown for completeness.

Figure C.7 shows a page from Chapter 9 of the report, where the balancing loads are first described and then calculated, and the results presented both in graphical and tabular forms. This page has been included in this thesis because it gives a quick glimpse of the clarity and thoroughness with which the report is generated by the digital certification framework.

Figure C.8 is from Chapter 10, describing the loads acting on the wing with a figure that presents the shear force spanwise distributions at various maneuvering and gust flight envelope points. Figure C.9 reports a page from Chapter 11, with tables and calculations addressing the loads acting on the horizontal tailplane of the aircraft.

Figures C.10 and C.11 conclude this Section, providing two pages of the report from the ending Chapter addressing loads on the powerplant supporting structures. Tables, equations, and textual references to the paragraphs of the ASTM documents are included in the contents of the report.

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Chapter 1. Introduction

This document defines the SUBPART C - Structure - Flight Loads of the aircraft P2012. The boundaries of the flight envelope will be defined within this document. All speeds are calibrated airspeeds (CAS) (requirement 49 [11, p. 22]) and given in knots if not stated otherwise. All other units used are metric (SI units). The Masses are given in mass units (kg), but the formulas require force units as input. These are calculated in place wherever they are used. Note: The speeds defined within this document should be used for the placards, speed markings, aeroplane flight manual (limitations), load calculations, and need to be verified by flight test.

This document is mainly focused on flight loads determination and structural sizing loads acting on the main wing and horizontal empennage, according to CS-23, AMDT. 6 airworthiness regulations. The structure of this document is derived from the actual Subpart C subdivision of the regulations. These regulations are applicable to aeroplanes in the normal category, with a passenger seating configuration of 19 or less and a Maximum Certified Takeoff Mass of 8618 kg (19000 pounds) or less. Aeroplane certification levels are:

1. Level 1 - For aeroplanes with a maximum seating config. of 0 to 1 passengers.
2. Level 2 - For aeroplanes with a maximum seating config. of 2 to 6 passengers.
3. Level 3 - For aeroplanes with a maximum seating config. of 7 to 9 passengers.
4. Level 4 - For aeroplanes with a maximum seating config. of 10 to 19 passengers.

Aeroplane performance levels are:

1. Low speed - For aeroplanes with a VNO or VMO less than 250 knots calibrated airspeeds (KCAS) or MMO less than 0.6.
2. High speed - For aeroplanes with a VNO or VMO greater than 250 KCAS or an MMO greater than 0.6.

1.1. Structure of this document

The structure of this document is the following:

1. Chapter 1 introduces the scope and applicability of this report.
2. Chapter 2 provides a list of references used throughout this report.
3. Chapter 3 lists all the abbreviations used throughout this report.
4. Chapter 4 lists all the aircraft data applicable to the scope of this report.
5. Chapter 5 addresses the calculation of the Design Airspeeds. It illustrates the methodology applied and the results.
6. Chapter 6 concerns the standard atmosphere data used for all the calculations.
7. Chapter 7 describes how the maneuvering and gust limit load factors have been calculated. Results are presented accordingly.
8. Chapter 8 illustrates and describes the resulting maneuvering and gust flight envelope. The resulting flight envelope is presented graphically. Two different altitudes have been considered.
9. Chapter 9 reviews the loads applied to the aircraft. The lift acting on the main aerodynamic surfaces of the aircraft is presented.

Figure C.4: The Flight Loads report Introduction Chapter first page.

Chapter 5. Design Airspeeds

This chapter defines the operating and design airspeed as required for certification CS -23, AMDT. 6.

Section 5.1 addresses the evaluation of the maximum airspeed in level flight VH, presenting the reference equation and illustrating the solution methodology. Section 5.2 presents the Design Stall Airspeeds, calculated in the clean, takeoff, and landing configurations. Section 5.3 presents the calculation of the Design Maneuvering Airspeeds, in accordance with the applicable regulations. Section 5.4 summarizes the calculation of the maximum operating airspeed with flaps deployed. Section 5.5 summarizes the calculation of the flaps maximum extension airspeed VFE. Section 5.6 illustrates the calculation of the Design Cruising Speed VC, in accordance with the applicable rule. Section 5.7 illustrates the calculation of the Design Dive Speed VD, in accordance with the applicable rule. Sections 5.8 and 5.9 illustrate the Demonstrated Dive Speed VDF and the Never Exceed Speed VNE, respectively. Finally, Section 5.10 summarizes all the airspeeds calculated in this chapter.

5.1. Maximum speed in level flight VH

According to flight performance analysis, aerodynamic drag, installed power at sea level conditions, the maximum speed in level flight has been determined: VH = 93.3659 m/s

The reference equation used to find the VH is the following:

$$P_{\text{req}} = N_{eng} \cdot P_{\text{max cont.}} = \frac{1}{2} \cdot f \cdot \rho \cdot V^3 + \frac{2\left(\frac{\text{MTOW}}{b}\right)^2}{\pi \cdot \rho \cdot e \cdot V}$$

where:

- P_{req} = Power required (W)
- f = friction area equal to $C_{D_0} \cdot S_{\text{wing}}$ (m^2)
- ρ = air density (kg/m^3)
- V = aircraft airspeed (m/s)
- MTOW = maximum takeoff weight (N);
- b = aircraft wingspan (m)
- e = Oswald efficiency factor
- $P_{\text{max cont}}$ = max continuous power (W)
- N_{eng} = number of engine

The equation has been solved assuming V as the unknown, and using a MATLAB solver capable of finding roots of a nonlinear function. The power input was equal to the maximum continuous power available for the examined aircraft.

5.2. Stall speeds VS, VS0, VS1

This Section addresses the calculations performed for evaluating the Design Stall Airspeeds of the aircraft in different configurations. These speeds will be verified by flight test according to certification requirements. In order to calculate the stall speed, the maximum lift coefficient of the aeroplane as a whole is determined first. The maximum lift coefficient of the aeroplane has been calculated from high -

Figure C.5: The Flight Loads report; example page from Chapter 5.

Chapter 5. Design Airspeeds

fidelity CFD. In landing configuration computed with full flap, CLMAX landing = 1.752 in take-off configuration, leading to CLMAX takeoff = 1.679, and in clean configuration, leading to CLMAX clean = 1.46, also considering the horizontal tail balancing force.

Flaps retracted (clean configuration):

$$V_S = \sqrt{\frac{2\,W_{\mathrm{MTOM}}}{\rho_0 C_{L_{\mathrm{MAX}_{\mathrm{clean}}}} S}} = \sqrt{\frac{2 \cdot 36088.1401}{1.225 \cdot 1.46 \cdot 25.2346}} = 38.8885 \text{ m/s}$$

Flaps extended (Landing configuration):

$$V_{S_0} = \sqrt{\frac{2\,W_{\mathrm{MTOM}}}{\rho_0 C_{L_{\mathrm{MAX}_{\mathrm{landing}}}} S}} = \sqrt{\frac{2 \cdot 36088.1401}{1.225 \cdot 1.752 \cdot 25.2346}} = 35.5817 \text{ m/s}$$

Flaps extended (Take-off configuration):

$$V_{S_1} = \sqrt{\frac{2\,W_{\mathrm{MTOM}}}{\rho_0 C_{L_{\mathrm{MAX}_{\mathrm{takeoff}}}} S}} = \sqrt{\frac{2 \cdot 36088.1401}{1.225 \cdot 1.679 \cdot 25.2346}} = 37.2912 \text{ m/s}$$

Add here comments if necessary

Note: These speeds are estimates. The methods for the estimation can be various. It is important that these estimations are as precise as possible. Flight tests will be used to validate the stall speeds. In case the flight tests show different values, this might have an impact on the speeds used for design and ultimately might impair the compliance to the CS-23, AMDT. 6

5.3. Design manoeuvring speed VA

According to CS-23, AMDT. 6 Paragraph 5.1.3.1 Design Airspeeds of ASTM F3116/F3116M, the maneuvering speed VA cannot be less than:

$$V_A \geq V_S \sqrt{n_{max}} = 38.8885 \cdot \sqrt{3.425} = 71.9701 \text{ m/s}$$

VS is a 1g computed stalling speed with flaps retracted (normally based on the maximum airplane normal force coefficients, CNA) at either (1) the particular weight under consideration or (2) the design maximum takeoff weight. n is the limit maneuvering load factor used in design. The value of VA need not exceed the value of VC used in design.

The calculations showed that VA is greater than VC, which is equal to 84.029 m/s.

5.4. Flaps maximum operating speed VF

According to CS-23, AMDT. 6 Paragraph 4.8.2 Loads of ASTM F3116/F3116M, VF must be assumed to be not less than 1.4 * VS or 1.8 * VSF, whichever is greater, where:

1. VS is the 1g computed stalling speed with flaps retracted at the design weight.
2. VSF is the 1g computed stalling speed with flaps fully extended at the design weight.

The flaps operating speeds is:

Figure C.6: The Flight Loads report; example page from Chapter 5.

The external forces and moments acting on the aeroplane in a balanced flight condition have been determined. The aeroplane is reduced to the wing and the horizontal tail. The symbols used are:

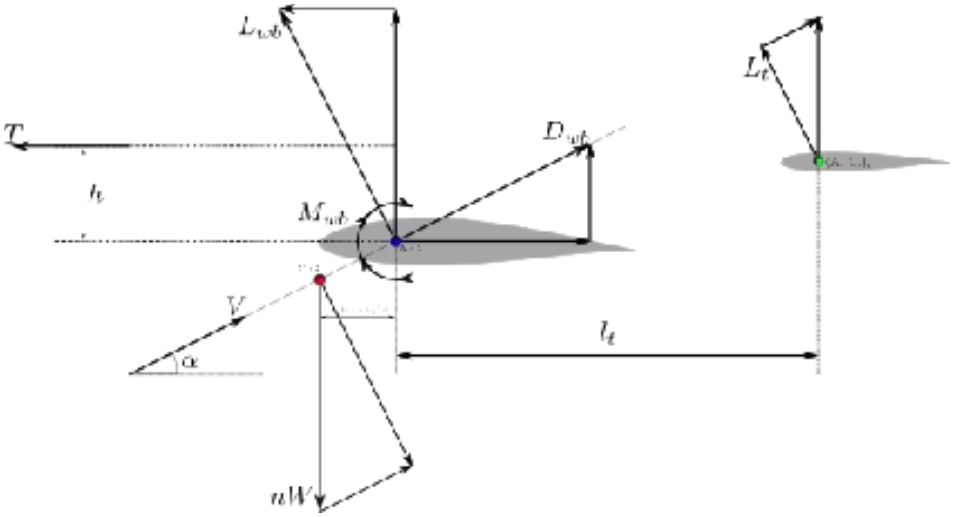

Figure 9.2. Simplified equilibrium of the aircraft

- x_{CG} = distance to aircraft centre of gravity;
- $x_{\mathrm{AC}f+w}$ = distance to wing fuselage combination aerodynamic centre;
- $x_{P_{f+w}}$ = distance to wing fuselage combination centre of pressure;
- nW = aircraft total weight force;
- x_{HT} = distance to HT quarter chord line;

Assuming positive forces and moment as depicted in the figures and remembering that equilibrium along the x and y axes is automatically fulfilled, it is possible to write the following equilibrium equations:

Table 9.1 summarizes the balancing loads parameters.

$$\text{X equilibrium}: \quad T = L_{\mathrm{wb}} \cdot \sin\alpha - D_{\mathrm{wb}} \cdot \cos\alpha$$

$$\text{Z equilibrium}: \quad L_{\mathrm{wb}} \cdot \cos\alpha + D_{\mathrm{wb}} \cdot \sin\alpha + \frac{L_t}{\cos\alpha} - nW \cdot \cos\alpha = 0$$

$$\text{Rotation about } Y:$$

$$[-L_{\mathrm{wb}} \cdot \cos\alpha \cdot (x_{\mathrm{AC}} - x_{\mathrm{CG}}) - L_{\mathrm{wb}} \cdot \sin\alpha \cdot z_{\mathrm{CG}}] + [D_{\mathrm{wb}} \cdot \cos\alpha \cdot z_{\mathrm{CG}} +$$

$$- D_{\mathrm{wb}} \cdot \sin\alpha \cdot \mathrm{MAC} \cdot (x_{\mathrm{AC}} - x_{\mathrm{CG}})] + \left[-\frac{L_t}{\cos\alpha} \cdot [l_t + \mathrm{MAC} \cdot (x_{\mathrm{AC}} - x_{\mathrm{CG}})]\right] +$$

$$- T \cdot h + M_{\mathrm{wb}} = 0$$

Figure C.7: The Flight Loads report; example page from Chapter 9.

Chapter 10. Loads on the wing

Table 10.1. Shear, bending, and torsion distribution along the semi-span.

Point	V (m/s)	n (g)	S (N/m)	M (N*m)	T (N*m)
Point S	38.89	1	16803	67751.7	75.233
Point A	71.97	3.425	57550.4	232051	257.674
Point C	84.03	3.425	58048.3	236772	2364.65
Point D	117.2	3.425	58356.5	245300	2892.28
Point Sinv	52.54	-1	-16929.6	-65356	-5119.47
Point G	61.49	-1.37	-23193.7	-89538.1	-7013.7
Point F	84.03	-1.37	-23299.4	-86017.3	-5379.12
Point E	117.2	-0.39449	-6727.61	-18677.8	-3924.79

Table 10.1 summarizes the shear, bending, and torsion distribution along the semi-span.

10.2.4. Critical load condition

- The critical shear is at point D and is equal to 58048.3 N/m.
- The critical bending is at point D and is equal to 245300 N*m.
- The critical torsion is at point G and is equal to -7013.7 N*m.

Table 10.2. Critical shear, bending, and torsion.

Point	Value	Units	Load
Point D	58356.5	N/m	Shear
Point D	245300	N*m	Bending
Point G	-7013.7	N*m	Torsion

Table 10.2 summarizes the critical shear, bending, and torsion values for convenience.

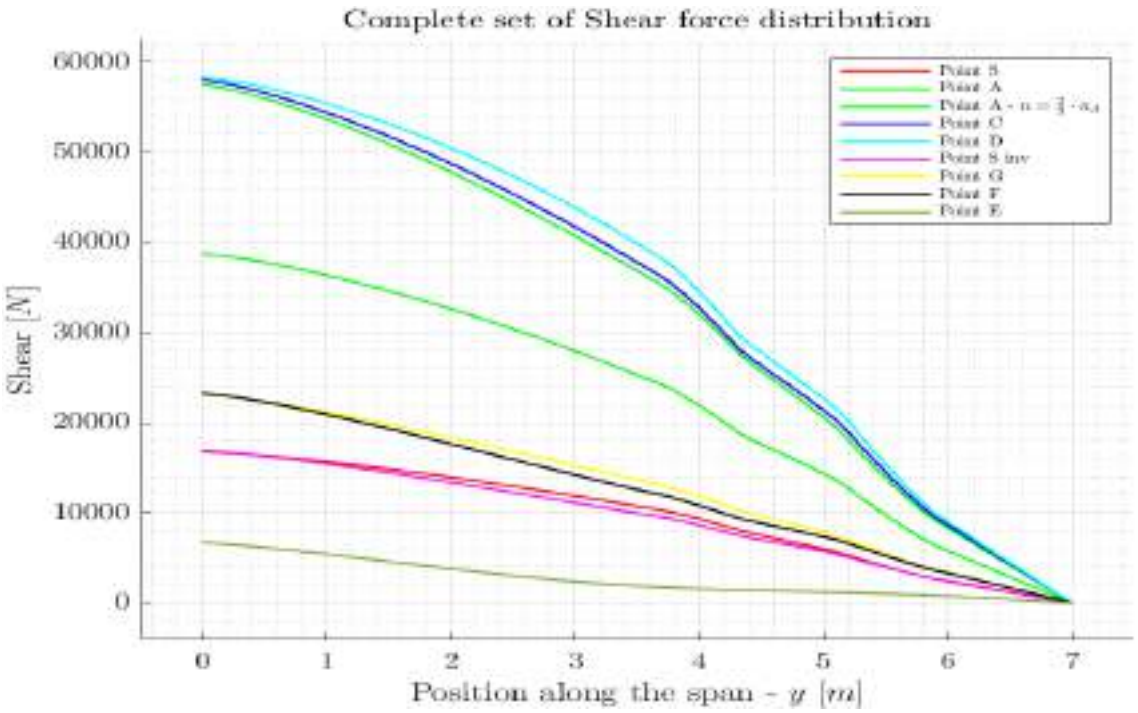

Figure C.8: The Flight Loads report; example page from Chapter 10.

Results at sea level altitude can be summarized as follows:

1. At VA = 71.97 m/s, the positive gust horizontal tailplane load is 8820.76 N when the incremental tail load is summed and -5283.1 N when the incremental tail load is subtracted.
2. At VC = 84.03 m/s, the positive gust horizontal tailplane load is 10199 N when the incremental tail load is summed and -6268.09 N when the incremental tail load is subtracted.
3. At VD = 117.2 m/s, the positive gust horizontal tailplane load is 6595.83 N when the incremental tail load is summed and -4888.7 N when the incremental tail load is subtracted.

11.5. Unsymmetrical loads

According to CS-23, AMDT. 6, in the absence of more rational data, for conventional aircraft must be applied: (1) 100% of the maximum loading from the symmetrical flight conditions on the surface on one side of the plane of symmetry, and (2) the following percentage of that loading to the opposite side:

$$\% \ = \ 100 - 10 * (n - 1) \ = \ 100 \ - \ 10 \cdot (3.425 \ - \ 1) \ = 75.75$$

Table 11.14 presents the resulting unsymmetrical loads compared with the critical loads. Refer to Section 11.6 for a discussion of the critical loads.

Table 11.14. Critical unsymmetrical loads applied on the horizontal tailplane.

Critical load (N)	Critical Fully-loaded side (N)	Critical Partially-loaded side (N)
-1681.27	-840.636	-636.781
5396.7	2698.35	2044
-3600.87	-1800.44	-1363.83
10625.7	5312.83	4024.46

The percentage evaluated is equal to 75.75, leading to the unsymmetrical application of the critical loads.

11.6. Horizontal tail loads summary

Table 11.15. Summary of critical loads acting on the horizontal tailplane.

Airspeed (m/s)	Critical Load (N)
38.8885	-1681.27
71.9701	5396.7
84.0293	-3600.87
117.208	10625.7

Table 11.15 presents the critical loads acting on the horizontal tailplane. This section provides a summary of the critical horizontal tailplane loads. The maximum values of the horizontal tailplane critical loads are listed below.

1. At VS = 38.89 m/s, the critical horizontal tailplane is -1681.27 N.
2. At VA = 71.97 m/s, the critical horizontal tailplane is 5396.7 N.
3. At VC = 84.03 m/s, the critical horizontal tailplane is -3600.87 N.
4. At VD = 117.2 m/s, the critical horizontal tailplane is 10625.7 N.

The critical horizontal tailplane load is, therefore, 10625.7 N.

Figure C.9: The Flight Loads report; example page from Chapter 11.

Chapter 15. Powerplant Loads

This chapter illustrates all the evaluated engine loads, in accordance with Paragraph 6.1.1 Engine Torque of ASTM F3116/F3116M. Each engine mount and its supporting structure must be designed for the effects of:

1. A limit engine torque corresponding to takeoff power and, if applicable, propeller speed acting simultaneously with 75% of the limit loads from flight condition A of 4.4.4.
2. The limit engine torque as specified in 6.1.3 acting simultaneously with the limit loads from flight condition A of 4.4.4

In addition to previous requirements, Paragraph 6.1.1.3 Engine Torque of ASTM F3116/F3116M illustrates further requirements for turbo-propeller aircraft: For turbo-propeller installations, in addition to the conditions specified in 6.1.1.1 and 6.1.1.2, a limit engine torque corresponding to takeoff power and propeller speed, multiplied by a factor accounting for propeller control system malfunction, including quick feathering, acting simultaneously with 1g level flight loads. In the absence of a rational an alysis, a factor of 1.6 must be used.

Finally, turbine engine installations are addressed in Paragraph 6.1.2 Engine Torque of ASTM F3116/F3116M: For turbine engine installations, the engine mounts and supporting structure must be designed to withstand each of the following:

1. A limit engine torque load imposed by sudden engine stoppage due to malfunction or structural failure (such as compressor jamming).
2. A limit engine torque load imposed by the maximum acceleration of the engine.

The limit engine torque to be considered under 6.1.1 must be obtained by multiplying the mean torque for maximum continuous power by a factor determined as follows:

1. 1.25 for turbo-propeller installations.
2. For four-stroke engines:
3. 1.33 for engines with five or more cylinders.
4. 2, 3, 4, or 8 for engines with four, three, two, or one cylinders, respectively.
5. For two-stroke engines:
6. 2 for engines with three or more cylinders.
7. 3 or 6, for engines with two or one cylinders respectively.

Table 15.1. Summary of propeller data.

Prop. mass (kg)	Prop. weight (N)	Prop. Diameter (m)	Prop. Polar mom. of inertia (kg * m^2)
24	235.357	2.1	1.78
32.9	322.636	2.083	1.763
30	294.197	1.95	1.8

Table 15.1 summarizes all the propeller data used in all the subsequent calculations.

Section 15.1 addresses the mean and limit torque per single engine. Section 15.2 illustrates the calculation concerning loads on the powerplant structure due to its inertia. Section 15.3 shows results

Figure C.10: The Flight Loads report; example page from Chapter 15.

concerning the sideload calculations. Finally, Section 15.4 summarizes and describes gyroscopic and aerodynamic loads acting on the powerplant structure.

15.1. Engine torque

The engine takeoff power is 279.64 kW at 2575 rpm. The rotational speed of the propeller is 2575/1 = 2575 rpm. The maximum continuous power is 279.64 kW per single engine. The limit torque acts simultaneously with the 75% of the inertia limit load. The mean engine torque at max continuous power is 9902.9 N*m. Using a factor of 1.33 for a five or more cylinder engine, the limit torque will be 13171 N*m which acts simultaneously with the 100% of the inertia limit load.

$$\mathrm{MT}_{\mathrm{continuous}} = P_{\mathrm{continuous}} \cdot \frac{1000}{\frac{2\pi \cdot \mathrm{RPM}_{\mathrm{prop}}}{60}} = 279.64 \cdot \frac{1000}{\frac{2 \cdot 3.14 \cdot 2575}{60}} = 9902.9 \ \mathrm{N} * \mathrm{m}$$

$$\mathrm{LT}_{\mathrm{continuous}} = \mathrm{MF}_{\mathrm{ASTM}} \cdot \mathrm{MT}_{\mathrm{continuous}} = 1.33 \cdot 9902.9 = 13171 \ \mathrm{N} * \mathrm{m}$$

Where:

- $\mathrm{MT}_{\mathrm{takeoff}}$ = mean torque at takeoff power (N · m); and
- $\mathrm{LT}_{\mathrm{takeoff}}$ = limit torque at takeoff power (N · m); and
- $\mathrm{MT}_{\mathrm{continuous}}$ = mean torque at max continuous power (N · m); and
- $\mathrm{LT}_{\mathrm{continuous}}$ = limit torque at max continuous power (N · m).
- $\mathrm{MF}_{\mathrm{ASTM}}$ = multiplying factor (Non dim.).

Table 15.2. Summary of data used for the Max Continuous calculations.

P (kW)	Prop. rot. speed (rpm)	Mean Torque (N*m)	Limit Torque (N*m)
279.637	2575	9902.87	13170.8

Table 15.2 summarizes all the numerical values involved in the mean and limit engine torques calculations at maximum continuous power.

$$\mathrm{MT}_{\mathrm{takeoff}} = P_{\mathrm{takeoff}} \cdot \frac{1000}{\frac{2\pi \cdot \mathrm{RPM}_{\mathrm{prop}}}{60}} = 279.64 \cdot \frac{1000}{\frac{2 \cdot 3.14 \cdot 2575}{60}} = 9902.9 \ \mathrm{N} * \mathrm{m}$$

$$\mathrm{LT}_{\mathrm{takeoff}} = \mathrm{MF}_{\mathrm{ASTM}} \cdot \mathrm{MT}_{\mathrm{takeoff}} = 1.33 \cdot 9902.9 = 13171 \ \mathrm{N} * \mathrm{m}$$

Table 15.3. Summary of data used for the takeoff maneuver.

P (kW)	Prop. rot. speed (rpm)	Mean Torque (N*m)	Limit Torque (N*m)
279.637	2575	9902.87	13170.8

Table 15.3 summarizes all the numerical values involved in the mean and limit engine torques calculations at takeoff power.

Point A of the Maneuvering and Gust Flight Envelope corresponds to the Maneuvering Design Airspeed VA, equal to 71.97 m/S, and to a limit load factor equal to 3.425. The additional load applicable will be evaluated as an inertial load, in a conservative fashion:

$$\%\ \mathrm{Load}_{\mathrm{Point\ A}} = n_A \cdot \mathrm{ASTM}_{\mathrm{coeff.}} \cdot \mathrm{MTOW} = 3.425 \cdot 0.75 \cdot 36088 = 92701.8 \ \mathrm{N}$$

Figure C.11: The Flight Loads report; example page from Chapter 15.

Appendix D
Additional information concerning the Certification documents

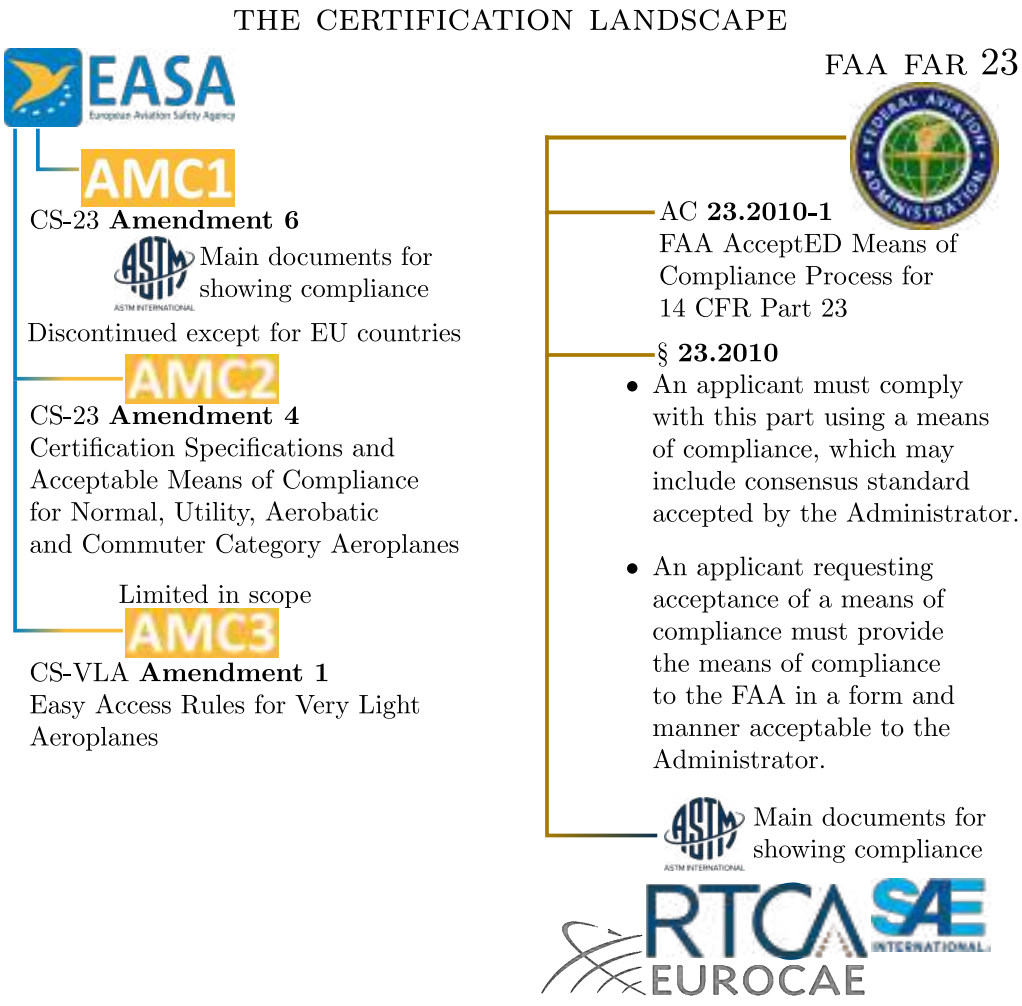

Figure D.1: A schematic representation of the present certification landscape.

This Appendix presents additional information and considerations about the present certification landscape in aeronautics. Figure D.1 describes the complexity that applicants must face when selecting the reference certification basis applicable to their new design. The process described above is often managed through a series of documents, the distribution of which across the design and manufacturing organization seeking certification is labor-intensive. When a document is revised, the updated version must be communicated promptly to all relevant departments. Without careful control, multiple versions of the same document can circulate simultaneously, increasing the risk of inconsistency and error. More broadly, a document-based approach can be inefficient and time-consuming: retrieving information is often cumbersome, and the volume of certification documentation can become substantial by the end of the certification program [236, 237].

The description provided in this Section applies to many certification processes, including those for chemical and nuclear plants [238, 239, 240, 241], large vehicles (e.g., ships and trains), specialized industrial vehicles, space launch systems [242], and aircraft. In aviation, certification is overseen by authorities such as the *Federal Aviation Administration* (FAA) in the United States [243] and the *European Union Aviation Safety Agency* (EASA) in the European Union [244, 245]. Other authorities or procurement institutions oversee the certification of dual-use or specialized platforms [246]. The aviation community regularly develops, revises, and maintains a set of high-level airworthiness requirements supported by extensive consensus standards issued by organizations including the *European Organization for Civil Aviation Equipment* (EUROCAE) [247], the *Society of Automotive Engineers* (SAE) International [248], the *Radio Technical Commission for Aeronautics* (RTCA) [249], and ASTM International (formerly the *American Society for Testing and Materials*), among others [250]. Usually, comprehensive agreements among different regulators are signed, ensuring and enforcing strong cooperation and collaboration on accepting standards applicable to a variety of aeronautical products [251, 252].

Within the European Union, EASA publishes *Certification Specifications* (CS) that apply to different aircraft categories and aeronautical products [253]. For example:

- EASA CS-23 establishes high-level requirements for general aviation aircraft in the normal category [254]. Since CS-23, Amendment 5, EASA has accepted ASTM International standards as *Acceptable Means of Compliance* (AMC), consistent with the FAA's *Advisory Circulars* (AC) and guidance material [255, 256] that recognize consensus standards for certification. In practice, CS-23 defines the applicability and limitations (e.g., seating configurations from 0 to 19, a maximum takeoff weight not exceeding 8618 kg (19 000 lb), relevant speed limitations, and whether aerobatic approval is sought) and identifies acceptable substantiation methods through the documents referenced under each paragraph [257]. Applicants may select an AMC consistent with their certification strategy and, when appropriate, propose alternative means of compliance through engagement with the competent authority. Earlier CS-23 amendments may still be accepted as AMCs for certification activities within the European Union. For instance, CS-23, Amendment 4, includes specific

means of compliance and does not explicitly rely on consensus standards; however, this approach is expected to be discontinued.

- EASA CS-25 establishes high-level requirements and associated AMCs for large, turbine-powered commercial aeroplanes (commonly referred to as transport-category aircraft) seeking certification in the European Union [258]. In practice, these aircraft typically exceed 9000 kg maximum takeoff weight, which is a commonly used industry delimiter rather than a defining CS-25 criterion [259]. The most recent major revision is Amendment 28. Applicants retain flexibility in demonstrating compliance with performance-based requirements, including those related to structural integrity [260], onboard systems reliability and redundancy, and other certification topics. In coordination with FAA practices [261], EASA accepts multiple standards and substantiation methods, which provide applicants with several compliance pathways.

The FAA issues and amends parallel performance-based regulations under the *Federal Aviation Regulations* (FAR), including Part 23 for general aviation aircraft and Part 25 for transport-category aircraft [262, 263]. For applicants, the associated effort can be substantial, particularly during the identification, selection, and acceptance of a minimum set of applicable certification and substantiation documents. This selection is not limited to established consensus standards and often requires additional guidance material and supporting evidence. As a result, domain specialists and certification personnel are typically engaged in an iterative process that can be time-consuming.

In the European Union, EASA manages this process by first establishing the appropriate *Level of Involvement* (LoI) for the applicant [264, 265, 266, 267]. The LoI is determined by considering factors such as [268]:

- The applicant's experience and technical capability in developing similar aeronautical products, including prior experience with novel design features and their operational implications.
- The applicant's demonstrated competence in the use of advanced engineering tools and methods, such as computational fluid dynamics (CFD), finite element methods (FEM), and computer-aided design and manufacturing (CAD/CAM).
- The applicant's experience in planning and conducting ground and laboratory tests (e.g., static strength, fatigue, deformation and stress measurements, and wind-tunnel testing).
- The facilities available to the applicant for design, manufacturing, testing, and quality assurance.
- The applicant's financial capacity to sustain the certification program.
- The quality assurance processes implemented by the applicant.

In operational terms, the LoI defines the set of compliance demonstration activities and data (CDA and CDD) that EASA retains for verification during certification, as well as the depth of that verification [269, 270]. A higher LoI generally implies a larger volume of documentation to be prepared, issued, and maintained over time to support verification and validation of compliance [271, 272]. The applicant's proposed LoI is incorporated into the certification program and summarizes the certification plan, including the analyses performed for each Compliance Demonstration Item (CDI), together with the applicable criteria and agreed assumptions [273]. EASA, for approved design organizations, uses relevant data to evaluate applicants' expected behavior and recorded performance on previous certification processes, and may accept the plan as submitted or request revisions and additional substantiation. It also records deviations from the agreed course of action and notifies the applicant accordingly [274]. Establishing the LoI typically requires iterative exchanges between the applicant and the authority: applicants often seek to limit the LoI to manage cost and schedule, whereas EASA must ensure that the certification basis and supporting evidence provide an adequate level of safety and compliance. In practice, EASA commonly applies a four-tier risk classification. At risk level 1, EASA has limited direct involvement in substantiation activities; at risk level 4, EASA personnel are more actively engaged, including detailed review of compliance data, close assessment of test results, witnessing of selected tests, and frequent audits in coordination with the applicant's staff. In some certification activities, EASA involvement is not subject to negotiation. Examples include:

- The approval, publication, and subsequent revision of the *Aircraft Flight Manual* (AFM) [275, 276].
- The assessment of handling qualities, performance, and structural failures following accidents or incidents. Such cases may require additional substantiation, including flight tests or simulator-based evaluations, and may result in new airworthiness limitations [277, 278, 279].

EASA CS-23 airworthiness requirements call for the determination of the structural loads that the aircraft is expected to encounter in service, accounting for the intended operating environment and the anticipated life cycle [280]. In particular, an applicant seeking a new type certificate must demonstrate compliance with Subpart C (Structure) by defining the structural design envelope, which establishes the boundaries of design and operating conditions associated with structural loads, strength and durability, and aeroelastic behavior. The relevant parameters include, but are not limited to, the following:

- Design airspeeds used to evaluate maneuvering and gust limit load factors. These requirements establish operational speed boundaries while providing margins against loss of control in adverse maneuvering conditions and in turbulent atmospheres.
- Aerodynamic loads expected during normal flight.
- Operating parameters such as center-of-gravity position, consistent with allowable mass distributions.

- Control inputs associated with pilot actions during normal flight.
- Load redistribution caused by deflection of aerodynamic surfaces that can significantly alter the distribution of external and internal loads.

The certification program should therefore identify appropriate means of compliance for evaluating these and other parameters and for demonstrating the structural safety and integrity of the proposed aircraft design [134] as early as possible into the design loop [281]. Applicants may propose and agree with EASA on alternative approaches for demonstrating compliance [282, 283]. However, EASA CS-23 explicitly references ASTM International F3116/F3116M, *Standard Specification for Design Loads and Conditions* [230], which provides established methods for addressing the relevant high-level requirements. With specific reference to Subpart C, ASTM International also publishes additional standards that may be used to demonstrate compliance with CS-23 requirements:

- F3396/F3396M, *Standard Practice for Aircraft Simplified Loads Criteria*, which is limited to aircraft with a stalling speed not exceeding 45 kn. Its applicability is therefore restricted to small, simple, conventional aircraft configurations [233].
- F3331-18, *Standard Practice for Aircraft Water Loads*, which provides methods and substantiation provisions for seaplanes [234].
- F2245-23, *Standard Specification for Design and Performance of a Light Sport Airplane*, which also includes substantiation provisions relevant to the structural requirements of light-sport aircraft [235].

In practice, applying these provisions offers a structured substantiation pathway and supports the demonstration of compliance for individual certification items with appropriate safety margins [284].

Applicants should recognize that *Acceptable Means of Compliance* (AMCs) are intended to reduce certification burden by providing substantiation methods that are presumed to satisfy the CS-23 airworthiness requirements. An AMC represents an accepted certification practice, but it is not the only means by which a high-level requirement may be met. Accordingly, individual certification items may be discussed with EASA, which may accept alternative methodologies, subject to an adequate substantiation plan. Such discussions typically consider the applicant's prior experience, including their demonstrated capability in the use of engineering tools and software and in conducting flight and ground tests.

When a proposed design includes novel or unproven features [285], the applicant may need to propose and justify alternative means of compliance if the published AMCs do not adequately address the design's characteristics [286, 287]. Conversely, when the applicant elects to use a consensus standard as an AMC, the applicant must demonstrate that the standard's provisions are applicable to the product and certification basis, align with the scope and objectives of the type-certificate program, and satisfy the intent of the corresponding high-level requirements [288]. When existing airworthiness requirements

do not provide sufficiently specific criteria for a novel design feature, the applicant and EASA may need to agree on *special conditions* [289], either by applying previously issued conditions or by defining new conditions in consultation with EASA technical specialists [290]. For example, special conditions may be required for aircraft employing electric propulsion or for electrical systems whose integration has a material effect on structural design and substantiation.

As discussed above, establishing an appropriate certification basis and developing a coherent certification program—including a complete set of compliance demonstration items—can be highly complex and typically requires months of intense work and expert effort [291]. In practice, applicants may rely on established certification practices and widely used substantiation pathways rather than tailoring the most efficient certification basis to a specific aircraft architecture, because a detailed assessment of the applicable requirements and consensus standards can be time-consuming and resource-intensive. These constraints, together with the scale and administrative burden of managing large certification document repositories, can discourage organizations from introducing novel design features, adopting emerging technologies, or implementing improved certification methods. The challenges are often more pronounced for small manufacturers, which typically operate with limited resources and reduced organizational capacity to absorb additional workload or process changes. At the same time, new designs should incorporate lessons learned from previous certification and operational experience [292, 293, 294, 295], particularly with respect to controllability and handling qualities, so as to strengthen safety-oriented features, promote more predictable behavior, and reduce the likelihood of accidents during critical phases of flight [296, 297, 298, 299, 300, 301].

Over time, these tendencies may reduce the attractiveness and marketability of aeronautical products and, in turn, weaken the economic position of the sector [302]. Market demand continues to evolve and increasingly emphasizes improved utility and fuel efficiency [303, 304]. Improvements in operations and flight management techniques can provide some economic relief in the industry [305, 306]. In the general aviation segment, customers often prioritize greater capacity and operational utility while seeking low fuel consumption on routes shorter than 200 nm ($\approx$ 370 km), with less emphasis on maximum performance [307].

Over the past decade, interest in new aircraft configurations appears to have increased, even though new models typically command higher purchase prices than comparable aircraft available on the second-hand market [308]. At the same time, the pace of new product introductions has not consistently matched this interest, and the limited availability of new designs has been cited as a concern within the sector [309, 310], with a focus on the final operational costs of these new designs. Certification burden is frequently identified as one of several contributing factors.

Notable exceptions exist, including manufacturers such as Pipistrel. In parallel, many small companies and start-ups are investigating unconventional aircraft architectures that incorporate technologies such as distributed electric propulsion (DEP), alternative fuels, and new operational concepts associated with *Urban Air Mobility* (UAM) [311, 312, 313, 314]. However, these concepts have not yet been deployed at scale in the market,

and two factors are frequently cited [315]:

- The high costs associated with certification and compliance demonstration [316, 317].
- The need for validated methods and evidence frameworks to assess an equivalent level of safety for the intended operating conditions of these new designs [318].

Although experts generally emphasize the importance of a thorough flight-test program to expand the operational envelope of new aircraft designs in a gradual and controlled manner, reducing certification burden can also have a substantial effect on program feasibility. More efficient certification processes can support timely market deployment while maintaining, and potentially improving, the equivalent level of safety of the aeronautical products [319, 320].

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